

Exact Sampling from Neural Networks: Depth, Attention, and Decoding

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Abstract

One random output can require much less work than numerical evaluation. We study exact neural sampling, counting input probes, random bits, arithmetic, and every precision refinement.

For dense tanh networks with absolute row sums at most one, weight preprocessing permits polylogarithmic expected sampling work at logarithmic depth. With zero biases, the sharp query scale is the square root of depth at every fixed first-layer rank, and also at arbitrary rank when a column envelope is bounded. The unrestricted gap between square-root and linear query cost remains open.

Prefill changes attention. We construct an appendable cache index and a complete exact decoder with positive RMSNorm stabilizers. Its expected work is polynomial in model size and $\log T$ when each layer has a common affine key space of fixed rank and the context has length T . The decoder’s polynomial degree grows with rank. These bounds give no useful speed prediction at the large ranks of ordinary models. A completely fixed full-rank model still has a formal polylogarithmic context bound, with potentially enormous constants.

The decoder maintains finite Taylor moments, prepares precision in the background, and charges rare complete refinements by their probability. Its cost includes neural key and value creation. Without a context index, standard scaling can force linear positional search; arbitrary indexed queries face a conditional barrier in growing dimension. We also prove exact normalization factories and a matching law for an explicit additive-residual family. A checkpoint section gives the certificates and cost formulas needed to apply each positive result.

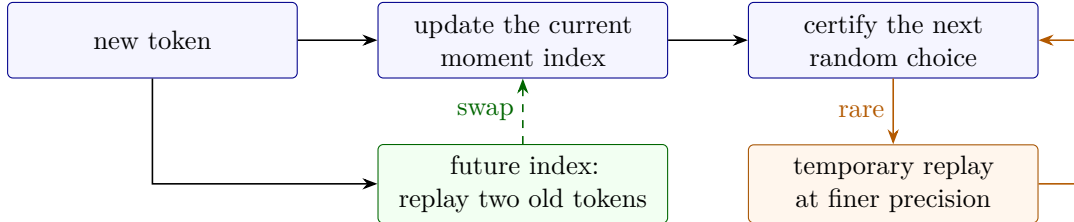


Figure 1: Maintaining an exact decoder. A routine step evaluates the new token in the current index and replays at most two old tokens in a future index. The future index takes over when it catches up. If the random choice is too close to a probability boundary, a temporary finer replay resolves it. All three computations use the same discrete token history.

1 Introduction

A sampler can reveal only the part of a computation needed for one random decision. Nonlinearity makes this difficult: independently randomizing activations generally changes the final distribution. Bernoulli factories provide exact transformations of unknown coin biases, but recursive composition can multiply their expected costs at every layer. Precision can create a second source of large work on rare branches. We control both effects.

The paper follows three preprocessing interfaces. First, weights are known and a fresh input arrives. Second, prefill may build an index over keys and values before a fresh query. Third, a decoder must construct and append the next keys and values itself. A lower bound in the first interface need not survive the second; a static index in the second leaves maintenance to be proved in the third.

We state the results in the order of these interfaces, for width n , depth D , and context length T . With a fresh input, tanh networks with row norms at most one have polylogarithmic expected sampling work at logarithmic depth, and every fixed larger row norm forces polynomial work on some networks (Theorem 3.2). Some zero-bias networks with row norms at most one need $\Omega(\sqrt{D})$ probes. Corrected Bernstein operators attain this order at every fixed first-layer rank (Theorem 3.3), and an empirical-mean construction attains it at arbitrary rank when the column envelope $C_{\text{col}} = \sum_j \max_i |w_{1,ij}|$ is bounded (Theorem A.69). Without a context index, attention can force linear positional search (Theorem 5.1). Under the randomized Orthogonal Vectors Hypothesis (OVH), polynomial prefill does not remove this barrier for arbitrary queries in growing dimension (Theorem 6.1). For keys of fixed affine rank r , an appendable index does remove it (Theorem 6.2). A complete decoder maintains Taylor moments and certified intervals, and background replay raises precision before it is needed. Its conditional expected work per token is $\text{poly}_r(n, \log T)$ at standard scaling with polynomially bounded model parameters, including all cache work (Theorem 7.1). For an explicit pre-norm additive family, we determine the query exponent on its entire input cube, with a matching bit bound (Theorem 8.3). Table 2 lists the cost scales. Section 10 credits the feature representations and interval-sampling principles used in the construction.

The results have two main limitations. The decoder bound needs a key-space dimension fixed independently of model width. Its degree increases with rank, and the feature count can become prohibitive even at modest ranks, so we obtain no practical runtime guarantee for large-rank checkpoints; Section 9 makes this quantitative. For zero-bias networks with row norms at most one and unrestricted first-layer rank, the gap between $\Omega(\sqrt{D})$ and $O(D)$ probes remains open (Section 11).

Here n is width, D depth, T current context length, T_0 prompt length, V vocabulary size, and b a bound on parameter encoding lengths. The common bit scale is

$$B = 16 + b + \lceil \log_2(n + D + T + V + 2) \rceil. \quad (1.1)$$

Irrelevant size parameters are omitted. Additional finite interface encodings are included as specified below. Table 1 lists the symbols used in more than one section.

Symbol	Meaning	Symbol	Meaning
n, D	Residual width and depth	T, T_0, V	Current and prompt lengths; vocabulary size
b, B	Parameter encoding length and total bit scale	s	Maximum absolute row sum
β, A	Attention temperature and score radius	r	A fixed rank bound
ε, λ	Normalization stabilizer and certified energy floor	H, N_m	Moment coefficient bound and feature count
Q	Number of distinct input probes		

Table 1: Notation used in more than one section. Each interface supplies the bounds used in its theorem. The precise bit scale B is defined in (1.1).

2 Models and exact computation

2.1 Networks, signs, and input probes

The scalar network has width $n \geq 2$ and depth $D \geq 1$:

$$h_{0,j} = x_j \in \{-1, 1\}, \quad h_{\ell,i} = \tanh \left(b_{\ell,i} + \sum_{j=1}^n w_{\ell,ij} h_{\ell-1,j} \right). \quad (2.1)$$

The parameters are dyadic, $|b_{\ell,i}| \leq 1$, and $\sum_j |w_{\ell,ij}| \leq s$. We reserve β for attention temperature and use $b_{\ell,i}$ for neural biases; the unindexed b denotes encoding length. The target bit has probability $(1 + h_{D,1})/2$ of being one. Equivalently, the target sign $Y \in \{-1, 1\}$ has mean $h_{D,1}$.

A *sign interface of radius $R > 0$* for a real value u returns independent signs of mean u/R , with the promise $|u| \leq R$. An identically zero value is handled by a fair sign. Independence is conditional on the fixed input. Two requests for the same neuron use fresh randomness. Multiplying independent signs multiplies their means. Choosing one of several interfaces with known probabilities takes the same convex combination of their means. These elementary identities are the basis of every construction here.

Weights and biases may be preprocessed before x is revealed. An input probe reads one specified coordinate of x . We charge its address, every fair random bit, integer arithmetic, and all precision refinement. A *source request* is an invocation of an unknown-mean sign or bit interface. A lower bound on distinct input probes applies even if repeated coordinates are memoized. Upper bounds may count repeated requests; this only makes them larger.

The bit scale in (1.1) charges finite scalar encodings. Known positive floors and nonrational algebraic temperatures require their specified encoding lengths as well. A factor B^4 is a conservative bound obtained with elementary fixed-point arithmetic. It is not a unit-cost real-arithmetic convention.

Problem	Preprocessing	Expected upper bound	Lower bound or boundary
One tanh row, norm s	Weights	$O(B^4[1 + \min(n, s + s^2)])$	Matching probe order
Critical tanh, arbitrary biases	$O(Dn^2B^6)$ weights	$O(B^4(D + 1)^{35/32})$	$\Omega(\sqrt{D})$ probes
Critical tanh, zero biases	Same budget	$O(B^4(D + 1))$	$\Omega(\sqrt{D})$ probes
Zero biases, fixed first rank r	Original budget at logarithmic depth	$O_r(B^{\max\{4, r+2\}}\sqrt{D + 1})$	Matching $\Theta_r(\sqrt{D})$ probes
Zero biases, $C_{\text{col}} \leq C_0$ fixed	Same budget; arbitrary rank	$O_{C_0}(B^{40}\sqrt{D + 1})$	Matching $\Theta(\sqrt{D})$ probes for $C_0 \geq 1$
Fresh-context binary attention	Weights only	$O(B^4[1 + \alpha \min(T, e^{2A})])$	Matching probe order
Arbitrary indexed attention query	Any fixed polynomial in cache size	$O(TnB^4)$ by a scan	No $T^{1-\epsilon} \text{poly}(n, B)$ under randomized OVH
Appendable fixed-rank cache	One pass over finite records	$\text{poly}_r(n, \log T)$ at $\beta = \sqrt{n}$	Fresh rank-one queries can require $\Omega(n)$
Full decoder, fixed key-map rank	One prompt pass; see Theorem 7.1	Same polynomial per token, including maintenance	Polynomial degree depends on rank
Additive pre-norm, relative floor	Weights and floor certificate	$O(B^4(2 + L)(1 + \log R)R^{m-1})$ for a stream sign	Sharp exponents remain open for the full class
Explicit carrier family	$O_s((D + 1)n^2B^6)$	$O_s(B^4 \min(R^{2\nu_\eta}, n))$	Matching probe order; η is the group fraction in Theorem 8.3

Table 2: Main cost scales. Query lower bounds also lower-bound bit work. Critical matching bounds require enough input coordinates and coefficient precision. Fixed-rank cache bounds assume bounded coordinates; Theorem 6.2 gives their full parameter dependence.

2.2 Attention and preprocessing interfaces

A causal attention row at position t has

$$a_j = \frac{\beta}{n} \langle q_t, k_j \rangle, \quad \pi_j = \frac{e^{a_j}}{\sum_{u \leq t} e^{a_u}}, \quad \text{Att}_t = \sum_{j \leq t} \pi_j v_j. \quad (2.2)$$

Queries, keys, and values are linear maps of the preceding residual stream, with each row having absolute weight sum at most s . If every stream coordinate has magnitude at most R , then $|a_j| \leq A := |\beta|s^2R^2$. Standard attention scaling is $\beta = \sqrt{n}$, because then $a_j = \langle q_t, k_j \rangle / \sqrt{n}$. A convex residual sublayer replaces h by $(1 - \alpha)h + \alpha \text{update}$.

Normalization uses the RMS map with stabilizer $\varepsilon \geq 0$,

$$\text{RMSNorm}_\varepsilon(h)_i = \frac{h_i}{\sqrt{\varepsilon + n^{-1} \sum_{j=1}^n h_j^2}}. \quad (2.3)$$

The denominator must be positive on the allowed inputs. Learned coordinate gains can be absorbed

into the following linear maps by using their resulting row norms. A pre-norm block normalizes before every additive update:

$$a_\ell = h_{\ell-1} + \alpha_{\ell,A} \text{Att}_\ell(\text{RMSNorm}_{\varepsilon_{\ell,A}}(h_{\ell-1})), \quad h_\ell = a_\ell + \alpha_{\ell,F} \tanh(b_\ell + W_\ell \text{RMSNorm}_{\varepsilon_{\ell,F}}(a_\ell)). \quad (2.4)$$

The step sizes $\alpha_{\ell,A}, \alpha_{\ell,F}$ are known nonnegative rationals. The common-step case has both equal to α .

In the *fresh-context model*, only weights are preprocessed. All context coordinates are revealed online through probes. In the *static-cache model*, the keys and values are known at prefill, and a data structure may depend on them. A query is then supplied online. The arbitrary-query lower bound uses this second model and allows every fixed polynomial prefill budget. In the *autoregressive model*, a fixed neural model and the sampled token history determine each new key and value. Online work includes their construction, insertion, and any refinement of the cache.

The positive index theorems use finitely represented cache entries. For example, an explicitly stored dyadic cache defines an exact mathematical target. An unrounded $\tanh$ computation generally produces transcendental values. Such a cache must retain a certified refinement procedure, whose cost is charged; it cannot be treated as a collection of exact real words. The finite-query-list theorem below allows certified computation of the complete candidate output laws, and therefore does not require storing their probabilities as exact reals.

Unless a theorem states otherwise, the final output is a softmax over V logits in $[-1, 1]$. The logit interfaces must be specified, such as designated coordinates or a known affine readout. A range promise alone gives no algorithm for accessing an otherwise unspecified function.

2.3 Exactness and stopping

For a known computable probability p , expose the binary digits of a fresh uniform random variable U . At precision t , compare a certified interval containing p with the current dyadic interval for U , and return the bit once they are disjoint. If both widths are at most 2^{-t} , failure at that stage requires U to lie in an interval of length at most 2^{2-t} around p . Thus the procedure halts almost surely, and stage work $C(B+t)^d$ gives expected work at most $2^{d+5}d!C(B+1)^d + O(B+1)$ (Lemmas A.18 and B.14). A precomputed table can supply these intervals. An unresolved comparison continues the same computation at higher precision, and the probability of this branch pays for its work (Lemma B.15). We bound this unconditional contribution, because conditioning on a rare miss can distort the remaining precision law.

Recursive work is charged predictably. If the decision to request the next child is made before that child's fresh random tape is drawn, a uniform expected child cost may be multiplied by the expected number of requests (Lemma A.10). Every recursive sampler here has finite depth.

3 From one row to a depth threshold

3.1 The exact law for one row

Consider $u = \tanh(c + \sum_i w_i x_i)$ with $|c| \leq 1$, and put $\rho = \sum_i |w_i|$. A weighted coordinate request returns a sign of mean $z = \rho^{-1} \sum_i w_i x_i$ when $\rho > 0$. This costs one input probe. The remaining problem is to turn independent z signs into a sign of mean $\tanh(c + \rho z)$.

Theorem 3.1 (One-row cost). *After weight preprocessing, a row of norm ρ has an exact sampler with at most $2\rho(1+\rho)$ expected input requests and $O(B^4[1+\min\{n, \rho+\rho^2\}])$ expected bit work, using full evaluation when it is cheaper. Conversely, whenever the averaging coefficients fit the dyadic grid, some zero-bias rows of norm ρ require $\Omega(\min\{n, \rho+\rho^2\})$ worst-input expected probes. The absolute constants and width padding are given in the proof.*

Theorem A.27 proves this statement, and Corollary A.25 gives the request bound for a row whose inputs are themselves supplied as signs.

The construction uses a positive mixture of odd majorities. If $M_{2k+1}(z)$ is the mean of the majority of $2k+1$ independent signs of mean z , then

$$M'_{2k+1}(z) = C_k(1 - z^2)^k, \quad C_k = (2k+1) \binom{2k}{k} 4^{-k}. \quad (3.1)$$

A positive expansion of the derivative of a normalized tanh in powers of $1 - z^2$ integrates to an exact mixture of these majority functions. A known gate restores the desired amplitude. A rational bias correction uses the tanh addition identity and an exact race. Its normalization cancels with the race acceptance, which keeps the expected number of children small.

For the upper bound, mix the bias into a known sign source before applying the scalar factory. With $a = |c| + \rho$, its expected arity is at most $2 \max\{a, a^2\}$, and only the fraction ρ/a of its requests reaches x ; the product is at most $2\rho(1+\rho)$. The lower bound combines a coupling argument when $\rho \leq 1$ with a stopped-score inequality when $\rho \geq 1$, and the transition from ρ to ρ^2 measures the difficulty of amplifying an unknown mean.

3.2 Why constant scalar cost is insufficient

For a whole network, each child sign can itself require a factory (Figure 3). A direct product of worst-case scalar costs gives an exponential bound even at unit row norm. The depth analysis instead maintains an input-independent interval

$$h_{\ell,i} \in c_{\ell,i} + [-r_\ell, r_\ell],$$

where the centers and outward-rounded radii are prepared from the weights. Sampling the normalized displacement costs close to one child as r_ℓ decreases. A final known mixture restores the center and radius.

At zero bias and unit norm, the ideal radius satisfies $r_{\ell+1} = \tanh r_\ell$, starting from $r_0 = 1$. The bounds $u^{-2} + 2/3 \leq \coth^2 u \leq 1 + u^{-2}$ give

$$\frac{1}{\ell+1} \leq r_\ell^2 \leq \frac{3}{2\ell+3} \quad (3.2)$$

(Lemma A.12 and Figure 2), so $r_\ell^2 = \Theta(1/\ell)$. Since $\sinh^2 u \geq u^2 + u^4/3$, also $\coth^2 u \leq u^{-2} + 2/3 + u^2/9$. Summing this over the recursion, with $r_i^2 \leq 3/(2i+3)$ and $\sum_{i < \ell} 1/(2i+3) \leq \sqrt{\ell}$, gives $r_\ell^{-2} \leq 1 + 2\ell/3 + \sqrt{\ell}/3$, and hence $r_\ell^2 \sim 3/(2\ell)$. Thus a per-layer cost $1 + cr_\ell^2 + O(r_\ell^4)$ accumulates polynomially in depth. For arbitrary biases, a concave potential charges the positions of normalized means within their intervals. Combining the first majority vote with the bias correction yields the exponent 35/32. The proof of Theorem A.32 gives the constants; its largest cutoff is an asymptotic proof device.

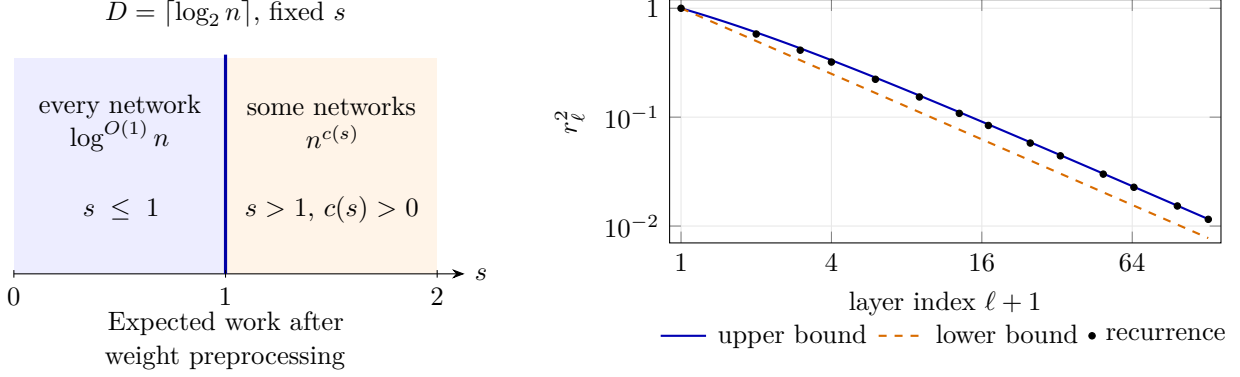


Figure 2: The row-norm threshold and the critical radius. Left: s bounds the absolute row sums. At logarithmic depth, the original preprocessing budget gives polylogarithmic sampling for $s \leq 1$; every fixed $s > 1$ permits a polynomial lower-bound family. The leading supercritical exponents remain unmatched. Right: both axes are logarithmic, displaying the proved bounds $1/(\ell + 1) \leq r_\ell^2 \leq 3/(2\ell + 3)$ across the depth range. The black dots are numerical values of $r_0 = 1, r_{\ell+1} = \tanh r_\ell$; no proof uses them.

Theorem 3.2 (Critical and supercritical networks). *For row norms at most one, weight preprocessing costs $O(Dn^2B^6)$ and storage costs $O(Dn^2B)$; exact output sampling uses $O(B^4(D+1)^{35/32})$ expected bit operations. With zero biases the upper bound is $O(B^4(D+1))$. For every $D \leq n/2$ on a sufficient dyadic grid, a zero-bias network requires $\Omega(\sqrt{D})$ expected input probes.*

Set $D = \lceil \log_2 n \rceil$ and $b = 20\lceil \log_2 n \rceil$. For every fixed $s \leq 1$, one uniform exact sampler has polylogarithmic expected online work under the original $O(Dn^2 \log^6 n)$ preprocessing budget. For every fixed $s > 1$, the worst-case online cost is $n^{\Theta(\min\{\log s, 1\})}$, with absolute constants in the exponent and fixed- s prefactors.

Theorem A.1 gives the preprocessing bounds, Theorems A.32 and A.34 the two upper bounds, Theorem A.2 the probe bound, and Corollary A.4 the threshold.

The supercritical statement determines the scale of the exponent. It does not determine its leading constant. Near $s = 1$, the best full-class upper rate proved here is $(51/16)(s-1) + O((s-1)^2)$ per layer; the lower rate is $2 \log s$, until it reaches the input-size cap. These rates leave a genuine gap. For convex residual tanh layers with step α , the fixed- $s > 1$ polylogarithmic threshold at polynomial depth is $\alpha D = O(\log \log n)$ (Corollary A.49), and at $s = 1$ the upper bound becomes $O(B^4(1 + \alpha D)^{35/32})$ (Theorem A.51).

Exact sampling is also separated from certified evaluation. In the critical mean-chain family, computing the activation to error $2^{-20\lceil \log_2 n \rceil}$ requires $\Omega(n)$ probes, while the exact sample uses polylogarithmic work (Proposition A.6); we do not claim that every individual network needs the $\Theta(Dn^2)$ operations of dense evaluation.

3.3 A sharp critical law at bounded input rank

If the first matrix has rank r , choose a bounded maximum-volume basis of r rows. Each basis row defines an unknown scalar mean of the input. The subsequent signed network is a known function of these few means. Preprocessing can therefore treat the composite function on a fixed-dimensional box.

Theorem 3.3 (Critical networks with fixed first-layer rank). *Fix $r \geq 1$. For a zero-bias, norm-one network whose first matrix has rank at most r , exact sampling uses $O_r(\sqrt{D+1})$ expected input probes and $O_r(B^{\max\{4,r+2\}}\sqrt{D+1})$ expected bit operations. At $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, preprocessing and storage fit $O_r(Dn^2 \log^6 n)$. If $n \geq D+1$ and $b \geq \lceil \log_2(D+1) \rceil$ on the coefficient grid $2^{-b}\mathbb{Z}$, the worst-case query order is $\Theta_r(\sqrt{D})$. The constants and online bit exponent depend on the fixed rank cap.*

Theorem A.67 proves this statement. The output magnitude is $O(D^{-1/2})$. Open a dyadic gate of that order and, on its open branch, sample a bounded normalized function. The corrected scalar Bernstein operator is

$$T_m f = B_m \left(f - \frac{p(1-p)}{2m} f'' \right),$$

where B_m averages a function over the empirical bias of m coin flips. Exact degree elevation makes successive dyadic corrections $O(m^{-2})$. In several coordinates, we telescope the product of the corrected operators one coordinate at a time, so coefficient work does not multiply across coordinates. Derivative bounds through order $2r+2$ give $O_r(D)$ sources on the open gate, hence the square-root bound. The coefficient oracle is compiled on $O_r(D^{r/2})$ Taylor patches and batched before each dense layer by rectangular matrix multiplication [26], within the original preprocessing budget (Appendix A.20); its constants may be enormous. Rare unresolved comparisons are refined directly and paid for by their probability. For first-layer rank at most three, after general-depth preprocessing, a rare full-input read also gives the finite-width law $\Theta(\min\{\sqrt{D}, n/\sqrt{D}\})$ in expected probes when $b \geq \lceil \log_2 \min\{n, D+1\} \rceil$ (Corollary A.66, with the lower bound from Theorem A.38).

A sufficient condition without a rank bound. The column envelope $C_{\text{col}} = \sum_j \max_i |w_{1,ij}|$ gives a second sharp subclass and a uniform upper bound. Put $K_{\text{col}} = \max\{1, C_{\text{col}}\}$. Theorem A.69 proves

$$\mathbb{E}Q = O(K_{\text{col}}^2 \sqrt{D+1}), \quad \mathbb{E}W = O(K_{\text{col}}^7 \sqrt{D+1} B^{40})$$

within the original preprocessing budget at logarithmic depth. For example, a Hadamard matrix divided by its width has full rank and column envelope one. A single weighted coordinate probe represents an entire bounded random vector with the required mean. A corrected empirical average and sampled fourth derivatives give the exact coin. The large bit exponent is a conservative implementation bound.

4 What an input lower bound measures

Fix a network and let $f(x)$ be its output sign mean. Supply its input coordinates independently with mean $t \in (-1, 1)$, and set $H(t) = \mathbb{E}_t f(X)$. An exact sampler returns a sign Y with this mean. Repeated input probes can be memoized without revealing new information. If Q is the number of distinct coordinates exposed, the transcript score is

$$Z = \sum_{\text{newly exposed } i} \frac{X_i - t}{1 - t^2}.$$

The finite input population gives exact identities

$$\mathbb{E}_t Z^2 = \frac{\mathbb{E}_t Q}{1 - t^2}, \quad H'(t) = \mathbb{E}_t (YZ). \quad (4.1)$$

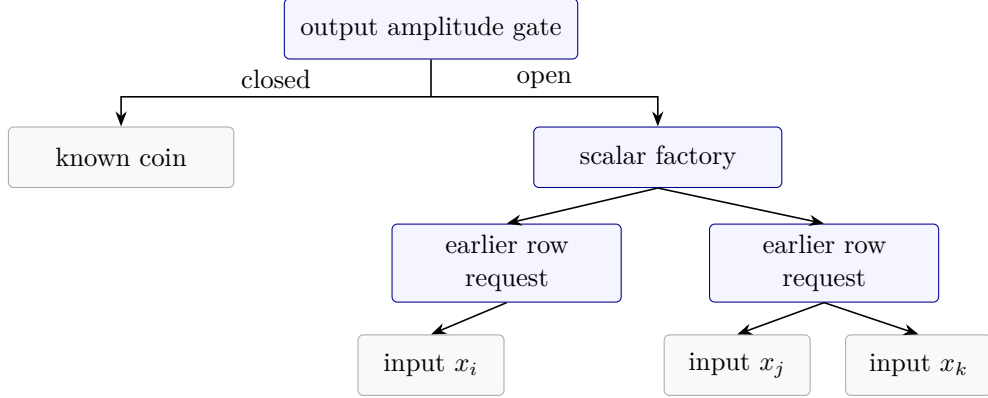


Figure 3: One possible request tree. Only reached calls incur work. A factory may request a random number of earlier signs; the two children shown here are one realization. Each call uses fresh randomness, so predictable charging bounds total work. At fixed input rank, the critical sampler uses the composite function through a precomputed coefficient oracle.

Each new input is independent of the preceding transcript. Consequently the score increments are orthogonal martingale differences. The output identity follows by differentiating the finite input distribution, followed by conditioning on the transcript. There is no assumption that the sampler’s internal arithmetic has an exponential tail.

Since $\mathbb{E}_t Z = 0$, also $H'(t) = \mathbb{E}_t[(Y - H(t))Z]$, and Cauchy–Schwarz gives, when $|H(t)| < 1$,

$$\mathbb{E}_t Q \geq \frac{(1 - t^2)H'(t)^2}{1 - H(t)^2}. \quad (4.2)$$

A second differentiation gives the curvature bound $|H''(t)| \leq 2\mathbb{E}_t Q/(1 - t^2)$. Lemma A.17 proves the identities (4.1), the bound (4.2) as (A.33), and the curvature bound. Lemma A.54 gives the first-score identity for unknown-coin factories from a finite first moment alone. The transcript-score method is prior work, with classical precedents in Wolfowitz’s sequential Cramér–Rao inequality and Mendo’s Bernoulli-factory lower bound [28, 43].

A complementary bound applies to two fixed inputs. Couple the algorithm’s random tapes on x and on $x^{(i)}$, which differs only at coordinate i . Until that coordinate is queried, the transcripts agree. Therefore

$$\mathbb{P}_x(i \text{ is queried}) \geq \text{TV}(\mathcal{L}_x(Y), \mathcal{L}_{x^{(i)}}(Y)), \quad (4.3)$$

as proved in Lemma B.7. Summing this inequality over many possible changes can force search through a context.

Preprocessing determines when these arguments apply. A weight-only index is identical on x and $x^{(i)}$, so the coupling remains valid. A cache index built from the context can differ before the query starts. The coupling then gives no online lower bound. For example, if all keys are identical, a prefill computation can store the average value. A lower bound obtained by hiding an exceptional value in a fresh context disappears under that preprocessing. The indexed lower bound in Section 6 uses a different reduction.

5 Attention before a cache index is available

Attention is already an average. The unknown part is its query-dependent sampling distribution. Suppose that $|a_j| \leq A$ and that an independent sign of mean a_j/A is available. Choose a position j uniformly, draw $N \sim \text{Pois}(2A)$, and test N independent bits of bias $(1 + a_j/A)/2$. Accept the position if every test succeeds; stop at the first failure otherwise. The acceptance probability is exactly

$$\mathbb{E} \left(\frac{1 + a_j/A}{2} \right)^N = \exp(a_j - A). \quad (5.1)$$

Conditioning on acceptance therefore gives the attention law π_j . A fresh value sign at the accepted position completes the sample.

This identity does not require evaluating a score. For $a_j = \beta \langle Qh_t, Kh_j \rangle / n$, choose a coordinate uniformly and multiply independent query and key signs. Each score test uses at most two predecessor requests. The stopped test count over the whole race is at most $e^{2A} - 1$ in expectation, so the complete attention sign uses at most

$$2e^{2A} - 1$$

expected predecessor requests and $O(B^4 e^{2A})$ auxiliary bit work (Theorem B.1). The Poisson variable is generated through finite-bit procedures; large numerical means are handled in blocks with stopping before unused blocks are drawn.

A softmax over logits $z_k \in [-1, 1]$ uses the same construction with $A = 1$. One proposal selects a token uniformly and accepts with probability $e^{z_k - 1}$. The expected proposal count is at most e^2 , and the expected total number of logit-sign requests is at most $e^2 - 1$ (Theorem B.6). Thus a bounded categorical output adds a constant expected number of logit requests and an address cost of order $\log V$.

Theorem 5.1 (Attention with bounded scores, and its search boundary). *If all needed score radii, value radii, and recursively supplied sign costs are bounded independently of T , one exact attention output sample has no linear dependence on T . The local expected factor is at most $2e^{2A} - 1$ source requests.*

For $\beta \geq 0$, there is a one-block binary family with row norms at most $s \geq 1$ and convex residual weight α whose optimal worst-context probe cost satisfies

$$Q_{\text{bin}}^* = \Theta(1 + \alpha \min\{T, e^{2\beta s^2}\}). \quad (5.2)$$

Its exact bit upper bound is $O(B^4[1 + \alpha \min\{T, e^{2\beta s^2}\}])$. These bounds allow weight-only preprocessing.

Theorem B.1 proves the first part and Theorem B.9 the second.

To see the lower mechanism, let a position contain a hidden sign x_j , give it score Ax_j , and use the same sign as its value. At the all-negative input, changing one position raises its attention mass to $e^{2A}/(T - 1 + e^{2A})$. The final token probability changes by a constant times this mass and α . Summing the coupling inequality (4.3) proves the lower bound. The full proof keeps both residual sublayers and the bounded softmax head.

Under standard scaling, $A = \sqrt{n} s^2$ in this family. Consequently $2\sqrt{n} s^2 \geq \log T$ can force $\Omega(\alpha T)$ expected context probes. A bounded coordinate range does not by itself give a context-length-independent guarantee at standard scaling. When n is fixed, this score bound is also fixed; a uniform statement in growing n must retain the dependence on A .

Several blocks compose through their expected source counts, as in Section 3. Shared centers can improve this product: the score term common to every position cancels, and the uncertainty radius shrinks through critical tanh layers.

6 Exact sampling after prefill

6.1 A lower bound for arbitrary fresh queries

The randomized Orthogonal Vectors Hypothesis, abbreviated randomized OVH, says that for every $\eta > 0$ there is a constant $c > 0$ such that no bounded-error randomized algorithm solves balanced Orthogonal Vectors on two sets of N vectors in $\{0, 1\}^{c \log N}$ in time $O(N^{2-\eta})$. Randomized SETH implies this hypothesis. We state the randomized assumption because the sampler and the prefill algorithm may both use randomness.

Haverbeck et al. [19, Theorem 4.1] already prove a constant-error attention lower bound after arbitrary polynomial KV preprocessing; the hardness principle and the dummy-key separation below are theirs.

Theorem 6.1 (A polynomial index does not remove the general barrier). *Assume randomized OVH. For every fixed polynomial prefill bound and every $\epsilon > 0$, there is no uniform algorithm which, for every static bounded cache and every fresh query, samples the bounded-logit final token in expected time $T^{1-\epsilon} \text{poly}(n, B)$. This holds at*

$$n = O(\log^2 T), \quad \beta = \sqrt{n},$$

with row norms at most one, sign or zero cache coordinates, and a binary output. It also holds if the returned law may have total-variation error at most $1/64$.

Theorem B.42 proves this statement, and Corollary B.43 realizes it with a row-norm-one block. An exact scan takes $O(TnB^4)$ expected bit work (Lemma B.16), so the conditional lower bound matches the power of T at these widths. The statement is a fine-grained conditional barrier. It is not an unconditional $\Omega(T)$ data-structure lower bound.

The reduction places candidate keys against dummy keys. A witness makes the total candidate weight large; without a witness it is small. A legal tanh layer and binary softmax preserve a constant probability gap. Coordinate repetition and Rubinstein’s gap-Hamming reduction [38] give the stated bounded-coordinate width, and bucketing one side of the hard instance handles every fixed polynomial prefill budget. Section 10 compares this bound with earlier attention and kernel-density lower bounds.

6.2 An index that accepts new keys

Fixed-dimensional key geometry permits both cheap queries and cheap maintenance. The next theorem assumes finite dyadic records. Section 7 includes their neural construction.

Theorem 6.2 (Appendable fixed-rank attention). *Fix $r \geq 1$ and dyadic coordinate bounds $K, Q_{\max} > 0$. Keys belong to $[-K, K]^n$, queries to $[-Q_{\max}, Q_{\max}]^n$, and every cache prefix has affine rank at most r . Records and queries have at most b bits per scalar. Let β be nonnegative*

dyadic or $\sqrt{n}$, and choose dyadic $\beta \leq \bar{\beta} \leq 2\beta$ when $\beta > 0$, and set $\bar{\beta} = 0$ when $\beta = 0$. Queries use a nonempty cache, $T \geq 1$. There is an exact index with

$$\begin{aligned} \text{worst-case append work} &= O_r((n + \log(T + 2))B^3), \\ \text{stored bits} &= O_r((T + 1)nB), \\ \text{expected query work} &= O_r((n + M_T + \log(T + 2))B^4), \end{aligned}$$

where

$$M_T \leq \min\{T, (r + 1)(5 + 32r\bar{\beta}Q_{\max}K)^r\}.$$

A query uses fewer than $17/8$ proposals in expectation and returns an index with the exact attention probabilities. The result holds conditionally on every adaptive legal history. At $\beta = 0$, uniform index sampling suffices.

Theorem B.17 proves this statement.

At standard scaling and fixed $r, K, Q_{\max}$, the query bound is $O_r((n + n^{r/2} + \log(T + 2))B^4)$. Appending a record never scans its predecessors. These bounds settle the maintenance question in the fixed-rank, bounded-coordinate regime.

To construct the index, choose a rational coordinate chart whose reconstruction coefficients have magnitude at most two. A row swap that increases a coefficient beyond two doubles a nonzero minor, so integer determinant bounds limit the number of swaps (Lemma B.18). On a grid where scores within a cell differ by at most $1/4$ for every legal query, cell representatives give a constant-factor rejection envelope (Lemma B.19). When an append increases the rank, we keep the earlier charts and their records and start a new chart for future records. Rank increases at most r times, so a query combines at most $r + 1$ chart indices and a large degenerate prefill causes no later migration. The keys may also lie within a small certified distance of a fixed subspace (Corollary B.21). For rank one, an ordered tree with blocks of sizes $1, 1, 2, 4, \dots$ and the classical monotone envelope [11] give polylogarithmic context cost at every numerical temperature (Theorem B.13 and Corollary B.20).

6.3 An implicit rank-one query and a worked example

Reading all n query coordinates is unnecessary in some regimes. Suppose the relevant scalar projection is available through independent coins of unknown bias p . A rank-one cache gives an attention-coordinate function

$$u(p) = \frac{\sum_{j=1}^T v_j e^{a_j p + c_j}}{\sum_{j=1}^T e^{a_j p + c_j}}, \quad |v_j| \leq 1, \quad |a_j| \leq \Lambda. \quad (6.1)$$

The parameters have rational encodings, or the slopes and offsets share the specified factor $1/\sqrt{n}$. The offsets need no magnitude bound. The scalar corrected Bernstein factory returns a sign of mean $u(p)/2$ in $O((1 + \Lambda^2)B^4)$ expected auxiliary work and $O(1 + \Lambda^2)$ source requests. Its certified table has polynomial prefill cost in T and the numerical slope bound Λ (Theorem B.24). This interface samples a value coordinate; an explicit attention index has a different output.

For raw sign queries, combine this construction with reading the whole query and using the explicit rank-one index. Consider dyadic rank-one caches with keys in $[-1, 1]^n$ and scalar values in $[-1, 1]$. For dyadic $\beta \geq 1$ and for $\beta = \sqrt{n}$, the worst-cache query law for the unit tanh and binary softmax head becomes

$$Q_{\text{rank } 1}^* = \Theta(\min\{n, \beta^2\}). \quad (6.2)$$

The upper bit cost is $O((1 + \min\{n, \beta^2\} + \log T)B^4)$ (Theorem B.26). The implicit table is built only when $12000\beta^2 < n$, which keeps preprocessing polynomial in T, n, b even when the encoded temperature is numerically very large.

Worked example. Let half the T keys equal $\mathbf{1}$ and half equal $-\mathbf{1}$, with corresponding values $+1$ and -1 . For a fresh sign query $x \in \{-1, 1\}^n$, put $M = n^{-1} \sum_i x_i$. Its attention value is exactly $u = \tanh(\beta M)$. A unit tanh update followed by logits $\pm \tanh u$ gives final sign mean

$$f(x) = \tanh(\tanh(\tanh(\beta M))).$$

Prefill stores two groups. Appending another key from these groups changes one group count; the index then handles any new query from those counts. Nevertheless, the score lower bound forces $\Omega(\min\{n, \beta^2\})$ fresh-query probes. At $\beta = \sqrt{n}$ this is $\Theta(n)$, with an upper bound $O(nB^4)$, for every even T . At fixed β , the expected probe count is constant (Theorem B.25). This example separates cache length from the amount of information needed from a new query.

6.4 A known finite query list

A second static positive case requires no rank bound. Suppose the permitted queries are a known list of size M , each specified online by its list identifier. If all V probabilities of one candidate can be enclosed to width 2^{-t} in $H(B + t)^d$ bit operations, with fixed $d \geq 2$ and $H \geq V$, then Theorem B.27 gives $O(MH[B + \log(VH + 2)]^d)$ prefill work and $O([B + \log(VH + 2)]^d)$ expected work per exact sample. The table leaves a residual mass of order $1/H$; a certified full calculation on that branch pays for itself in expectation. Such a table is specific to one cache; the decoder in the next section maintains one causal evaluator as the history and the required precision change.

7 A complete autoregressive decoder

This section states the decoder bound announced in the introduction and the ideas of its proof. The index of Section 6 assumes finite records. The decoder also constructs each new key and value with the neural model and charges that work.

Fix a dyadic embedding table with entries of magnitude at most s and a causal network of the pre-norm form (2.4). Use dyadic residual steps in $[0, 1]$; convex residual mixtures with coefficients in $[0, 1]$ are also allowed. Allow an affine attention output map and an affine vocabulary head. Parameters have at most b bits, row norms and biases are at most $s \geq 1$, the temperature is nonnegative dyadic or $\sqrt{n}$, and every normalization has positive stabilizer at least 2^{-b} . Each layer's fixed linear key map has rank at most r . Its keys at all positions must lie in one common affine space; separate rank bounds on position-dependent maps do not imply that promise.

Theorem 7.1 (Exact decoding with maintained neural keys). *For each fixed r , there is a constructive polynomial F_r such that a prompt of length $T_0 \geq 1$ can be prefilled in*

$$T_0 F_r(n, D, V, b, s, \beta, \log(T_0 + 2))$$

bit operations. Every subsequent token has conditional expected cost at most

$$F_r(n, D, V, b, s, \beta, \log(T + 2)),$$

given every previously sampled token history. This includes key and value creation, insertion, random bits, and all refinements. The output has the exact original autoregressive law. At $\beta = \sqrt{n}$, with D, V, b, s polynomially bounded in n , the per-token bound is $\text{poly}_r(n, \log T)$. The affine head's logit range is determined from the parameter bounds. The polynomial degree depends on r .

Theorem B.28 proves this statement for logits promised to lie in $[-1, 1]$, and Corollary B.32 removes that promise for the affine head.

In a bounded chart, write a score, up to a common offset, as $\langle t, z_j \rangle$, where $z_j \in [-1, 1]^r$ and $\|t\|_1 \leq H = \text{poly}_r(n, s, \beta)$. A degree- m Taylor expansion uses $\binom{m+r}{r}$ monomials. Store their sums and their value-weighted sums as exact integers over a common dyadic denominator. Each new record adds to these counters. Degree $m = O(H + p)$ yields certified attention accuracy 2^{-p} ; the average denominator is at least e^{-H} , so this includes the error of division. At $\beta = \sqrt{n}$, rational even and odd degree sums keep the specified square root exact.

Attention is Lipschitz from the maximum score error to the total probability error with constant two, independently of the number of keys. For positive RMS stabilizers, the logarithm of the Lipschitz constant is of order $b + \log n$. Consequently errors propagate through the D layers, with no linear factor in the number of positions. A precision of $p + O((D + 1)[b + \log(n + s + \beta + 2)] + \log(D + V + 2))$ suffices. Rounding the normalized stream before its exact key-map product preserves the promised rank. This detail prevents a finite approximation from silently changing the structural hypothesis.

Routine precision is a sufficiently large multiple of $\log T$. If a uniform draw is too close to a categorical boundary, replay the actual token history at higher precision until the sample is certified. An unresolved precision- p comparison has probability at most $4V2^{-p}$. Choosing routine error of order T^{-3}/V pays for a complete replay on that rare branch. A second routine index replays two old tokens after each append and catches up before the current precision expires. Ordinary maintenance uses at most three deterministic token evaluations per step. Figure 1 shows this maintenance loop.

For a basic tanh model with an affine head that is fixed in all parameters, the key ranks are fixed even when the key matrices have full rank. The explicit implementation gives $O_{\text{model}}(\log^{2r+12}(T+2))$ expected work per token, with model-dependent constants that can be enormous (Corollary B.34). Fixed coordinate-pair positional rotations also admit a $\log^{O_{\text{model}}(1)} T$ bound: certified trigonometric evaluation costs polynomial time in precision and $\log T$ (Corollary B.35). The same finite-evaluation argument covers specified GELU and SwiGLU updates using their polynomial range and Lipschitz bounds on bounded normalized inputs (Corollary B.33).

8 Normalization and additive residuals

8.1 Two exact interfaces for RMS normalization

Let $a \in [-R, R]^n$, let $\varepsilon \geq 0$, and consider $\text{RMSNorm}_\varepsilon(a)$ from (2.3). RMS normalization removes the mean-subtraction step of LayerNorm [46]. The Gaussian construction below also applies to regularized LayerNorm after centering.

Theorem 8.1 (Exact RMS normalization). *Suppose $R > 0$, $\varepsilon \geq 0$, and $\lambda > 0$ are known rational parameters, and λ certifies $\varepsilon + n^{-1} \sum_j a_j^2 \geq \lambda$ on every allowed input, and put $\kappa = \max\{1, R^2/\lambda\}$. Independent signs of mean a_j/R suffice for an exact sign interface for $\text{RMSNorm}_\varepsilon(a)_i$ at radius $2R/\sqrt{\lambda}$, using at most $O(1 + \kappa^2)$ expected source requests and $O(\kappa^2 B^4)$ auxiliary bit operations.*

If $\varepsilon > 0$, a second construction has radius $4\sqrt{n}$. With $\kappa_\varepsilon = R^2/\varepsilon$, it uses at most $1 + n\kappa_\varepsilon$ expected source requests and $O(n(1 + \kappa_\varepsilon)B^4)$ auxiliary work.

Theorems C.1 and C.9 prove the bounds for the two constructions. The first samples a coordinate and two independent signs, whose disagreement probability represents $(1 - n^{-1} \sum_j (a_j/R)^2)/2$. A linear Bernoulli factory uses the certified floor, and a negative-binomial expansion supplies the inverse square root. The second projects onto a random Gaussian direction; an error-function factory gives an arcsine transform of $a_i/\sqrt{\|a\|_2^2 + n\varepsilon}$, which a signed sine-series mixture inverts. Applied to signs of mean $(a_i - \bar{a})/(2R)$, with $\bar{a} = n^{-1} \sum_j a_j$, it also gives regularized LayerNorm at radius $4\sqrt{n}$ (Corollary C.10). Error-function factories and trigonometric stochastic constructions have prior precedents [14, 16], and Gaussian sign identities and inverse-power corrections occur in rounding theory [7]. The contribution here is the explicit normalization interface, its source and bit bounds, and its use inside the network recurrence.

A positive stabilizer alone gives no uniform small cost. For the scalar family

$$f_{\kappa,A}(z) = \frac{z}{A\sqrt{\kappa^{-1} + z^2}}, \quad A \geq 1,$$

the optimal uniform source count obeys

$$\frac{\sqrt{\kappa} + \kappa}{8A} \leq Q^*(\kappa, A) \leq \frac{1 + 2\kappa}{A\sqrt{1 + \kappa^{-1}}} \leq \frac{3(\sqrt{\kappa} + \kappa)}{A} \quad (8.1)$$

(Theorem C.14). At fixed A and large κ the cost is of order κ , and increasing the radius to order $\sqrt{\kappa}$ reduces it to order $\sqrt{\kappa}$. The lower bound needs only a uniform first moment.

8.2 The additive recursion

Consider sublayers

$$H_j = H_{j-1} + F_j(H_{j-1}), \quad |H_0| \leq 1, \quad |F_j(H)_{t,i}| \leq U_j,$$

where $U_j \geq 0$ is a known bound. Set $R_j = 1 + \sum_{k \leq j} U_k$. The quantity B includes the full finite encodings of these radius prefixes. Suppose an exact sign of mean $F_j(H)_{t,i}/U_j$ uses at most m_j expected predecessor signs of mean $H_{t',i}/R_{j-1}$ and at most $C_0 B^4 L_j$ expected local bit work. Zero updates are omitted. These are uniform conditional bounds, so they remain valid when the factory is called adaptively.

Theorem 8.2 (An additive cost product). *Put*

$$P_j = \prod_{k=1}^j \left(1 + \frac{U_k m_k}{R_{k-1}}\right).$$

There is an exact sampler for $H_{J,t,i}/R_J$ with at most P_J/R_J expected input leaves and at most

$$C_0 B^4 \frac{P_J}{R_J} \left(2 + \sum_{j=1}^J \frac{U_j(1 + L_j)}{P_j}\right) \quad (8.2)$$

expected bit work. If $m_j \leq m$ for $m \geq 1$ and $L_j \leq L$, these bounds become R_J^{m-1} leaves and

$$C_0 B^4 (2 + L)(1 + \log R_J) R_J^{m-1}$$

bit operations.

Theorem C.15 proves this statement. To sample H_j/R_j , choose the input contribution with probability $1/R_j$ or update $k \leq j$ with probability U_k/R_j . A prefix table makes this choice directly, so the algorithm never traverses a long sequence of identity branches; an update's children restart the procedure at an earlier level. A linear recurrence for the bit work and the inequality $\sum_j U_j/R_j \leq \log R_J$ show that auxiliary arithmetic adds only a logarithm in the residual radius.

8.3 Pre-norm attention and feed-forward updates

Consider the pre-norm blocks (2.4). A *relative energy certificate* with constant $\gamma > 0$ asserts, at every requested normalization and for every allowed input and position,

$$\varepsilon_j + \frac{1}{n} \sum_i H_{j-1,t,i}^2 \geq \gamma R_{j-1}^2. \quad (8.3)$$

This is an additional hypothesis on the network and its allowed inputs. It does not follow from a fixed stabilizer: with stabilizer three, networks of logarithmic depth can force linear query cost (Theorem C.18 and Corollary C.19).

With the power of two $G \in [2/\sqrt{\gamma}, 4/\sqrt{\gamma})$ and $K_\gamma = 1 + 228 \max\{1, \gamma^{-2}\}$ of Theorem C.16, set

$$E = e^{2|\beta|s^2G^2}, \quad \rho = sG, \quad m = \max\{1, (2E - 1)K_\gamma, 2\rho(1 + \rho)K_\gamma\}. \quad (8.4)$$

These constants bound the number of predecessor requests for normalized attention and tanh updates. Like m , the local arithmetic factor L of that theorem depends only on γ, s, β . The unscaled update radii are sG and 1, and a step size multiplies an update's radius without changing its sign factory. Theorem 8.2 then gives polynomial dependence on $1 + \tau$, where τ is total step size, and only address dependence on T (Corollary C.17). For a common step α and D blocks, one may use $R = 1 + \alpha D(sG + 1)$. A final RMSNorm satisfying the same relative certificate, followed by a bounded affine softmax head, adds a constant factor when its row norm is fixed (Theorem C.16). A readout of the unnormalized stream costs at most a further factor R^2 (Theorem B.10).

The polynomial scale is necessary in a uniform theorem. There are tanh families with a full-input relative certificate and a fixed bounded readout that require

$$\Omega_s(\min\{(1 + \tau)^{2(s-1)}, n\}) \quad (8.5)$$

probes for every fixed legal gain $s > 1$. This holds for arbitrary prescribed dyadic steps in $[0, 1]$, with τ their sum (Theorem C.25, whose exponent is larger, and Corollary C.23). Thus a polylogarithmic accumulated step size is sufficient and necessary for a uniform polylogarithmic guarantee in these fixed-parameter classes (Corollary C.24). Convex supercritical residuals instead have the $\log \log n$ step-mass scale. The next theorem sharpens the additive exponent for one explicit family.

Theorem 8.3 (A sharp additive family). *Put $c = \tanh 1$ and fix a legal dyadic $s > c/\sqrt{2}$. For $n \geq 4$, let N be the largest power of two at most $n/2$ and put $\eta = N/n$. For prescribed dyadic steps $a_0, \dots, a_{D-1} \in [0, 1]$, write*

$$R_j = 1 + \sum_{k < j} a_k, \quad R = R_D, \quad \nu_\eta = \frac{s}{c\sqrt{\eta}} - 1.$$

There is an explicit additive tanh family with stabilizer 3, row norms at most s , and a fixed bounded binary head whose optimal worst-input query cost is

$$\Theta_s(\min\{R^{2\nu_\eta}, n\}).$$

After $O_s((D+1)n^2B^6)$ weight preprocessing, exact sampling uses $O_s(B^4 \min\{R^{2\nu_\eta}, n\})$ expected bit operations. The entire input sign cube satisfies a relative energy floor stronger than $R_j^2/12$.

Theorem C.26 proves the query law and Theorem C.27 the bit bound. One group carries the unknown feature mean. An equal-sized group moves deterministically at speed c , driven by a bias of -1 . After division by the residual radius, the feature gain approaches $s/(c\sqrt{\eta})$, and the transcript score gives the lower bound. For the upper bound, two unknown means and one observed sign determine the head, and a Taylor oracle on a complex neighborhood of radius proportional to $R^{-\nu_\eta}$ supplies its corrected Bernstein factory. Rounding is amplified only through factors $1+O_s(a_j/R_{j+1})$, so the required precision grows logarithmically in depth. The full additive class may have a different worst-case exponent.

The additive product also covers common feed-forward activations. Write $\text{GELU}(v) = v\Phi(v)$ and $\text{SwiGLU}(u, v) = uv\sigma(v)$, where Φ is the standard normal distribution function and $\sigma(v) = (1 + e^{-v})^{-1}$. For bounded affine inputs, error-function, product, and logistic factories give exact interfaces with polynomial source cost in their radius (Lemma C.28). For every fixed legal dyadic gain $s \geq 2$, affine GELU and SwiGLU lower constructions with an explicit relative floor and a final RMSNorm also satisfy (8.5) (Theorem C.31 and Corollary C.32). The identities $\text{GELU}(v) - \text{GELU}(-v) = v$ and $\text{SiLU}(v) - \text{SiLU}(-v) = v$, where $\text{SiLU}(v) = v\sigma(v)$, embed the required linear amplification using three hidden units.

The relative-floor product gives a sufficient condition for an architecture. Its constants can be large when row norms grow or the floor is small. In particular, the score factor in (8.4) grows exponentially in $\sqrt{n}$ at standard scaling. The decoder of Section 7 instead uses the prefilled cache and charges full width-dependent arithmetic at one position, while controlling its dependence on context length.

9 Quantities to check on a model

The positive results give sufficient conditions that can be checked against a specified model. Their current dependence on rank is severe. The maintained decoder is polynomial in width when its rank bound is fixed independently of width; both its feature count and the polynomial degree grow with that rank. These bounds currently supply no useful speed prediction at large key ranks.

For this section, n is residual width, d is one head’s key dimension, and a is its value dimension. Scores are $\beta\langle q, k_j \rangle/d$, with $\beta = \sqrt{d}$ at standard scaling. RMSNorm still averages over n coordinates. If its learned gain is g , use the effective following matrix $W^{\text{eff}} = W \text{diag}(g)$. Its row budgets are $s_i = \sum_j |W_{ij}g_j|$. These conventions match the inside-root stabilizer, learned gain, and head-dimension scaling of the reference implementation [29]. Corollary E.1 proves the rectangular-dimension extension used here. Table 3 lists the quantities to record.

Rank and an appendable cache. Without positional transforms, $b_K + \text{image}(W_K^{\text{eff}})$ is a common key space. With rotations, the containing space must include every rotated image and the differences of rotated biases. Separate ranks of the transformed matrices do not certify its dimension. Even a rank-one unrotated key image can acquire a full common span (Proposition E.3). The unrotated score rank $\text{rank}[(W_Q^{\text{eff}})^T W_K^{\text{eff}}; b_Q^T W_K^{\text{eff}}]$ can sometimes replace the key rank (Corollary E.2).

Quantity	What to record	Where it enters
Affine key rank r	An exact common containing space over allowed positions and histories; the full effective key matrix includes all learned updates.	Cell count $(5 + O(r\beta Q_{\max}K))^r$; moment count $\binom{m+r}{r}$.
Score and coordinate ranges	Certified query/key bounds $Q_{\max}, K$, value bound U , and score radius. Record empirical extrema separately.	The unindexed source factor is $2e^{2A_{\text{coin}}} - 1$; the moment degree uses the chart bound H .
Row budgets	Effective absolute row sums, biases, and weighted radii $\rho_i = \sum_j W_{ij} R_j$.	A tanh row with $ b_i \leq 1$ uses at most $2\rho_i(1 + \rho_i)$ predecessor requests (Corollary A.25).
Normalization floor	The inside-root ε , input radius R_{j-1} , and a certificate $\lambda_j \leq \varepsilon_j + \ H_{j-1}\ _2^2/n$.	$\kappa_j = \max\{1, R_{j-1}^2/\lambda_j\}$ for the floor factory; $(1 + \sqrt{n})/\sqrt{\lambda_j}$ in the precision guard.

Table 3: Quantities to record for a model and where they enter the bounds.

Formulas to evaluate. For supplied finite records with M_T occupied index cells, the index of Theorem 6.2 has append work $O_r((d+a+\log(T+2))B^3)$ and query work $O_r((d+M_T+\log(T+2))B^4)$, with $M_T \leq \min\{T, (r+1)(5 + 32r\beta Q_{\max}K)^r\}$. For the moment decoder with a chart bound H (Appendix E) and attention precision p , one valid degree and its exact feature count are $m = \lceil 8(p + 3H + 7) \rceil$ and $N_m = \binom{m+r}{r}$, with $(a+1)N_m$ integer counters per head and precision index. At illustrative degree $m = 256$, N_m is 2,862,209 for rank three and exceeds $5 \cdot 10^{14}$ for rank eight. For additive updates with radius increments u_j (including step sizes), source bounds m_j , local arithmetic factors L_j , an initial radius R_0 , $R_j = R_0 + \sum_{k \leq j} u_k$, and $P_j = \prod_{k \leq j} (1 + u_k m_k / R_{k-1})$, Corollary E.7 gives

$$W_{\text{add}} \leq C_0 B^4 \frac{P_J}{R_J} \left(2R_0 + \sum_{j=1}^J \frac{u_j(1 + L_j)}{P_j} \right).$$

Appendix E also defines the coin radius A_{coin} , gives the stage-by-stage precision guard (Lemma E.6), and treats floors that cover only some histories. For the critical tanh subclasses, the exact first-layer rank and the column envelope enter Theorems 3.3 and A.69.

Interpreting measurements. Exact weight ranks and row budgets can be certified from the stored parameters. A score maximum or energy minimum measured on selected texts describes those texts. A universal claim requires a bound for all allowed histories, or a certificate checked as records enter the cache. Verification that reads the current context is charged to online work. The accompanying worksheet reports construction sizes and symbolic bit-work bounds. Timing, energy, and speedup require an implementation and calibration at the same target law.

10 Relation to attention approximation and prior factories

A runtime comparison must match the output and preprocessing: an approximate attention vector, an attention index, and a final token impose different requirements.

KDEformer and HyperAttention give operator-norm approximations of the attention product that hold with high probability [17, 44]. Such a guarantee supplies no certified interval for every

output probability on every random run. MagicPIG samples through a prefill-built locality-sensitive hash index [10]; its implemented self-normalized estimate is generally biased. Appendix F states these three guarantees with their parameters. Randomly perturbed nearest-neighbor methods give exact categorical samples when a valid top- k oracle is available [33], and approximate retrieval then needs additional certification or an error allowance. Smooth-kernel data structures, including the quasi-Monte Carlo construction of Charikar, Kapralov, and Waingarten [9], offer another positive route for their specified kernel class. Their smoothness condition does not include the Gaussian kernel at arbitrary scale. These results therefore do not remove the constant-gap indexed lower bound proved here.

Alman and Song’s earlier high-precision theorem does not by itself give the constant-gap sampling conclusion of Theorem 6.1. They study approximation of a complete attention matrix-vector computation, with dimension, entry bounds, and inverse-polynomial accuracy linked together [3]. Kernel-density lower bounds also depend on the batch model and the requested accuracy [1, 2]. Estimating an inverse-polynomial probability gap by repeated samples can erase a desired exponent saving; the constant final-token gap avoids that loss.

Maintaining feature sums for causal attention is established work. Katharopoulos et al. [24] give recurrent prefix states for feature attention. Heinsen and Kozachkov [20] give the Taylor monomials, feature count, and accumulated states at arbitrary truncation order. Our decoder uses those representations with certified error, rank-preserving rounding, and a precision schedule that supports the exact original law through all layers. Exact decisions from finite probability approximations also have prior constructions. Han and Hoshi [18] study interval sampling, and Brassard, Devroye, and Gravel [6] give floored rejection envelopes and geometric refinement tails. The decoder combines these principles with a causal neural evaluator and background replay to prove a complete finite-bit maintenance bound. Dynamic attention maintenance [42] and dynamic KDE [27] support entry or point updates with numeric outputs, which differ from an appended neural record followed by one exact categorical sample.

Stochastic computing represents values by event probabilities [8, 15, 16]. Bernoulli-factory theory supplies analytic constructions, positive series, multivariate races, and composition principles [13, 21, 28, 32, 34]. We attribute these tools where they enter and make no novelty claim for scalar factory representations. Koskela, Karvonen, Łatuszyński, and Spanò [25] give a scalar debiased Bernoulli construction based on smooth Bernstein approximants. Its outcome-independent sample count is useful for composition, but does not supply the dense multivariate coefficient oracle needed by the unrestricted critical problem. Randomized telescoping and antithetic empirical means are also established tools [4, 5, 37]. The column-envelope theorem makes its correction estimates bounded on every branch. Its common derivative majorants, exact quadratic and quartic tilts, and retained observations then give a sign with the required law and the stated input cost.

11 Scope, remaining questions, and evidence

11.1 What follows for decoding

Before prefill, standard score scaling can force linear positional search. After prefill, fixed-rank neural key maps with a positive normalization stabilizer give $\text{poly}_r(n, \log T)$ conditional expected work per token. The OVH obstruction concerns ranks and widths that grow with the context. Pre-norm additive residuals, RMSNorm, and SwiGLU occur in widely used architectures [39, 41, 46],

but no theorem here establishes favorable rank or numerical constants for a trained checkpoint. Section 9 lists the quantities to certify on a model and shows how fast the moment dimension grows with rank.

11.2 Open problems

Main open problem: the unrestricted critical exponent. For zero-bias networks with every row norm at most one, the worst-case query gap remains $\Omega(\sqrt{D})$ versus $O(D)$. Does an exact sampler use $O(\sqrt{D+1})$ expected input probes at arbitrary rank, under the same weight preprocessing budget and with polylogarithmic bit overhead? Even $\sqrt{D+1} \log^{O(1)}(D+2)$ probes would close the depth exponent. No factor $\log^{O(1)} n$ is hidden in this query claim: at logarithmic depth such a factor could absorb the whole gap. The fixed-rank theorem has constants and a bit exponent that depend on the rank, and the column envelope $\sum_j \max_i |w_{1,ij}|$ may be n , so these two certificates leave the general critical exponent unsettled.

The final analysis sharpens the limitations of two proof routes. The integer stopped-score obstruction also holds for bounded real estimators, so delayed Bernoulli rounding does not reduce the leading local amplification cost (Corollary A.55). A common derivative majorant shows that the sum of one-coordinate coupling witnesses is at most one. Even grouped, matrix-valued second-order score witnesses are at most $\sqrt{6D}$ (Propositions A.58 and A.59). These are ceilings on the stated lower-bound methods. They neither prove a faster global sampler nor rule out a stronger lower bound by another method. Near unit row norm, the supercritical gap remains between $2 \log s$ and $(51/16)(s-1) + O((s-1)^2)$ per layer.

The full relative-floor class has an open additive exponent, as does the carrier family’s boundary $\nu_\eta = 0$. A general instance law also faces computational obstacles. Let $Q^*(W)$ be the optimal worst-input expected number of distinct probes for a network W . At fixed $s > 1$, a deterministic polynomial-time algorithm that approximates $1 + Q^*(W)$ within a polylogarithmic factor on width- n networks of depth $\lceil \log_2 n \rceil$ with row norms at most s and precision $b = 20 \lceil \log_2 n \rceil$ would imply $P = NP$. A randomized one that succeeds with probability at least $2/3$ would imply $NP \subseteq BPP$ (Theorem D.6).

For attention, useful remaining questions concern deletions, weaker geometric promises, and algorithms whose rank dependence is useful at model dimensions. Sharp normalization laws under input-dependent energy certificates also remain open.

11.3 Proofs, numerical checks, and Lean

The appendices prove the theorem claims and identify imported results and conditional hypotheses. The source includes numerical and exact-arithmetic checks of coefficient identities, degree elevation, score embeddings, and residual recurrences. These finite checks provide supplementary evidence. No sampler benchmark or trained-model evaluation was performed.

A Lean 4 formalization with Mathlib in the repository <https://github.com/SamMausberg/exact-sampling-networks> checks the probability laws, exactness identities and inequalities of the proofs, using only the standard axioms (Appendix G.2).

AI disclosure. GPT-6 Astra Pro and Claude models assisted with the mathematical proofs and the Lean formalization.

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Guide to the proofs

The scalar factories establish exact local laws. Radius and potential bounds control their composition. Finite transcript identities give input lower bounds. The attention appendix separates a fresh context, an explicit cache, and a maintained neural decoder. Normalization then supplies the additive product and its sharp carrier example. Each formal proof begins with its main idea.

Neural biases are $b_{\ell,i}$; the unindexed b bounds their encoding length. The scalar factory uses $c_{\ell,i}$ for interval centers. Attention temperature is β . Appendix A uses the bit scale $\mathcal{B} = b + \lceil \log_2(n + D + 2) \rceil + 16$ of (A.2), which is at most B of (1.1); its bounds therefore also hold with B . When a table contains M groups or candidate queries, B also includes $\lceil \log_2(M + 1) \rceil$. For rational residual schedules, it includes the actual lengths of the stored radius prefixes. A common dyadic grid keeps those lengths within the displayed parameter and address scale; unrelated rational denominators may require a larger B .

Main result	Detailed statement and proof
One tanh row	Theorem A.27
Critical 35/32 exponent	Theorem A.32
Zero-bias linear upper	Theorem A.34
Query lower bounds and the norm threshold	Theorem A.2; Corollaries A.4 and A.5
Every fixed first-layer rank	Theorem A.67; elementary low-rank implementations in Theorems A.61 and A.65
Bounded first-layer column envelope	Theorem A.69; arbitrary algebraic rank
Convex residual scale	Theorem A.51 and Corollary A.49
Attention race and positional search	Theorems B.1 and B.9
Indexed lower bound at standard scaling	Theorem B.42 and Corollary B.43
Appendable affine-rank index	Theorem B.17
Full exact decoder	Theorem B.28 and Corollary B.32
Checkpoint certificates and dimensions	Appendix E
Limits of second-order score bounds	Propositions A.58 and A.59
Integer stopped-score obstruction	Corollary A.55
Fixed model and positional rotations	Corollaries B.34 and B.35
Implicit rank-one query	Theorem B.24
Worst-case rank-one query law	Theorem B.26
Finite candidate list	Theorem B.27
RMS normalization	Theorems C.1 and C.9
Scalar RMS law at a fixed amplitude	Theorem C.14
Additive product	Theorem C.15
Sharp carrier-family cost	Theorems C.26 and C.27
Relative-floor lower families	Theorem C.25; Corollary C.23
GELU and SwiGLU lower families	Theorem C.31 and Corollary C.32
Instance complexity	Appendix D

A known scalar coin always denotes finite certified computation. An unknown-mean coin denotes a supplied independent source interface. The proofs below do not depend on the Lean formalization described in Appendix G.2.

A Tanh networks: scalar factories, depth, and lower bounds

A.1 The tanh network model and quantitative bounds

Let $n \geq 2$, $D \geq 1$, and $b \geq 1$. A network has activations

$$h_{0,j} = x_j \in \{-1, 1\}, \quad h_{\ell,i} = \tanh\left(b_{\ell,i} + \sum_{j=1}^n w_{\ell,ij} h_{\ell-1,j}\right), \quad 1 \leq \ell \leq D. \quad (\text{A.1})$$

All parameters are multiples of 2^{-b} , and

$$|b_{\ell,i}| \leq 1, \quad S := \max_{\ell,i} \sum_j |w_{\ell,ij}| \leq s.$$

The exact row norm S is computed from the weights. The target bit has probability $(1 + h_{D,1}(x))/2$.

Preprocessing, probes, and online work are charged as in Section 2, including every precision refinement. A *recursive call* is an invocation of the normalized sampler for a neuron or input, including a repeated invocation of the same coordinate. The root is counted as well. Write

$$\mathcal{B} = b + \lceil \log_2(n + D + 2) \rceil + 16. \quad (\text{A.2})$$

This is at most the bit scale B of (1.1), so every bound stated here with $\mathcal{B}$ also holds with B . A plain B in a lemma of this appendix bounds the encoding lengths of the parameters and addresses that the lemma uses.

Theorem A.1 (Exact sampling upper bound). *One uniform algorithm samples the target exactly for every network with $S \leq 2$, and terminates almost surely. Deterministic preprocessing uses $O(Dn^2 \mathcal{B}^6)$ bit operations and $O(Dn^2 \mathcal{B})$ stored bits. Uniformly over all contexts, with $\delta = (S - 1)_+$,*

$$\mathbb{E} N_{\text{calls}} \leq 80(D + 1)^2 e^{51\delta D}. \quad (\text{A.3})$$

Expected online bit work is

$$O(\mathcal{B}^4 (D + 1)^2 e^{51\delta D}). \quad (\text{A.4})$$

The constants in these O -bounds are absolute.

The explicit power 2 in (A.3) counts recursive calls. Equation (A.4) also charges the manipulation of finite parameters and refinable scalar probabilities. A stable fixed-precision coefficient recurrence proves the fourth-power overhead in Section A.7.2. For $S \leq 1$, Theorem A.32 lowers the power of depth in (A.3) and (A.4) to $35/32$, and Theorem A.34 gives a linear bound when all biases are zero.

Our lower bounds hold even with unlimited preprocessing. Set

$$m = 2^{\lceil \log_2 n \rceil}, \quad F_{a,0}(z) = z, \quad F_{a,d+1}(z) = \tanh(a F_{a,d}(z)).$$

If a/m is on the parameter grid, a legal network realizes $F_{a,D}(m^{-1} \sum_{j=1}^m x_j)$: the first distinguished row has weights a/m , and each later distinguished row has one incoming weight a . All other neurons and all biases are zero.

Theorem A.2 (Universal query lower bounds). *For this network and $a > 0$, every exact sampler satisfies*

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{a^{2D}}{1 + (3m - 2)a^{2D} H_D / m^2}, \quad H_D = \sum_{j=0}^{D-1} a^{-2j}, \quad (\text{A.5})$$

where Q counts distinct input probes. If $a > 1$, then

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{a^2 - 1}{4a^2 - 1} \min\{a^{2D}, m\}. \quad (\text{A.6})$$

At $a = 1$ and $1 \leq D \leq m$,

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{\sqrt{D}}{100}. \quad (\text{A.7})$$

These bounds apply to every almost surely terminating exact algorithm.

Remark A.3 (Other averaging widths). The proof in Section A.6 uses m only through the output $F_{a,D}(m^{-1} \sum_{j=1}^m x_j)$ and the condition $D \leq m$ in (A.7). Theorem A.2 therefore holds for every integer $1 \leq m \leq n$, for example every power of two $m \leq n$, whenever a/m is on the parameter grid. It also holds for every legal network with this output, whatever its other weights.

Specialize now to $L = \lceil \log_2 n \rceil$, $D = L$, and $b = 20L$. Let $\mathcal{T}_n(s)$ be the infimum of worst-network, worst-context expected online bit work among uniform exact samplers with $O(Ln^2 \log^6 n)$ preprocessing time and storage. The norm s is fixed as n grows.

Corollary A.4 (The norm threshold). *For every fixed $0 \leq s \leq 1$, $\mathcal{T}_n(s) = \log^{O(1)} n$. For every fixed $s > 1$,*

$$\mathcal{T}_n(s) = n^{\Theta(\min\{\log s, 1\})}. \quad (\text{A.8})$$

Explicitly, for $1 < s \leq 2$,

$$\Omega_s\left(n^{\min\{2 \log_2 s, 1\}}\right) \leq \mathcal{T}_n(s) \leq \tilde{O}_s\left(\min\{n^2, n^{51(s-1)/\log 2}\}\right). \quad (\text{A.9})$$

For larger fixed s , $\tilde{O}_s(n^2)$ is an upper bound.

Corollary A.5 (The transition window). *Let $s_n = 1 + \delta_n$, $0 \leq \delta_n \leq 1$, with $D = L$ and $b = 20L$. If $\delta_n L = O(\log L)$, the upper bound is polylogarithmic. If $\delta_n L / \log L \rightarrow \infty$, no polylogarithmic expected online bound is possible. Quantitatively, if $\delta L \geq 4 \log(30L)$, then for all sufficiently large L some legal network and context require at least $e^{\delta L/4}$ expected input probes.*

Proposition A.6 (Separation from numerical evaluation). *For the critical mean-chain network $F_{1,L}(m^{-1} \sum_j x_j)$, every always-correct algorithm that returns its activation with absolute error at most 2^{-20L} must probe all $m \geq n/2$ relevant inputs. Even an evaluator with success probability at least $2/3$ on every context requires $m/3$ probes in expectation under uniform inputs. Exact output sampling has polylogarithmic expected online bit work.*

A.2 Intervals and the recursive sampler

A.2.1 Dyadic centers and radii

Choose

$$P = b + 4 \lceil \log_2(D+1) \rceil + 8, \quad \varepsilon = 2^{-P}. \quad (\text{A.10})$$

Use the reference gain $g = \max\{1, S\}$, computed exactly from the weights. Set $r_0 = 1$ and $c_{0,j} = 0$. Obtain a dyadic q^* with $|q^* - \tanh(gr_{\ell-1})| \leq \varepsilon/16$, and set $r_\ell = q^* + 5\varepsilon/2$. Then

$$\tanh(gr_{\ell-1}) + 2\varepsilon \leq r_\ell \leq \tanh(gr_{\ell-1}) + 3\varepsilon. \quad (\text{A.11})$$

A certified approximation followed by rational rounding supplies this choice without finding the exact floor of a transcendental number.

For each row store

$$\sigma = \sum_j |w_{\ell,ij}|, \quad a = b_{\ell,i} + \sum_j w_{\ell,ij} c_{\ell-1,j}, \quad \rho = \sigma r_{\ell-1}, \quad (\text{A.12})$$

together with prefix sums of the absolute integer weight numerators. Define the mathematical endpoints, midpoint, and image radius by

$$y_\pm = \tanh(a \pm \rho), \quad m = \frac{y_+ + y_-}{2}, \quad R = \frac{y_+ - y_-}{2}.$$

Choose a dyadic $c_{\ell,i} \in [-1, 1]$ with $|c_{\ell,i} - m| \leq \varepsilon$. Clamping a rounded center to $[-1, 1]$ preserves its error bound. The exact m, R are scalar functions of stored rationals. Online refinement does not recompute the row sum.

Lemma A.7 (Interval invariant). *For every context, $|h_{\ell,i} - c_{\ell,i}| \leq r_\ell$. With $t = \tanh a$, $q = \tanh \rho$,*

$$m = \frac{t(1 - q^2)}{1 - t^2 q^2}, \quad R = \frac{q(1 - t^2)}{1 - t^2 q^2} \leq q. \quad (\text{A.13})$$

Proof idea. Propagate the endpoints of each row interval through monotone tanh. The row norm controls its width, while a small rounding allowance keeps the stored center and radius valid at the next layer.

Proof. The preactivation differs from a by at most $\sum_j |w_{\ell,ij}|r_{\ell-1} = \rho$, so monotonicity puts the output in $[y_-, y_+]$. The tanh addition formula proves (A.13). Its denominator is positive, and $1 - t^2 \leq 1 - t^2 q^2$. Hence

$$R + |m - c_{\ell,i}| \leq \tanh(\sigma r_{\ell-1}) + \varepsilon \leq r_\ell.$$

Induction starts with $|x_j| = 1$. □

A.2.2 Normalized signs and the top output

The internal sampler $S_{\ell,i}$ returns a sign of mean $v_{\ell,i} = (h_{\ell,i} - c_{\ell,i})/r_\ell$. At layer zero it returns x_i . For a nonzero row, choose j with probability $|w_{\ell,ij}|/\sigma$, request a fresh $S_{\ell-1,j}$, and multiply by $\text{sgn}(w_{\ell,ij})$. This source has mean

$$z = \sum_j \frac{w_{\ell,ij}}{\sigma} v_{\ell-1,j}, \quad b_{\ell,i} + \sum_j w_{\ell,ij} h_{\ell-1,j} = a + \rho z.$$

The next two sections construct a centered factory of mean

$$G = \frac{\tanh(a + \rho z) - m}{R}. \quad (\text{A.14})$$

Put $\alpha = R/r_\ell$, $\kappa = (m - c_{\ell,i})/r_\ell$. The internal sampler chooses the centered factory with probability α , and constant signs with probabilities

$$\mathbb{P}(+1 \text{ constant}) = \frac{1 - \alpha + \kappa}{2}, \quad \mathbb{P}(-1 \text{ constant}) = \frac{1 - \alpha - \kappa}{2}. \quad (\text{A.15})$$

The interval invariant gives $\alpha + |\kappa| \leq 1$. The resulting mean is $\alpha G + \kappa = v_{\ell,i}$.

At the top use the actual midpoint and image radius:

$$\mathbb{P}(\text{centered factory}) = R, \quad \mathbb{P}(+1 \text{ constant}) = \frac{1 - R + m}{2}, \quad \mathbb{P}(-1 \text{ constant}) = \frac{1 - R - m}{2}. \quad (\text{A.16})$$

These probabilities are valid since $|m| + R \leq 1$. The sign has mean $h_{D,1}$; return $(1 + Y)/2$. A zero row uses its known scalar probability directly.

A.3 A positive majority factory

For $\rho > 0$, let $G_\rho(z) = \tanh(\rho z)/\tanh \rho$. The unit-norm construction only needs $\rho \leq 2$, but the representation also supports the large-norm extension. If $M_k(z)$ is the mean of majority on $2k + 1$ independent signs of mean z , differentiating its binomial sum gives

$$M'_k(z) = C_k(1 - z^2)^k, \quad C_k = \frac{(2k + 1)\binom{2k}{k}}{4^k} \geq 1, \quad M_k(0) = 0, \quad M_k(1) = 1. \quad (\text{A.17})$$

Lemma A.8 (Majority expansion). *Write*

$$F_\rho(v) = \text{sech}^2(\rho\sqrt{1-v}) = \sum_{k \geq 0} d_k v^k. \quad (\text{A.18})$$

The coefficients are nonnegative and sum to one. Moreover

$$\pi_k = \frac{\rho}{\tanh \rho} \frac{d_k}{C_k}$$

is a probability distribution and

$$G_\rho(z) = \sum_{k \geq 0} \pi_k M_k(z), \quad -1 \leq z \leq 1. \quad (\text{A.19})$$

Proof idea. The cosine product gives nonnegative coefficients for a generating function. Integrating the derivatives of odd-majority polynomials recovers $\tanh$; evaluation at the endpoint proves that the mixture weights sum to one.

Proof. The cosh product [35] gives

$$F_\rho(v) = \prod_{j \geq 0} \left(\frac{1 - \eta_j}{1 - \eta_j v} \right)^2, \quad \eta_j = \frac{\lambda_j}{1 + \lambda_j}, \quad \lambda_j = \frac{4\rho^2}{\pi^2(2j+1)^2}.$$

Each factor has nonnegative coefficients. The product converges locally uniformly before its first pole, and $\sum_j \eta_j / (1 - \eta_j) = \sum_j \lambda_j < \infty$. It is a probability generating function with $F_\rho(1) = 1$. For $z \in [0, 1]$, Tonelli's theorem and (A.17) give

$$\sum_k \pi_k M_k(z) = \int_0^z \frac{\rho}{\tanh \rho} F_\rho(1 - u^2) du = G_\rho(z).$$

At $z = 1$, this proves $\sum_k \pi_k = 1$. Oddness gives negative z . $\square$

Sample K from π , then request signs until one sign has occurred $K + 1$ times. This gives the full majority's output while sometimes using fewer than $2K + 1$ signs.

Lemma A.9 (Sequential majority cost). *Its expected source count satisfies*

$$A_{\text{seq}}(\rho, z) \leq 1 + \left(1 - \frac{z^2}{3}\right) \rho^2 + \rho^4. \quad (\text{A.20})$$

Proof idea. For each odd majority, sum the probabilities that a further vote is needed. An integral identity expresses this cost in the same nonnegative coefficients used by the factory and gives a uniform radius bound.

Proof. The identity $C_k^{-1} = \int_0^1 (1 - v^2)^k dv$ follows by integration by parts. The expected full-majority cost is therefore

$$A(\rho) = \frac{\rho}{q} \int_0^1 [F_\rho(1 - v^2) + 2(1 - v^2)F'_\rho(1 - v^2)] dv.$$

Put $g(v) = \text{sech}^2(\rho v)$. Integration by parts applied to $g(v) - 1 = O(v^2)$ gives

$$A(\rho) = \frac{\rho}{q} \left(1 + \int_0^1 \frac{\tanh^2(\rho v)}{v^2} dv \right) \leq \frac{\rho}{q} (1 + \rho^2). \quad (\text{A.21})$$

The singularity at zero is removable. For $K = 1$, sequential majority costs $5/2 - z^2/2$, saving $(1 + z^2)/2$ source requests. Also $\pi_1 = (2/3)\rho^2 \text{sech}^2 \rho$. Saving only this contribution gives

$$A_{\text{seq}} \leq \frac{\rho}{q} (1 + \rho^2) - \frac{1 + z^2}{3} \rho^2 \text{sech}^2 \rho.$$

Use $\text{sech}^2 \rho \geq 1 - \rho^2$ and

$$u \coth u \leq 1 + \frac{u^2}{3}, \quad u > 0. \quad (\text{A.22})$$

For the latter, $(3 + u^2) \sinh u - 3u \cosh u$ vanishes at zero and has derivative $u(u \cosh u - \sinh u) \geq 0$. Substitution proves

$$A_{\text{seq}} \leq 1 + \left(1 - \frac{z^2}{3}\right) \rho^2 + \frac{2 + z^2}{3} \rho^4,$$

which implies the claim. $\square$

A.4 Biases and exact stopping

Define $T_\theta(u) = (u + \theta)/(1 + \theta u)$. The addition formula implies

$$\tanh(a + \rho z) = m + R T_{tq}(G_\rho(z)). \quad (\text{A.23})$$

For $\theta \geq 0$, put $\lambda = (1 - \theta)/(1 + \theta)$. Request a source sign. Return on $+1$; on -1 , return that sign with probability λ , otherwise repeat. If the source's plus probability is p , a trial succeeds with $\sigma = p + (1 - p)\lambda$. The accepted plus probability is p/σ , with signed mean $T_\theta(2p - 1)$. For negative θ , complement the source and the output. Its sign is the sign of the stored rational a .

Lemma A.10 (Work of reached trials). *Suppose reaching trial j is determined before its fresh randomness. If its conditional expected nonnegative cost is at most B , then*

$$\mathbb{E} \sum_{j=1}^T C_j \leq B \sum_{j \geq 1} \mathbb{P}(T \geq j) = B \mathbb{E}T. \quad (\text{A.24})$$

For independent trials of success probability $\sigma > 0$, $\mathbb{E}T = 1/\sigma$.

Proof idea. A trial is reached before its own randomness is exposed. Its expected work can therefore be charged using the probability of reaching it, even when its duration and its returned value are correlated.

Proof. Condition on the history before each trial, sum over finite prefixes, and use monotone convergence. In the independent case $\mathbb{P}(T \geq j) = (1 - \sigma)^{j-1}$. A trial's cost may be correlated with its outcome. $\square$

Lemma A.11 (Joint majority and race cost). *A normalized call has expected children at most*

$$\frac{\tanh \rho}{r_\ell} \exp\left(\frac{5}{3}\rho^2 + \rho^4\right). \quad (\text{A.25})$$

A uniform layer bound is $B_\ell = \exp(\frac{5}{3}(gr_{\ell-1})^2 + (gr_{\ell-1})^4)$. The root's expected children are at most $r_D B_D$.

Proof idea. Multiply the mean cost of one reached majority by the mean number of race trials. The acceptance probability cancels the bias denominator, leaving a bound that depends on the row radius.

Proof. Assume $t \geq 0$, put $u = \tanh(\rho z)$, and combine the gate, the exact race mean, and Lemma A.10:

$$\mathbb{E}[\text{children}] = \frac{q}{r_\ell} A_{\text{seq}}(\rho, z) \frac{1 - t^2}{(1 - tq)(1 + tu)}. \quad (\text{A.26})$$

The last factor is

$$K = \frac{\cosh \rho \cosh(\rho z)}{\cosh(a - \rho) \cosh(a + \rho z)} \leq \frac{\cosh \rho \cosh(\rho z)}{\cosh^2(\rho(1 + z)/2)}.$$

The denominator is minimized at $a = \rho(1 - z)/2$, by the product-to-sum formula. Since $(\log \cosh)'' = \text{sech}^2 \leq 1$, its midpoint second-difference bound yields $\log K \leq \rho^2(1 - z)^2/4$. For negative t , replace z by $-z$. The arity is even in z , hence

$$\log(A_{\text{seq}}K) \leq \left(1 - \frac{z^2}{3} + \frac{(1 + |z|)^2}{4}\right) \rho^2 + \rho^4 \leq \frac{5}{3}\rho^2 + \rho^4.$$

The prefactor in (A.25) is at most one by (A.11). At the root the gate is R , retaining the factor r_D . $\square$

Almost sure termination follows by induction on the layer. The majority index is finite almost surely, lower-layer requests terminate almost surely, and the race has positive success probability. Known scalar coins terminate almost surely by Section A.7. This reasoning precedes the expected-work bound, so finiteness of the expectation is not assumed.

A.5 Radius decay and the recursion size

The radius lemmas below hold for any reference gain $0 \leq g \leq 2$; the uniform algorithm uses $g = \max\{1, S\}$. Let $R_0 = 1$, $R_{j+1} = \tanh(gR_j)$ be ideal reference radii used only in the analysis, and put $q_j = 3/(2j+3)$.

Lemma A.12 (Radius decay). *For $u > 0$,*

$$\coth^2 u \geq u^{-2} + \frac{2}{3}. \quad (\text{A.27})$$

If $g \leq 1$, $R_j^2 \leq q_j$. If $1 \leq g \leq 2$,

$$R_j^2 \leq \frac{3}{2}(1 - g^{-2}) + q_j. \quad (\text{A.28})$$

Also $R_j \leq g^j R_j^{(1)}$ when $g \geq 1$, where $R_j^{(1)}$ is the critical radius.

Proof idea. Use the reciprocal square of the radius as a potential. A coefficientwise hyperbolic-function inequality gives a fixed increase at criticality and a geometric recurrence above it.

Proof. Multiplication by $u^2 \sinh^2 u$ reduces (A.27) to $u^2 - \sinh^2 u + (u^2/3) \sinh^2 u \geq 0$. Write $\sinh^2 u = \sum_{j \geq 1} a_j u^{2j}$, where $a_j = 2^{2j-1}/(2j)!$. The coefficients of u^2, u^4 vanish. For $j \geq 3$, the coefficient is $a_{j-1}/3 - a_j \geq 0$, since $a_j/a_{j-1} = 4/[(2j)(2j-1)] \leq 1/3$. Absolute convergence justifies the calculation.

At $g = 1$ the reciprocal square telescopes to $R_j^{-2} \geq 1 + 2j/3$. Monotonicity handles $g < 1$. For $g \geq 1$, put $A = \frac{3}{2}(1 - g^{-2})$ and $\Phi_g(v) = g^2 v / (1 + \frac{2}{3}g^2 v)$. We have $R_{j+1}^2 \leq \Phi_g(R_j^2)$, and

$$\Phi_g(A + q) - A = \frac{q}{g^2(1 + \frac{2}{3}q)} \leq \frac{q}{1 + \frac{2}{3}q}.$$

Since $q_{j+1} = q_j / (1 + \frac{2}{3}q_j)$, induction proves (A.28). Concavity gives $\tanh(cu) \leq c \tanh u$ for $c \geq 1$; another induction proves the final assertion. $\square$

Lemma A.13 (Uniform rounding stability). *The stored radii lie in $(0, 1]$ and*

$$0 \leq r_j - R_j \leq 3j\varepsilon.$$

Replacing ideal radii by stored radii changes the logarithm of any offspring product by at most $29/200$. Moreover,

$$r_D \leq \frac{2}{\sqrt{D+1}} \quad (g \leq 1), \quad r_D \leq \frac{2e^{(g-1)D}}{\sqrt{D+1}} \quad (1 \leq g \leq 2).$$

Proof idea. Concavity makes the ideal radius trajectory contract perturbations from above. Thus the rounding errors add over layers, and their effects on squared radii and reciprocal radii remain summable.

Proof. Let $f(u) = \tanh(gu)$. The ideal sequence decreases: $R_1 < 1$, and f is increasing. For $g > 0$, concavity and $f(0) = 0$ imply

$$f'(R_j) \leq \frac{f(R_j)}{R_j} = \frac{R_{j+1}}{R_j} \leq 1.$$

Since f' decreases on $[0, \infty)$, f is 1-Lipschitz on $[R_j, \infty)$. The upward rounding ensures $r_j \geq R_j$. Thus the error at each step increases by at most 3ε , including above unit norm. The case $g = 0$ is immediate. This observation removes any need for precision proportional to D .

The precision choice gives $\varepsilon \leq 1/[100(D+1)^4]$. Also $\tanh 2 + 3\varepsilon < 1$, so $r_j \leq 1$. For example, $e < 3$ implies $\tanh 2 < 40/41$, whereas $\varepsilon \leq 1/1600$. Put $e_0 = 6D\varepsilon$. Then $0 \leq r_j^2 - R_j^2 \leq e_0$. On $[0, 4]$, the derivative of $\phi(v) = 5v/3 + v^2$ is at most $29/3$. Since $g^2 \leq 4$, the total log perturbation is at most

$$D \frac{29}{3} 4e_0 \leq \frac{232D^2}{100(D+1)^4} \leq \frac{29}{200},$$

using $D/(D+1)^2 \leq 1/4$. The final-radius bounds follow from Lemma A.12, $3/(2D+3) \leq 3/[2(D+1)]$, and the rounding error. The coefficient 2 absorbs that error. $\square$

Proof idea. Bound each generation by the product of its local offspring factors. The initial output gate retains the final radius, and the resulting harmonic sums determine the power of depth.

Proof of the call bound in Theorem A.1. Conditional expectation at each reached call bounds the expected number at layer $k < D$ by $r_D \prod_{\ell=k+1}^D B_\ell$. Therefore

$$\mathbb{E}N_{\text{calls}} \leq 1 + r_D \sum_{k=0}^{D-1} \exp \left(\sum_{j=k}^{D-1} \left[\frac{5}{3} (gr_j)^2 + (gr_j)^4 \right] \right). \quad (\text{A.29})$$

For $g \leq 1$, the exponent is at most $29/200 + \sum_{j=k}^{D-1} (5q_j/3 + q_j^2)$. Convexity of $1/x$ and $1/x^2$, integrated over intervals centered at their evaluation points, gives

$$\sum_{j=k}^{D-1} q_j \leq \frac{3}{2} \log \frac{D+1}{k+1}, \quad \sum_{j=k}^{D-1} q_j^2 \leq \frac{9}{4(k+1)}. \quad (\text{A.30})$$

Since $29/200 + 9/4 < 3$,

$$\prod_{\ell=k+1}^D B_\ell \leq e^3 \left(\frac{D+1}{k+1} \right)^{5/2}.$$

Keeping the top factor and summing generations,

$$\mathbb{E}N_{\text{calls}} \leq 1 + 2e^3 (D+1)^2 \sum_{k=0}^{D-1} (k+1)^{-5/2} \leq 1 + \frac{10}{3} e^3 (D+1)^2 \leq 80(D+1)^2.$$

Here $\sum_{j \geq 1} j^{-5/2} \leq 5/3$, by an integral bound; $e < 11/4$ suffices for the last numerical constant.

For $1 \leq g \leq 2$, put $\delta = g - 1 = (S - 1)_+$. Equation (A.28) yields

$$(gR_j)^2 \leq q_j + \frac{5}{2}(g^2 - 1) \leq q_j + \frac{15}{2}\delta.$$

Also $(gR_j)^2 \leq 4$, $q_j \leq 1$, so $(gR_j)^4 \leq q_j^2 + (75/2)\delta$. Consequently

$$\frac{5}{3}(gR_j)^2 + (gR_j)^4 \leq \frac{5}{3}q_j + q_j^2 + 50\delta.$$

The product bound gains at most $e^{50\delta D}$, and the top factor gains at most $e^{\delta D}$. The same sum proves (A.3). $\square$

Remark A.14 (Subcritical work). For fixed $0 < S < 1$, an optional variant uses reference gain $g = S$. For this variant let

$$C_S = \frac{5S^2}{3(1-S^2)} + \frac{S^4}{1-S^4}.$$

Using $R_j \leq S^j$, the rounding bound, and $r_D \leq S^D + 3\varepsilon/(1-S)$, (A.29) gives

$$\mathbb{E}N_{\text{calls}} \leq 1 + e^{C_S+1} D \left(S^D + \frac{3\varepsilon}{1-S} \right).$$

The expected number of nonroot calls tends to zero for each fixed subcritical norm. The root still performs scalar random computation.

Proposition A.15 (A refined fixed-norm rate). *Fix $1 < s \leq 2$, and let $r_* \in (0, 1)$ solve $r_* = \tanh(sr_*)$. Set*

$$\lambda = s(1 - r_*^2), \quad \Psi_s = \frac{5}{3}(sr_*)^2 + (sr_*)^4, \quad A_s = \frac{116s(1 - r_*)}{3(1 - \lambda)},$$

and

$$C_s = 1 + \frac{\exp(29/200 + A_s)}{1 - e^{-\Psi_s}}.$$

Then $0 < \lambda < 1$, and for every network with $S \leq s$,

$$\mathbb{E}N_{\text{calls}} \leq C_s e^{D\Psi_s}.$$

If $\delta = s - 1 \in (0, 1]$, then

$$\Psi_s \leq 5\delta + \frac{91}{4}\delta^2. \quad (\text{A.31})$$

Thus the upper exponent in Corollary A.4 can be replaced by $\min\{2, \Psi_s/\log 2\}$.

Proof idea. Above criticality the ideal radius approaches a positive fixed point geometrically. The total excess over the limiting offspring rate is finite, giving a fixed prefactor and an exponential rate.

Proof. Strict concavity gives a unique positive fixed point and derivative $\lambda < 1$ there. The ideal radius sequence for norm s decreases to r_* , and

$$0 \leq R_j - r_* \leq \lambda^j(1 - r_*).$$

The derivative of $5\rho^2/3 + \rho^4$ is at most 116/3 on $[0, 2]$. Hence its excess above Ψ_s , summed over all layers, is at most A_s . Lemmas A.11 and A.13, followed by a geometric generation sum with $r_D \leq 1$, prove the bound. For a network with $S < s$, its reference gain satisfies $g = \max\{1, S\} \leq s$. Monotonicity therefore majorizes its ideal radii and offspring factors by those at norm s .

Letting $j \rightarrow \infty$ in (A.28) gives $(sr_*)^2 \leq \frac{3}{2}(s^2 - 1) = 3\delta + \frac{3}{2}\delta^2$. Substitution yields

$$\Psi_s \leq 5\delta + \frac{23}{2}\delta^2 + 9\delta^3 + \frac{9}{4}\delta^4 \leq 5\delta + \frac{91}{4}\delta^2.$$

Conversely, put $u = sr_*$. The fixed-point equation and (A.22) give $s \leq 1 + u^2/3$, so $u^2 \geq 3\delta$. Thus $5\delta + 9\delta^2 \leq \Psi_s$, proving $\Psi_s = 5\delta + O(\delta^2)$ directly from explicit inequalities. The prefactor C_s diverges as $s \downarrow 1$, so the transition window uses the uniform theorem. $\square$

Lemma A.16 (The nearcritical fixed-point gap). *Let $s = 1 + \delta$, $0 < \delta \leq 1/2$, and let $r_* = \tanh(sr_*) > 0$. Put $u_* = sr_*$ and $\lambda = s(1 - r_*^2)$. Then*

$$3\delta \leq u_*^2 \leq 3\delta + \frac{3}{2}\delta^2, \quad 1 - \lambda \geq \delta.$$

Proof idea. Rewrite the fixed-point equation using $u \coth u$. Inequality (A.22) bounds its distance from one, and differentiation converts that bound into the contraction gap.

Proof. Since $s = u_* \coth u_*$, inequality (A.22) gives the lower bound. The radius estimate at the fixed point gives $u_*^2 \leq \frac{3}{2}(s^2 - 1) = 3\delta + \frac{3}{2}\delta^2$. Finally $1 - \lambda = -\delta + u_*^2/s \geq \delta(3/s - 1) \geq \delta$. $\square$

A.6 Universal lower bounds

A.6.1 Adaptive transcripts

Let $X_1, \dots, X_m$ be independent signs under $\mathbb{P}_t$, with $\mathbb{P}_t(X_i = 1) = (1+t)/2$, $-1 < t < 1$. Fix the weights and unused input coordinates. Include all randomness used in preprocessing in the same independent random tape as the online randomness; it is averaged throughout this proof. An arbitrary exact sampler returns $Y \in \{-1, 1\}$ with $\mathbb{E}[Y \mid X = x] = f(x)$. Put $H(t) = \mathbb{E}_t f(X) = \mathbb{E}_t Y$. Queries to fixed or previously read coordinates may be answered without cost in the lower-bound argument; this only strengthens the algorithm.

Lemma A.17 (Transcript derivatives). *If Q counts distinct relevant probes, then*

$$\mathbb{E}_t Q \geq (1 - t^2)H'(t)^2, \quad \mathbb{E}_t Q \geq \frac{1 - t^2}{2} |H''(t)|. \quad (\text{A.32})$$

If $|H(t)| < 1$, the first bound improves to

$$\mathbb{E}_t Q \geq \frac{(1 - t^2)H'(t)^2}{1 - H(t)^2}. \quad (\text{A.33})$$

Proof idea. Under a product distribution on contexts, each newly revealed sign contributes a centered score increment. The stopped score records how sensitively the output law changes with the common input bias; its variance is the expected number of distinct probes.

Proof. Fix a random tape and collapse computation between distinct probes into a decision tree. There are finitely many contexts, so almost sure termination on each supplies one common set of tapes of probability one on which the tree has depth at most m . This reduction imposes no bound on the original arithmetic time.

A leaf with queried values $x_1, \dots, x_q$ has likelihood $2^{-q} \prod_{j=1}^q (1 + tx_j)$. Define

$$Z = \sum_{j=1}^Q \frac{X_{I_j}}{1 + tX_{I_j}}, \quad A = \sum_{j=1}^Q \frac{1}{(1 + tX_{I_j})^2}.$$

Leaf differentiation gives

$$H'(t) = \mathbb{E}_t[YZ], \quad H''(t) = \mathbb{E}_t[Y(Z^2 - A)].$$

Differentiation commutes with the random-tape average: on compact subintervals of $(-1, 1)$, all relevant derivatives are bounded by constants depending only on m .

Conditional on the past, a new sign retains law $\mathbb{P}_t$. Its score has mean zero and second moment $(1 - t^2)^{-1}$. The corresponding summand of A has the same expectation. Finite stopped-martingale identities give

$$\mathbb{E}_t Z = 0, \quad \mathbb{E}_t Z^2 = \mathbb{E}_t A = \frac{\mathbb{E}_t Q}{1 - t^2}.$$

Cauchy–Schwarz proves the first bound. Since $\mathbb{E}_t Z = 0$, also $H'(t) = \mathbb{E}_t[(Y - H(t))Z]$, and $\mathbb{E}_t(Y - H(t))^2 = 1 - H(t)^2$ because $Y^2 = 1$. Cauchy–Schwarz now gives (A.33). For the second bound in (A.32),

$$|H''(t)| \leq \mathbb{E}_t(Z^2 + A) = \frac{2\mathbb{E}_t Q}{1 - t^2}.$$

□

The finite set of inputs makes the proof valid without a tail assumption on the original runtime. Section 4 recalls the classical precedents of the first bound.

A coupling gives a second elementary bound. Run the algorithm on two contexts with the same random tape. The two transcripts agree until a coordinate on which the contexts differ is probed. The proof of Lemma B.7 therefore shows that such a probe has probability at least the total variation distance between the two exact output laws. We call the case of the all-positive and all-negative contexts *endpoint coupling*. It applies in the same way to all-plus and all-minus source signs.

A.6.2 Amplification above unit norm

Write $F_D = F_{a,D}$ and $T_m = \sum_{i=1}^m X_i$. For $u \geq 0$,

$$\tanh u \geq \frac{u}{\sqrt{1 + u^2}}, \quad (\text{A.34})$$

since $\sinh u \geq u$. Induction in the reciprocal square gives, for $z > 0$,

$$F_D(z) \geq \frac{a^D z}{\sqrt{1 + a^{2D} H_D z^2}}, \quad H_D = \sum_{j=0}^{D-1} a^{-2j}. \quad (\text{A.35})$$

Oddness handles negative z . Under $\mathbb{P}_0$, $\mathbb{E}T_m^2 = m$ and $\mathbb{E}T_m^4 = 3m^2 - 2m$. The derivative of the input likelihood at zero is T_m , so

$$H'(0) = \mathbb{E}_0[F_D(T_m/m)T_m].$$

Put $c = a^{2D} H_D / m^2$. Equation (A.35) and Jensen's inequality yield

$$\begin{aligned} H'(0) &\geq \frac{a^D}{m} \mathbb{E}_0 \frac{T_m^2}{\sqrt{1 + cT_m^2}} \\ &\geq \frac{a^D}{\sqrt{1 + c(3m - 2)}}. \end{aligned} \quad (\text{A.36})$$

For the last step, weight the original measure by T_m^2/m . Its mean value of T_m^2 is $3m - 2$, and $v \mapsto (1 + cv)^{-1/2}$ is convex. Lemma A.17 proves (A.5). For $a > 1$, use $H_D \leq a^2/(a^2 - 1)$ and

$$\frac{A}{1 + cA/m} \geq \frac{1}{1 + c} \min\{A, m\}, \quad A, c, m > 0,$$

to obtain (A.6).

A.6.3 Curvature at unit norm

We use the following bending argument several times. Let f be twice continuously differentiable on $[0, 1)$, with $f(0) = 0$, $f'(0) = h > 0$, and $f \leq R$, and suppose that $T = 2R/h \leq 1/2$. Taylor's theorem gives

$$f(T) = hT + \int_0^T (T - t)f''(t) dt.$$

Since $f(T) \leq R$ and $hT = 2R$, some $t \in [0, T]$ satisfies $|f''(t)| \geq h^2/(2R)$. If a cost M obeys $M \geq (1 - t^2)|f''(t)|/2$ for every $t \in [0, T]$, then

$$M \geq \frac{3h^2}{16R}. \quad (\text{A.37})$$

Take $a = 1$ and $D \leq m$. The function H is odd, $|H(t)| \leq R_D$, and (A.36) gives

$$h := H'(0) \geq (1 + 3D/m)^{-1/2} \geq \frac{1}{2}.$$

Use the weaker radius estimate $R_D \leq \sqrt{6/(D + 6)}$ for convenient constants. If $D \geq 384$, then $R_D \leq 1/8$ and $T = 2R_D/h \leq 1/2$. An average over contexts is at most their maximum, so $M = \sup_x \mathbb{E}[Q \mid x]$ satisfies the second transcript bound at every t . Equation (A.37) yields

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{3h^2}{16R_D} \geq \frac{3\sqrt{D}}{64\sqrt{6}}.$$

For $D < 384$, the first bound at zero gives $\mathbb{E}_0 Q \geq 1/4 \geq \sqrt{D}/(32\sqrt{6})$. This proves (A.7) and Theorem A.2.

A.6.4 Dyadic parameters and the transition window

Let $L = \lceil \log_2 n \rceil$, $k = \lfloor \log_2 n \rfloor$, and $\eta = 2^{-(20L - k)} \leq 2^{-19L}$. Choose

$$a = \eta \lfloor s/\eta \rfloor. \quad (\text{A.38})$$

Then $a \leq s$, a/m is on the required grid, and $0 \leq s - a < \eta$. For fixed $s > 1$, $a^{2L} = s^{2L}(1 + o(1))$; (A.6) proves the lower side of (A.9).

If $s = 1 + \delta \leq 2$ and $\delta \geq 2\eta$, then

$$a - 1 \geq \frac{\delta}{2}, \quad \frac{a^2 - 1}{4a^2 - 1} \geq \frac{\delta}{15}, \quad \log a \geq \frac{\delta}{3}.$$

For the last inequality use $\log(1 + u) \geq u/(1 + u)$. Since $m \geq 2^{L-1} \geq \frac{1}{2}e^{\delta L/2}$,

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{\delta}{30} e^{\delta L/2}. \quad (\text{A.39})$$

Put $w = \delta L$. The logarithm of the last expression is at least $w/2 - \log(30L) + \log w$. For $w \geq 4 \log(30L)$ this is at least $w/4$, and the grid condition holds for sufficiently large L . Theorem A.1 gives the other direction of Corollary A.5.

A.6.5 Numerical evaluation needs the inputs

Proof idea. An unread input coordinate can be changed without changing the algorithm's transcript. A row for which every coordinate affects the required numerical value therefore forces the stated evaluation cost.

Proof of Proposition A.6. For $-1 \leq z \leq 1$,

$$F'_{1,L}(z) = \prod_{j=0}^{L-1} \text{sech}^2(F_{1,j}(z)) \geq 4^{-L},$$

because $\cosh 1 < 2$. A single input flip changes the mean by $2/m$, hence changes the activation by at least $2/(m4^L) > 2 \cdot 2^{-20L}$, since $m4^L \leq 2^{3L} < 2^{20L}$. Stopping with an unqueried relevant coordinate leaves two contexts with the same transcript and disjoint allowed output intervals. An always-correct evaluator therefore reads all m coordinates. The argument holds on a common set of random tapes of probability one for randomized evaluators.

For the bounded-error statement use uniform inputs. Distinct sums have disjoint valid-output intervals, so a successful approximation identifies the sum. Conditional on a transcript with $Q < m$, the unqueried signs are still independent and fair. Their sum has maximum point mass at most $1/2$: convolution does not increase the maximum point mass of the first fair sign. Thus success is at most $\mathbb{P}(Q = m) + \frac{1}{2}\mathbb{P}(Q < m)$. Success at least $2/3$ forces $\mathbb{P}(Q = m) \geq 1/3$ and $\mathbb{E}Q \geq m/3$. $\square$

A.7 Finite-bit realization

The recursive description contains known real probabilities and one countable distribution. This section gives finite algorithms for both. All interval endpoints used by the implementation are rational.

Lemma A.18 (Exact comparison with a computable probability). *Suppose an oracle gives a rational interval containing $p \in [0, 1]$ of width at most 2^{-m} at stage m , in $O((B + m)^c)$ bit operations. A fair-bit algorithm samples $\text{Bern}(p)$ exactly with $O(B^c + 1)$ expected bit work. It terminates almost surely, including when $p = 0$, $p = 1$, or p is dyadic.*

Proof idea. Compare one uniform binary expansion with successively narrower certified intervals for the probability. A comparison remains undecided only when the uniform value lies close to the target, so its geometric probability tail pays for precision refinement.

Proof. Reveal a uniform binary expansion U , maintaining its length- m dyadic interval. Stop when the two intervals are disjoint, and output whether $U < p$. If no decision has been made at stage m , then

$|U - p| \leq 2^{1-m}$. This event has probability at most $4 \cdot 2^{-m}$. The equality event has probability zero. Summing the stage costs against this geometric tail proves the expectation bound. Rational choices among finitely many categories are handled either by an integer rejection draw or by successive conditional coins. $\square$

On bounded real arguments, certified Taylor sums with rounding after each operation give $\tanh$ intervals of width 2^{-m} in $O((B+m)^4)$ bit operations. Section A.7.2 gives the error analysis, first for the less immediate countable distribution and then for the elementary functions.

For a nonzero row, $\rho = \sigma r_{\ell-1}$ is a positive rational of $O(\mathcal{B})$ bits. In particular, $\rho \geq 2^{-O(\mathcal{B})}$. Dividing by $\tanh \rho$, r_ℓ , or R requires only $O(\mathcal{B})$ guard bits: on the bounded argument range, $\tanh \rho$ and R are at least a constant multiple of ρ . For R , use $R = \frac{1}{2} \int_{a-\rho}^{a+\rho} \operatorname{sech}^2 u \, du$ and $|a| + \rho \leq 5$. Zero denominators are avoided by the explicit zero-row branch.

A.7.1 An exact arity draw

For $0 < \rho \leq 2$, positivity gives

$$F_\rho(5/4) = \sec^2(\rho/2) \leq 4, \quad d_k \leq 4(4/5)^k.$$

Here $\cos 1 \geq 1/2$. Also $\rho / \tanh \rho \leq 7/3 < 3$ and $C_k \geq 1$, so

$$\pi_k \leq 12(4/5)^k. \tag{A.40}$$

Propose $k \geq 0$ with probability $\gamma_k = (1/5)(4/5)^k$, and accept with probability $\pi_k / (60\gamma_k)$. Each trial succeeds with probability $1/60$. Its accepted index has law π . No source signs are requested before acceptance.

A.7.2 A fourth-power bound for scalar bit work

The scalar probabilities can be refined without carrying a growing common rational denominator. We give the details because this reduces the bit overhead in the network theorem.

Lemma A.19 (Fixed-precision coefficient evaluation). *Let $0 < \rho \leq 2$ be a dyadic rational whose description has at most B bits. For integers $K \geq 0$ and $m \geq 1$, the coefficient d_K of*

$$\operatorname{sech}^2(\rho\sqrt{1-z}) = \sum_{j \geq 0} d_j z^j$$

has a certified dyadic enclosure of width 2^{-m} , computable in $O((B + K + m + 1)^4)$ bit operations. The constant is absolute, and schoolbook multiplication and division suffice.

Proof idea. Compute the coefficients of the reciprocal denominator first, then invert its triangular coefficient system. Explicit error bounds allow rounding after every operation, so the rational denominators never grow with the number of operations.

Proof. It suffices to construct an approximation with error 2^{-m-2} . In the following calculation replace the requested precision m by $m + 2$ for this purpose. Write

$$A(z) = \cosh^2(\rho\sqrt{1-z}) = \sum_{j \geq 0} a_j z^j.$$

The coefficients have alternating signs before the constant term, so the power series for $\cosh$ gives

$$a_0 \geq 1, \quad \sum_{j \geq 0} |a_j| = \cosh^2(\rho\sqrt{2}) < 1024. \tag{A.41}$$

Indeed, $\rho\sqrt{2} < 3$, $e < 3$, and $\cosh^2 3 < e^6 < 729$. To bound the reciprocal, write $\sqrt{1-z} = u + iv$. On $|z| \leq 1$,

$$|z|^2 = (u^2 + v^2 - 1)^2 + 4v^2, \quad |v| \leq 1/2.$$

Consequently

$$|A(z)| = \sinh^2(\rho u) + \cos^2(\rho v) \geq \cos^2 1 \geq 1/4,$$

and hence

$$|d_j| \leq 4 \quad (j \geq 0) \quad (\text{A.42})$$

by Cauchy's coefficient estimate.

Define integer precision parameters

$$T = m + 12(K + 1) + \lceil \log_2(K + 1) \rceil + 20,$$

$$M = T + 32,$$

$$P = T + 8M + 10\lceil \log_2(M + 1) \rceil + 64.$$

All operations below round to P binary fractional places, with absolute error at most 2^{-P} per rounding. Their integer parts have $O(M)$ bits. The input ρ is retained exactly.

For $0 \leq j \leq M$, compute

$$c_j = \frac{(2\rho)^{2j}}{(2j)!}, \quad c_0 = 1, \quad c_{j+1} = c_j \frac{4\rho^2}{(2j+1)(2j+2)}.$$

Use these values to form the coefficients through degree K of

$$A_M(z) = \frac{1}{2} \left(1 + \sum_{j=0}^M c_j (1-z)^j \right).$$

The binomial coefficients are exact integers, obtained by Pascal's recurrence; each has at most $M + 1$ bits. Terms of degree above K may be discarded, but this is unnecessary for the stated complexity.

The multipliers in the recurrence for c_j are at most 16. Its accumulated rounding error is at most $16^{M+1}2^{-P}$ at every index. Expanding the binomials adds at most a factor $(M + 1)2^M$. The remaining additions and roundings add at most $(M + 1)^2 2^{-P}$ to a coefficient error. The specified value of P bounds their sum by 2^{-T-4} .

The truncation error on $|z| \leq 1$ is at most

$$E_M = \frac{1}{2} \sum_{j>M} \frac{32^j}{(2j)!} \leq 2^{32-4M} \leq 2^{-T-4}.$$

For the middle inequality, $M \geq 16$; successive terms have ratio below $1/16$, and $34! \geq 2^{118}$ follows from $16! \geq 2^{44}$, $\prod_{j=17}^{32} j \geq 2^{64}$, and $33 \cdot 34 \geq 2^{10}$. Cauchy's estimate applies to each coefficient error. Thus the computed coefficients satisfy

$$|\hat{a}_j - a_j| \leq \Delta := 2^{-T}, \quad 0 \leq j \leq K, \quad \hat{a}_0 \geq 1/2. \quad (\text{A.43})$$

Now use the reciprocal recurrence with the same precision:

$$\hat{d}_0 = \text{round}_P(1/\hat{a}_0), \quad \hat{d}_k = \text{round}_P \left(-\frac{\sum_{j=1}^k \hat{a}_j \hat{d}_{k-j}}{\hat{a}_0} \right).$$

Each product is rounded before summation. Put $E_k = \max_{0 \leq j \leq k} |\hat{d}_j - d_j|$. Provided $E_{k-1} \leq 1$, equations (A.41)–(A.43) give

$$E_0 \leq 3\Delta, \quad E_k \leq 2048E_{k-1} + (12K + 8193)\Delta \quad (1 \leq k \leq K).$$

To see the second estimate, the error in the numerator is at most $1024E_{k-1} + 5k\Delta + k2^{-P}$. Division by $\hat{a}_0$ multiplies this by at most two. Changing $1/a_0$ to $1/\hat{a}_0$ contributes at most 8192Δ , since the exact numerator is bounded by 4096. The final rounding contributes at most 2^{-P} .

Iterating the affine recurrence gives

$$E_k \leq \left(3 + \frac{12K + 8193}{2047}\right) 2^{11k} \Delta \leq 2^{11k+14} (K+1) \Delta \quad (0 \leq k \leq K).$$

The choice of T makes this less than one at every index and gives $E_K < 2^{-m}$, validating the assumed bound throughout. The known error bounds supply a certified interval about $\hat{d}_K$.

There are $O(M^2 + K^2)$ arithmetic operations. Every operand has $O(B + P + M) = O(B + K + m + 1)$ bits, including exact binomial coefficients, products with the input dyadic rational, and rounded quotients. Schoolbook arithmetic therefore costs $O((B + K + m + 1)^4)$ bit operations in total. $\square$

The elementary exponential series on bounded real arguments can be evaluated by the same fixed-precision method in cubic bit work: use linearly many terms, linearly many guard bits, and round after each multiplication or division. Their total amplification is bounded by a constant on the argument range used here. A fourth-power bound therefore covers $\tanh$, the bias probabilities, and their rational combinations. Nonzero denominators require $O(\mathcal{B})$ guard bits as established above. The exact integers defining C_K , and the factor $(5/4)^K$ in the arity acceptance probability, require $O(K)$ additional bits.

It follows from Lemma A.19 that the accepted arity can be drawn with $O(\mathcal{B}^4)$ expected scalar bit work. Both the proposal arity and the precision stage have geometric tails, uniformly in the row and the context. The mean number of proposals is 60. All other scalar work at a reached call obeys the same bound. Lemma A.10 then gives

$$\text{expected online bit work} = O(\mathcal{B}^4 \mathbb{E} N_{\text{calls}}). \quad (\text{A.44})$$

The preprocessing and storage bounds are unchanged.

Proof idea. Preprocessing supplies short rational row descriptions and sampling tables. At every reached call, the scalar routines have a uniform expected cost. Predictable charging then sums those costs over the random recursion tree.

Completion of Theorem A.1. Every stored parameter has $O(\mathcal{B})$ bits because $P = O(\mathcal{B})$. A dense center sum costs $O(n\mathcal{B}^2)$; the bounded scalar endpoints cost $O(\mathcal{B}^4)$. The prefix tables and all row data are obtained within $O(Dn^2\mathcal{B}^6)$ time and $O(Dn^2\mathcal{B})$ storage.

A child index is drawn from the absolute integer weights by an integer rejection draw of success probability at least $1/2$ and binary search in the stored prefix sums. This costs $O(\mathcal{B}^2)$ expected bit operations, including the address. All other scalar work per reached call is $O(\mathcal{B}^4)$ in expectation, uniformly over its context and the prior history. Lemma A.10, summed over the recursion, bounds total bit work by $O(\mathcal{B}^4 \mathbb{E} N_{\text{calls}})$. No independence between the duration and output of a child is used. The exactness identities in Sections A.2–A.4 and almost sure termination prove the required output law. $\square$

For completeness, a dense exact sampler is available for every fixed s . Evaluate the network with certified intervals and compare its output probability to one uniform binary expansion. To obtain output width 2^{-m} , it suffices to use internal precision

$$m + O(b + \log(nD) + D \log^+ s).$$

This follows by propagating interval errors through the s -Lipschitz rows. At $D = L$, $b = 20L$, Lemma A.18 gives $\tilde{O}_s(n^2)$ expected bit work.

One uniform choice of sampler. At $D = L$, the refined bounds and the dense upper bound can be attained by one weight-only selection rule. If $S > 2$, choose the dense algorithm. Otherwise, after finding S , consider the rational comparison gains $q = 1 + k/D$, $1 \leq k \leq D$, and the additional gain $1 + 2^{-16}$, retaining those at least $\max\{1, S\}$. Evaluate the bounds of Propositions A.15 and A.33 where applicable, together with the uniform recursive bound, the critical bound when $S \leq 1$, and the dense envelope n^2 . Approximate the logarithm of each envelope by a certified upper value within one, and select the recursive or dense algorithm

corresponding to the smallest value, using the original or fused factory as required. The common finite-bit factors are powers of $\log n$, so this choice achieves the minimum of the displayed envelopes up to such a factor. The recursive preprocessing is performed once, with reference gain $\max\{1, S\}$; the other gains are used only to compare bounds. The fused factory uses the same stored data and a fixed cutoff.

This selection costs $O(D\mathcal{B}^6)$ preprocessing operations. For completeness, write $w = u_*^2$ and invert

$$\Phi(w) = \sqrt{w} \coth \sqrt{w} = \frac{\sum_{j \geq 0} w^j / (2j)!}{\sum_{j \geq 0} w^j / (2j+1)!}, \quad 0 \leq w \leq 4.$$

Its derivative is at least $1/12$: with $u = \sqrt{w}$,

$$\Phi'(w) = \frac{\sinh(2u) - 2u}{4u \sinh^2 u} \geq \frac{1}{12},$$

using $\sinh(2u) - 2u \geq 4u^3/3$ and $\sinh u/u \leq \sinh 2/2 < 2$; the value at zero is removable. Certified interval bisection therefore encloses w to the required accuracy without an exact equality test. When the midpoint value interval overlaps the target, the derivative bound encloses the root in a shorter interval about that midpoint. The smallest positive candidate gap is at least $\min\{D^{-1}, 2^{-16}\}$, and the fixed-point estimates give $w \geq 3(q-1)$ and $1-\lambda \geq (q-1)/2$. Thus all inverse and logarithmic operations need only $O(\mathcal{B})$ guard bits. The geometric denominators are bounded below by constant multiples of the same gap. Logarithms of the large prefactors are evaluated directly, using stable log-sum formulas and bounds on small negative exponentials; the prefactors themselves are never formed.

For any fixed $1 < s \leq 2$, the candidate $q_D = \lceil sD \rceil / D$ is eligible and satisfies $s \leq q_D < s + 1/D$. The fixed-point rates are smooth for $s > 1$, and their prefactors are locally bounded. Thus replacing s by q_D changes the rate multiplied by D by $O_s(1)$. The same argument applies to the fused rate for $s < 1 + 2^{-16}$; the added endpoint covers equality. Consequently the refined minima with the dense bound hold for this single uniform algorithm. The selection is completed before the context is revealed.

The chosen sampler therefore retains its exact output law. This proves the upper side of Corollary A.4, including the refined fixed-norm minima.

A.8 A finite-bit scalar tanh backend

The scalar construction also admits a quadratic cost bound at arbitrary radius. The following product sampler supplies the finite arithmetic implementation needed by attention and normalization.

Lemma A.20 (Majority representation and raw arity). *Let $u > 0$, $q = \tanh u$, and*

$$\operatorname{sech}^2(u\sqrt{1-v}) = \sum_{k \geq 0} d_k v^k, \quad C_k = (2k+1) \binom{2k}{k} 4^{-k}.$$

The coefficients d_k are nonnegative and sum to one. The numbers $\pi_k = (u/q)d_k/C_k$ form a probability distribution, and

$$\frac{\tanh(uz)}{q} = \sum_{k \geq 0} \pi_k M_k(z), \quad -1 \leq z \leq 1,$$

where M_k is the mean of majority on $2k+1$ independent signs of mean z . A gate of probability q , followed by this majority, and a fair sign on the closed branch, returns a sign of mean $\tanh(uz)$. Its expected full-majority source count is

$$Q_u = u + \int_0^1 \frac{u \tanh^2(uz)}{z^2} dz \leq 2 \max\{u, u^2\}.$$

Proof idea. The expansion and its cost integral are those of Lemmas A.8 and A.9, whose proofs hold at every radius. Splitting the cost integral gives a quadratic bound at large radius.

Proof. The coefficient and mixture claims are Lemma A.8 with $\rho = u$; its proof uses no bound on the radius. Lemma A.21 below gives a second proof of the coefficient claims. The gate and the fair sign give mean $q \cdot \tanh(uz)/q = \tanh(uz)$. The derivation of (A.21) also holds at every radius. It gives the expected full-majority count on the open gate, and multiplication by the gate probability q gives the stated formula for Q_u . If $u \leq 1$, the inequality $|\tanh(uz)| \leq uz$ bounds it by $u + u^3 \leq 2u$. If $u \geq 1$, split the integral at $1/u$:

$$Q_u \leq u + \int_0^{1/u} u^3 dz + \int_{1/u}^1 uz^{-2} dz = 2u^2.$$

□

A.8.1 Sampling the arity through the cosh product

The classical product for cosh gives a direct sampler for the majority index. It also gives a quadratic scalar bit bound at arbitrary row norm. The product is recorded in DLMF Eq. 4.36.2, and its relation to independent gamma sums is part of the Pólya–Gamma construction of Polson, Scott, and Windle [35, 36]. The discrete construction below is a Poissonization of that product.

Lemma A.21 (A product law for the arity proposal). *Let $u > 0$, and write*

$$\operatorname{sech}^2(u\sqrt{1-v}) = \sum_{k \geq 0} d_k v^k, \quad a_j = \frac{4u^2}{\pi^2(2j-1)^2}, \quad p_j = \frac{1}{1+a_j}.$$

Let $G_{j,1}, G_{j,2}$ be independent geometric random variables with $\Pr(G_{j,r} = k) = p_j(1-p_j)^k$, $k \geq 0$. Then

$$K = \sum_{j \geq 1} (G_{j,1} + G_{j,2}) \quad \text{satisfies} \quad \Pr(K = k) = d_k, \quad \mathbb{E}K = u^2.$$

The sum is finite almost surely.

Proof idea. Each factor of the classical cosine product is a geometric probability generating function. Their total mean is finite, so the infinite sum is almost surely finite and has the required coefficient law.

Proof. The hyperbolic cosine product gives

$$\operatorname{sech}^2(u\sqrt{1-v}) = \prod_{j \geq 1} \{1 + a_j(1-v)\}^{-2}.$$

Each factor is the probability generating function of $G_{j,1} + G_{j,2}$. Also $2 \sum_j a_j = u^2$, either by comparing quadratic coefficients in the hyperbolic cosine product or by the sum of the reciprocal odd squares. Monotone convergence gives $\mathbb{E}K = u^2 < \infty$, so $K < \infty$ almost surely. For $0 \leq v \leq 1$, dominated convergence identifies the generating function of the sum with the displayed product. Uniqueness of probability generating functions proves the law. □

Here is a finite algorithm for this infinite product. Put $H = \max\{1, u\}$ and $J = \lceil H \rceil$. First sample both geometric variables at every index $1 \leq j \leq J$. For $j > J$, put

$$\mu_j = 2 \log(1 + a_j), \quad \nu_j = \frac{2u^2}{j(j-1)}, \quad \Lambda = \sum_{j > J} \nu_j = \frac{2u^2}{J}.$$

Generate $N \sim \text{Poisson}(\Lambda)$ as the sum of J independent Poisson variables of parameter $2u^2/J^2 \leq 2$. For each of its N points, independently choose

$$j = 1 + \lfloor J/U \rfloor, \quad U \sim \text{Uniform}(0, 1), \quad (\text{A.45})$$

and retain the point with probability μ_j/ν_j . Remove duplicate retained indices. At each remaining index, sample $G_{j,1} + G_{j,2}$ conditioned to be positive. All other tail indices contribute zero.

Lemma A.22 (Exact tail sampling). *The finite algorithm just described has the law in Lemma A.21. It proposes at most $2H$ tail points in expectation and visits at most $2H$ prefix indices. For every retained index, its conditional positive value can be sampled as follows. With probability $1/(1 + p_j)$ return $1 + G_1 + G_2$; otherwise return $1 + G_2$, using fresh independent $\text{Geom}(p_j)$ variables.*

Proof idea. A summable Poisson envelope selects the tail indices that can contribute. Thinning gives the correct activation probability for each index, and a two-case geometric mixture generates its positive value.

Proof. Since $j > J \geq u$, $a_j < 1/\pi^2 < 1/8$. In particular,

$$\mu_j \leq 2a_j \leq \nu_j.$$

The second inequality follows from $4j(j-1) \leq \pi^2(2j-1)^2$. The mark in (A.45) has probability $J/[j(j-1)] = \nu_j/\Lambda$ at j , with $j \geq J+1$. Poisson marking and independent thinning therefore leave independent counts $Z_j \sim \text{Poisson}(\mu_j)$. The probability of an active index is

$$\Pr(Z_j > 0) = 1 - e^{-\mu_j} = 1 - p_j^2,$$

which equals the probability that the sum of its two geometric variables is positive. The activation events are independent. Conditional on positivity, the event $G_{j,1} > 0$ has probability $(1-p_j)/(1-p_j^2) = 1/(1+p_j)$. The memoryless identity for a geometric variable now proves the stated conditional sampler. Its fresh randomness also gives the correct joint law across indices. Finally, $\Lambda \leq 2H$ and $J \leq 2H$. $\square$

Lemma A.23 (Finite-bit implementation of the product proposal). *Suppose $u > 0$ is rational. Let $\mathcal{B}_u \geq 8$ bound its binary input length and the bits needed to address a source request. The preceding proposal K can be sampled exactly with*

$$O(H\mathcal{B}_u^4)$$

expected bit operations. This count includes all random bits, arithmetic, duplicate removal, and precision refinement. The constant is absolute and independent of u .

Proof idea. Implement every count and comparison with certified scalar intervals. Geometric and Poisson moment bounds control both the number of generated objects and the precision needed to decide their comparisons.

Proof. We give the numerical primitives and the moment bounds that make this assertion uniform. All elementary functions in the construction have certified rational intervals. For a precision parameter V , use $8V$ fractional working bits. The Machin formula $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ gives π from $O(V)$ alternating-series terms. Binary range reduction followed by

$$\log y = 2 \sum_{r \geq 0} \frac{t^{2r+1}}{2r+1}, \quad t = (y-1)/(y+1),$$

evaluates logarithms: one may reduce to $1/4 \leq y \leq 4$, so $|t| \leq 3/5$. Taking $8V$ terms leaves an error less than 2^{-10V} ; the accumulated rounding error is $O(V^2 2^{-8V})$. Exponentials on a bounded interval are evaluated as in Section A.7.2, with range reduction and repeated squaring. These algorithms take $O(V)$ arithmetic operations on $O(V)$ -bit numbers and hence $O(V^3)$ bit work with schoolbook multiplication and division.

For completeness, the unbounded discrete inversions can be performed with these primitives. To sample $G \sim \text{Geom}(p_j)$, put

$$d = -\log(1 - p_j) = \log(1 + a_j) - \log a_j, \quad G = \lfloor (-\log U)/d \rfloor.$$

Expose the leading zero bits of U . If their number is T , write $U = 2^{-T}V_0$ with $1/2 \leq V_0 < 1$. Thus $-\log U = T \log 2 - \log V_0$. Here T has a geometric tail,

$$d \geq (1 + a_j)^{-1} \geq (1 + u^2)^{-1}, \quad d \leq 8(\mathcal{B}_u + \log_2(j+2) + 1).$$

The first bound uses $a_j < u^2$ and $\log(1+x) \geq x/(1+x)$. At refinement stage m , the choice

$$V = \lceil 64\{\mathcal{B}_u + \log_2(j+2) + T + m + 1\} \rceil$$

absorbs the division by d , all argument magnitudes, and the series and arithmetic errors above. At that stage reveal enough further bits of V_0 to know it to $8V$ fractional bits; this also charges all random bits used in the inversion. In particular it gives an interval for $(-\log U)/d$ of width at most $2^{-m-8}/(1+d)$. Intersect the interval with $[0, \infty)$, and stop when all its values have the same integer part.

Only positive integers can obstruct this stopping rule. Since $(-\log U)/d$ is exponential with rate d , for $0 < \epsilon \leq 1/2$ and $d\epsilon \leq 1$ the probability of lying within ϵ of a positive integer is at most

$$\sum_{k \geq 1} \left(e^{-d(k-\epsilon)} - e^{-d(k+\epsilon)} \right) = \frac{2 \sinh(d\epsilon)}{e^d - 1} \leq 6\epsilon.$$

Consequently the refinement stage has a geometric tail, uniformly in j and u . The expected bit cost of one such geometric draw is $O((\mathcal{B}_u + \log_2(j+2))^3)$. This argument does not require independence between its leading-zero count and its final refinement stage: the moments of their sum are bounded by the moments of each separately.

For (A.45), expose T leading zero bits of its fresh uniform variable. Conditional on T , J/U lies in $[J2^T, J2^{T+1}]$, and its density is at most $2/(J2^T)$. The interval contains at most $2J2^T$ integer boundaries. A union bound therefore bounds the probability of being within ϵ of a boundary by 8ϵ . Reading $\lceil \log_2(J+1) \rceil + 2T + m + 8$ bits of U gives an interval for J/U of width at most 2^{-m-4} by rational division. The floor thus stops with a geometric refinement tail conditional on T . Its expected bit cost is $O(\mathcal{B}_u^2)$, and its output obeys

$$\log_2(j+2) \leq \log_2(4J) + T + 1. \quad (\text{A.46})$$

The thinning probability is numerically stable in the form

$$\frac{\mu_j}{\nu_j} = \frac{4j(j-1)}{\pi^2(2j-1)^2} \frac{\log(1+a_j)}{a_j}.$$

At a tail index $a_j < 1/8$, so the last factor is the series $\sum_{r \geq 0} (-a_j)^r / (r+1)$, with uniformly geometric convergence. An interval of width 2^{-m} costs $O((\mathcal{B}_u + \log_2(j+2) + m)^3)$ bit operations. A fresh lazy uniform comparison gives a geometrically decaying precision tail. The conditional-positive case coin $1/(1+p_j)$ has the same bound. Its subsequent geometric variables have the bound already proved. All these statements are conditional on the chosen index, so their averages can use (A.46) directly.

One may generate each Poisson variable of parameter $0 \leq \lambda \leq 2$ without an unbounded-mean primitive. Propose k with probability 2^{-k-1} and accept it with probability

$$\frac{e^{-\lambda} \lambda^k / k!}{16 \cdot 2^{-k-1}}.$$

This is at most one because the numerator-to-proposal ratio is at most $2e^\lambda \leq 2e^2 < 16$. The mean number of trials is exactly 16. At refinement stage m , the preceding exponential algorithm and k sequential multiplications and divisions give the acceptance interval with $O((\mathcal{B}_u + k + m)^3)$ bit work. The proposal k and the lazy comparison both have geometric tails. The expected cost of this Poisson primitive is therefore $O(\mathcal{B}_u^3)$.

There are $J \leq 2H$ prefix indices and Poisson draws, each of expected cost $O(\mathcal{B}_u^3)$. The total proposal count N is Poisson with mean $\Lambda \leq 2H$, and its independent marks satisfy (A.46). Sorting the retained indices removes duplicates. For example, pad all proposed indices to their maximum bit length and use mergesort. Conditional on $N = n$, that maximum has moments

$$O_r((\mathcal{B}_u + \log_2(n+2))^r),$$

because the maximum of n geometric leading-zero counts has tail at most $n2^{-t}$. The sorting cost is at most a constant times $n \log_2(n+2)$ times the sum of this maximum length and $\log_2(n+2)$, including address manipulation. For a Poisson variable, $\mathbb{E}[Nf(N)] = \Lambda \mathbb{E}[f(N+1)]$ for every nonnegative f . Markov's

inequality at thresholds $(1 + \Lambda)2^t$ bounds every fixed logarithmic moment of $N + 1$ by a constant times $(1 + \log_2(2 + \Lambda))^r$. Since $\log_2(H + 2) = O(\mathcal{B}_u)$, the expected sorting cost is $O(H\mathcal{B}_u^2)$. The same moment bounds charge the scalar work at every proposed or retained mark by $O(H\mathcal{B}_u^3)$ in expectation. Adding the sampled counts to form K also fits $O(H\mathcal{B}_u^4)$: its counter length is bounded by $O(1 + \log_2(K + 1))$, and $\mathbb{E}K = u^2$ gives all fixed logarithmic moments by the same Markov argument. For this last charge, Cauchy–Schwarz handles the dependence between the count and the counter length: the number of summands is at most $J + N$, with $\mathbb{E}(J + N)^2 = O(H^2)$, while $\mathbb{E}\log_2^2(K + 2) = O(\mathcal{B}_u^2)$. These bounds prove the lemma. $\square$

Theorem A.24 (Quadratic bit cost for an arbitrary radius). *Let $u > 0$ be rational, let $\mathcal{B}_u \geq 8$ have the meaning in Lemma A.23, and put $H = \max\{1, u\}$. The positive-majority factory for an exact sign of mean $\tanh(uz)$ has expected scalar bit cost*

$$O(H^2\mathcal{B}_u^4).$$

This count excludes the computations producing its source signs, and includes all source-selection bookkeeping and majority processing. Its expected number of source requests, including its outer gate, is at most $2\max\{u, u^2\}$.

Proof idea. Sample the product-law proposal and accept according to the reciprocal majority normalizer. A finite fair-coin experiment realizes that acceptance probability, while its expected repetitions cancel the proposal normalization.

Proof. Write $q = \tanh u$ and $C_k = (2k + 1)\binom{2k}{k}/4^k$. The majority-index law is

$$\pi_k = \frac{u}{q} \frac{d_k}{C_k}.$$

Generate K with law d and accept it with probability $1/C_K$. Its acceptance coin is the following finite sequence: for $r = 1, \dots, K$, flip a coin of probability $2r/(2r + 1)$; reject at its first failure and accept if all succeed. Indeed

$$\frac{1}{C_k} = \prod_{r=1}^k \frac{2r}{2r + 1}.$$

Also $C_k \geq \sqrt{k + 1}$, by induction using $((2k + 1)/(2k))^2 \geq (k + 1)/k$. The expected number of tests conditional on $K = k$ is thus

$$\sum_{r=0}^{k-1} \frac{1}{C_r} \leq 2\sqrt{k}.$$

Each rational test can be performed by drawing a uniform integer with denominator $2r + 1$, using binary rejection. Its expected bit cost is $O((1 + \log_2(k + 2))^2)$, including the counter and comparison. Consequently the expected scalar cost of this acceptance loop, conditional on $K = k$, is $O(\sqrt{k}(\mathcal{B}_u + \log_2(k + 2))^2)$.

For each fixed integer $r \geq 1$, $\mathbb{E}K = u^2$ and Markov's inequality give

$$\mathbb{E}_d(\mathcal{B}_u + \log_2(K + 2))^r = O_r(\mathcal{B}_u^r).$$

Cauchy–Schwarz therefore bounds the average acceptance-loop cost by $O(u\mathcal{B}_u^2)$. A proposal trial, including generation of K , costs $O(H\mathcal{B}_u^4)$ in expectation. The identity

$$\sum_{k \geq 0} \frac{d_k}{C_k} = \int_0^1 \operatorname{sech}^2(uz) dz = \frac{q}{u}$$

follows either from the majority normalization or from $\int_0^1 (1 - z^2)^k dz = 1/C_k$ and monotone convergence. Thus the accepted index has law π , and the mean number of independent trials is $u/q \leq 1 + u \leq 2H$. The predictable trial-cost lemma gives total expected arity cost $O(H^2\mathcal{B}_u^4)$ even though a trial's cost may depend on whether it succeeds.

The accepted majority loop has the same order of bit work. Indeed, using $C_k \geq \sqrt{k+1}$, its expected cost is at most a constant times

$$\frac{u}{q} \mathbb{E}_d \left[\sqrt{K+1} (\mathcal{B}_u + \log_2(K+2))^2 \right] = O(H^2 \mathcal{B}_u^2).$$

The source-selection tables have $O(\mathcal{B}_u)$ -bit entries, so their work is included by increasing the logarithmic power to four. The expected number of source requests is the bound $Q_u \leq 2 \max\{u, u^2\}$ in Lemma A.20. Finally, the gate $q = \tanh u$ can be evaluated with certified exponentials. To accuracy 2^{-m} , values $u \geq m+2$ are handled by $0 \leq 1 - \tanh u \leq 2e^{-2u}$; otherwise the argument is $O(m)$. Range reduction and repeated squaring then cost $O((\mathcal{B}_u + m)^3)$ bits. Its fresh uniform comparison has a geometric refinement tail. This fits the claimed bound. $\square$

Corollary A.25 (Biased linear tanh row). *Suppose independent source signs have means h_j/R , where $|h_j| \leq R$, $|b| \leq 1$, and $\sum_j |w_j|R = \rho$. There is an exact sign sampler of mean $\tanh(b + \sum_j w_j h_j)$ with at most $2\rho(1 + \rho)$ expected source requests and $O((1 + \rho)^2 B^4)$ expected local bit work.*

Proof idea. Fold the known bias into the source distribution and apply the arbitrary-radius scalar factory. Only the part of each source request assigned to the row requires a child sample.

Proof. For $u = |b| + \rho > 0$, make a sign of mean $(b + \sum_j w_j h_j)/u$ by selecting the known bias sign with mass $|b|/u$ and a signed source coordinate with the remaining row masses. Each request for this mixed sign uses a source with probability ρ/u . Apply the preceding majority construction at radius u . Its expected source count is at most

$$\frac{\rho}{u} Q_u \leq 2\rho \max\{1, u\} \leq 2\rho(1 + \rho).$$

The last inequality uses $|b| \leq 1$. The product proposal theorem bounds its local bit cost by $O(\max\{1, u\}^2 B^4)$, including the known gate and row selection. For $u = 0$, a fair sign suffices. A zero-weight row can instead sample its known tanh bias directly. Conditional charging is valid because the decision to request a mixed sign precedes its fresh bias-versus-source choice. $\square$

Corollary A.26 (A bounded-radius tanh factory). *Given independent signs of unknown mean $z \in [-1, 1]$ and known finite parameters $|a| \leq 1$, $0 \leq r \leq 4$, a sign of mean $\tanh(a + rz)$ can be sampled exactly with at most 40 expected source requests and $O(B^4)$ auxiliary bit operations. Here B includes the encodings of a, r and the source interface; each source call uses fresh randomness. The same assertion holds for a weighted mixture of predecessor signs of total absolute weight at most four. If $a = 0$ and $r \leq 1$, at most two expected source requests suffice.*

Proof idea. Apply the preceding arbitrary-radius row bound on a fixed bounded interval. Its source cost and all scalar precision costs are then bounded by absolute constants apart from the encoding factor.

Proof. Apply Corollary A.25, whose source bound is $2r(1 + r) \leq 40$. A zero row uses only its known bias coin. Absolute-weight sampling and sign complementation implement the weighted source. The bounded parameters leave the fourth-power bit factor unchanged. In the unbiased unit-radius case, Lemma A.20 gives the sharper bound $Q_r \leq 2$. $\square$

A.9 The sharp cost of one tanh row

The distinction between a value and a sample already appears in one row. It is useful to state its complete query law before considering depth. Auxiliary bit work is at least constant even when no input coordinate is requested.

Theorem A.27 (One-row law). *Let $x \in \{-1, 1\}^n$, let $|b_0| \leq 1$, and let w be a dyadic row with $r = \sum_i |w_i|$. After an integer alias table has been built from the row, an exact sign of mean*

$$f(x) = \tanh\left(b_0 + \sum_i w_i x_i\right)$$

can be generated using at most $2r(1+r)$ expected source-coordinate requests. Reading the entire input gives the alternative bound n . Including random bits, row choices, and transcendental refinement, the expected bit work is

$$O(B^4[1 + \min\{n, r + r^2\}]),$$

where B includes the maximum encoding length of one coefficient, the bias encoding, addresses, and $\log(2+r)$.

Conversely, when n is a power of two and $r > 0$ is dyadic, the zero-bias row $w_i = r/n$ has worst-input expected distinct-probe complexity $\Omega(\min\{n, r + r^2\})$, with an absolute implied constant. This lower bound permits arbitrary preprocessing independent of x . The displayed row must, as usual, fit the stipulated coefficient grid. For general widths, using the largest power of two at most n changes only the absolute constant.

Proof idea. Absolute-weight sampling turns the row into one unknown source mean, giving the upper bound. For the lower bound, use a uniform row and compare its sensitivity to the information obtained from distinct input probes.

Proof. If $r = 0$, a known scalar coin suffices. Otherwise apply Corollary A.25 with $R = 1$ and $h_i = x_i$: a source request for coordinate i is one input probe. This gives at most $2r(1+r)$ expected input requests and $O((1+r)^2 B^4)$ expected local bit work, with coordinates selected through the integer alias table. The dense alternative computes the dyadic preactivation after n input probes and makes a known tanh coin by a certified uniform comparison. Choose the cheaper alternative using the known row norm. Both are exact and terminate almost surely.

For the lower bound write $M = n^{-1} \sum_i x_i$. If $0 < r \leq 1$, endpoint coupling (Section A.6) shows that some probe occurs with probability at least the total variation distance between the two endpoint output laws, namely $\tanh r \geq r/2$. Since $r + r^2 \leq 2r$, this proves the claim in this range.

Suppose $r \geq 1$. Give the input signs independent common mean t , and put $H(t) = \mathbb{E}_t \tanh(rM)$. At $t = 0$, let $Z = \sqrt{n} M$. Then

$$\mathbb{E}Z^2 = 1, \quad \mathbb{E}Z^4 \leq 3, \quad \Pr(|Z| \geq 1/2) \geq 3/16.$$

The last inequality is Paley–Zygmund applied to Z^2 at threshold $1/4$. The finite-input score identity gives

$$H'(0) = \sqrt{n} \mathbb{E}[Z \tanh(rZ/\sqrt{n})].$$

For $u \geq 0$, $\tanh u \geq \frac{1}{2} \min\{u, 1\}$: concavity gives the assertion on $[0, 1]$ using $\tanh 1 > 1/2$, and monotonicity gives it above one. Restricting the expectation to $|Z| \geq 1/2$ therefore gives

$$H'(0) \geq \frac{3}{128} \min\{r, \sqrt{n}\}.$$

The transcript information inequality gives

$$\mathbb{E}_0 Q \geq H'(0)^2 \geq \frac{9}{16384} \min\{r^2, n\} \geq \frac{1}{65536} \min\{r + r^2, n\}.$$

An input attaining at least this average expected cost exists. The argument includes all preprocessing randomness in the independent auxiliary tape and imposes no bound on preprocessing time. $\square$

For fixed positive r , this law gives bounded expected input work at arbitrarily large width. In contrast, the exact dyadic preactivation of a row with every coefficient nonzero depends on every input coordinate. Its deterministic evaluation must read all of them: changing an unqueried sign changes the preactivation. The network question is whether this saving survives composition.

A.10 Fusing the first vote with the bias correction

For row norms at most one, the uniform algorithm uses reference gain one in the radius recurrence. The exact row norms still determine the row mixtures and their radii ρ . This choice preserves the sampler, its exactness proof, and the bound $80(D+1)^2$. We prove a smaller power of depth in two steps. First we assign a positive weight to each reached call; the weights are used only in the analysis. Then, at sufficiently small row radius, we replace the bias race by a factory whose first vote also selects the bias correction. This reduces the critical power without changing the represented function.

A.10.1 Weighted local estimates for the critical sampler

Fix $0 < \eta \leq 1/4$, write $\alpha = 1 + \eta$, and define

$$\varphi(v) = \sqrt{\alpha - v^2}, \quad \ell(v) = \log \varphi(v), \quad -1 \leq v \leq 1.$$

The function φ is even and concave, with $\sqrt{\eta} \leq \varphi \leq \sqrt{\alpha}$. Direct differentiation gives

$$\ell'(v) = -\frac{v}{\alpha - v^2}, \quad \ell''(v) = -\frac{\alpha + v^2}{(\alpha - v^2)^2}, \quad |\ell'| \leq \eta^{-1}, \quad |\ell''| \leq \frac{9}{4\eta^2}. \quad (\text{A.47})$$

At a nonzero row put

$$q = \tanh \rho, \quad g = \frac{\tanh(\rho z)}{q}, \quad t = \tanh a.$$

If $t < 0$, complement source and output signs; evenness of φ leaves the analysis unchanged. Assume henceforth that $t \geq 0$, and set

$$c = tq, \quad k = \frac{1 - t^2}{1 - c^2}, \quad T_c(g) = \frac{g + c}{1 + cg}.$$

The exact midpoint and radius satisfy $R = qk$ and $h = m + RT_c(g)$. For the stored center c_{row} and radius r , let

$$\theta = q/r \leq 1, \quad \delta = (m - c_{\text{row}})/r.$$

The parent's normalized mean is $v = \delta + \theta k T_c(g) \in [-1, 1]$.

Assign a child of normalized mean v_j the analysis weight $\varphi(v_j)$. If its row-mixture probability and sign are p_j and σ_j , Jensen's inequality gives

$$\sum_j p_j \varphi(v_j) = \sum_j p_j \varphi(\sigma_j v_j) \leq \varphi \left(\sum_j p_j \sigma_j v_j \right) = \varphi(z).$$

Applying Lemma A.10 first to the majority and then to the race therefore gives

$$\mathbb{E} \sum_{\text{children}} \varphi(v_j) \leq \theta k A_{\text{seq}}(\rho, z) \frac{1 + c}{1 + cg} \varphi(z). \quad (\text{A.48})$$

This use of the lemma permits correlation between a source's cost and returned sign. The event that requests a source is determined before its fresh randomness is used.

For fixed $|u| \leq 1$, the function $\theta/\varphi(\theta u)$ is increasing on $[0, 1]$. Both $\theta k T_c(g)$ and v lie in $[-1, 1]$, so (A.47) bounds the effect of the center shift by $\exp(|\delta|/\eta)$. Consequently the right side of (A.48), divided by $\varphi(v)$, is at most

$$\exp\left(\frac{\varepsilon}{\eta r}\right) W, \quad W = A_{\text{seq}}(\rho, z) k \frac{1 + c}{1 + cg} \frac{\varphi(z)}{\varphi(k T_c(g))}. \quad (\text{A.49})$$

Lemma A.28 (Majority cost and the bias contribution). *For $0 < \eta \leq 1/4$ and $0 < \rho \leq 1/16$, the majority term satisfies*

$$\log A_{\text{seq}} + \ell(z) - \ell(g) \leq \left(1 - \frac{\eta z^2}{3(\alpha - z^2)}\right) \rho^2 + \frac{4}{\eta^2} \rho^4.$$

The remaining bias term $B = \log k + \log((1+c)/(1+cg)) + \ell(g) - \ell(kT_c(g))$ obeys

$$B \leq \frac{-(1-2q^2)t^2 + 3tq}{1-q^2}.$$

In particular $B \leq 0$ when $t \geq (7/2)q$.

Proof idea. A real Taylor expansion controls the majority contribution. For the bias term, express the potential along the coordinate $\text{artanh } g$. Its derivative is bounded by one, which controls the whole translation even near the endpoints.

Proof. First record Taylor bounds that hold throughout $\rho \leq 1/16$:

$$\left|g - z - \frac{\rho^2 z(1-z^2)}{3}\right| \leq 11\rho^4, \quad |g - z| \leq \rho^2/2. \quad (\text{A.50})$$

Indeed, for complex $|w| \leq 1$, $|\cosh w - 1| \leq \cosh 1 - 1 < 3/5$ and $|\sinh w| \leq \sinh 1 < 6/5$, by $8/3 < e < 11/4$; hence $|\tanh w| < 3$. Cauchy's coefficient estimate and oddness give

$$|\tanh w - w + w^3/3| \leq 5|w|^5 \quad (w \in \mathbb{C}, |w| \leq 1/16). \quad (\text{A.51})$$

Also $q/\rho \geq 1 - \rho^2/3 \geq 767/768$, by integrating $\tanh'(u) \geq 1 - u^2$. Subtracting zq/ρ from the numerator expansion leaves an error at most $10\rho^4$; dividing by q/ρ adds less than ρ^4 . Finally, $1/3 + 11/256 < 1/2$.

Lemma A.9, Taylor's theorem for ℓ , and (A.50) give, with $d_z = \alpha - z^2$,

$$\begin{aligned} \log A_{\text{seq}} + \ell(z) - \ell(g) &\leq \left(1 - \frac{\eta z^2}{3d_z}\right) \rho^2 + \left(1 + \frac{11}{\eta} + \frac{9}{32\eta^2}\right) \rho^4 \\ &\leq \left(1 - \frac{\eta z^2}{3d_z}\right) \rho^2 + \frac{4}{\eta^2} \rho^4. \end{aligned} \quad (\text{A.52})$$

The second line uses $\eta^2 + 11\eta + 9/32 \leq 99/32 < 4$.

It remains to bound the bias contribution

$$B = \log k + \log \frac{1+c}{1+cg} + \ell(g) - \ell(kT_c(g)).$$

Since $k \leq 1$, the last difference is at most $\ell(g) - \ell(T_c(g))$. For $v = \tanh u$,

$$\left|\frac{d}{du} \ell(\tanh u)\right| = \frac{|v|(1-v^2)}{\alpha - v^2} \leq 1.$$

Thus that difference is at most $\text{artanh } c$, including $g = \pm 1$ by continuity. The race logarithm is at most $2 \text{artanh } c$. Since

$$\log k \leq -\frac{1-2q^2}{1-q^2} t^2, \quad \text{artanh } c \leq \frac{tq}{1-q^2},$$

we obtain

$$B \leq \frac{-(1-2q^2)t^2 + 3tq}{1-q^2}.$$

If $t \geq (7/2)q$, its numerator is at most $tq(-1/2 + 7q^2) \leq 0$, because $q \leq \rho \leq 1/16$.

□

Bounds at a fixed cutoff. For gain one, $\coth^2 u = 1 + \sinh^{-2} u \leq 1 + u^{-2}$ gives $R_j^2 \geq 1/(j+1)$. Lemmas A.12 and A.13 therefore give

$$(j+1)^{-1} \leq R_j^2 \leq 3/(2j+3), \quad 0 \leq r_j - R_j \leq 3j\varepsilon, \quad r_j \leq 1.$$

The original normalized sampler at level J obeys

$$M_J \leq 1 + e^3 (J+1)^{5/2} \sum_{k \geq 1} k^{-5/2} \leq 36(J+1)^{5/2}. \quad (\text{A.53})$$

This follows from the suffix proof of Theorem A.1, with the root gate omitted, and $\sum k^{-5/2} \leq 5/3$, $e^3 < 21$. For $\eta = 1/64$, $J \geq 2^{25}$, $D > J$, the center terms in (A.49) sum to at most $64\varepsilon D \sqrt{D+1} < 1/100$. If a squared-radius potential has derivative at most two, the proof of Lemma A.13 shows that rounding changes a suffix logarithm by at most $12D^2\varepsilon < 1/100$. These estimates use $\varepsilon \leq [100(D+1)^4]^{-1}$.

A.10.2 An exact scalar factory

From here on fix the absolute constants

$$\eta = 1/64, \quad \alpha = 65/64, \quad \varphi(z) = \sqrt{\alpha - z^2}, \quad c_0 = 17/16, \quad M_0 = 2^{22}, \quad J_0 = 2^{25}.$$

Consider a row with $0 < \rho \leq 1/32$ and $0 \leq a \leq 4\rho$. Negative a is handled by complementing source and output signs. Write

$$q = \tanh \rho, \quad t = \tanh a, \quad c = tq, \quad G(z) = \tanh(\rho z)/q, \quad F(z) = \frac{G(z) + c}{1 + cG(z)}.$$

The function

$$H(z) = \frac{F(z) - z}{1 - z^2} = \sum_{k \geq 0} h_k z^k$$

has removable singularities at $z = \pm 1$.

Lemma A.29 (Coefficients of the correction). *The function H is analytic on a neighborhood of the closed complex disk $|z| \leq 2$. Its first coefficients obey*

$$h_0 = c \geq 0, \quad h_1 = (1 - c^2)\rho/q - 1 \geq 0, \quad h_0 \leq 4\rho^2, \quad h_1 \leq \rho^2/3.$$

For $k \geq 2$,

$$|h_k| \leq 200\rho^4 2^{-k}. \quad (\text{A.54})$$

Also, with $u = t/\rho \in [0, 4]$,

$$|h_0 - u\rho^2| \leq 2\rho^4, \quad |h_1 - \rho^2/3| \leq 100\rho^4.$$

Proof idea. Expand the scalar correction on a complex disk where all denominators stay away from zero. Cauchy's estimate bounds the entire coefficient tail, while direct differentiation identifies the two coefficients handled separately by the factory.

Proof. We use the complex bound (A.51). On $|z| = 2$, use $q/\rho \geq 1 - \rho^2/3$ to obtain

$$\left| G(z) - z - \frac{\rho^2}{3} z(1 - z^2) \right| \leq 200\rho^4, \quad |G(z) - z| \leq 4\rho^2, \quad |G(z)| < 3.$$

Here the two fifth-order Taylor remainders, after division by ρ , total at most $170\rho^4$; the denominator and the quadratic correction increase this to less than $200\rho^4$. The denominator $1 + cG$ has modulus at least $1 - 12\rho^2 > 49/50$. The sharper numerator bound $|G + c| \leq 2 + 8\rho^2$ also gives

$$|F(z)| \leq \frac{2 + 8\rho^2}{1 - 12\rho^2} < 3 \quad (|z| \leq 2).$$

Moreover

$$F - z - \frac{\rho^2}{3}z(1 - z^2) - c(1 - z^2) = G - z - \frac{\rho^2}{3}z(1 - z^2) + \frac{c\{z^2 - G^2 - cG(1 - z^2)\}}{1 + cG}.$$

The last fraction has modulus at most $4\rho^2(20\rho^2 + 60\rho^2)/(49/50)$. The entire expression is therefore bounded by $600\rho^4$. On this circle $|1 - z^2| \geq 3$, so

$$|H(z) - c - (\rho^2/3)z| \leq 200\rho^4. \quad (\text{A.55})$$

The denominators above are nonzero throughout the disk. Since $F(1) = 1$ and $F(-1) = -1$, its apparent singularities in H are removable. Cauchy's estimate now proves (A.54) and the assertion about h_1 .

Direct differentiation at zero gives the displayed values of h_0, h_1 . The real Taylor bound implies $q \leq \rho(1 - \rho^2/4)$, whereas $\rho/q \leq 1 + \rho^2/3$ by (A.22). Consequently

$$h_1 \geq \rho^2/4 - 17\rho^4 > 0,$$

using $c^2 \leq 16\rho^4$ and $\rho/q \leq 17/16$. Finally $|tq - t\rho| \leq 4\rho^4/3$, proving the remaining estimate. $\square$

Here is an explicit factory for F . Read one source sign X_1 . Put $A = h_0$, $B = h_1$, and $e_0 = 200\rho^4$. After $X_1 = +1$, choose a majority-of-three continuation with probability $2(B - A)_+$. After $X_1 = -1$, choose such a continuation with probability $2B + 2\min(A, B)$, or an OR continuation with probability $2(A - B)_+$. A majority continuation stops as soon as two signs agree. An OR continuation after the negative first sign reads one further sign and returns it.

In either case, reserve the additional probability e_0 for a tail proposal. Otherwise return X_1 . For a tail proposal, draw $K \geq 2$ with $\Pr(K = k) = 2^{1-k}$, and choose its positive or negative correction with respective probabilities

$$\frac{(h_k)_+}{200\rho^4 2^{-k}}, \quad \frac{(-h_k)_+}{200\rho^4 2^{-k}}.$$

On rejection return X_1 . On acceptance, read X_2 . If $X_1 = X_2$, return their common sign. Otherwise return the chosen sign times the product of k fresh source signs. All category probabilities are valid: their total is at most $9\rho^2 + 200\rho^4 < 1/100$. The positive-part formulas avoid deciding an exact equality between computable real numbers.

Lemma A.30 (Exactness and source cost). *The factory terminates almost surely and has mean $F(z)$. Define*

$$E(A, B, z) = B(3 - z^2) - 2Az + (A - B)_+(1 + z).$$

Its expected number of source signs is

$$A_f(\rho, t, z) = 1 + E(h_0, h_1, z) + \sum_{k \geq 2} |h_k| \{2 + k(1 - z^2)\}.$$

In particular,

$$A_f \leq 1 + \rho^2 e(u, z) + 1000\rho^4, \quad (\text{A.56})$$

where

$$e(u, z) = \begin{cases} 1 - z^2/3 - 2uz, & 0 \leq u \leq 1/3, \\ 2/3 - z/3 - z^2/3 + u(1 - z), & u \geq 1/3. \end{cases}$$

Proof idea. Condition on the first source sign to compute the majority and OR corrections. Opposite first signs expose the product correction, whose geometric mixture supplies the remaining Taylor series. The same conditioning counts every additional source request.

Proof. Conditioning on X_1 , the majority and OR continuations change its mean by precisely $(1-z^2)(A+Bz)$. The signed tail component has mean $z \pm (1-z^2)z^k/2$, since two source signs disagree with probability $(1-z^2)/2$. Its unconditional probability is $2(h_k)_+$ or $2(-h_k)_+$, independently of X_1 . Absolute convergence therefore gives mean $z + (1-z^2)H(z) = F(z)$.

Conditional on a positive first sign, a majority continuation uses $(3-z)/2$ additional signs on average; after a negative one it uses $(3+z)/2$. The OR continuation uses one. This yields $E(A, B, z)$. A tail component uses, beyond the first sign, $1 + k(1-z^2)/2$ signs on average, proving the exact cost formula. Its geometric index is finite almost surely.

The function E is continuous, with separate Lipschitz constants 2 in A and 3 in B . Hence the first two coefficient errors cost at most $304\rho^4$. The tail costs at most

$$200\rho^4 \sum_{k \geq 2} (k+2)2^{-k} = 500\rho^4.$$

Substitution of $(A, B) = (u\rho^2, \rho^2/3)$ gives (A.56). Known scalar decisions terminate almost surely by the same interval-comparison lemma as the original factory. $\square$

A.10.3 The weighted estimate

Let $k = (1-t^2)/(1-t^2q^2)$. After the internal gate, the normalized mean before the center error is $v = kF(z)$. As in Section A.10.1, concavity gives $\sum_j p_j \varphi(v_j) \leq \varphi(z)$, and the ratio $\theta/\varphi(\theta v)$ increases with θ . It therefore suffices to bound

$$W_f = A_f k \frac{\varphi(z)}{\varphi(kF(z))}.$$

Lemma A.31 (A fixed potential bound). *For $0 < \rho \leq 1/32$ and $0 \leq a \leq 4\rho$,*

$$\log W_f \leq \frac{17}{16}\rho^2 + 2^{22}\rho^4. \quad (\text{A.57})$$

For $a > 4\rho$, the original majority and race factory obeys the same weighted estimate. Negative biases are complemented. The center error multiplies the estimate by $\exp(64\varepsilon/r_\ell)$.

Proof idea. Combine the local source expansion with the logarithm of the potential ratio. Its leading quadratic form can be maximized explicitly over the normalized bias; a uniform fourth-order remainder controls the error.

Proof. Put $d = \alpha - z^2$, and

$$b(u, z) = (1-z^2)(z/3 + u) - u^2z.$$

The bounds in Lemma A.29 imply

$$|kF(z) - z - \rho^2 b(u, z)| \leq 1000\rho^4, \quad |kF(z) - z| \leq 22\rho^2.$$

For completeness, the first remainder is at most $602\rho^4 + 80\rho^4 + 17\rho^4 < 700\rho^4$: these are respectively the remainder in $F - z$, the term $t^2(F - z)$, and the difference $k - (1 - t^2)$. Here $|F - z| \leq 5\rho^2$, $|b| < 21$, and $0 \leq k - (1 - t^2) \leq 17\rho^4$. Also $\log k \leq -u^2\rho^2 + 17\rho^4$. Taylor's theorem for $\log \varphi$, whose first and second derivatives have absolute bounds $1/\eta$ and $9/(4\eta^2)$, gives

$$\log W_f \leq C_\eta(u, z)\rho^2 + \left(1017 + \frac{1000}{\eta} + \frac{1089}{2\eta^2}\right)\rho^4, \quad C_\eta = e(u, z) - u^2 + \frac{zb(u, z)}{d}.$$

At $\eta = 1/64$, the coefficient of ρ^4 is $2295289 < 2^{22}$.

We show $C_\eta \leq 17/16$ throughout $0 < \eta \leq 1/64$. For $u \leq 1/3$,

$$C_\eta = 1 - \frac{\eta z^2}{3d} - z \left(1 + \frac{\eta}{d}\right) u - \frac{\alpha}{d} u^2.$$

If $z \geq 0$, this is at most one. Otherwise maximize the quadratic over all real u . With $y = z^2$, the desired bound is equivalent to

$$P_\eta(y) = 3(1-y)(2y-1)^2 + \eta(6-35y+48y^2) + \eta^2(3-32y) \geq 0.$$

For $y = 1/2 + v$, $0 \leq v \leq 1/2$, the first term is nonnegative, the coefficient of η is $1/2 + 13v + 48v^2 \geq 1/2$, and that of η^2 is at least -29 . Thus $P_\eta \geq \eta/2 - 29\eta^2 > 0$. For $y = 1/2 - v$, $0 \leq v \leq 1/2$,

$$P_\eta \geq 6v^2 - 13\eta v + \eta/2 - 13\eta^2 = 6(v - 13\eta/12)^2 + \eta/2 - 481\eta^2/24 > 0.$$

Both inequalities use only $\eta \leq 1/64$.

For $u \geq 1/3$,

$$C_\eta = 2/3 - z/3 - \frac{\eta z^2}{3d} + \left(1 - \frac{\eta z}{d}\right)u - \frac{\alpha}{d}u^2.$$

If its unconstrained maximum occurs below $1/3$, the maximum on the half-line is attained at the boundary, already covered. Otherwise $z^2 + \eta z \leq \alpha/3$, implying $z \geq -3/5$. Completing the square and discarding the nonpositive term $\eta z^2(\eta/(4\alpha) - 1/3)/d$ gives

$$C_\eta \leq 11/12 - \left(1/3 + \frac{\eta}{2\alpha}\right)z - \frac{z^2}{4\alpha}.$$

This quadratic decreases for $z \geq -3/5$. Its value at $-3/5$ is $67/60 + (30\eta - 9)/(100\alpha) \leq 21/20 < 17/16$, since $\eta \leq 1/64 < 7/110$. This proves the small-bias assertion.

For $a > 4\rho$, monotonicity and $\tanh(4\rho) \geq 4\rho - (4\rho)^3/3$ give $t \geq (7/2)q$. Lemma A.28 gives

$$B \leq \frac{-(1-2q^2)t^2 + 3tq}{1-q^2} \leq 0,$$

because $-1/2 + 7q^2 < 0$. Its remaining majority contribution is at most $\rho^2 + 4\rho^4/\eta^2$, covered by (A.57). The center shift and the factor $\theta = q/r_\ell \leq 1$ are treated exactly as in (A.49). $\square$

A.10.4 Consequences for depth

Use the original factory at levels $\ell \leq J_0$, and the new scalar factory above J_0 when $\rho \leq 1/32$ and $|a| \leq 4\rho$. All tests use stored rational quantities. Outside this case retain the original factory.

Theorem A.32 (Improved critical bound). *For row norms at most one, this sampler is exact and satisfies*

$$\mathbb{E}N_{\text{calls}} = O((D+1)^{35/32}).$$

Its preprocessing obeys the original bound, and its expected online bit work is $O(\mathcal{B}^4(D+1)^{35/32})$. Preprocessing may select between this bound and the quadratic critical bound of Theorem A.1.

Proof idea. Use the fused estimate only above a fixed cutoff. Weighted offspring products across those layers give a power of depth, and the final output gate reduces that power by one half. The original sampler bounds the finite prefix.

Proof. When $D > J_0$, every squared row radius above level J_0 is at most $2/J_0 < 1/1024$. On this interval the derivative of $c_0v + M_0v^2$ is at most $17/16 + 4M_0/J_0 = 25/16 < 2$. The cumulative rounding and center terms are each less than $1/100$, as in Section A.10.1. The ideal fourth-power contribution to a suffix starting at $k \geq J_0$ is at most

$$\frac{9M_0}{4(k+1)} \leq \frac{9}{32} < 1.$$

Consequently its weighted offspring product is at most

$$e^2 \left(\frac{D+1}{k+1} \right)^{51/32}.$$

The original normalized subtree bound $36(J_0 + 1)^{5/2}$ applies below the cutoff because those levels retain the original factory. Keep the root factor $r_D \leq 2/\sqrt{D+1}$. Since $\max \varphi / \min \varphi = \sqrt{65} < 9$, $e^2 < 8$, and $\sum_{j \geq 1} j^{-51/32} < 3$,

$$\mathbb{E}N_{\text{calls}} \leq 5617(J_0 + 1)^{29/32}(D + 1)^{35/32} < 10^{11}(D + 1)^{35/32}.$$

The final inequality uses $J_0 + 1 \leq 2^{26}$, so $(J_0 + 1)^{29/32} < 2^{24}$. For $D \leq J_0$, the original $80(D + 1)^2$ bound is smaller than the same displayed estimate. Conditional charging and almost-sure termination are unchanged. $\square$

Proposition A.33 (Improved fixed supercritical rate). *For $s = 1 + \delta$, $0 < \delta \leq 2^{-16}$, let r_* be the positive fixed point of $r = \tanh(sr)$, and put $u_* = sr_*$. There is an explicitly computable rate $\Gamma_s = (17/16)u_*^2 + O(u_*^4)$ such that, for every network with actual maximum row norm $S \leq s$, the sampler with reference gain $\max\{1, S\}$ obeys*

$$\mathbb{E}N_{\text{calls}} = O_s(e^{D\Gamma_s}), \quad \frac{51}{16}\delta \leq \Gamma_s \leq \frac{51}{16}\delta + O(\delta^2).$$

The constants in the rate estimates are absolute; explicit choices for the rate and prefactor are given in the proof. The prefactor is for fixed s ; the original uniform transition-window theorem remains in force.

Proof idea. Compare all sufficiently late radii with the positive fixed point. Contraction bounds the accumulated transient error, so the limiting weighted offspring factor determines the exponential rate.

Proof. Use

$$\Gamma_s = \frac{17}{16}u_*^2 + 2^{22}u_*^4, \quad H_0 = 2^{14},$$

and the computable prefactor

$$C_\delta = 1 + \sqrt{65} \exp\left(1 + \frac{3H_0}{200} + \frac{8H_0}{\delta}\right) \left(\exp(23(J_0 + 1)) + \frac{1}{1 - \exp(-51\delta/16)}\right).$$

For $D > J_0$ and $j \geq J_0$, the comparison radii satisfy

$$s^2 r_j^2 \leq 4\delta + 6/J_0 + 24/(100J_0) < 1/1024.$$

On $[0, 1/1024]$, the derivative of $17v/16 + 2^{22}v^2$ is at most $17/16 + 8192 < H_0$. Write $\lambda = s(1 - r_*^2)$. The ideal-radius contraction and the rounding proof give the suffix bound

$$\exp(1 + 3H_0/200 + 8H_0/\delta) \exp((D - k)\Gamma_s).$$

Here $1 - \lambda \geq \delta$, as in Lemma A.16. Below J_0 , the original normalized subtree has mean at most $\exp(23(J_0 + 1))$. Weighted charging and a geometric sum prove the stated bound, including $D \leq J_0$. Finally $3\delta \leq u_*^2 \leq 3\delta + (3/2)\delta^2$, and $3 + (3/2)\delta \leq 25/8$. Thus the quadratic error is at most

$$(51/32 + 2^{22}(25/8)^2) \delta^2 < 40960002\delta^2.$$

$\square$

A.10.5 Finite-bit implementation

Only the small-radius branch is new. The probabilities h_0, h_1 , their positive parts, and their minimum are computable with the existing scalar interval oracles; all these operations are Lipschitz. No sign or equality oracle is required. For the tail coefficients there is a direct formal-series oracle. Set $A = \rho/q$ and

$$X(z) = \frac{e^{2\rho z} - 1}{2\rho} = \sum_{k \geq 1} \frac{(2\rho)^{k-1}}{k!} z^k.$$

Then

$$F(z) = \frac{c + (A + c\rho)X(z)}{1 + (\rho + cA)X(z)}.$$

Its denominator has constant coefficient one. Truncating at degree K , applying the quotient recurrence, and then dividing $F - z$ formally by $1 - z^2$ computes h_K . The computation rounds after each arithmetic operation, as in Lemma A.19; it never carries a growing common rational denominator.

Here are explicit error bounds for this new quotient. Let $\zeta = 2^{-V}$ be the rounding unit and first enclose A, c to absolute error ζ . Computing A needs only $O(\mathcal{B})$ additional guard bits because $q \geq \rho/2$ and a nonzero dyadic row has $\log(1/\rho) = O(\mathcal{B})$. The recurrence $x_1 = 1$, $x_{j+1} = 2\rho x_j/(j+1)$ has multipliers at most $1/32$, so each computed x_j has error at most 2ζ . The numerator and denominator coefficients of the displayed quotient then have error at most 10ζ apiece. Its true nonconstant denominator norm is less than $1/8$:

$$\rho + cA < 1/16, \quad \sum_{j \geq 1} |x_j| < 2.$$

Also $|F(z)| < 3$ on $|z| = 2$, so every true coefficient of F has absolute value at most three.

If the previous coefficient errors are at most $E \leq 1$, one quotient step has error at most $E/8 + (50K + 20)\zeta$, including the rounded products and sum. Induction bounds every error through degree K by $100(K + 1)\zeta$. The final recurrence $h_j = F_j - \mathbf{1}_{\{j=1\}} + h_{j-2}$, with missing negative-index terms zero, gives error at most $101(K + 1)^2\zeta$. Taking $V = m + 4\mathcal{B} + 4K + 64$, and replacing m by $m + 2$ for an enclosure of width 2^{-m} , validates the preceding induction and supplies the requested certificate. There are $O(K^2)$ operations on $O(\mathcal{B} + K + m)$ -bit integers. Schoolbook multiplication and division therefore give $O((\mathcal{B} + K + m)^4)$ bit work. The acceptance denominator $200\rho^4 2^{-K}$ requires only $O(\mathcal{B} + K)$ further precision bits, preserving that bound.

The geometric tail proposal gives finite moments of every order for k . Interval comparisons have geometric precision tails. The conditional distribution of a coefficient sign is implemented through its two nonnegative parts in the displayed probabilities. An accepted branch never requires resolving whether a zero coefficient is positive. The original reached-trial lemma charges all arithmetic, rejected proposals, and subsequent source work. Thus a reached invocation costs $O(\mathcal{B}^4)$ expected scalar bit work, including every precision refinement. No table of coefficients is added to preprocessing storage.

A.11 A linear depth bound with zero biases

Theorem A.34 (Unrestricted zero-bias critical networks). *For any zero-bias network with absolute row sums at most one, there is an exact sampler with the preprocessing bounds of Theorem A.1 and*

$$\mathbb{E}N_{\text{calls}} \leq 128(D + 1), \quad \mathbb{E}[\text{online bit work}] = O(\mathcal{B}^4(D + 1)).$$

Weights can have arbitrary signs. The bound is uniform in the context.

Proof idea. With zero biases, every stored center is zero and no bias race is needed. The output-radius gate and the majority costs then telescope to a linear bound in depth.

Proof. Use reference gain one and the stored radii of (A.11). Store every center as exactly zero, which is valid because all biases are zero and $\tanh$ is odd. At a row of radius ρ , the normalized call has an active gate $\tanh \rho/r_\ell \leq 1$, followed by the majority factory of Lemma A.9; there is no bias race. Its expected number of reached children is at most

$$1 + \rho^2 + \rho^4 \leq \exp(r_{\ell-1}^2 + r_{\ell-1}^4).$$

The actual top output retains its gate $\tanh \rho \leq r_D$.

Write $q_j = 3/(2j + 3)$. Lemma A.13 gives $0 \leq r_j^2 - R_j^2 \leq 6D\varepsilon$ and $r_j \leq 1$. The derivative of $v + v^2$ on $[0, 1]$ is at most three, so the rounding contribution to any suffix logarithm is at most $18D^2\varepsilon < 1/50$. The integral estimates in (A.30) give, for $0 \leq k < D$,

$$\sum_{j=k}^{D-1} (r_j^2 + r_j^4) \leq \frac{3}{2} \log \frac{D+1}{k+1} + \frac{9}{4(k+1)} + \frac{1}{50} \leq \frac{3}{2} \log \frac{D+1}{k+1} + 3.$$

The conditional charging lemma therefore bounds the expected total number of calls, with the top call charged separately, by

$$1 + r_D \sum_{k=0}^{D-1} e^3 \left(\frac{D+1}{k+1} \right)^{3/2} \leq 1 + 2e^3(D+1) \sum_{k \geq 1} k^{-3/2} \leq 128(D+1).$$

Here $r_D \leq 2/\sqrt{D+1}$, $\sum k^{-3/2} \leq 3$, and $e^3 < 21$. The scalar and index implementations in Section A.7 cost $O(\mathcal{B}^4)$ conditionally per reached call. Their geometric refinement tails and Lemma A.10 give the bit bound. The majority identity proves exactness; finite depth and almost sure termination of each scalar call prove almost sure termination of the entire sampler. $\square$

A.12 Sharp bounds for a scalar mean chain

A whole-chain factory improves on independent layer composition when the network depends on the input through one scalar mean. This section proves matching query bounds for that family.

Let $F_{s,0}(z) = z$, $F_{s,d+1}(z) = \tanh(sF_{s,d}(z))$, where $s > 0$ is dyadic. The analytic identities below hold for every positive real s ; the dyadic assumption supplies an executable parameter description. A source request returns an independent sign of unknown mean $z \in [-1, 1]$. Let $U_D(s)$ be the infimum, over exact factories for a sign of mean $F_{s,D}(z)$, of the worst-mean expected source count. Auxiliary randomness and arithmetic are excluded from this particular query measure.

Theorem A.35 (Uniform scalar complexity). *Set $\lambda = s^D$ and $A = \sum_{j=1}^D s^{2j}$. Then, with universal constants,*

$$\frac{1}{25} \lambda \sqrt{1+A} \leq U_D(s) \leq 2\lambda \sqrt{1+A}. \quad (\text{A.58})$$

Thus the dependence on the gain and depth is determined jointly, including when s approaches one with D .

Theorem A.36 (The scalar phase diagram). *For $D \geq 1$,*

$$(1-s)s^D \leq U_D(s) \leq \frac{s^D}{1-s^2}, \quad 0 < s < 1, \quad (\text{A.59})$$

$$\frac{\sqrt{D}}{8} \leq U_D(1) \leq 2\sqrt{D}, \quad (\text{A.60})$$

$$s^{2D} \leq U_D(s) \leq \frac{2s}{\sqrt{s^2-1}} s^{2D}, \quad s > 1. \quad (\text{A.61})$$

The upper bounds are realized by an exact majority mixture for the entire composition.

A.12.1 Positivity for the entire composition

A smooth function f on $(0, \infty)$ is *completely monotone* if $(-1)^k f^{(k)} \geq 0$ for all k . A nonnegative smooth function is a *Bernstein function* if its derivative is completely monotone. Two elementary closure facts will suffice. Products of completely monotone functions are completely monotone, by Leibniz's rule. If C is completely monotone and g is Bernstein, then $C \circ g$ is completely monotone. Indeed, a term in the Faà di Bruno formula has the form

$$C^{(k)}(g) \prod_{j \geq 1} (g^{(j)})^{m_j}, \quad \sum_j m_j = k, \quad \sum_j j m_j = q,$$

and its sign is $(-1)^{k+\sum_j (j-1)m_j} = (-1)^q$. The coefficients in that formula are nonnegative. Differentiating a composition then shows that Bernstein functions are closed under composition.

Put

$$C(w) = \operatorname{sech}^2 \sqrt{w}, \quad T(w) = \frac{\tanh \sqrt{w}}{\sqrt{w}}, \quad g(w) = \tanh^2 \sqrt{w}.$$

The cosh product shows that C is completely monotone: each finite product of factors $(1 + \alpha w)^{-2}$ has this property, and local uniform convergence permits differentiation. Its logarithmic derivative gives

$$T(w) = \sum_{j \geq 0} \frac{8}{\pi^2(2j+1)^2 + 4w},$$

which is completely monotone as well. Since $g' = TC$, g is Bernstein. Positive scaling preserves these properties. Writing $g_s(w) = g(s^2 w)$, $C_s(w) = C(s^2 w)$, we obtain

$$H_{D,s}(w) := F'_{s,D}(\sqrt{w}) = s^D \prod_{j=0}^{D-1} C_s(g_s^{\circ j}(w)), \quad (\text{A.62})$$

a completely monotone function.

This derivative has the analytic domain needed for its Taylor series. If $\sqrt{w} = x + iy$ with $\Re w > 0$, then $x > |y|$, and

$$\Re \tanh^2(x + iy) = \frac{\sinh^2(2x) - \sin^2(2y)}{(\cosh(2x) + \cos(2y))^2} > 0.$$

Thus g_s maps the right half-plane into itself, and $H_{D,s}$ is holomorphic there. Its removable value at zero is s^D . Consequently

$$H_{D,s}(1 - v) = \sum_{k \geq 0} d_k v^k, \quad d_k \geq 0, \quad \sum_k d_k = s^D.$$

The coefficient signs follow from complete monotonicity at $w = 1$; the sum follows by monotone convergence as $v \uparrow 1$.

Let $R = F_{s,D}(1) > 0$. The same derivative integration as in Lemma A.8 proves

$$\frac{F_{s,D}(z)}{R} = \sum_{k \geq 0} \pi_k M_k(z), \quad \pi_k = \frac{d_k}{RC_k}, \quad \sum_k \pi_k = 1. \quad (\text{A.63})$$

Open a gate of probability R . On a closed gate return a fair sign; on an open gate draw $K \sim \pi$ and return its majority. This sign has mean $F_{s,D}(z)$.

A.12.2 The cost of the global mixture

Define $A_D(s) = \sum_{j=1}^D s^{2j}$. We count every vote of the full majority; stopping early can only lower the cost. Integration by parts gives the exact full-majority mean

$$Q_{D,s} = s^D + \int_0^1 \frac{s^D - F'_{s,D}(z)}{z^2} dz. \quad (\text{A.64})$$

The gate probability R cancels the normalization in (A.63). For $z \geq 0$, $F_{s,j}(z) \leq s^j z$, and

$$\frac{F'_{s,D}(z)}{s^D} = \prod_{j=1}^D (1 - F_{s,j}(z)^2).$$

Using $1 - \prod_j (1 - u_j) \leq \sum_j u_j$ for $0 \leq u_j \leq 1$,

$$0 \leq s^D - F'_{s,D}(z) \leq s^D \min\{1, A_D(s)z^2\}.$$

Hence

$$Q_{D,s} \leq s^D(1 + A_D(s)), \quad Q_{D,s} \leq 2s^D \sqrt{A_D(s)} \quad \text{if } A_D(s) \geq 1. \quad (\text{A.65})$$

The first bound holds for every $A_D(s)$. It implies the upper bound in (A.58): if $A \leq 1$, $1 + A \leq 2\sqrt{1 + A}$, and the second bound handles $A \geq 1$. For $s < 1$, $1 + A_D(s) \leq (1 - s^2)^{-1}$. At $s = 1$, $A_D(1) = D$. For $s > 1$, $A_D(s) \leq s^2 s^{2D} / (s^2 - 1)$. These inequalities prove all upper bounds in Theorem A.36.

A.12.3 Lower bounds without a stopping-time regularity assumption

Consider any almost surely terminating scalar factory whose worst-mean expected source count is $M < \infty$. Truncate it after T source requests and return a fair sign if it has not terminated. Its mean f_T is a polynomial and

$$\sup_{z \in [-1, 1]} |f_T(z) - f(z)| \leq M/T.$$

The finite transcript proof of Lemma A.17 gives

$$|f'_T(z)| \leq \sqrt{\frac{M}{1-z^2}}, \quad |f''_T(z)| \leq \frac{2M}{1-z^2}.$$

Integrate the first bound on a short interval and pass to the uniform limit. For the second, use

$$f_T(z+h) - 2f_T(z) + f_T(z-h) = \int_{-h}^h (h-|u|) f''_T(z+u) du.$$

Pass to the limit in T , divide by h^2 , and then let $h \rightarrow 0$. For twice differentiable f ,

$$M \geq (1-z^2) f'(z)^2, \quad M \geq \frac{1-z^2}{2} |f''(z)|. \quad (\text{A.66})$$

These are bounds on the global worst-mean cost M . No differentiation through an unbounded runtime is needed.

Proof idea. The reciprocal-square recurrence compares the chain's range with its derivative at zero. An input interval on which the derivative stays large then forces any exact factory to use enough source samples.

Lower bound in Theorem A.35. The reciprocal-square inequalities

$$u^{-2} + \frac{2}{3} \leq \coth^2 u \leq u^{-2} + 1,$$

from (A.27) and $\sinh u \geq u$, give the range comparison

$$\frac{\lambda}{\sqrt{1+A}} \leq R = F_{s,D}(1) \leq \frac{\lambda}{\sqrt{1+\frac{2}{3}A}}. \quad (\text{A.67})$$

Indeed, apply the inequalities to each step of the radius recurrence and multiply the final reciprocal inequality by s^{2D} . The accumulated geometric sum is exactly A .

Endpoint coupling (Section A.6) gives $M \geq R$. If $A < 24$, this is at least $\lambda\sqrt{1+A}/25$. If $A \geq 24$, equation (A.67) gives $T = 2R/\lambda \leq 2/\sqrt{17} < 1/2$. The target has derivative λ at zero and range $[-R, R]$, so the bending bound (A.37), with the cost bound (A.66), gives

$$M \geq \frac{3\lambda^2}{16R} \geq \frac{3}{16} \sqrt{\frac{2}{3}} \lambda \sqrt{1+A} \geq \frac{1}{25} \lambda \sqrt{1+A}.$$

□

For $f = F_{s,D}$, $f'(0) = s^D$, proving the supercritical lower bound. For $s = 1$, (A.34) gives $F_{1,j}(z)^2 \geq (z^{-2} + j)^{-1}$. Take $z_0 = 1/(2\sqrt{D})$. Then

$$F'_{1,D}(z_0) = \prod_{j=1}^D (1 - F_{1,j}(z_0)^2) \leq (1 - 1/(5D))^D \leq e^{-1/5} \leq 5/6.$$

Since $F'_{1,D}(0) = 1$, the mean value theorem gives some $z_* \in (0, z_0)$ with $|F''_{1,D}(z_*)| \geq \sqrt{D}/3$. Here $1 - z_*^2 \geq 3/4$, so (A.66) yields $M \geq \sqrt{D}/8$.

For $0 < s < 1$, endpoint coupling with all-plus and all-minus source signs shows that some source request occurs with probability at least $R = F_{s,D}(1)$, the difference of the two target Bernoulli probabilities. For $u \geq 0$, the bound $\tanh u \geq u/(1+u)$ follows from $e^{2u} \geq 1 + 2u$. Writing $R_j = F_{s,j}(1)$,

$$\frac{s^{j+1}}{R_{j+1}} \leq \frac{s^j}{R_j} + s^{j+1}.$$

Thus $s^D/R \leq 1 + \sum_{j=1}^D s^j \leq (1-s)^{-1}$. This proves the remaining lower bound.

A.12.4 Finite inputs and arbitrary depth

For a vector of m signs, independently choosing a uniform coordinate on each request supplies the unknown-mean source with $z = m^{-1} \sum_j x_j$. Repeated coordinate choices remain independent conditional on the fixed input, because the indices are independent. A second option opens the same gate R , then reads all m signs and evaluates the scalar function if the gate opens. It uses mR expected probes.

Theorem A.37 (Uniform finite-input complexity). *Let $m \geq 2$ be even, $D \geq 1$, $s > 0$ dyadic, $\lambda = s^D$, $A = \sum_{j=1}^D s^{2j}$, and*

$$\Xi_{s,D,m} = \lambda \min \left\{ \sqrt{1+A}, \frac{m}{\sqrt{1+A}} \right\}.$$

For the mean-chain target on m input signs, the optimal worst-input expected number $Q_{s,D,m}^$ of distinct probes obeys*

$$\frac{1}{128} \Xi_{s,D,m} \leq Q_{s,D,m}^* \leq 2 \Xi_{s,D,m}. \quad (\text{A.68})$$

The constants are independent of s, D, m .

Proof idea. Choose between a global majority factory and a rare complete input read. For the lower bound, apply the transcript score to the empirical mean and use its fourth moment to retain the dependence on the finite input size.

Proof. Choose the global factory when $1 + A \leq m$, and the gated scan otherwise. Equations (A.65) and (A.67) give the upper bound in both cases. This choice uses an exact rational comparison before seeing the input.

For the lower bound, endpoint coupling gives $Q^* \geq R$. If $A < 100$, then

$$R \geq \frac{\lambda}{\sqrt{1+A}} \geq \frac{\Xi_{s,D,m}}{101}.$$

Suppose now that $A, m \geq 100$. Equation (A.36) gives $h = H'(0) \geq \lambda/\sqrt{1+3A/m}$. Using (A.67), the interval $T = 2R/h$ has

$$T^2 \leq \frac{4(1+3A/m)}{1+\frac{2}{3}A} \leq 6/A + 18/m \leq 0.24 < 1/4.$$

The bending bound (A.37) and the second transcript bound yield

$$Q^* \geq \frac{3h^2}{16R} \geq \frac{3\lambda\sqrt{1+\frac{2}{3}A}}{16(1+3A/m)} \geq \frac{3}{64} \sqrt{\frac{2}{3}} \Xi_{s,D,m}.$$

For the last inequality, $\sqrt{1+2A/3} \geq \sqrt{2/3}\sqrt{1+A}$ and $1+3A/m \leq 4 \max\{1, (1+A)/m\}$.

It remains to consider $m < 100$. At a balanced input, flipping one coordinate changes the plus probability by $F_{s,D}(2/m)/2$. Lemma B.7, summed over the m coordinates, gives

$$Q^* \geq \frac{m}{2} F_{s,D}(2/m) \geq \frac{\lambda}{\sqrt{1+4A/m^2}}.$$

If $A \leq m^2$, this is at least $\lambda/\sqrt{5}$, while $\Xi_{s,D,m} \leq \lambda\sqrt{m} < 10\lambda$. If $A > m^2$, it is at least $m\lambda/\sqrt{5A} \geq \Xi_{s,D,m}/\sqrt{5}$. Each displayed constant is larger than $1/128$. $\square$

Theorem A.38 (Finite critical mean chain). *For even $m \geq 2$, $D \geq 1$, and $f(x) = F_{1,D}(m^{-1} \sum_j x_j)$, the optimal worst-input expected number of distinct probes lies between*

$$\frac{1}{100} \min\{\sqrt{D}, m/\sqrt{D}\} \quad \text{and} \quad 2 \min\{\sqrt{D}, m/\sqrt{D}\}.$$

For every fixed $s > 1$, the corresponding order is $\Theta_s(\min\{s^{2D}, m\})$.

Proof idea. Combine the two available upper bounds. For the lower bound, run the case split of the preceding proof at $s = 1$ and track the constants against $\sqrt{D}$, including the regime where the number of inputs is smaller than the depth.

Proof. The critical upper bound follows by choosing between $Q_{D,1} \leq 2\sqrt{D}$ and $mR_D \leq 2m/\sqrt{D}$. The supercritical claim follows from (A.6), (A.65), and $mR \leq m$.

For the critical lower bound set $\psi = \min\{\sqrt{D}, m/\sqrt{D}\}$, and use the proof of Theorem A.37 at $s = 1$, where $\lambda = 1$ and $A = D$. If $D < 100$, endpoint coupling and (A.67) give $\sup_x \mathbb{E}[Q | x] \geq R_D \geq (1+D)^{-1/2} \geq 1/10$, while $\psi \leq \sqrt{D} < 10$. If $D, m \geq 100$, its bending case gives

$$\sup_x \mathbb{E}[Q | x] \geq \frac{3\sqrt{1+2D/3}}{16(1+3D/m)} \geq \frac{3}{64\sqrt{3/2}} \psi > \frac{\psi}{100},$$

since $\sqrt{1+2D/3} \geq \sqrt{2/3}\sqrt{D}$ and $\sqrt{D}/(1+3D/m) \geq \psi/4$. If $m < 100$, its balanced-input bound reads

$$\mathbb{E}[Q | x_{\text{balanced}}] \geq \frac{m}{2} F_{1,D}(2/m) \geq (1+4D/m^2)^{-1/2}. \quad (\text{A.69})$$

If $D \leq m^2$, this is at least $1/\sqrt{5}$, while $\psi \leq \sqrt{m} < 10$. If $D > m^2$, it is at least $m/\sqrt{5D} = \psi/\sqrt{5}$. All cases imply the stated constant. $\square$

For arbitrary critical networks one may also enter the root factory with probability $R \leq r_D$, then read the whole context and evaluate its conditional scalar probability. This gives at most $2n/\sqrt{D}+1$ expected input probes. Together with Theorem A.38, it proves full-class query order $\Theta(n/\sqrt{D})$ for $D \geq n$, whenever the uniform first row is on the parameter grid. This statement concerns queries; evaluating on the rare open branch may still be arithmetically expensive.

A.13 Residual layers and effective depth

Consider the architecture

$$h_{\ell,i} = (1-\alpha)h_{\ell-1,i} + \alpha \tanh\left(b_{\ell,i} + \sum_j w_{\ell,ij} h_{\ell-1,j}\right), \quad 0 \leq \alpha \leq 1. \quad (\text{A.70})$$

The scalar α is known and belongs to the same $2^{-b}\mathbb{Z}$ grid as the network parameters. For a more finely encoded α , replace b by the larger fractional bit length in the definition of $\mathcal{B}$ and in the preprocessing bound. Put $\tau = \alpha D$. A skip step preserves the coordinate, so a geometrically distributed jump can bypass many layers. This observation is essential when D is large and α is small.

Theorem A.39 (Residual bound). *Suppose that the actual maximum row norm is $S \leq 2$ and the biases have absolute value at most one. The exact sampler can be extended to (A.70) with $O(Dn^2\mathcal{B}^6)$ preprocessing and $O(Dn^2\mathcal{B})$ stored bits. Let J count the root operation, the geometric-jump proposals, and the input leaves in the compressed recursion. Then*

$$\mathbb{E}J = O((1+\tau)^2) \quad (S \leq 1), \quad (\text{A.71})$$

$$\mathbb{E}J = O((1+\tau)^2 e^{7000(S-1)\tau}) \quad (1 < S \leq 2). \quad (\text{A.72})$$

With the scalar and geometric implementations described below, the expected online bit work is $O(\mathcal{B}^4 \mathbb{E}J)$. All implied constants are absolute; explicit choices are given in the proof. The product bound (A.78) is more informative for a particular network.

For fixed $1 < s \leq 2$, the lower bound proved below has exponential factor $(1 + \alpha(s - 1))^{2D}$. Since

$$\frac{\alpha(s - 1)}{2} \leq \log(1 + \alpha(s - 1)) \leq \alpha(s - 1),$$

the upper and lower exponential scales agree up to absolute constant factors. At unit norm, the upper cost is polynomial in $1 + \alpha D$, and Theorem A.51 lowers its exponent to $35/32$. The critical exponent remains open for the full class.

A.13.1 Intervals whose endpoints stay inside the output range

Use $g = \max\{1, S\}$ and $\varepsilon = 2^{-P}$ with the precision P in (A.10). In particular, $\varepsilon \leq [100(D + 1)^4]^{-1}$. Set $r_0 = 1$, $c_{0,i} = 0$, and choose dyadic radii satisfying

$$f_{\alpha,g}(r_{\ell-1}) + 2\alpha\varepsilon \leq r_\ell \leq f_{\alpha,g}(r_{\ell-1}) + 3\alpha\varepsilon, \quad f_{\alpha,g}(r) = (1 - \alpha)r + \alpha \tanh(gr). \quad (\text{A.73})$$

All stored radii and centers are rounded onto the fixed grid $2^{-(P+b+6)}\mathbb{Z}$. Since a nonzero α is at least 2^{-b} , this grid is finer than $\alpha\varepsilon/64$. Approximating the displayed scalar expression and then adding $5\alpha\varepsilon/2$ supplies (A.73) on this grid. Denominators therefore do not grow with the layer index. When $\alpha = 0$, the network is the identity and the sampler returns x_1 immediately. Hence assume $\alpha > 0$ below.

For a row, put $p = 1 - \alpha$ and retain the definitions σ, a, ρ, m, R from (A.12) and (A.13). The residual interval has midpoint and radius

$$M = pc_{\ell-1,i} + \alpha m, \quad V = pr_{\ell-1} + \alpha R.$$

Approximate M to error at most $\alpha\varepsilon$, then project that approximation onto $[-1 + r_\ell, 1 - r_\ell]$ to obtain $c_{\ell,i}$. The resulting invariant is

$$|h_{\ell,i} - c_{\ell,i}| \leq r_\ell, \quad |c_{\ell,i}| + r_\ell \leq 1. \quad (\text{A.74})$$

Indeed, both endpoints $M \pm V$ lie in $[-1, 1]$, because the previous interval and both $\tanh$ endpoints do. Also $V + \alpha\varepsilon \leq r_\ell$. Before projection the center belongs to $[M + V - r_\ell, M - V + r_\ell]$. Projection onto $[-1 + r_\ell, 1 - r_\ell]$ preserves membership in this interval: its upper endpoint is at least $-1 + r_\ell$, and its lower endpoint is at most $1 - r_\ell$. This proves both inequalities in (A.74). The projection may move the center by more than the approximation error. The unweighted offspring proof below does not require a small displacement from the midpoint.

The normalized residual sampler chooses the same-coordinate skip with probability

$$p_\ell = \frac{pr_{\ell-1}}{r_\ell}, \quad (\text{A.75})$$

and the centered $\tanh$ factory with probability $\alpha R/r_\ell$. Its two constant masses complete the mixture and contribute mean $(M - c_{\ell,i})/r_\ell$. The interval invariant makes these masses nonnegative. Its mean is therefore $(h_{\ell,i} - c_{\ell,i})/r_\ell$. At the root, open this normalized sampler with probability r_D and otherwise use constant masses $(1 - r_D \pm c_{D,1})/2$. The second invariant in (A.74) makes this root mixture valid.

A.13.2 Compressing the skips

Let $R_0 = 1$ and $R_{j+1} = f_{\alpha,g}(R_j)$. This ideal sequence is decreasing. Concavity gives $f'_{\alpha,g}(R_j) \leq R_{j+1}/R_j \leq 1$, so the proof of Lemma A.13 applies with per-step error $3\alpha\varepsilon$:

$$0 < R_j \leq r_j \leq 1, \quad 0 \leq r_j - R_j \leq 3j\alpha\varepsilon. \quad (\text{A.76})$$

The upper bound one follows from $f_{\alpha,g}(1) + 3\alpha\varepsilon < 1$. Moreover $\tanh(gr_j) \leq r_j$: the ratio $\tanh(gu)/u$ decreases with u , and this inequality holds at R_j because the ideal sequence decreases.

For $g = 1$, Jensen's inequality applied to $u \mapsto (1 + u)^{-1/2}$ and the $\tanh$ lower bound gives $f_{\alpha,1}(r) \geq r/\sqrt{1 + \alpha r^2}$. Monotonicity in g consequently gives

$$r_j \geq R_j \geq (1 + \alpha j)^{-1/2} \quad (g \geq 1). \quad (\text{A.77})$$

Writing $r_\ell = f_{\alpha,g}(r_{\ell-1}) + e_\ell$, $0 \leq e_\ell \leq 3\alpha\varepsilon$, one obtains

$$1 - p_\ell = \alpha + \frac{p(r_\ell - r_{\ell-1})}{r_\ell} \leq \alpha + \frac{3\alpha\varepsilon}{r_\ell} \leq 2\alpha.$$

Thus $h = \min\{2\alpha, 1\}$ dominates every non-skip hazard. From the current layer, draw a truncated geometric waiting time with success probability h . Bypass the intervening layers. At a proposal layer ℓ , retain the proposal with probability $(1 - p_\ell)/h$. A rejected proposal takes one skip step; a retained proposal uses the conditional non-skip mixture. If the waiting time passes layer zero, probe the current input coordinate. Independent geometric waiting times and thinning reproduce the original skip probabilities exactly.

Truncated geometric sampling does not scan the skipped layers. For a proposed gap $k \leq D$, evaluate $(1 - h)^k$ by fixed-point binary powering and compare it with a lazy uniform random number. Binary search locates the inverse distribution function. There are at most $D + 1$ boundaries. At precision $m + 3\lceil \log_2(D + 2) \rceil$ the probability that any boundary remains ambiguous is $O(2^{-m})$. Binary powering needs $O(\log D)$ multiplications, and binary search needs $O(\log D)$ comparisons. Schoolbook arithmetic with $O(\mathcal{B} + m)$ guarded bits therefore gives $O((\mathcal{B} + m)^4)$ work per refinement round and $O(\mathcal{B}^4)$ expected work. The arithmetic computes certified intervals and never decides an equality of real numbers.

A.13.3 A product bound for the compressed recursion

For $0 < r \leq 1$ and $1 \leq g \leq 2$, define

$$\chi_g(r) = \min \left\{ \frac{\tanh(gr)}{r} \exp \left(\frac{5}{3} g^2 r^2 + g^4 r^4 \right), g(1 + g^2 r^2) e^{g^2 r^2} \right\},$$

$$G_j = 1 - \alpha + \alpha \chi_g(r_j).$$

The first expression is Lemma A.11 before dividing by the new radius. For the second expression, the full majority has expected source count at most $\rho(1 + \rho^2)/\tanh \rho$, and the joint race factor is at most e^{ρ^2} . Both bounds increase with the actual row norm, so using g is valid. The expected number of normalized children at layer $j + 1$ is at most $(r_j/r_{j+1})G_j$. After the root gate, the expected number of reached uncompressed calls at layer k is at most $r_k \prod_{j=k}^{D-1} G_j$. Each such call is a proposal with probability $h \leq 2\alpha$. Nonnegative summation and the predictable trial-cost lemma give

$$\mathbb{E}(J - 1) \leq \prod_{j=0}^{D-1} G_j + 2\alpha \sum_{k=1}^D r_k \prod_{j=k}^{D-1} G_j. \quad (\text{A.78})$$

The empty product is one. This derivation also counts branches that reach an input without any proposal.

Lemma A.40 (A radial comparison for the offspring factor). *Put $d_g(r) = 1 - \tanh(gr)/r$. Whenever $0 < r \leq 1$, $1 \leq g \leq 2$, and $d_g(r) \geq 0$,*

$$\chi_1(r) \leq 1 + 4d_1(r) + 5r^4, \quad (\text{A.79})$$

$$\chi_g(r) \leq 1 + 4d_g(r) + 5r^4 + 6000(g - 1). \quad (\text{A.80})$$

Proof idea. Relate the excess offspring cost to the amount by which one residual step decreases its radius. Elementary Taylor bounds control the critical term and its change when the row norm exceeds one.

Proof. Since $r - \tanh r = \int_0^r \tanh^2 u \, du$ and $\tanh^2 u \geq u^2/(1 + u^2) \geq u^2 - u^4$,

$$d_1(r) \geq r^2/3 - r^4/5.$$

Write $t = r^2$. All Taylor coefficients of $E(t) = \exp(5t/3 + t^2)$ are nonnegative. On $0 \leq t \leq 2/5$, they give $E(t) \leq 1 + 5t/3 + 4t^2$, because $E(2/5) = e^{62/75} < 173/75$. Here is one rational verification of this numerical inequality: $e < 68/25$ by the exponential series and its geometric tail, $e^{1/6} \geq 85/72$, and therefore

$e^{62/75} < e^{5/6} < 4896/2125 < 173/75$. Multiplication by $1 - d_1(r)$ and $d_1(r) \geq t/3 - t^2/5$ now gives $\chi_1(r) \leq 1 + 4d_1(r) + 5t^2$. For $2/5 \leq t \leq 1$, the second expression defining χ_1 and positivity of the series for $(1+t)e^t$ give $\chi_1(r) \leq 1 + 2t + (5/2)t^2$. This is at most $1 + 4d_1(r) + 5t^2$, since $(17/10)t^2 - (2/3)t \geq 0$ for $t \geq 2/5$.

Let $\delta = g - 1$. If $\delta \leq 1/10$, differentiating the first expression defining χ_g with respect to g bounds its derivative by 400. Indeed, $gr \leq 11/10$, $5g^2r^2/3 + g^4r^4 < 7/2$, and the remaining derivative factor is less than 11; $e^{7/2} < 36$ suffices. The derivative of the second expression is

$$e^{g^2r^2}(1 + 5g^2r^2 + 2g^4r^4) < 40$$

on the same interval. Both functions are therefore 400-Lipschitz in g , and so is their minimum. Also $d_1(r) \leq d_g(r) + \delta$, by the unit Lipschitz bound for $\tanh$. This proves (A.80) with 404 in place of 6000 in this case. If $\delta \geq 1/10$, the second expression gives $\chi_g(r) \leq 10e^4 < 600$, which proves the stated bound. $\square$

Proof idea. The residual radius supplies a telescoping potential for the offspring product. Summing the remaining fourth-order and rounding terms over the effective depth gives the stated bound for compressed skips.

Proof of Theorem A.39. We prove the explicit prefactors $9e^{21}$ in the critical case and $10e^{30}$ in the supercritical case. Write $d_j = d_g(r_j)$ and $F_j = f_{\alpha,g}(r_j) = r_j(1 - \alpha d_j)$. The preceding lemma, $\log(1+u) \leq u$, and $-\log(1-v) \geq v$ give

$$\log G_j \leq 4 \log \frac{r_j}{r_{j+1}} + 5\alpha r_j^4 + 6000\alpha(g-1) + \frac{12\alpha\varepsilon}{F_j}. \quad (\text{A.81})$$

Here the last term accounts for $r_{j+1} = F_j + e_{j+1}$ and $4 \log(1 + e_{j+1}/F_j) \leq 12\alpha\varepsilon/F_j$. By (A.77), its sum over all layers is less than one.

At $g = 1$, Lemma A.12 and convexity imply

$$\frac{\tanh r}{r} \leq (1 + 2r^2/3)^{-1/2} \leq 1 - r^2/5, \quad 0 \leq r \leq 1.$$

The last inequality follows from the chord between zero and one and $\sqrt{3/5} < 4/5$. Thus the ideal critical recurrence obeys $R_j \leq (1 + 2\alpha j/5)^{-1/2}$. Integral bounds and (A.76) yield

$$\alpha \sum_{j=0}^{D-1} r_j^4 \leq 4, \quad \alpha \sum_{j=1}^D r_j^5 \leq 3. \quad (\text{A.82})$$

For clarity, the ideal sums are at most $1 + \int_0^\infty (1 + 2t/5)^{-2} dt = 7/2$ and $1 + \int_0^\infty (1 + 2t/5)^{-5/2} dt = 8/3$. The rounding contributions are at most $6\varepsilon D^2$ and $15\varepsilon D(D+1)/2$, respectively. Summing (A.81) therefore gives

$$\prod_{j=k}^{D-1} G_j \leq e^{21}(r_k/r_D)^4.$$

Insert this inequality into (A.78), use (A.82), and finally use $r_D^{-4} \leq (1 + \tau)^2$. Adding the root operation proves (A.71).

For $g = 1 + \delta$, the unit Lipschitz bound gives $d_g(r) \geq r^2/5 - \delta$. Since $0 \leq d_j \leq 1$,

$$\alpha r_j^2 d_j \leq r_j^2 - F_j^2 \leq r_j^2 - r_{j+1}^2 + 2e_{j+1}.$$

Consequently

$$\alpha \sum_{j=k}^{D-1} r_j^4 \leq 5 + 5\delta\alpha(D-k) + 30\alpha(D-k)\varepsilon.$$

Equation (A.81) now gives

$$\prod_{j=k}^{D-1} G_j \leq e^{30} (r_k/r_D)^4 e^{6025\delta\alpha(D-k)}.$$

Put $a = 1 + \alpha\delta$. Concavity and $\tanh(gr) \leq g \tanh r$ show $f_{\alpha,g}(r) \leq a f_{\alpha,1}(r)$. Induction gives $R_j^{(g)} \leq a^j R_j^{(1)}$. Hence, using the same rounding estimates,

$$\alpha \sum_{j=1}^D r_j^5 \leq 4e^{5\delta\tau}.$$

Substitution in (A.78) proves (A.72), with room in the displayed constants.

The normalized construction is exact by induction on the depth. Geometric thinning preserves its transition probabilities. Almost sure termination follows first for the finite-depth recursion and then for each scalar comparison; no expected-work bound is assumed in this argument. Each reached proposal has conditional expected scalar cost $O(\mathcal{B}^4)$ for $g \leq 2$. The hazard ratio introduces no small- α division in the cost: $\tanh(gr)/r \geq \tanh 1 > 3/4$ implies $1 - p_\ell \geq \alpha/2$ under the chosen rounding precision. The predictable trial-cost lemma charges scalar operations, weighted index selection, and geometric jumps by $O(\mathcal{B}^4 \mathbb{E}J)$. Centers, radii, and row data use $O(\mathcal{B})$ bits; their preprocessing fits the stated budget. $\square$

A.13.4 A simple construction for large row norms

The preceding estimates use small uncertainty radii to exploit near-criticality. A simpler construction gives a useful branch bound for every finite norm s .

Proposition A.41 (Quadratic single-row branch cost). *For a row of norm σ and bias b_0 , set $u = \sigma + |b_0|$. There is an exact $\tanh$ -row sampler using at most $2\sigma(1+\sigma)$ expected recursive child requests. If every residual row has norm at most s , put $C_s = 2s(1+s)$ and $b_s = 1 - \alpha + \alpha C_s$. There is an exact residual sampler whose expected number of input leaves plus active residual rows is at most*

$$b_s^D + \alpha \sum_{j=0}^{D-1} b_s^j. \quad (\text{A.83})$$

In particular, for $s \geq 1$ this is at most $4b_s^D/3$. For each fixed s , expected bit work is $O_s(\mathcal{B}^4[1+b_s^D + \alpha \sum_{j<D} b_s^j])$. For arbitrary $s \geq 1$, Corollary A.43 gives the uniform bit bound $O(\mathcal{B}_s^4 b_s^D)$, where $\mathcal{B}_s = \mathcal{B} + \lceil \log_2(s+2) \rceil$. Its preprocessing consists of rational row sums and prefix tables and takes $O(Dn^2 \mathcal{B}_s^2)$ bit operations.

Proof idea. At each active update, sample the bias and row through one scalar source. Consecutive skip choices can be generated in a single geometric jump, so the accounting counts active updates only.

Proof. The row sampler is Corollary A.25 with $R = 1$, the children being the source signs. Its proof bounds the expected child requests by $2\sigma \max\{1, u\} \leq 2\sigma(1+\sigma)$, and its conditional charging covers a majority that stops early.

At each residual row, first select the nonlinear branch with probability α . Otherwise skip to the preceding layer. Geometric waiting times implement these skips directly. The expected number of descendant input leaves is at most b_s^D , and the expected number of active rows is at most $\alpha \sum_{j<D} b_s^j$. This proves (A.83). For $s \geq 1$, $C_s \geq 4$ and the geometric sum contributes at most $(b_s^D - 1)/(C_s - 1) \leq b_s^D/3$. Theorem A.24 gives scalar cost $O((1+s)^2 \mathcal{B}_s^4)$ at each active row, excluding child computations. Compressed skips and input leaves cost $O(\mathcal{B}_s^4)$ each. For fixed s , the predictable trial-cost lemma proves the stated bit bound. Corollary A.43 absorbs the quadratic row factor into the branching sum. $\square$

Proposition A.42 (The quadratic norm scale is necessary in one layer). *Let $s \geq 2$ and $0 \leq \alpha \leq 1$. For an unknown source sign of mean z , the optimal worst-mean source count for the target $f(z) = (1-\alpha)z + \alpha \tanh(sz)$ satisfies*

$$\frac{3}{64}(1 + \alpha s^2) \leq U \leq 2(1 + \alpha s^2).$$

For one residual layer with zero biases, uniform row weights s/m , and even $m \geq 2$, the optimal worst-input expected number of distinct probes satisfies

$$\frac{1}{130}(1 + \alpha \min\{s^2, m\}) \leq Q^* \leq 2(1 + \alpha \min\{s^2, m\}). \quad (\text{A.84})$$

The uniform weights are assumed legal on the parameter grid. Thus the one-layer scale αs^2 is necessary as well as sufficient, until the finite input size becomes the limiting resource. This statement does not assert that the factor multiplies optimally across arbitrary depth.

Proof idea. A skip-or-update choice gives the upper bound. For the lower bound, a mean network retains a large derivative at zero; the transcript inequality converts this derivative into a probe requirement.

Proof. The upper bound chooses the skip with probability $1 - \alpha$ and the global tanh majority with probability α . Its source count is at most $1 - \alpha + 2\alpha s^2$. For a finite input, one may instead scan all m coordinates on the nonlinear branch. Choosing the cheaper construction gives $1 - \alpha + \alpha \min\{2s^2, m\}$ expected probes and proves the finite upper bound. Endpoint coupling gives both lower-bound quantities at least $(1 - \alpha) + \alpha \tanh s \geq 1/2$. For the unknown-source problem, the second transcript inequality, justified by the truncation argument preceding (A.66), applies at $z = 1/s$. Since $\tanh 1 \geq 1/2$ and $\text{sech}^2 1 \geq 1/4$,

$$U \geq \frac{1 - s^{-2}}{2} 2\alpha s^2 \tanh 1 \text{sech}^2 1 \geq 3\alpha s^2/32.$$

Combining this with $U \geq 1/2$ proves the stated bound.

For the finite network, under independent inputs of mean t , put $G(t) = \mathbb{E}_t \tanh(sM)$, $R = \tanh s$, and $q_0 = \min\{s^2, m\}$. The averaged output is $H(t) = (1 - \alpha)t + \alpha G(t)$. The same weighted Jensen argument as in (A.36) gives

$$h := G'(0) \geq \frac{s}{\sqrt{1 + 3s^2/m}}, \quad h^2 \geq q_0/4.$$

If $q_0 \geq 64$, set $T = 2R/h \leq 1/2$. Since $G(0) = 0$, $G'(0) = h$, and $G(T) \leq R$, the bending argument before (A.37) gives some $t \in [0, T]$ with $|G''(t)| \geq h^2/(2R)$. The linear skip contributes no second derivative, so the finite transcript inequality yields

$$Q^* \geq \frac{3\alpha h^2}{16R} \geq \frac{3\alpha q_0}{64}.$$

Together with $Q^* \geq 1/2$, this is stronger than the lower bound in (A.84). If $q_0 < 64$, the endpoint bound alone gives $Q^* \geq 1/2 \geq (1 + \alpha q_0)/130$. $\square$

Corollary A.43 (Residual bit bound at arbitrary norm). *Let $s \geq 1$, with C_s , b_s , and $\mathcal{B}_s$ as in Proposition A.41. The raw residual sampler has expected online bit work*

$$O(\mathcal{B}_s^4 b_s^D). \quad (\text{A.85})$$

Its preprocessing remains rational row sums and prefix tables, at cost $O(Dn^2 \mathcal{B}_s^2)$. In particular, $\alpha s^2 D = O(\log \log n)$ and $\mathcal{B}_s = \log^{O(1)} n$ suffice for polylogarithmic expected online bit work. This permits s itself to exceed every fixed power of $\log n$. To fit the original $O(Dn^2 \log^6 n)$ preprocessing allowance, it is sufficient that $\mathcal{B}_s = O(\log^3 n)$.

Proof idea. Use the sharp large-radius scalar routine at each active residual update. The same compressed-skip accounting now charges its quadratic radius cost together with all scalar precision work.

Proof. An active row has $u = |b_0| + \sigma \leq 1 + s$ and therefore costs $O((1 + s)^2 \mathcal{B}_s^4)$ by Theorem A.24. Its expected number of child requests remains at most C_s . The expected number of leaves is at most b_s^D , and

the expected active-row count is at most $\alpha \sum_{j < D} b_s^j$. Compressed geometric skips have the same logarithmic bit cost per visited branch. The stopping-cost lemma gives

$$O\left(\mathcal{B}_s^4 \left[1 + b_s^D + (1+s)^2 \alpha \sum_{j < D} b_s^j\right]\right).$$

For $s \geq 1$,

$$\frac{(1+s)^2}{C_s - 1} \leq \frac{4}{3}, \quad (1+s)^2 \alpha \sum_{j < D} b_s^j \leq \frac{4}{3}(b_s^D - 1),$$

including $\alpha = 0$ by continuity or direct inspection. This proves (A.85). Finally $C_s - 1 \leq 4s^2$ and $b_s^D \leq \exp(4\alpha s^2 D)$. $\square$

Corollary A.44 (One residual layer). *For any signed, biased row of width n , row norm at most $s \geq 1$, and $|b_0| \leq 1$, one residual output admits an exact sampler using*

$$O(\mathcal{B}_s^4[1 + \alpha \min\{s^2, n\}])$$

expected online bit operations after the same preprocessing. For the legal uniform zero-bias rows of Proposition A.42, with $s \geq 2$ and even width, its input-query dependence is optimal up to an absolute factor.

Proof idea. There is only one active nonlinear row. Mix its exact sample with the directly accessible skip coordinate, and compare this algorithm with reading the entire row.

Proof. Choose the identity input on the skip branch. On the nonlinear branch, put $H = \max\{1, \sigma + |b_0|\}$ and use the quadratic-radius sampler if $H^2 \leq n$; otherwise probe and sum the entire rational row and sample its tanh output with a certified comparison. The latter costs $O(n\mathcal{B}_s^4)$ in expectation. Since $H \leq 2s$, the chosen cost is bounded by $O(\mathcal{B}_s^4 \min\{s^2, n\})$. The query lower bound is (A.84). $\square$

A.13.5 A matching law for residual mean networks

Let $m \geq 2$ be even. Use zero biases and put $w_{\ell,ij} = s/m$ on the same m relevant coordinates in every row and layer. The displayed weights are assumed to be legal on the dyadic grid. For class lower bounds at fixed $s > 1$, a downward dyadic approximation to s makes them legal and preserves the stated exponential order. Define

$$\begin{aligned} p &= 1 - \alpha, & a &= p + \alpha s, & \lambda &= a^D, \\ F_0(z) &= z, & F_{j+1}(z) &= pF_j(z) + \alpha \tanh(sF_j(z)), \\ A &= \frac{\alpha s^3}{a} \sum_{j=0}^{D-1} a^{2j}. \end{aligned} \tag{A.86}$$

Here $0 < s \leq 2$, so $a > 0$; $\alpha = 0$ is permitted. Writing $M = m^{-1} \sum_i x_i$, direct induction gives

$$h_{D,1} = F_D(M) + p^D(x_1 - M). \tag{A.87}$$

Theorem A.45 (Uniform residual mean-network law). *For the network (A.87), define*

$$\Xi = \lambda \min \left\{ \sqrt{1 + A}, \frac{m}{\sqrt{1 + A}} \right\}.$$

Its optimal worst-input expected number of distinct input probes satisfies

$$\frac{\Xi}{512} \leq Q^* \leq 3\Xi. \tag{A.88}$$

The constants are uniform in s, α, D, m in the stated range. For an unknown source coin and target mean $F_D(z)$, the optimal worst-mean source count U satisfies

$$\frac{1}{81}\lambda\sqrt{1+A} \leq U \leq 2\lambda\sqrt{1+A}. \quad (\text{A.89})$$

These statements concern source or input queries; auxiliary arithmetic is excluded from this particular measure.

Lemma A.46 (Residual range comparison). For $0 \leq z \leq 1$,

$$\frac{\lambda z}{\sqrt{1+Az^2}} \leq F_D(z) \leq \frac{\lambda z}{\sqrt{1+Az^2/5}}. \quad (\text{A.90})$$

Proof idea. Compare one residual step with reciprocal-square maps above and below it. Composing these inequalities gives explicit bounds for the range of the full scalar chain.

Proof. Set $v = \alpha s/a$ and $\theta = \alpha s^3/a$. The one-step ratio is

$$\frac{pz + \alpha \tanh(sz)}{az} = (1-v) + v \frac{\tanh(sz)}{sz}.$$

The $\tanh$ lower bound and convexity of $(1+u)^{-1/2}$ give the lower bound $(1+\theta z^2)^{-1/2}$ for this ratio. For $0 \leq y \leq 2$, the $\coth$ inequality (A.27) and a chord bound give

$$\frac{\tanh y}{y} \leq (1+2y^2/3)^{-1/2} \leq 1-y^2/10.$$

The last chord uses $(1+8/3)^{-1/2} < 3/5$. Since $1-w/2 \leq (1+w)^{-1/2}$ for $w \geq 0$, the ratio is at most $(1+\theta z^2/5)^{-1/2}$. Every iterate remains in $[0, 1]$. Iterating these reciprocal-square bounds gives (A.90); the accumulated sum is precisely A . $\square$

Lemma A.47 (A positive majority expansion for residual compositions). The function $H(w) = F'_D(\sqrt{w})$ is completely monotone on $(0, \infty)$ and holomorphic on the right half-plane. It therefore has a representation

$$H(1-v) = \sum_{k \geq 0} d_k v^k, \quad d_k \geq 0, \quad \sum_k d_k = \lambda.$$

If $R = F_D(1)$, then $F_D(z) = \sum_k (d_k/C_k) M_k(z)$ and $\sum_k d_k/C_k = R$.

Proof idea. The derivative of the composed scalar map has a nonnegative expansion in the majority basis. Closure properties of the squared residual map preserve this positivity through every layer.

Proof. Retain $C(w) = \text{sech}^2 \sqrt{w}$ and $T(w) = \tanh \sqrt{w}/\sqrt{w}$ from the scalar proof. The square of the residual map is

$$g(w) = p^2 w + 2p\alpha\sqrt{w} \tanh(s\sqrt{w}) + \alpha^2 \tanh^2(s\sqrt{w}).$$

It is Bernstein: each summand is Bernstein, and

$$\frac{d}{dw} [\sqrt{w} \tanh(s\sqrt{w})] = \frac{s}{2} \{T(s^2 w) + C(s^2 w)\}$$

is completely monotone. Also the one-step derivative $p + \alpha s C(s^2 w)$ is completely monotone. The closure facts already proved for the scalar composition show that the product expressing H is completely monotone.

If $\sqrt{w} = x + iy$ with $\Re w > 0$, then $x > |y|$. Both $x + iy$ and $\tanh(s(x + iy))$ lie in the cone $\Re z > |\Im z|$: for $\tanh$ this follows from $\sinh(2sx) > |\sin(2sy)|$ and its positive denominator. Their nonnegative linear combination lies in the same cone. Thus g maps the right half-plane into itself and the composed derivative is holomorphic there. The Taylor expansion and derivative integration from the scalar case now prove all remaining claims. $\square$

Proof idea. Sample one global majority for the whole residual chain and retain its output-range gate. A rare scan of all inputs supplies the alternative bound when the input is short.

Upper bounds in Theorem A.45. Use the global majority mixture and its gate R . At a positive input z , the derivative product satisfies

$$\frac{F'_D(z)}{\lambda} = \prod_{j=0}^{D-1} \left(1 - \frac{\alpha s}{a} \tanh^2(sF_j(z))\right).$$

Since $F_j(z) \leq a^j z$, the loss from one is at most $\min\{1, Az^2\}$. The computation of (A.64) and (A.65), with s^D replaced by λ , bounds the full-majority source count by $2\lambda\sqrt{1+A}$. This proves the unknown-source upper bound.

For the actual network output in (A.87), the unconditional mass of the majority-of-one component is $d_0 = F'_D(1) \geq p^D$. Replace mass p^D of that component's uniform-coordinate request by a request for x_1 . This changes its mean by exactly $p^D(x_1 - M)$ and leaves its cost unchanged. All mixture masses remain nonnegative. A second strategy uses the same gate R , then scans all m coordinates and evaluates the resulting scalar target. The modified positive majority representation has total mass R , so it also proves $|h_{D,1}| \leq R$ for every context. This verifies that the gated scan uses a valid conditional sign. The range lemma gives $mR \leq \sqrt{5}\lambda m/\sqrt{1+A}$. Select the first strategy when $1+A \leq m$, and the second otherwise. This proves $Q^* \leq 3\Xi$. $\square$

Proof idea. Separate the unknown-source lower bound from the finite-input lower bound. Endpoint coupling controls the range, while the score of the empirical input mean captures the derivative growth and its saturation.

Lower bounds in Theorem A.45. Endpoint coupling gives $U \geq R$ and $Q^* \geq R$. For the unknown-source problem, if $A < 80$, the lower range bound gives $R \geq \lambda\sqrt{1+A}/81$. If $A \geq 80$, then $T = 2R/\lambda \leq 2/\sqrt{17} < 1/2$, and $F'_D(0) = \lambda$. The bending bound (A.37) with the scalar cost bound (A.66) yields

$$U \geq \frac{3\lambda^2}{16R} \geq \frac{3}{16\sqrt{5}}\lambda\sqrt{1+A} \geq \frac{1}{81}\lambda\sqrt{1+A}.$$

For the finite network, under independent input signs of mean t , the expected correction $p^D(X_1 - M)$ in (A.87) is zero. Thus its averaged output is $H(t) = \mathbb{E}_t F_D(M)$. The weighted Jensen argument used in (A.36), applied to the lower range bound, gives

$$h := H'(0) \geq \frac{\lambda}{\sqrt{1+(3m-2)A/m^2}} \geq \frac{\lambda}{\sqrt{1+3A/m}}. \quad (\text{A.91})$$

If $A < 400$, endpoint coupling gives $Q^* \geq \Xi/401$. If $A, m \geq 400$, put $T = 2R/h$. Then

$$T^2 \leq \frac{4(1+3A/m)}{1+A/5} \leq 20/A + 60/m \leq 1/5 < 1/4.$$

The bending bound (A.37) and the second transcript bound give

$$Q^* \geq \frac{3h^2}{16R} \geq \frac{3\lambda\sqrt{1+A/5}}{16(1+3A/m)} \geq \frac{3}{64\sqrt{5}}\Xi.$$

It remains to consider $m < 400$ and $A \geq 400$. At a balanced input, flipping coordinate i changes the output mean by absolute amount

$$F_D(2/m) + 2p^D \mathbf{1}_{\{i=1\}} - 2p^D/m.$$

These amounts are nonnegative because $F_D(z) \geq p^D z$. Their sum is $mF_D(2/m)$. Lemma B.7, summed over the coordinates, gives

$$Q^* \geq \frac{m}{2} F_D(2/m) \geq \frac{\lambda}{\sqrt{1+4A/m^2}}.$$

If $A \leq m^2$, this is at least $\lambda/\sqrt{5}$, while $\Xi \leq \lambda\sqrt{m} < 20\lambda$. If $A > m^2$, it is at least $m\lambda/\sqrt{5A} \geq \Xi/\sqrt{5}$. Every resulting constant is at least $1/512$. $\square$

Corollary A.48 (Residual amplification lower bound). *Every exact sampler for the residual mean network satisfies*

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{\lambda^2}{1 + (3m - 2)A/m^2}. \quad (\text{A.92})$$

For $1 < s \leq 2$, put $\delta = s - 1$. Then, uniformly in $0 \leq \alpha \leq 1$,

$$\sup_x \mathbb{E}[Q \mid x] \geq \frac{2\delta}{2\delta + 3s^3} \min\{(1 + \alpha\delta)^{2D}, m\}. \quad (\text{A.93})$$

At $s = 1$, the matching mean-network law becomes

$$\frac{1}{512} \min\left\{\sqrt{1 + \alpha D}, \frac{m}{\sqrt{1 + \alpha D}}\right\} \leq Q^* \leq 3 \min\left\{\sqrt{1 + \alpha D}, \frac{m}{\sqrt{1 + \alpha D}}\right\}. \quad (\text{A.94})$$

Proof idea. Insert the finite-input derivative estimate into the transcript inequality. Elementary bounds on the accumulated residual gain yield the supercritical and critical consequences.

Proof. Square (A.91) and apply the first transcript bound. If $\alpha > 0$ and $s > 1$, then

$$A = \frac{\alpha s^3}{a} \frac{a^{2D} - 1}{a^2 - 1} \leq \frac{s^3}{2\delta} \lambda^2.$$

Use $1 + (3m - 2)A/m^2 \leq 1 + 3A/m$ and the elementary minimum inequality from the feed-forward lower bound. For $\alpha = 0$, the target is x_1 and the displayed inequality holds directly. At $s = 1$, $a = \lambda = 1$ and $A = \alpha D$. $\square$

These bounds identify the residual scaling that permits polylogarithmic work at polynomial depth.

Corollary A.49 (Residual depth for polylogarithmic work). *Fix $s > 1$. Consider residual networks (A.70) of width n with row norms at most s , biases of absolute value at most one, and α and all parameters on the grid $2^{-b}\mathbb{Z}$, where $20\lceil\log_2 n\rceil \leq b \leq \log^{O(1)} n$ and $D \leq n^{O(1)}$. Let $\alpha = \alpha_n$ and $D = D_n$ vary with n . If $\alpha D = O(\log \log n)$, one uniform exact sampler has expected online bit work $\log^{O(1)} n$ on every such network and context. Conversely, if some exact sampler has worst-network, worst-context expected online bit work $\log^{O(1)} n$, then $\alpha D = O(\log \log n)$.*

Proof idea. The raw residual sampler costs at most an exponential of a fixed multiple of αD . The mean-network lower bound grows exponentially in αD until it saturates at the width.

Proof. Here $\mathcal{B} = \log^{O(1)} n$. For sufficiency, Corollary A.43 gives expected online bit work $O(\mathcal{B}_s^4 b_s^D)$ with $b_s^D \leq \exp(4\alpha s^2 D)$. This is polylogarithmic when $\alpha D = O(\log \log n)$.

For necessity, choose a fixed dyadic gain $s_0 \in (1, \min\{s, 2\}]$, put $\delta = s_0 - 1$, and let $m = 2^{\lceil \log_2 n \rceil} \geq n/2$. For large n the weights s_0/m lie on the grid, so the residual mean network of gain s_0 belongs to the class. By (A.93) and $\log(1 + \alpha\delta) \geq \alpha\delta/2$, every exact sampler makes at least $c \min\{e^{\delta\alpha D}, m\}$ expected probes on some input, where $c = 2\delta/(2\delta + 3s_0^3)$. Each probe costs at least one bit operation. Suppose the work is at most $\log^K n$ for a fixed K . Since $cm > \log^K n$ for all sufficiently large n , the minimum is then the exponential term, and $e^{\delta\alpha D} \leq c^{-1} \log^K n$. Taking logarithms gives $\alpha D = O(\log \log n)$. $\square$

A.13.6 The improved critical exponent survives residual mixing

The fixed potential of Appendix A.10 also controls residual networks. The skip branch costs one child when it is examined, but its effect on the potential has a useful concavity term. That term prevents an additional factor from appearing in the quadratic growth coefficient.

Lemma A.50 (Mixing a potential estimate with an identity step). *Let $0 < \eta \leq 1/4$, $\phi(v) = \sqrt{1 + \eta - v^2}$, $u, w \in [-1, 1]$, $0 \leq \omega \leq 1$, and $0 \leq \xi \leq 1$. Then*

$$\log \frac{(1 - \omega)\phi(u) + \omega e^\xi \phi(w)}{\phi((1 - \omega)u + \omega w)} \leq \omega \xi + 4\eta^{-3/2} \omega \xi^2. \quad (\text{A.95})$$

Proof idea. The concavity gap of the potential absorbs the mismatch between skip and update means. A quadratic completion bounds the remaining product term uniformly over the mixing probability.

Proof. Put $A = 1 + \eta$, $v = (1 - \omega)u + \omega w$, and $\Delta = |u - w|$. Direct differentiation gives $|\phi'| \leq \eta^{-1/2}$ and $-\phi'' \geq A^{-1/2}$. Thus the Jensen gap is at least

$$\phi(v) - (1 - \omega)\phi(u) - \omega\phi(w) \geq \frac{\omega(1 - \omega)}{2\sqrt{A}} \Delta^2.$$

For example, this follows by applying concavity to $\phi(x) + x^2/(2\sqrt{A})$. Also $|\phi(w) - \phi(v)| \leq (1 - \omega)\Delta/\sqrt{\eta}$. Writing $t = e^\xi - 1$, subtraction of $(1 + \omega t)\phi(v)$ from the numerator in (A.95) therefore gives at most

$$\omega(1 - \omega) \left(\frac{t\Delta}{\sqrt{\eta}} - \frac{\Delta^2}{2\sqrt{A}} \right) \leq \frac{\omega t^2 \sqrt{A}}{2\eta}.$$

Divide by $\phi(v) \geq \sqrt{\eta}$. For $0 \leq \xi \leq 1$, positivity of the exponential series gives $e^\xi - 1 \leq \xi + \xi^2 \leq 2\xi$. The resulting ratio is at most $1 + \omega \xi + 4\eta^{-3/2} \omega \xi^2$. Taking logarithms and using $\log(1 + x) \leq x$ proves the claim. $\square$

For this variant, retain the radii from (A.73), but use centers within $\alpha\epsilon$ of the residual midpoint M . Project these centers only onto $[-1, 1]$. The normalized interval invariant still holds, and the center shift is now small enough for the potential estimate. The root is implemented separately: draw the last active nonlinear layer by a reverse geometric waiting time with parameter α . If there is no such layer, return x_i . Otherwise use the raw tanh-row sampler at that layer. This implements the exact identity

$$h_{D,i} = (1 - \alpha)^D x_i + \alpha \sum_{j=1}^D (1 - \alpha)^{D-j} \tanh \left(b_{j,i} + \sum_k w_{j,ik} h_{j-1,k} \right). \quad (\text{A.96})$$

All internal requests still use normalized residual signs and compressed skips.

Theorem A.51 (Improved critical residual bound). *For row norms at most one, there is one uniform exact residual sampler with the stated preprocessing budget and expected online bit cost*

$$O\left(\mathcal{B}^4(1 + \alpha D)^{35/32}\right).$$

Its expected number of compressed proposals and input leaves, together with one root operation, is

$$O((1 + \alpha D)^{35/32}). \quad (\text{A.97})$$

The parameter $\alpha = 0$ returns x_1 directly.

Proof idea. Expand the raw output along its last active residual update, preserving its output gate. Above an effective-depth cutoff, the mixed potential estimate controls each active branch; below it, the original residual sampler bounds the remaining subtree.

Proof. Fix

$$\eta = 1/64, \quad \phi(v) = \sqrt{65/64 - v^2}, \quad c_0 = 17/16, \quad M_0 = 2^{22}, \quad \kappa = 3c_0 = 51/16, \quad J = 2^{28}.$$

For $\alpha > 0$, let $K = \min\{D, \lceil J/\alpha \rceil\}$. Use the original majority and race below or at layer K . Above K , use the fused factory when its rational eligibility conditions hold and the original large-bias factory otherwise. The cutoff depends only on the weights and α .

Consider a normalized request at layer $j + 1$ above the cutoff. Write $r = r_j$, $q_* = \tanh r$, and $f = (1 - \alpha)r + \alpha q_*$. For the actual row let $q = \tanh \rho \leq q_*$ and write $v_0 = kF(z)$ for its normalized tanh mean before radius scaling. Put

$$\omega = \alpha q_*/f \leq \alpha, \quad \theta = f/r_{j+1} \leq 1, \quad w = (q/q_*)v_0, \quad u = (h_{j,i} - c_{j,i})/r_j.$$

The normalized residual mean is $v = \theta[(1 - \omega)u + \omega w] + \delta_j$ with $|\delta_j| \leq \alpha\varepsilon/r_{j+1}$. Lemma A.31, concavity, and evenness give an expected sum of child potentials at most

$$\theta\{(1 - \omega)\phi(u) + \omega e^{\xi_j}\phi(w)\}, \quad \xi_j = c_0 r_j^2 + M_0 r_j^4.$$

Indeed, the actual-row factor q/q_* can be absorbed using $(q/q_*)\phi(v_0) \leq \phi((q/q_*)v_0)$. A zero row has no nonlinear child requests and satisfies the same upper bound by setting $w = 0$. The ratio $\theta/\phi(\theta v)$ increases with θ , and $|\frac{d}{dv} \log \phi(v)| \leq 64$. Thus (A.95) bounds the weighted offspring factor by the exponential of

$$\alpha c_0 r_j^2 + \alpha(M_0 + 8192)r_j^4 + 64\alpha\varepsilon/r_{j+1}, \quad (\text{A.98})$$

provided $\xi_j \leq 2r_j^2 \leq 1$.

If the upper region is nonempty, then $D \geq K \geq J$. The ideal critical radius comparison and the rounding bound give, for $j \geq K$,

$$r_j^2 \leq 3/J, \quad \alpha \sum_{j=K}^{D-1} r_j^4 \leq 14/J. \quad (\text{A.99})$$

For the second inequality, the ideal sum is at most $25/(4J^2) + 25/(4J)$. The rounding contribution is at most $6\varepsilon D^2 < 1/J$. The first follows from $R_j^2 \leq (1 + 2\alpha j/5)^{-1}$ and $r_j^2 - R_j^2 \leq 6D\alpha\varepsilon$. These bounds imply $\rho \leq 1/32$ and $\xi_j \leq (c_0 + 3M_0/J)r_j^2 < 2r_j^2 < 1$.

Set $d_j = 1 - \tanh(r_j)/r_j$. Since $d_j \geq r_j^2/3 - r_j^4/5$,

$$\alpha c_0 r_j^2 \leq \kappa \alpha d_j + (\kappa/5)\alpha r_j^4 \leq \kappa \log \frac{r_j}{f_{\alpha,1}(r_j)} + (\kappa/5)\alpha r_j^4.$$

Replacing $f_{\alpha,1}(r_j)$ by r_{j+1} costs at most $3\kappa\alpha\varepsilon/f_{\alpha,1}(r_j)$. Consequently a suffix of weighted offspring factors, from level ℓ to level $k \geq K$, is at most

$$e^2(r_k/r_\ell)^\kappa. \quad (\text{A.100})$$

To check the constant, the fourth-order coefficient in (A.98) and the preceding display is less than 2^{23} . Its total is at most $2^{23} \cdot 14/J = 7/16 < 1$. The center and radius rounding terms total at most

$$74\alpha D\varepsilon\sqrt{1 + \alpha D} < 1.$$

Let $C_\ell(i, x)$ denote the expected number of proposals and input leaves from one normalized request. The original sampler below the cutoff obeys

$$C_\ell(i, x) \leq 7e^{21}r_\ell^{-5} \quad (\ell \leq K), \quad (\text{A.101})$$

by the proof of (A.71) before its root gate. The proof depends only on interval validity, so it also applies to these midpoint centers. Since $1/8 \leq \phi \leq \sqrt{65}/8$, weighted charging above the cutoff gives

$$\frac{C_\ell(i, x)}{\phi(v_{\ell, i})} \leq e^2 r_\ell^{-\kappa} \left(56e^{21} r_K^{\kappa-5} + 16\alpha \sum_{k=K+1}^{\ell} r_k^{\kappa} \right).$$

Here the factor 16α charges a proposal at a reached call: its proposal probability is at most 2α and its potential is at least $1/8$. The sum obeys $\alpha \sum r_k^{\kappa} \leq 7$. One may bound it by the third-power sum, whose ideal integral is five, and then use the rounding estimate. Also $r_K^{\kappa-5} \geq 1$ and

$$r_K^{\kappa-5} \leq (1 + \alpha K)^{29/32} \leq (J + 2)^{29/32}.$$

Together with (A.101), this proves, for every layer including those below the cutoff,

$$C_\ell(i, x) \leq A_0 \phi(v_{\ell, i}) r_\ell^{-\kappa}, \quad A_0 = 64e^{24} (J + 2)^{29/32}. \quad (\text{A.102})$$

Finally condition on the raw root's last active layer j . Its source radius is $r = r_{j-1}$. Above the cutoff, Lemma A.31 bounds the expected sum of child potentials by $re^{\xi_{j-1}} \max \phi < 4r$. Below the cutoff the original joint bound gives at most $r\chi_1(r) \max \phi < 12r$, since $\chi_1(r) \leq 1 + 4d_1(r) + 5r^4 \leq 10$. Thus $16r$ is valid in both regions. Equation (A.102) bounds the expected descendant cost of this root by

$$16A_0 r^{1-\kappa} \leq 16A_0 (1 + \alpha D)^{35/32}.$$

The case with no active layer costs one input probe. Averaging over the reverse geometric choice and adding the root operation proves (A.97): $J + 2 \leq 2^{29}$, $(J + 2)^{29/32} < 2^{27}$, $e^{24} < 2^{36}$, and $16 \cdot 64 = 2^{10}$ leave room in the constant 2^{76} .

The exactness induction, geometric jump implementation, and predictable scalar-cost charging are unchanged. The fixed-precision oracle for the fused tail has geometric index and precision tails, so its expected scalar work per reached proposal is $O(\mathcal{B}^4)$. No extra coefficient table is stored. $\square$

Lemma A.52 (A residual radius envelope above unit norm). *For $s = 1 + \delta$ with $0 \leq \delta \leq 1$, let $R_0 = 1$ and $R_{j+1} = f_{\alpha, s}(R_j)$. Then*

$$R_j^2 \leq 10\delta + \frac{1}{1 + \alpha j/5}. \quad (\text{A.103})$$

Proof idea. Compare the squared radius with its supercritical equilibrium scale. Above that scale each residual step decreases its excess by a reciprocal-square recurrence, giving the stated effective-depth decay.

Proof. The ideal radius sequence decreases and remains in $[0, 1]$. For $0 \leq r \leq 1$, the unit Lipschitz bound and the critical radius inequality give

$$f_{\alpha, s}(r) \leq r(1 + \alpha\delta - \alpha r^2/5).$$

If $v = r^2 \geq 10\delta$, the right side is at most $r(1 - \alpha v/10)$. Since $(1 - x)^2(1 + 2x) \leq 1$ for $0 \leq x \leq 1/10$, the next squared radius is at most $v/(1 + \alpha v/5)$. Put $w_j = (R_j^2 - 10\delta)_+$. If $w_j = 0$, monotonicity of the radius sequence gives $w_{j+1} = 0$. Otherwise the preceding bound and direct rational algebra give $w_{j+1} \leq w_j/(1 + \alpha w_j/5)$. Since $w_0 \leq 1$, induction proves $w_j \leq (1 + \alpha j/5)^{-1}$. $\square$

Proposition A.53 (A fixed supercritical residual rate). *Fix $s = 1 + \delta$ with $0 < \delta \leq 2^{-16}$, let $r_* \in (0, 1)$ solve $r_* = \tanh(sr_*)$, and set $u_* = sr_*$. There is an explicitly computable rate $\Gamma_s = (17/16)u_*^2 + O(u_*^4)$ such that the midpoint-centered residual sampler satisfies, for every actual row norm bound $S \leq s$,*

$$\mathbb{E}J_f = O_s(e^{\alpha D \Gamma_s}), \quad \frac{51}{16}\delta \leq \Gamma_s \leq \frac{51}{16}\delta + O(\delta^2). \quad (\text{A.104})$$

Its expected bit cost is $O_s(\mathcal{B}^4 e^{\alpha D \Gamma_s})$. The constants in the rate estimate are absolute; the prefactor is uniform in α . Explicit choices are given in the proof. The sampler computes the actual reference gain and need not receive the real comparison parameter s .

Proof idea. Above the cutoff, compare the weighted cost polynomial with its value at the limiting radius. Contraction makes the excess summable, and the compressed residual branches contribute the effective depth to the exponential rate.

Proof. Take $J = 2^{28}$ and

$$\begin{aligned} \Gamma_s &= \frac{17}{16}u_*^2 + 2^{32}u_*^4, & H_0 &= 2^{22}, & E_\delta &= 1 + \frac{3H_0}{200} + \frac{8H_0}{\delta}, \\ C_\delta &= 2 + 2^{13}e^{E_\delta} \left(\exp(2^{26}) + 2 + \frac{32}{51\delta} \right). \end{aligned} \quad (\text{A.105})$$

Write $M_* = \exp(2^{26})$. We prove the explicit bound $\mathbb{E}J_f \leq C_\delta e^{\alpha D \Gamma_s}$. Let $K = \min\{D, \lceil J/\alpha \rceil\}$ for $\alpha > 0$. The case $\alpha = 0$ is immediate. By comparison with the ideal radii for gain s , Lemma A.52, and the rounding estimate, every squared row radius above the cutoff is bounded by

$$x_j = s^2 r_j^2 \leq 11\delta + 7/J < 2^{-12}.$$

Here $s^2 < 11/10$, $\alpha j \geq J$, and $D \geq J$; the radius-rounding term is less than $1/J$. The fused factory is eligible whenever its bias is small, and the original large-bias branch has the same weighted estimate. For $x \leq 2^{-12}$, put $\xi = (17/16)x + 2^{22}x^2$. Then $\xi \leq 1026x < 1$, so Lemma A.50 gives a residual logarithmic offspring bound

$$\alpha \left(\frac{17}{16}x_j + 2^{32}x_j^2 \right) + 64\alpha\varepsilon/r_{j+1}. \quad (\text{A.106})$$

Indeed, $2^{22} + 2048 \cdot 1026^2 < 2^{32}$. The derivative of $P(x) = (17/16)x + 2^{32}x^2$ on this interval is less than $H_0 = 2^{22}$.

Write $\lambda_s = s(1 - r_*)^2$ and $\lambda_\alpha = 1 - \alpha + \alpha\lambda_s$. The ideal comparison radii satisfy

$$0 \leq R_j - r_* \leq (1 - r_*)\lambda_\alpha^j.$$

The fixed-point inequalities proved earlier give $1 - \lambda_s \geq \delta$. Consequently the sum of the excess of (A.106) over $\alpha\Gamma_s$, on any suffix above the cutoff, is at most

$$\frac{2s^2 H_0 (1 - r_*)}{1 - \lambda_s} + 24H_0 \alpha^2 D^2 \varepsilon + 64\alpha D \varepsilon \sqrt{1 + \alpha D} \leq E_\delta.$$

The cancellation $\alpha/(1 - \lambda_\alpha) = 1/(1 - \lambda_s)$ is the reason the transient bound is uniform in α . Thus the weighted suffix factor from level ℓ to $k \geq K$ is at most $e^{E_\delta} e^{\alpha(\ell-k)\Gamma_s}$.

Below the cutoff, the original normalized residual sampler has expected cost at most

$$10e^{30}(J+2)^{5/2}e^{7000\delta(J+1)} \leq M_*.$$

This is the unweighted bound before its formal radius gate, using $r_K^{-1} \leq \sqrt{1 + \alpha K}$. The numerical inequality follows from $\delta \leq 2^{-16}$, $J = 2^{28}$, and $7000 \cdot 2^{-16}(J+1) + 30 + \log 10 + (5/2)\log(J+2) < 2^{26}$.

Weighted charging now gives, uniformly in the normalized output mean at level ℓ ,

$$C_\ell(i, x) \leq 8\phi(v_{\ell,i})e^{E_\delta}e^{\alpha\ell\Gamma_s} \left(M_* + 2 + \frac{2}{\Gamma_s} \right).$$

To see that the geometric generation sum is uniform in α , use

$$\frac{\alpha}{1 - e^{-\alpha\Gamma_s}} \leq \alpha + \frac{1}{\Gamma_s} \leq 1 + \frac{1}{\Gamma_s}.$$

The proposal probability contributes its factor 2α . For levels below K , the same bound follows directly from M_* .

At a raw root, the expected sum of child potentials is at most 1024. Below the cutoff this follows from $\chi_g(r) \leq 10e^4 < 600$ and $\max \phi < 9/8$; above it the weighted bound and $\xi < 1$ give a smaller constant. Conditioning on the reverse geometric choice of the last active layer, using $\Gamma_s \geq 51\delta/16$, and adding the root operation proves (A.105) and the work bound.

Finally, $3\delta \leq u_*^2 \leq 3\delta + (3/2)\delta^2$ gives

$$\Gamma_s \leq \frac{51}{16}\delta + \left(\frac{51}{32} + 2^{32}(25/8)^2\right)\delta^2 \leq \frac{51}{16}\delta + 2^{36}\delta^2.$$

Its lower bound follows from $u_*^2 \geq 3\delta$. The finite-bit implementation and exactness induction are those of Theorem A.51. $\square$

A.14 Finite expected work suffices for the first score identity

The usual first-order score argument needs no exponential tail for a Bernoulli factory with bounded output. This observation removes the extra regularity hypothesis from the independent-recursion obstruction. It is a statement about sequential experiments, and we make no novelty claim for the information principle. We give a proof because the finite-mean hypothesis is the one used by the sampling problem.

Lemma A.54 (A first derivative under a finite-mean stopping rule). *Let $I \subset (-1, 1)$ be an open interval containing zero. For $z \in I$, let $X_1, X_2, \dots$ be independent signs of mean z . A fixed randomized algorithm uses an auxiliary random tape whose law does not depend on z , requests these signs sequentially, and terminates almost surely for every $z \in I$. Let T be its number of source requests and let $Y \in [-1, 1]$ be its output. If $\mathbb{E}_0 T < \infty$, then $f(z) = \mathbb{E}_z Y$ is differentiable at zero, and*

$$f'(0) = \mathbb{E}_0[Y S_T], \quad S_T = \sum_{j=1}^T X_j, \quad \mathbb{E}_0 S_T^2 = \mathbb{E}_0 T.$$

No bound on $\mathbb{E}_z T$ away from zero is assumed.

Proof idea. Truncate the stopped transcript at a finite time, where likelihood differentiation is elementary. A relative-entropy tail bound and the stopped score isometry let the truncation tend to infinity under only a finite mean stopping time at the bias of interest.

Proof. All random tapes may be taken in a standard Borel space, for example a countable sequence of fair bits. Include the complete auxiliary tape in the initial sigma-field. Conditional on this tape, T is a stopping time for the source signs. For an integer $N \geq 0$, let $\mathcal{F}_N$ be the stopped prefix sigma-field at $T \wedge N$, and put $g_N = \mathbb{E}_0[Y \mid \mathcal{F}_N]$. The finite-prefix likelihood ratio under mean z , relative to mean zero, is

$$L_{z,N} = \prod_{j=1}^{T \wedge N} (1 + zX_j) \leq (1 + |z|)^N.$$

Consequently $f_N(z) = \mathbb{E}_z g_N$ is a polynomial of degree at most N , with

$$f_N(0) = f(0), \quad f'_N(0) = \mathbb{E}_0[g_N S_{T \wedge N}] = \mathbb{E}_0[Y S_{T \wedge N}].$$

The integration over the auxiliary tape preserves this finite-degree polynomial identity.

Write $d(z) = -\frac{1}{2} \log(1 - z^2)$, the relative entropy of a fair sign with respect to a sign of mean z . Conditional on a reached prefix, the relative entropy of the remaining stopped transcript under mean zero with respect to mean z is

$$d(z) \mathbb{E}_0[(T - N)_+ \mid \mathcal{F}_N].$$

To justify this formula directly, the log likelihood ratio is a sum of source increments. Their absolute values are bounded for each fixed $z \in (-1, 1)$, and the remaining expected number of increments is finite almost

surely under mean zero. Conditional Wald summation therefore applies. The auxiliary tape contributes zero relative entropy. Almost-sure termination under both parameters makes these stopped-transcript laws probability measures.

Pinsker's inequality and $|Y| \leq 1$ bound the conditional change of its mean by the square root of twice that relative entropy. Averaging under the mean- z prefix law and using the displayed bound on $L_{z,N}$ gives

$$\begin{aligned} |f(z) - f_N(z)| &\leq (1 + |z|)^N \sqrt{2d(z)} \mathbb{E}_0 \left[\sqrt{\mathbb{E}_0[(T - N)_+ \mid \mathcal{F}_N]} \right] \\ &\leq (1 + |z|)^N \sqrt{2d(z)} \sqrt{\mathbb{E}_0(T - N)_+}. \end{aligned}$$

The second line is Jensen's inequality. Since $\sqrt{2d(z)}/|z| \rightarrow 1$, it follows that

$$\limsup_{z \rightarrow 0} \left| \frac{f(z) - f(0)}{z} - f'_N(0) \right| \leq \sqrt{\mathbb{E}_0(T - N)_+}.$$

The fair-sign martingale isometry gives $\mathbb{E}_0 S_{T \wedge N}^2 = \mathbb{E}_0(T \wedge N)$. For $M > N$ it also gives

$$\mathbb{E}_0 |S_{T \wedge M} - S_{T \wedge N}|^2 = \mathbb{E}_0[(T \wedge M) - (T \wedge N)] \leq \mathbb{E}_0(T - N)_+.$$

Thus the stopped sums converge to S_T in L^2 , proving the stated isometry and $f'_N(0) \rightarrow \mathbb{E}_0[YS_T]$. Letting $N \rightarrow \infty$ in the previous limsup bound proves differentiability and the score identity. $\square$

Corollary A.55 (The integer obstruction requires only finite mean). *Under Lemma A.54, retain the assumption $Y \in [-1, 1]$. For every integer $k \geq 0$,*

$$\mathbb{E}_0 T \geq (2k + 1)|f'(0)| - k(k + 1).$$

In particular, every finite-mean factory for $G_r(z) = \tanh(rz)/\tanh r$ obeys

$$\mathbb{E}_0 T \geq 3r \coth r - 2 = 1 + r^2 + O(r^4).$$

Proof idea. The stopped score is integer-valued. The gap between two consecutive integer magnitudes gives a stronger pointwise inequality than Cauchy-Schwarz; expectation and the score identity turn it into a source-cost bound.

Proof. For every integer v , the consecutive integers k and $k + 1$ give $(|v| - k)(|v| - k - 1) \geq 0$. Hence

$$(2k + 1)|v| \leq v^2 + k(k + 1).$$

Apply this to $v = S_T$, use the score identity, and use the stopped isometry. The last assertion takes $k = 1$ and $G'_r(0) = r \coth r$. $\square$

For $a = |f'(0)|$ and $k = \lfloor a \rfloor$, the strongest line in this family is

$$(2k + 1)a - k(k + 1) = a^2 + (a - k)(k + 1 - a) \in [a^2, a^2 + 1/4].$$

The proof uses only $|Y| \leq 1$, so postponing the final Bernoulli rounding of a bounded real estimator does not avoid this local obstruction. The small amplification $a = 1 + u$ therefore costs at least $1 + 3u$, whereas a single large amplification of size a has the lower bound $a^2 + O(1)$. This distinction explains the loss under repeated fresh composition; it supplies no upper bound for a composite factory.

Proposition A.56 (A finite-mean independent-recursion obstruction). *Put $R_0 = 1$ and $R_{j+1} = \tanh R_j$. Consider the scalar critical mean chain, represented by independently invoking a scalar factory for $G_{R_{j-1}}$ at each layer. Every source request initiates a fresh recursive call, and the top normalized factory is entered with probability R_D . If all local factories terminate almost surely for all source means and have finite expected work at mean zero, their expected number of bottom source requests at mean zero is at least*

$$e^{-3/4} R_D^{-2} \geq e^{-3/4} (1 + 2D/3) > D/4.$$

Proof idea. At mean zero, every local invocation faces the same derivative constraint. Predictable charging multiplies the resulting local lower bounds through the chain, preventing a purely local improvement of the critical exponent.

Proof. Write $a_j = R_{j-1}/R_j$. Corollary A.55 gives local mean offspring at least $3a_j - 2$. Each conditional source mean is zero in the mean-zero chain. Since every source is fresh, predictable charging and Tonelli's theorem give the bottom request bound

$$R_D \prod_{j=1}^D (3a_j - 2).$$

For $b \geq 0$,

$$3 \log(1+b) - \log(1+3b) = \int_0^b \frac{6u}{(1+u)(1+3u)} du \leq 3b^2.$$

Inequality (A.22) gives $a_j - 1 \leq R_{j-1}^2/3$, and Lemma A.12 at gain one gives

$$R_j^2 \leq \frac{3}{2j+3}, \quad R_D^{-2} \geq 1 + 2D/3.$$

Convexity and midpoint integration give

$$\sum_{j \geq 0} \frac{9}{(2j+3)^2} \leq \int_{-1/2}^{\infty} \frac{9}{(2t+3)^2} dt = \frac{9}{4}.$$

It follows that

$$\begin{aligned} \prod_{j=1}^D (3a_j - 2) &\geq \left(\prod_{j=1}^D a_j \right)^3 \exp \left(-3 \sum_{j=1}^D (a_j - 1)^2 \right) \\ &\geq R_D^{-3} e^{-3/4}. \end{aligned}$$

Multiplication by the root gate proves the result. Finally $e^{-3/4}(2/3) > 1/4$, for example from $e^{3/4} < 8/3$. $\square$

The obstruction counts fresh recursive source calls. It permits every finite-mean scalar factory, but it does not restrict methods that sample a composite function directly, share information across layers, or use a different probabilistic representation. The global scalar mean-chain factory already attains order $\sqrt{D}$. The general zero-bias norm-one network exponent remains between $1/2$ and 1 .

A.15 A dimension-free smoothness certificate

A second-order approach to a composite factory has a useful analytic bound. The following proposition concerns derivatives of the real network extension. It is not an exact-sampling algorithm. For a derivative tensor, write $\|\cdot\|_{1,\text{ent}}$ for the sum of the absolute values of all its entries.

Proposition A.57 (Derivatives of zero-bias contractive tanh networks). *Let $h_0(x) = x \in [-1, 1]^n$, let $h_\ell(x) = \tanh(W_\ell h_{\ell-1}(x))$ coordinatewise, and suppose all rows of every W_ℓ have ℓ_1 norm at most one. For every output coordinate $f = h_{D,i}$, every $x \in [-1, 1]^n$, and $D \geq 1$,*

$$\|Df\|_{1,\text{ent}} \leq 1, \quad \|D^2 f\|_{1,\text{ent}} \leq 2\sqrt{6}\sqrt{D}, \quad \|D^3 f\|_{1,\text{ent}} \leq 38D, \quad \|D^4 f\|_{1,\text{ent}} \leq 624D^{3/2}.$$

The derivatives at boundary points are those of the analytic real extension of the network.

Proof idea. Differentiate the network through fourth order while retaining the shrinking activation radius. Even derivatives of $\tanh$ carry an extra radius factor, which improves the usual chain-rule growth estimates.

Proof. Let $R_0 = 1$, $R_\ell = \tanh R_{\ell-1}$. Then $|h_{\ell,i}| \leq R_\ell$ and $R_\ell \leq \sqrt{3/(2\ell+3)}$. Let $A_\ell, B_\ell, C_\ell, E_\ell$ bound, respectively, the largest entrywise ℓ_1 norms of the first through fourth derivative tensors of a coordinate at depth ℓ , uniformly over x . At depth zero these bounds are $1, 0, 0, 0$. Linear row contraction preserves each bound. Tensor products multiply entrywise ℓ_1 norms, and the scalar chain rule therefore applies without a dimension factor.

Writing $t = \tanh u$, direct differentiation gives

$$|\tanh' u| \leq 1, \quad |\tanh'' u| \leq 2|t|, \quad |\tanh''' u| \leq 2, \quad |\tanh'''' u| \leq 16|t|.$$

For the last bound, $\tanh'''' u = t(16 - 40t^2 + 24t^4)$, whose bracket has absolute value at most 16 on $[-1, 1]$. The chain rule gives $A_\ell \leq 1$ and

$$\begin{aligned} B_\ell &\leq B_{\ell-1} + 2R_\ell, \\ C_\ell &\leq C_{\ell-1} + 6R_\ell B_{\ell-1} + 2, \\ E_\ell &\leq E_{\ell-1} + 8R_\ell C_{\ell-1} + 6R_\ell B_{\ell-1}^2 + 12B_{\ell-1} + 16R_\ell. \end{aligned}$$

The decreasing-function integral bound gives $\sum_{j=1}^\ell R_j \leq \sqrt{6}\sqrt{\ell}$, and hence $B_\ell \leq 2\sqrt{6}\sqrt{\ell}$. Since $R_\ell\sqrt{\ell} - 1 \leq \sqrt{3/2}$, the increment of C_ℓ is at most $36 + 2 = 38$. Using these two bounds, the increment of E_ℓ is at most

$$(448\sqrt{3/2} + 24\sqrt{6} + 16)\sqrt{\ell} < 624\sqrt{\ell}.$$

The strict numerical inequality follows from $\sqrt{3/2} < 49/40$ and $\sqrt{6} < 49/20$. Summing $\sqrt{\ell} \leq \sqrt{D}$ for $1 \leq \ell \leq D$ completes the proof. $\square$

A generic factory whose expected work were controlled by the total second-derivative mass, with access costs charged through the network's rows, could plausibly use this proposition to reach order $\sqrt{D}$. The known tensor Bernstein construction does not supply that interface: it requests coordinate vectors or evaluates the composite function at empirical inputs. Both operations can contain a dimension factor or a dense network evaluation. No such factory is proved here. The derivative proposition must therefore be kept separate from the sampling upper bounds.

A.16 The reach of second-order score bounds

The curvature lower bound has order $\sqrt{D}$. Allowing a different input bias in every group and retaining the full Hessian does not increase its exponent. The point is a common entrywise derivative majorant. A bound on the Hessian norm separately at each point would not suffice, because different finite differences average derivatives at different points.

Proposition A.58 (A common Hessian majorant). *Let $D \geq 1$ and let f be one output of a zero-bias, depth- D $\tanh$ network whose absolute row sums are at most one. Put $r_0 = 1$ and $r_\ell = \tanh r_{\ell-1}$. There is an entrywise nonnegative matrix C , determined by the weights, such that*

$$|\partial_{ij} f(x)| \leq C_{ij} \quad (x \in [-1, 1]^n), \quad \sum_{i,j} C_{ij} \leq 2 \sum_{\ell=1}^D r_\ell \leq 2\sqrt{6D}.$$

Partition the input coordinates into groups. Give coordinates in group a independent signs of common mean t_a , and write $H(t) = \mathbb{E}_t f(X)$. For $\Sigma(t) = \text{diag}(\sqrt{1 - t_a^2})$, one has

$$\|\Sigma(t) D^2 H(t) \Sigma(t)\|_* \leq 2 \sum_{\ell=1}^D r_\ell \leq 2\sqrt{6D}, \quad t \in (-1, 1)^k,$$

where $\|\cdot\|_*$ is the nuclear norm, the sum of singular values. For every sign input x , with $x^{(i)}$ denoting a flip of coordinate i , the sum of the one-coordinate coupling witnesses obeys

$$\frac{1}{2} \sum_i |f(x) - f(x^{(i)})| \leq 1.$$

The corresponding first-score product witness satisfies $\|\Sigma(t)\nabla H(t)\|_2^2/(1 - H(t)^2) < 3$.

Proof idea. Propagate an absolute gradient bound and a nonnegative Hessian bound through each row. The row norm prevents their total masses from acquiring a width factor. Mixed finite differences preserve the entrywise bound, even though they evaluate different coordinate squares. Grouping coordinates only regroups its mass.

Proof. Let $P_0 = I$ and $P_\ell = |W_\ell| \cdots |W_1|$, where the absolute value is entrywise. Every row of P_ℓ is nonnegative and sums to at most one. The absolute gradient of the preactivation of neuron (ℓ, a) is bounded by the row vector $v_{\ell,a} = P_\ell[a, :]$. Set $C_{0,a} = 0$ and define

$$C_{\ell,a} = \sum_j |w_{\ell,a,j}| C_{\ell-1,j} + 2r_\ell v_{\ell,a}^\top v_{\ell,a}.$$

Since $|\tanh'| \leq 1$ and $|\tanh'' u| \leq 2|\tanh u| \leq 2r_\ell$, the scalar chain rule shows inductively that $C_{\ell,a}$ bounds the absolute Hessian of that neuron at every point of the cube. Its entry sum is at most the largest preceding entry sum plus $2r_\ell$, because

$$\sum_{i,j} v_i v_j = \left(\sum_i v_i \right)^2 \leq 1.$$

Taking $C = C_{D,a}$ proves the first assertion. The bound $\sum_{\ell=1}^D r_\ell \leq \sqrt{6D}$ is the one in the proof of Proposition A.57.

For individual input means z_i , let $\Phi(z) = \mathbb{E}_z f(X)$. This is the multilinear extension of the values of f on the sign cube. Its diagonal second derivatives vanish. For $i \neq j$, its mixed derivative is the average, over the other coordinates, of the corresponding second finite difference of f , divided by four. The fundamental theorem of calculus writes that finite difference as the integral of $\partial_{ij} f$ over $[-1, 1]^2$. Therefore $|\partial_{ij} \Phi(z)| \leq C_{ij}$. If $g(i)$ denotes the group of coordinate i , substitution $z_i = t_{g(i)}$ gives

$$|\partial_{ab} H(t)| \leq \sum_{i:g(i)=a} \sum_{j:g(j)=b} C_{ij}.$$

Summing over a, b does not add mass. Multiplication by the diagonal entries of $\Sigma(t)$ cannot increase that bound, since those entries belong to $[0, 1]$. Finally the nuclear norm is at most the sum of the absolute matrix entries: expand a matrix in the rank-one coordinate matrices, each of nuclear norm one, and use the triangle inequality.

The same argument gives the common absolute gradient bound $|\partial_i f| \leq P_D[a, i]$, whose entries sum to at most one. Integrating over a flipped coordinate proves the coupling assertion. Finite differences and grouping also give $\|\nabla H(t)\|_1 \leq 1$. Thus $\|\Sigma \nabla H\|_2^2 \leq 1$, while $|H(t)| \leq r_D \leq \tanh 1$ for $D \geq 1$. The first-score witness is at most $\cosh^2(1) < 3$. $\square$

Proposition A.59 (Matrix-valued second-order transcript bound). *In the grouped product experiment above, every exact sampler satisfies*

$$\mathbb{E}_t Q \geq \frac{1}{2} \|\Sigma(t) D^2 H(t) \Sigma(t)\|_*,$$

where Q counts distinct input probes. Consequently this particular family of lower-bound witnesses is always at most $\sum_{\ell=1}^D r_\ell \leq \sqrt{6D}$. This conclusion places no upper bound on the optimal sampling cost.

Proof idea. Keep one score coordinate for every group. The Hessian of the stopped transcript likelihood is a rank-one score matrix minus a nonnegative diagonal term. After whitening, both terms have expected nuclear norm $\mathbb{E}_t Q$. The preceding proposition bounds the target Hessian that this inequality can detect.

Proof. Reduce to a finite decision tree as in the proof of Lemma A.17, keeping one score coordinate for every group. At a new probe of a coordinate in group a , put the score increment $e_a X_i / (1 + t_a X_i)$. Let Z be their sum. Let A be diagonal, with A_{aa} the sum of $(1 + t_a X_i)^{-2}$ over probes in group a . Finite-tree likelihood differentiation gives

$$D^2 H(t) = \mathbb{E}_t [Y(ZZ^\top - A)].$$

The martingale identities of that proof, applied in each group, give

$$\mathbb{E}_t \|\Sigma Z\|_2^2 = \mathbb{E}_t Q, \quad \mathbb{E}_t \text{tr}(\Sigma A \Sigma) = \mathbb{E}_t Q.$$

Since $|Y| = 1$, the triangle inequality and convexity of the nuclear norm imply

$$\|\Sigma D^2 H \Sigma\|_* \leq \mathbb{E}_t [\|(\Sigma Z)(\Sigma Z)^\top\|_* + \|\Sigma A \Sigma\|_*] = 2\mathbb{E}_t Q.$$

The first matrix has nuclear norm $\|\Sigma Z\|_2^2$; the second is nonnegative diagonal and has nuclear norm equal to its trace. The witness ceiling follows from Proposition A.58. $\square$

A.17 Corrected factories in a fixed number of variables

This subsection and the next three prove Theorem 3.3. Lemma A.60 below is the analytic factory. Sections A.18 and A.19 apply it at ranks two and three. Section A.20 adds rectangular batching and reaches every fixed rank. The finite-width law for rank at most three is Corollary A.66.

Here and in Section A.18 we use two results from Section B.7: the scalar corrected factory of Lemma B.22, and the hypergeometric recurrence of Theorem B.24.

Let $B_{i,m}$ be the degree- m Bernstein operator in coordinate i . For a derivative tensor, $\|\cdot\|_{1,\text{ent}}$ means the sum of the absolute values of all its entries. The bounds below are uniform on the unit cube. Changing one coordinate operator at a time keeps each correction one-dimensional, even when the known function depends on several source biases.

Lemma A.60 (A fixed-dimensional corrected factory). *Fix $r \geq 1$. Let $f \in C^{2r+2}([0, 1]^r)$ with $\|f\|_\infty \leq 4/5$, and let nonnegative rational M_k bound the uniform total derivative norm $\|D^k f\|_{1,\text{ent}}$, $2 \leq k \leq 2r + 2$. Suppose certified evaluations of its mixed derivatives through order $2r$ are available at rational arguments. For $0 \leq \ell < r$, put*

$$S_\ell = M_{2+2\ell} + M_{3+2\ell} + M_{4+2\ell}.$$

Choose the least power of two $m_0 \geq \max\{2, 2M_2\}$ satisfying

$$m_0^{\ell+2} \geq 8^{1-\ell} r^2 S_\ell, \quad 0 \leq \ell < r.$$

There is an exact factory for a sign of mean $f(p_1, \dots, p_r)$ using at most rm_0 independent source bits in expectation. At level m , each known correction coin requires one scalar hypergeometric average with $O_r(m)$ derivative evaluations.

Proof idea. Correct the scalar Bernstein operator in every coordinate and telescope their product one factor at a time. Binomial draws handle the unchanged coordinates. Only the changed coordinate requires degree elevation and a hypergeometric average. Total derivative norms absorb the sum over subsets of unchanged coordinates.

Proof. Put $v_i = p_i(1 - p_i)$, $A_i = \frac{1}{2}v_i\partial_{ii}$, and $T_{i,m} = B_{i,m}(I - A_i/m)$. Operators in distinct coordinates commute. For

$$P_m = \left(\prod_{i=1}^r T_{i,m} \right) f, \quad G_m = \left(\prod_{i=1}^r (I - A_i/m) \right) f,$$

the base coefficients are the grid values of G_m . Since a product of k distinct A_i has coefficient at most 8^{-k} ,

$$\|G_{m_0} - f\|_\infty \leq \sum_{k=1}^r \frac{M_{2k}}{(8m_0)^k} \leq \frac{1}{16} + \frac{r-1}{512r^2} < \frac{3}{40}.$$

For $k \geq 2$, use $M_{2k} \leq S_{k-2}$ and the corresponding starting-level inequality. The sum of mixed derivatives indexed by subsets is a sub-sum of the full derivative tensor norm, so no additional binomial coefficient is needed. Thus $\|G_{m_0}\|_\infty < 7/8$.

The exact product identity is

$$P_{2m} - P_m = \sum_{j=1}^r \left(\prod_{i < j} T_{i,m} \right) (T_{j,2m} - T_{j,m}) \left(\prod_{i > j} T_{i,2m} \right) f. \quad (\text{A.107})$$

For term j , draw a binomial total using m trials in coordinates $i < j$ and $2m$ trials in coordinates $i \geq j$. Hold the other coordinates at their empirical values and apply the scalar coefficient construction of Lemma B.22 to

$$h = \left(\prod_{i \neq j} (I - A_i/t_i) \right) f, \quad t_i = m \ (i < j), \quad t_i = 2m \ (i > j).$$

The weights in this product do not depend on coordinate j . For $k = 2, 3, 4$,

$$\|\partial_j^k h\|_\infty \leq \sum_{\ell=0}^{r-1} \frac{M_{k+2\ell}}{(8m)^\ell}.$$

Consequently the absolute coefficient is at most $\bar{A}/(4m^2)$, where

$$\bar{A} = \sum_{\ell=0}^{r-1} \frac{S_\ell}{(8m_0)^\ell} \leq \frac{m_0^2}{8r}.$$

Evaluation uses mixed derivatives of order at most $2r$ and one hypergeometric average; its magnitude bound uses order $2r + 2$.

Give the base probability $7/8$ and give each of the r correction types at each level $m = 2^a m_0$ probability $c_m = \bar{A}/(4m^2)$. Their total probability is $r\bar{A}/(3m_0^2) \leq 1/24$. Return a fair sign on the remaining mass. Every coefficient normalized by its branch mass belongs to $[-1, 1]$. Tensor Bernstein convergence and $G_m \rightarrow f$ show $P_m \rightarrow f$ uniformly, so the absolutely convergent identity (A.107) proves exactness. The level law is geometric and all known coefficient comparisons terminate almost surely. If $\bar{A} = 0$, omit all correction branches; no normalization by a zero branch mass is performed.

The base uses rm_0 source bits. Each correction uses at most $2rm$, and there are r types. Hence

$$\mathbb{E}N \leq \frac{7r}{8}m_0 + 2r^2 \sum_{a \geq 0} (2^a m_0) c_{2^a m_0} = \frac{7r}{8}m_0 + \frac{r^2 \bar{A}}{m_0} \leq rm_0.$$

All mixture parameters are rational. Integer-power comparisons compute m_0 without real root oracles. $\square$

A.18 Two scalar inputs suffice for the sharp critical law

The scalar bottleneck argument extends to a first layer of rank two. The extra coordinate requires care. A direct tensor Bernstein coefficient averages over a two-dimensional hypergeometric grid. Computing that average at every selected level would have infinite expected work. The construction below changes one coordinate operator at a time, so every correction contains one scalar average.

Theorem A.61 (Critical sampling after a rank-two first layer). *Consider a zero-bias width- n , depth- $D \geq 1$ tanh network with absolute row sums at most one and a first weight matrix of rank at most two. Its weights are dyadic with at most b bits. Put $B = 16 + b + \lceil \log_2(nD + 2) \rceil$. There is a weight-only data structure that samples the exact output using $O(\sqrt{D+1})$ expected input probes and $O(\sqrt{D+1} B^4)$ expected bit operations. At $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, preprocessing and storage are $O(Dn^2 \log^6 n)$. For arbitrary D, b , preprocessing is bounded by*

$$O(n^2 B^3 + D^2[n^2 K^2 + nK^4 + nM]M^2), \quad K = O(B), \quad M = O(B + D).$$

The constants are absolute. The worst-case query order is $\Theta(\sqrt{D})$ when $n \geq D + 1$ and $b \geq \lceil \log_2(D + 1) \rceil$.

We first give the two-coordinate factory, with the norms of Section A.17.

Lemma A.62 (A coordinatewise Bernstein correction). *Let $f \in C^6([0, 1]^2)$ satisfy $\|f\|_\infty \leq 4/5$. For $2 \leq k \leq 6$, let nonnegative rational M_k bound $\|D^k f\|_{1,\text{ent}}$. Suppose certified intervals for $f, f_{11}, f_{22}, f_{1122}$ are available at rational arguments. Put*

$$S = M_2 + M_3 + M_4, \quad U = M_4 + M_5 + M_6,$$

and choose the least power of two

$$m_0 \geq \max\{2, 2M_2, \sqrt{32S}, (4U)^{1/3}\}.$$

There is an exact sign factory of mean $f(p_1, p_2)$, from two independent Bernoulli sources, using at most $2m_0$ source bits in expectation. Every known correction coin requires one scalar hypergeometric average with $O(m)$ summands at level m .

Proof idea. Specialize the coordinatewise product telescope to two source biases. The unchanged coordinate contributes only a second derivative, so each correction needs four known derivatives and one scalar average.

Proof. Apply Lemma A.60 with $r = 2$. Its starting conditions are $m_0^2 \geq 32S$ and $m_0^3 \geq 4U$, which are precisely the displayed conditions. The common correction envelope is $\bar{A} = S + U/(8m_0) \leq m_0^2/16$. The product $(I - A_1/m)(I - A_2/m)$ uses only $f, f_{11}, f_{22}, f_{1122}$. The same is true of each coordinate correction, with its scalar hypergeometric average. The general source bound becomes $2m_0$. Integer-power comparisons select the starting level without testing equality of real roots. $\square$

The finite-bit implementation uses a complex strip. Suppose the first preactivations are $a_i z$, with $|a_i| \leq 1$, and every later row has absolute sum at most one. Write $F(z)$ for the final activation. A rank-one first layer has this form: choose a row of largest absolute sum as the scalar source row. The strip estimate also applies to several scalar inputs after bounding the imaginary part of their first linear combination.

Lemma A.63 (A uniform complex strip). *Every scalar neuron, and in particular F , is holomorphic on $|\Im z| < 1/(4\sqrt{D})$. Throughout this strip every neuron output has modulus at most two.*

Proof idea. The imaginary part of tanh is bounded by a scalar tangent recurrence. Its reciprocal-square bound keeps every preactivation away from the poles throughout the depth.

Proof. For real u, v with $|v| < \pi/4$,

$$|\Im \tanh(u + iv)| \leq \frac{|\sin(2v)|}{1 + \cos(2v)} = \tan |v|, \quad |\Re \tanh(u + iv)| \leq 1.$$

The denominator is positive, so there is no pole. Also $\tan t \leq t/\sqrt{1-t^2}$ for $0 \leq t < 1$, by $\sin^2 t \leq t^2$. Absolute row sums at most one show inductively that, at input imaginary magnitude y , every layer- ℓ imaginary magnitude is at most

$$\frac{|y|}{\sqrt{1 - \ell y^2}}.$$

For $|y| \leq 1/(4\sqrt{D})$ this is at most $1/\sqrt{15D} < 1/2$. Consequently all intermediate complex preactivations avoid the poles of $\tanh$, establishing holomorphy by composition. Their real outputs have absolute value at most one and their imaginary outputs have absolute value below $1/2$, proving the modulus bound two. The argument uses absolute weights and does not assume positivity. $\square$

Proof idea for Theorem A.61. Two basis rows turn the first layer into two unknown scalar means. The composite network is a known function on their enlarged square. Its output radius decays as $D^{-1/2}$, while its normalized second, fourth, and sixth derivative bounds grow as D, D^2, D^3 . The lemma therefore costs $O(D)$ conditionally on a gate of probability $O(D^{-1/2})$. A two-dimensional Taylor table supplies every known coefficient request within the preprocessing budget.

Proof. A zero first matrix gives a fair output. A single layer is handled by the existing unit-row factory. We henceforth assume $D \geq 2$ and first-layer rank $r \in \{1, 2\}$. Find r independent rows and r columns on which their minor is nonsingular. If a row's representation in the current row basis has a coefficient of absolute value greater than two, replace that basis row by this row. The fixed-column determinant grows by a factor greater than two. Every nonzero such determinant is at least 2^{-br} and at most $r!$ in absolute value. Thus at most $br + \lceil \log_2(r!) \rceil + 1$ replacements occur. Exact elimination and row checks cost $O(n^2 B^3)$ bits for $r \leq 2$.

The resulting basis rows v_a have absolute sums at most one, and every first-layer row is $\sum_a a_{ia} v_a$ with $|a_{ia}| \leq 2$. Their coefficients are ratios of fixed-order dyadic minors and have $O(b)$ bits. Fresh weighted-coordinate draws supply independent signs of means $z_a = v_a \cdot x$, each with at most one input probe. The network output is now a known function $F(z_1, z_2)$; a missing second coordinate is ignored. On the enlarged square, first-layer preactivations have magnitude at most $c = 4$. First-layer outputs have magnitude at most one, and subsequent rows remain contractive. Consequently

$$|F| \leq \tanh^{o(D-1)}(1) \leq \rho_D := \sqrt{3/(2D+1)}.$$

If $2\rho_D \geq 1$, set $q = 1$. Otherwise let q be the smallest number of the form 2^{-j} , $j \geq 0$, with $q \geq 2\rho_D$. Comparisons of squares determine this dyadic number exactly. Then

$$q \leq 6/\sqrt{D+1}, \quad q^{-1} \leq \sqrt{D}, \quad \|F/q\|_\infty \leq 4/5.$$

The range bound for $q = 1$ uses $D \geq 2$ and $\tanh 1 < 4/5$.

Here are explicit derivative constants. Put $T_k = 2^k k!$ and $C_1 = 1$, and define

$$C_k = T_k + 4 \sum_{\nu=2}^k T_\nu \mathbf{B}_{k,\nu}(C_1, \dots, C_{k-\nu+1}), \quad k \geq 2,$$

where $\mathbf{B}_{k,\nu}$ is the partial exponential Bell polynomial, with coefficient $k! / \prod_j (m_j! (j!)^{m_j})$ for $\sum_j m_j = \nu$ and $\sum_j j m_j = k$. The derivative polynomial of $\tanh$ has coefficient sum at most T_k . For even k it is odd, giving the bound $T_k |\tanh u|$; for odd k the bound is T_k . Use $|\tanh'| \leq 1$ for the first derivative.

Let $E_{k,\ell}$ be the largest uniform total derivative norm at layer ℓ on the enlarged square. The first layer has $E_{k,1} \leq T_k c^k$. Later scalar chain rules and row contraction give

$$E_{k,D} \leq C_k c^k D^{(k-1)/2}.$$

To check this induction, the radius at layer ℓ is at most $2/\sqrt{\ell}$. A nonlinear Bell term with $\nu \geq 2$ factors is bounded by $2T_\nu \mathbf{B}_{k,\nu}(C) c^k \ell^{(k-3)/2}$. Even ν uses the radius factor; odd $\nu \geq 3$ supplies the same power from its number of factors. Finally, $\sum_{\ell=2}^D \ell^{(k-3)/2} \leq 2D^{(k-1)/2}$. This proves the displayed recurrence, including its first-layer term. The induction uses only the bound c on the first preactivations and holds at every order k .

For $f(p) = F(2p-1)/q$, we may therefore take

$$M_k = 2^k C_k c^k (D+1)^{\lceil k/2 \rceil}, \quad 2 \leq k \leq 6.$$

In Lemma A.62, these give $m_0 = O(D + 1)$ and $\bar{A} = O((D + 1)^2)$. Open the gate q and run that factory on its open branch, returning a fair sign otherwise. Its expected input count is $O(\sqrt{D + 1})$.

We provide the finite-bit implementation. The complex-strip argument of Lemma A.63 applies when $\max_a |\Im z_a| < 1/(16\sqrt{D})$, since the first affine row has absolute sum at most four. Choose dyadic $1/(256\sqrt{D}) < a \leq 1/(128\sqrt{D})$ and grid centers on the square at spacing a . There are $O(D)$ centers. In scaled variables $z_a = c_a + 2au_a$, the polydisk $|u_a| \leq 2$ lies strictly inside the tube. Every neuron has modulus at most two, so its coefficient of u^ν is bounded by $2^{1-|\nu|}$.

Let $H = n^2D$, choose a dyadic $\varepsilon = 2^{-P} \leq H^{-2}$ with $P = O(B)$, and put

$$S_0 = P + 3\lceil \log_2(D + 1) \rceil + 64, \quad K = 16(P + \lceil \log_2(D + 1) \rceil + 64).$$

Store the total-degree- K expansions with coefficient uncertainty at most 2^{-S_0} . On the half-radius polydisk, coefficient errors in the mixed fourth derivative sum to at most $256 \cdot 2^{-S_0}$. Its scale conversion is at most $1/(qa^4) \leq 2^{32}D^{5/2}$. The geometric tail before conversion is at most $2^{12}(K + 2)^5 4^{-K}$. These choices provide certified width- ε intervals for $f, f_{11}, f_{22}, f_{1122}$. There are $O(K^2)$ coefficients. Rounded Horner evaluation, with $O(\log(K + D + 2))$ guard bits, costs $O((B + \text{bits}(p))^4)$.

We now bound the finite precision of the two-variable scaled-series computation. In the total-degree quotient algebra, the coefficient norm is submultiplicative. Exponentials satisfy the Euler recurrence

$$|\nu|e_\nu = \sum_{0 < \mu \leq \nu} |\mu|a_\mu e_{\nu-\mu}.$$

Write $\mathcal{E} = \sum_i u_i \partial_i$. If a computed exponential $\tilde{E}$ has residual $R = \mathcal{E}\tilde{E} - (\mathcal{E}A)\tilde{E}$, then $\mathcal{E}(e^{-A}\tilde{E}) = e^{-A}R$ in the truncated algebra. The inverse of $\mathcal{E}$ on series with zero constant term divides a coefficient by its total degree and has norm at most one. Multiplication by e^{-A} , this integration, and multiplication by e^A therefore bound the error, including the separately enclosed constant coefficient. There is no additional degree factor in this inverse operation. Each complex preactivation has imaginary part below one half, so $\Re e^{2s} > 0$ and $|1 + e^{2s}| \geq 1$. Cauchy's bound consequently gives coefficient norms eight for neurons, $4e^{10}$ for exponentials, and four for reciprocals. The first preactivation is affine with coefficient norm below five; later preactivations have norm at most eight. A perturbed exponential input therefore has norm at most seventeen. The exponential and inverse identities bound layer amplification by 2^{40} and local error by $2^{90}n(K + 2)^8 2^{-M}$. One sufficient choice is

$$M = S_0 + 128D + \lceil \log_2(256nD(K + 2)^{12}) \rceil + 4b + 256.$$

It verifies all perturbation margins and leaves error below 2^{-S_0-16} . Bounded rounded intermediates follow by applying the same residual identity in every further quotient that discards terms above a specified total degree. Induction over that degree keeps the computed coefficient norm within one of the exact bound; all numerator partial sums are then polynomially bounded in K . This closes the intermediate-size assumption in the rounding count. Dense rows cost $O(n^2K^2)$ coefficient operations, nonlinear recurrences $O(nK^4)$, and constant exponentials $O(nM^3)$ bit operations per layer and center. This proves the preprocessing bound in the theorem and the stated storage budget.

Each selected correction requires one mode-normalized hypergeometric average of $O(m)$ oracle values. The recurrence in Theorem B.24 evaluates it with $O(\log(m + 2))$ guard bits. Its cheap cost is $O(m(B + \log(m + 2))^4)$, and its absolute uncertainty before dividing by its branch mass is $O(\varepsilon)$. The base uses the same oracle without an average.

An augmented forward pass for derivatives through any fixed order supplies arbitrary further precision in $O(H(B + \text{bits}(p) + t)^4)$ bit operations, with a constant depending on that order. Here is an explicit guard-bit argument. If each completed derivative update has rounding error at most ζ , write $e_{k,j}$ for the largest error of order k at layer j . The coefficient of the highest derivative is $\tanh'$ at the approximate real preactivation and has magnitude at most one. Every remaining chain-rule term is a polynomial in lower derivatives. The derivative bounds already proved, with a one-unit perturbation margin, therefore give for every fixed k

$$e_{0,j+1} \leq e_{0,j} + \zeta, \quad e_{k,j+1} \leq e_{k,j} + A_k(D + 1)^k \sum_{a < k} e_{a,j} + \zeta.$$

Induction on k and summation over $j \leq D$ give $e_{k,j} \leq A'_k \zeta (D+1)^{(k+1)^2}$. Choosing ζ below the target accuracy divided by this polynomial also verifies the perturbation margin. Scale conversion by q and the dyadic cell radius costs only further powers of D . Assigning each row operation its share of ζ adds $O(\log n)$ guard bits. Thus all guard lengths are logarithmic in D, n . On a branch of mass c_m , the unresolved uniform interval has measure $O(\min\{1, \varepsilon/c_m\})$. For $c_m = O(\varepsilon)$, start direct evaluation at absolute precision proportional to c_m . For a cached miss, start at absolute precision proportional to ε . After t extra accuracy bits, the unresolved set of uniform values has measure $O((\varepsilon/c_m)2^{-t})$. Multiplying by the branch mass before summing refinement stages therefore charges $O(Hm \min\{c_m, \varepsilon\}(B + \log(m+2) + P)^4)$ to that branch. This uses unconditional interval measures and imposes no independence between work and the miss event. The total expensive refinement work is consequently bounded by

$$O\left(H \sum_{m=2^t m_0} \min\{\bar{A}/m^2, \varepsilon\} m(B + \log(m+2) + P)^4\right) = O\left(H\sqrt{\bar{A}\varepsilon} B^4\right) = O((D+1)B^4).$$

The cheap averages cost $O((\bar{A}/m_0)B^4) = O((D+1)B^4)$, including their geometric logarithmic moments. Source choices and counters fit this bound. Multiplication by the gate proves the online bit estimate. All comparisons terminate almost surely, all selected levels are finite, and the exact telescope proves the target law.

At logarithmic depth and parameter precision, the dense preprocessing term is $O(n^2 \log^6 n)$ and the nonlinear term is $O(n \log^8 n)$; both fit $O(Dn^2 \log^6 n)$. The rank-one critical mean-chain lower bound is contained in this class, proving the final claim under the stated width and grid conditions. $\square$

Rank one is a special case of rank two with a simpler basis.

Corollary A.64 (A sharp critical law with a rank-one first layer). *For a zero-bias width- n , depth- D tanh network with row norms at most one and first-layer rank at most one, exact sampling uses $O(\sqrt{D+1})$ expected input probes and $O(\sqrt{D+1} B^4)$ expected online bit operations, with B as in Theorem A.61. At $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, preprocessing and storage are $O(Dn^2 \log^6 n)$. Hidden weights may have either sign. The query order is sharp when $n \geq D+1$ and $b \geq \lceil \log_2(D+1) \rceil$.*

Proof idea. One scalar source describes the first layer, and a mean chain supplies the matching lower bound.

Proof. Apply Theorem A.61 with first-layer rank one. Its basis construction reduces here to choosing a nonzero row of largest absolute sum. The mean-chain lower bound in Theorem A.38 has a rank-one first matrix and fits the stated grid. $\square$

A.19 A third scalar input and batched preprocessing

The preceding analytic argument extends to every fixed first-layer rank. Preprocessing is the restriction: the known function must be compiled on a multidimensional grid. Batching all grid centers before each dense linear layer brings rank three within the original budget.

Theorem A.65 (Critical sampling after a rank-three first layer). *Under the assumptions of Theorem A.61, replace the first-layer rank bound by three. Exact sampling uses $O(\sqrt{D+1})$ expected input probes and $O(\sqrt{D+1} B^5)$ expected online bit operations. At $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, preprocessing and storage remain $O(Dn^2 \log^6 n)$. The worst-case query order is $\Theta(\sqrt{D})$ under the same width and grid conditions. For first-layer rank at most two, the sharper B^4 online bound of Theorem A.61 still applies.*

Proof idea. The analytic estimates and source law extend to three variables. Their direct dense preprocessing would exceed the chosen logarithmic budget. Each linear layer, however, acts on the Taylor coefficients at every grid center by the same weight matrix. Compute all those products together using exact integer matrix multiplication. The seven-product recursion supplies enough saving even with schoolbook integer arithmetic.

Proof. We give the construction for any fixed r , then specialize its preprocessing count to $r = 3$. The determinant-doubling basis construction gives r original rows of absolute sum at most one and representation coefficients of magnitude at most two. Its fixed-order rational arithmetic has $O_r(B)$ -bit operands and $O_r(n^2 B^3)$ work. Thus r independent scalar source streams represent the first layer, with at most one input probe per draw. The first preactivation on the enlarged box is bounded by $c = 2r$. After the first tanh, every output has magnitude at most one, so the subsequent critical radius bound is unchanged.

For $D \geq 2$, use a dyadic gate q with $q \leq 6/\sqrt{D+1}$, $q^{-1} \leq \sqrt{D}$, and $\|F/q\|_\infty \leq 4/5$. Depth one has the direct unit-row factory. The derivative induction in the rank-two proof holds at every order with first-preactivation bound $c = 2r$. With the same constants C_k , it gives $\|D^k F\|_{1,\text{ent}} \leq C_k(2r)^k D^{(k-1)/2}$ through order $2r+2$. For $f(p) = F(2p-1)/q$, take

$$M_k = 2^k C_k (2r)^k (D+1)^{\lceil k/2 \rceil}.$$

In Lemma A.60, $S_\ell = O_r(D^{\ell+2})$, $m_0 = O_r(D+1)$, and $\bar{A} = O_r((D+1)^2)$. The gate therefore gives $O_r(\sqrt{D+1})$ expected probes.

The uniform Taylor oracle uses a complex tube of width $1/(8r\sqrt{D})$, grid spacing $\Theta_r(D^{-1/2})$, and $O_r(D^{r/2})$ centers. A radius-two polydisk in scaled variables lies inside this tube. Each output coefficient satisfies $|b_\nu| \leq 2^{1-|\nu|}$. A total-degree- K truncation, with $K = O_r(B)$, provides all required derivatives through order $2r$. Its geometric tails have a fixed polynomial factor in K ; the only scale conversion is at most $a^{-2r}q^{-1}$, polynomial in D for fixed r . Choose $P = \lceil 2\log_2(n^2 D) \rceil = O(B)$ and $\varepsilon = 2^{-P} \leq (n^2 D)^{-2}$. Thus $O_r(B)$ stored bits per coefficient suffice for this absolute accuracy ε . The total-degree residual calculation also has explicit finite precision margins. On the unit polydisk, neuron coefficient norms are at most 2^{r+1} . First-layer preactivations are affine with coefficient norm at most $2r+1$, and later preactivations have norm at most 2^{r+1} . Thus a perturbed exponent input has norm at most $2^{r+2}+1$. Every complex preactivation has imaginary part below one half, so $|1+e^{2s}| \geq 1$ and reciprocal coefficient norm is at most 2^r . Put

$$a_r = 2^{r+4} + 4r + 16.$$

The exponential residual identity from the rank-two proof bounds nonlinear-layer amplification by 2^{a_r} and local rounding error by $2^{2a_r} n(K+2)^{4r+8} 2^{-M}$. Indeed the Euler recurrence has at most $(K+1)^{2r}$ products; its degree factors and divisions contribute only polynomial powers of K . Solving its residual equation has norm at most $\exp(2(2^{r+2}+1))$. An inverse perturbation of coefficient norm below 2^{-r-1} keeps the inverse norm at most 2^{r+1} . These bounds fit the displayed constants. If the stored coefficient precision is S_0 , it suffices to take

$$M = S_0 + 4a_r(D+1) + \lceil \log_2(256nD(K+2)^{4r+10}) \rceil + 4rb.$$

The resulting error induction verifies all perturbation margins and leaves width below 2^{-S_0} . Here $S_0 = O_r(B)$ and hence $M = O_r(B+D)$. There are

$$N_K = \binom{K+r}{r} = O_r(K^r)$$

coefficients per center. Rounded evaluation of the required fixed list of derivatives costs $O_r((B+\text{bits}(p))^{r+2})$.

Each factory correction contains only one $O(m)$ hypergeometric average. The rank-two uniform-comparison argument, with the evaluation power $r+2$ in place of four, therefore gives $O_r((D+1)B^{\max\{4,r+2\}})$ conditional expected work, including source selection, coefficient evaluation, and arbitrary precision refinement. For $r \geq 2$, the direct augmented-forward cost with its fourth power of precision is included in the exponent $r+2$. Multiplying by the gate gives the stated online bound for $r = 3$.

It remains to charge preprocessing. Batch every Taylor coefficient at every center into one matrix X with n rows and

$$C = O_r(D^{r/2} K^r)$$

columns. At each subsequent linear layer compute WX together. The entries of X have a common working dyadic denominator 2^M , and the original weights have denominator 2^b . Clear these denominators, multiply the resulting integer matrices exactly, and round the completed entries back to the working grid. There is no rounding inside the matrix multiplication.

We use Strassen’s seven-product recursion [40]. For clarity, its exact arithmetic identities are recorded here. For two block matrices A, B , form

$$\begin{aligned} Q_1 &= (A_{11} + A_{22})(B_{11} + B_{22}), & Q_2 &= (A_{21} + A_{22})B_{11}, \\ Q_3 &= A_{11}(B_{12} - B_{22}), & Q_4 &= A_{22}(B_{21} - B_{11}), \\ Q_5 &= (A_{11} + A_{12})B_{22}, & Q_6 &= (A_{21} - A_{11})(B_{11} + B_{12}), \\ Q_7 &= (A_{12} - A_{22})(B_{21} + B_{22}). \end{aligned}$$

The output blocks are

$$C_{11} = Q_1 + Q_4 - Q_5 + Q_7, \quad C_{12} = Q_3 + Q_5, \quad C_{21} = Q_2 + Q_4, \quad C_{22} = Q_1 - Q_2 + Q_3 + Q_6.$$

Direct expansion proves these identities over the integers. The recurrence $T(k) = 7T(k/2) + O(k^2)$ gives $O(k^\omega)$ scalar operations, $\omega = \log_2 7$.

Choose a power of two $k \in [C, 2C)$, pad n to a multiple of k , and pad the coefficient columns to k . There are $\lceil n/k \rceil^2$ square block products. The scalar operation count is

$$O(n^2 C^{\omega-2} + C^\omega).$$

This estimate also covers $C > n$. Recursive input sums have at most $O(\log C)$ levels, and output recombination adds at most four terms per level. Even the crude bound of C^4 on intermediate magnitude growth gives bit length $M + b + O(\log n + \log C) = O_r(B + D)$. Schoolbook scalar arithmetic therefore costs $O_r(M^2)$ per operation. The first affine layer in basis coordinates has only r inputs and is computed directly; its rational coefficients introduce no dense matrix factor.

Nonlinear coefficient recurrences use $O_r(K^{2r})$ scalar operations per neuron and center. Constant exponentials use $O_r(M^3)$ bits. The full preprocessing bound is consequently

$$O_r\left(n^2 B^3 + D \left[n^2 C^{\omega-2} + C^\omega + n D^{r/2} K^{2r} + n D^{r/2} M\right] M^2\right),$$

where the first term is the basis computation. Working matrices and stored tables fit this bound. Only the final Taylor table, the network weights, and the scalar source selectors are retained online.

At $r = 3$ and logarithmic depth and parameter precision, $C = O(\log^{9/2} n)$. The dense term is

$$O\left(n^2 \log^{3+(9/2)(\omega-2)} n\right).$$

Since $7^6 < 2^{17}$, one has $\omega < 17/6$, and its logarithmic exponent is less than $27/4 < 7$. The nonlinear term is $O(n \log^{21/2} n)$ and the padding term is a fixed power of $\log n$. Both fit $O(n^2 \log^7 n)$ because every fixed power of $\log n$ is dominated by any fixed positive power of n . This is $O(Dn^2 \log^6 n)$ for the stated depth. The rank-one mean chain supplies the matching query lower bound. $\square$

Corollary A.66 (The finite-input critical law for low input rank). *For first-layer rank at most $r \in \{1, 2, 3\}$, the samplers can be chosen to use*

$$O\left(\min\{\sqrt{D}, n/\sqrt{D}\}\right)$$

expected input probes and

$$O\left((1 + \min\{\sqrt{D}, n/\sqrt{D}\}) B^{\max\{4, r+2\}}\right)$$

expected bit operations, after the stated general-depth preprocessing. The worst-case query order over this network class is sharp when $b \geq \lceil \log_2 \min\{n, D + 1\} \rceil$.

Proof idea. The same function oracle also supports a rare full-input read. On the open amplitude gate, reading the context determines the few basis means exactly. The cached oracle then samples the known output probability without a dense forward pass except on a sufficiently rare precision-refinement event.

Proof. In addition to the source factory, consider opening its dyadic gate $q = O(D^{-1/2})$ and reading all n signs only when that gate opens. Compute the r basis means exactly with $O(nB^2)$ bit operations. The known sign mean on the open branch is $F(z)/q$. Its cached interval costs $O(B^{\max\{4, r+2\}})$ bits. Direct refinement has unconditional conditional-on-gate cost $O(H\varepsilon B^4) \leq O(B^4)$, because $\varepsilon \leq H^{-2}$ and $H = n^2 D$. This follows from the same fresh-uniform interval-measure argument as before. The expected probe count is at most nq and the expected bit cost is $O((1 + n/\sqrt{D})B^{\max\{4, r+2\}})$. Choose this algorithm when $n \leq D$ and the source factory otherwise.

For the lower bound choose the largest power of two $m \leq \min\{n, D+1\}$, average those m coordinates in the first layer, and use the scalar critical mean chain. Its finite-input lower bound is $\Omega(\min\{\sqrt{D}, m/\sqrt{D}\})$, which is within an absolute factor of the claimed expression. The remaining inputs and neurons are ignored. This first matrix has rank one, and $1/m$ belongs to the stated dyadic grid. $\square$

A.20 The sharp law at every fixed first-layer rank

Rectangular multiplication removes the rank-three restriction in the preceding preprocessing argument. The analytic construction already works at every fixed rank. Its Taylor tables have a number of columns that is polynomial in the logarithm of the width. A sufficiently unbalanced exact matrix multiplication computes all those columns within the original preprocessing budget.

Theorem A.67 (Critical sampling at every fixed first-layer rank). *Fix an integer $r \geq 1$. Consider a zero-bias width- n , depth- $D \geq 1$ tanh network whose absolute row sums are at most one and whose first weight matrix has rank at most r . Its weights are dyadic with at most b bits. Put $B = 16 + b + \lceil \log_2(nD + 2) \rceil$. One uniform algorithm, with constants depending on r , samples the exact output using*

$$O_r(\sqrt{D+1}) \text{ expected input probes,} \quad O_r\left(\sqrt{D+1} B^{\max\{4, r+2\}}\right) \text{ expected bit operations.}$$

At $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, its weight-only preprocessing and storage are $O_r(Dn^2 \log^6 n)$. The worst-case query order is $\Theta_r(\sqrt{D})$ when $n \geq D+1$ and $b \geq \lceil \log_2(D+1) \rceil$.

The rank is fixed before taking the asymptotic bound. Both the constants and the degree of the online precision polynomial depend on it. The theorem supplies no uniform polynomial bound when r grows with n . In particular, rank four is not a hardness threshold.

Lemma A.68 (Exact batching of a thin coefficient matrix). *For every fixed integer $r \geq 1$, an n by n integer matrix can be multiplied by an n by C integer matrix, $C \geq 1$, using*

$$O_r\left(\log(C+2) \left[n^2 C^{1/(2r)} + C^{8+1/(2r)}\right]\right) \quad (*)$$

scalar arithmetic operations. If the input entries have at most m bits, the computation can use only integers with $O_r(m + \log n + \log(C+2))$ bits. It is a uniform exact algorithm; no oracle for real arithmetic or for matrix multiplication constants is required.

Proof idea. Choose one rational bilinear formula with a square side q and a short side $w \geq q^{1/4}$. Its tensor powers supply a block large enough to hold all C coefficient columns. Tiling the large square matrix with these blocks yields the first term in $(*)$; padding yields the second.

Proof. Set $\epsilon = 1/(8r)$. Le Gall's rectangular multiplication bound gives exponent two at every aspect exponent below 0.3029805 [26]. We use the strictly smaller exponent 0.30. The theorem and the definition of the exponent by exact tensor rank apply over the rational field. Permuting the three tensor indices gives multiplication of a q by q matrix by a q by w matrix. Consequently there are integers $q \geq 2$, $2 \leq w \leq q$, and a finite exact rational bilinear formula with R products such that

$$w^4 \geq q, \quad q^2 \leq R \leq q^{2+\epsilon}. \quad (**)$$

To see that the inequalities can hold simultaneously, apply the exponent bound with excess $\epsilon/2$, then take a sufficiently large base size and $w = \lfloor q^{0.30} \rfloor$. The constant in the rank bound is absorbed by $q^{\epsilon/2}$. The tensor flattening along the q^2 entries of the first input has rank q^2 , which gives $R \geq q^2$.

There is also a direct uniform way to choose this finite formula. Enumerate integer triples and rational coefficient arrays by increasing encoding length. For each candidate, check $w^4 \geq q$ and $R^{8r} \leq q^{16r+1}$, and verify every coefficient in its bilinear tensor identity by rational arithmetic. This search terminates by the preceding existence theorem. It is performed once for a given r ; its finite time and stored certificate are constants in O_r . Thus this use of the exponent theorem does not hide nonuniform advice or a real coefficient test.

Fix the resulting formula. A tensor power of depth j multiplies a u by u matrix by a u by c matrix, where $u = q^j$ and $c = w^j$. Its operation count T_j satisfies

$$T_j \leq RT_{j-1} + O_r(q^{2j} + q^j w^j) \leq RT_{j-1} + O_r(q^{2j}).$$

Since $q^2 \leq R \leq q^{2+\epsilon}$, this implies

$$T_j = O_r((j+1)u^{2+\epsilon}).$$

The factor $j+1$ includes the possible equality $R = q^2$.

Choose the least $j \geq 0$ such that $w^j \geq C$. Then $c \leq wC$, $j+1 = O_r(\log(C+2))$, and

$$u \leq qC^{\log q / \log w} \leq qC^4.$$

Pad the row dimension to a multiple of u and the column dimension to c . With $h = \lceil n/u \rceil$, there are h^2 block products. Their output sums and the creation of zero padding fit the same operation count. Using $h \leq n/u + 1$ gives

$$O_r((j+1)[n^2 u^\epsilon + u^{2+\epsilon}]) = O_r(\log(C+2)[n^2 C^{4\epsilon} + C^{8+4\epsilon}]).$$

This argument also covers $C > n$. Substitution of $4\epsilon = 1/(2r)$ proves (*).

It remains to justify the bit lengths. Let Δ be a positive common denominator of the finitely many rational coefficients in the chosen base formula. Every input transformation through j levels has denominator dividing Δ^j . Leaf products therefore have denominator dividing Δ^{2j} , and the j output recombination levels raise this exponent by at most j . All intermediate quantities can be represented over predetermined powers of Δ of exponent at most $3j$. No reduction by a greatest common divisor is necessary.

The base formula contains only a fixed number of fixed coefficients. Their absolute linear-form sums give a fixed bound on magnitude growth at one level. After input transformation, multiplication, and output recombination, the logarithm of the magnitude has grown by $O_r(j)$, in addition to the input bit lengths. Summation over the h inner blocks adds at most $O(\log n)$ bits. Hence integer numerators, fixed-denominator rescalings, and completed output entries all have $O_r(m + j + \log n)$ bits. Applying schoolbook arithmetic to these integers proves the stated finite-bit implementation. The identities hold exactly before any later rounding of the completed matrix product. $\square$

Proof idea. The proof of Theorem A.65 builds its factory and analytic table for every fixed number of scalar source streams; only its matrix multiplication estimate uses rank three. Keep the factory, its amplitude gate, and its precision stopping rule. Apply the batching lemma at each dense linear layer, then count the remaining coefficient arithmetic directly. For fixed rank, the extra powers of $\log n$ in the latter work are absorbed by its smaller factor of n .

Proof of Theorem A.67. If the first matrix has rank zero, the output has mean zero and a fair bit suffices. Otherwise use its actual rank, padding the scalar source description to r if desired. The construction before the matrix multiplication step in the proof of Theorem A.65 applies to every fixed r . It gives an amplitude gate of probability $O(D^{-1/2})$, a conditional factory using $O_r(D+1)$ expected probes, and a Taylor oracle whose conditional expected bit work is $O_r((D+1)B^{\max\{4, r+2\}})$. That proof includes the unconditional probability bounds for all arbitrary precision refinements. These facts yield the claimed online bounds. Depth one uses the direct scalar factory.

The table of that proof has $J = O_r(D^{r/2})$ grid centers, Taylor degree $K = O_r(B)$, $N_K = \binom{K+r}{r} = O_r(K^r)$ coefficients per center, $C = JN_K = O_r(D^{r/2}K^r)$ coefficient columns, and working precision $M = O_r(B+D)$. Increasing M by $O_r(b + \log n + \log(C+2))$ preserves this order.

At each linear layer, clear denominators as in that proof and apply Lemma A.68 to the resulting integer matrices. All intermediate integers have $O_r(M)$ bits, so each scalar arithmetic operation costs $O_r(M^2)$ bit operations. The product is exact before its completed entries are rounded, so the analytic error induction applies without change. The basis computation and the nonlinear recurrences have the costs given in that proof. A general-depth preprocessing bound is thus

$$O_r\left(n^2B^3 + D\left[\log(C+2)(n^2C^{1/(2r)} + C^{8+1/(2r)}) + nD^{r/2}K^{2r} + nD^{r/2}M\right]M^2\right). \quad (***)$$

This bound includes the first affine layer, all working storage, and the stored Taylor table.

Let $v = \log_2 n \geq 1$. When $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, the chosen parameters satisfy $D, K, M = O_r(v)$ and $C = O_r(v^{3r/2})$. The dense term in (***) is at most

$$O_r\left(n^2v^{3+(3r/2)/(2r)+1}\right) = O_r(n^2v^{19/4}),$$

where the final factor v bounds $\log(C+2)$. The nonlinear term is $O_r(nv^{3+5r/2})$, the constant exponentials cost $O_r(nv^{4+r/2})$, and the padding term is $O_r(v^{12r+19/4})$. For each fixed r , each of the last three terms is $O_r(n^2v^7)$, because a fixed power of $\log n$ is bounded by any fixed positive power of n . The basis term also fits this bound. Consequently preprocessing and storage are $O_r(n^2 \log^7 n) = O_r(Dn^2 \log^6 n)$.

Finally, the rank-one mean-chain family lies in every class with rank bound $r \geq 1$. Its lower bound from Theorem A.38 is $\Omega(\sqrt{D})$ under the displayed width and dyadic-grid conditions. This proves the worst-case claim. $\square$

The rare full-input branch in Corollary A.66 uses the same Taylor oracle and hence also extends to every fixed r . Under (***), its expected probe bound is $O_r(\min\{\sqrt{D}, n/\sqrt{D}\})$, with the same finite-input lower bound and with expected bit work $O_r((1 + \min\{\sqrt{D}, n/\sqrt{D}\})B^{\max\{4, r+2\}})$. The unrestricted-rank critical exponent remains open.

A.21 A critical sampler from a column envelope

The first-layer rank can grow with the width in the result below. A different condition replaces the rank bound. Define the column envelope of the first weight matrix by

$$C_{\text{col}} = \sum_{j=1}^n \max_i |w_{1,ij}|.$$

This quantity is computed directly from the weights. It is one for a Hadamard sign matrix divided by its width, although that matrix has full rank. It can also be as large as n . The theorem therefore leaves the unrestricted critical problem open.

The empirical telescope uses the multilevel debiasing principles of Rhee and Glynn [37] and the antithetic vector-mean construction of Blanchet and Glynn [5]. We give explicit bounded correction branches, sampled derivatives, and a finite-bit implementation for the present network.

Theorem A.69 (Column-envelope sampling). *Put $K_{\text{col}} = \max\{1, C_{\text{col}}\}$. Consider zero-bias tanh networks with row sums at most one. At $D = \lceil \log_2 n \rceil$ and $b = O(\log n)$, weight-only preprocessing and storage $O(Dn^2 \log^6 n)$ suffice for an exact sampler with*

$$\mathbb{E}Q = O(K_{\text{col}}^2 \sqrt{D+1}), \quad \mathbb{E}W = O(K_{\text{col}}^7 \sqrt{D+1} B^{40}), \quad B = 16 + b + \lceil \log_2(nD+2) \rceil.$$

Here Q counts input probes and W counts all online bit operations. The constants in the online bounds are absolute. The preprocessing bound uses the stated precision convention $b = O(\log n)$; the additional preprocessing in this proof is explicitly $O(Dn^2 B^4)$. The worst-case probe order on $C_{\text{col}} \leq C_0$ for every fixed $C_0 \geq 1$ is $\Theta(\sqrt{D})$ under the width and grid conditions of the critical mean-chain lower bound.

Proof idea. One context probe supplies an implicit bounded vector whose expectation is the entire first preactivation, up to a known scale. A corrected empirical-mean factory needs $O(K_{\text{col}}^2 D)$ such vectors on its open amplitude branch. Its correction terms involve fourth derivatives. We sample their chain-rule expansions along at most four paths through the network. A small amount of slack in each first-derivative factor keeps their product bounded by an absolute constant. Fourth-order antithetic remainders then admit an exact importance sampler. Every subsequent neuron coin uses the already probed empirical observations; no dense evaluation is performed.

The proof occupies the rest of this subsection. Its first part builds the source vector, the amplitude gate, and the common derivative majorants. The second gives the coin for one chain-rule diagram, the third the empirical correction and its antithetic identity, and the fourth the exact tilting that samples the fourth-order remainders. The last part assembles the telescope and counts sources and bit work.

A.21.1 A source vector and common derivative bounds

For a nonzero first-layer column put $d_j = \max_i |w_{1,ij}|$, and let c_j be the least power of two at least d_j . Set $c_j = 0$ on a zero column and $C = \sum_j c_j$. Then

$$C_{\text{col}} \leq C \leq 2C_{\text{col}}.$$

The zero matrix has fair output and is handled separately. Otherwise choose J with probability c_J/C , query x_J , and retain the pair (J, x_J) . It represents the vector

$$Y_i = w_{1,iJ} x_J / c_J, \quad |Y_i| \leq 1, \quad \mathbb{E}Y = W_1 x / C. \quad (\text{A.108})$$

Only requested coordinates are formed. Each is a known dyadic number. Indeed $2^{-b} \leq c_J \leq 1$ is a power of two, so its denominator divides the original grid denominator 2^b . Averaging m such observations therefore introduces only $O(\log m)$ additional bits.

Let $F(y)$ be the scalar output obtained by first applying $\tanh(Cy)$ coordinatewise and then applying layers $2, \dots, D$ of the given network. Thus $F(\mathbb{E}Y)$ is the desired output mean. Put $r_1 = 1$. For $\ell \geq 2$, choose dyadic radii with

$$\tanh r_{\ell-1} \leq r_\ell \leq \tanh r_{\ell-1} + \epsilon_0, \quad 0 < \epsilon_0 \leq (16D^2)^{-1}.$$

Certified scalar evaluation and upward rounding construct these with $O(\log(D+2))$ bits. The one-Lipschitz property of $\tanh$ gives

$$r_\ell \leq R_{\ell-1} + D\epsilon_0, \quad R_j = \tanh^{\circ j}(1), \quad r_\ell^2 \leq 2/\ell, \quad r_\ell \leq 1.$$

For the square bound, expand $(\sqrt{3/(2\ell+1)} + 1/(16D))^2$ and use $\ell \leq D$; its three terms sum to less than $2/\ell$. The upper bound one follows inductively from $\tanh 1 + 1/16 < 1$. Also $r_D \geq R_{D-1} \geq (2D-1)^{-1/2}$, using the scalar radius lower bound. We work with $D \geq 32$; the finitely many smaller depths use the existing universal critical sampler, whose cost is absorbed by the absolute constants. Let q be the least power of two at least $2r_D$. Then

$$0 < q \leq 1, \quad q \leq 6D^{-1/2}, \quad q^{-1} \leq \sqrt{D}, \quad |f| \leq \frac{1}{2}, \quad f = F/q. \quad (\text{A.109})$$

A *common derivative majorant* is a nonnegative symmetric tensor $C^{(k)}$ such that $|\partial_{i_1} \cdots \partial_{i_k} f(y)| \leq C_{i_1 \dots i_k}^{(k)}$ for every $y \in [-1, 1]^n$. Its mass is $M_k = \sum C^{(k)}$. The tensor is independent of the point y . This is stronger than a pointwise bound on the sum of derivative magnitudes.

The positive chain-rule recursion, with scalar bounds

$$c_{\ell,1} = 1, \quad c_{\ell,2} = 2r_\ell, \quad c_{\ell,3} = 2, \quad c_{\ell,4} = 16r_\ell, \quad (\text{A.110})$$

constructs common majorants through order four. To check their masses, divide order k by C^k and let $A_\ell, B_\ell, E_\ell, H_\ell$ denote the resulting masses at orders one through four. The usual chain rule gives

$$\begin{aligned} A_\ell &\leq 1, & B_\ell &\leq B_{\ell-1} + 2r_\ell, \\ E_\ell &\leq E_{\ell-1} + 6r_\ell B_{\ell-1} + 2, & H_\ell &\leq H_{\ell-1} + 8r_\ell E_{\ell-1} + 6r_\ell B_{\ell-1}^2 + 12B_{\ell-1} + 16r_\ell. \end{aligned}$$

Consequently

$$B_D \leq 4\sqrt{2}\sqrt{D} < 6\sqrt{D}, \quad E_D \leq 50D, \quad H_D \leq 928D^{3/2}.$$

For the last inequality, the increment is at most $656\sqrt{2}\sqrt{\ell} < 928\sqrt{\ell}$. Writing $\kappa = \max\{1, C\}$, sufficient rational upper bounds for $f = F/q$ are

$$M_2 \leq 6\kappa^2 D, \quad M_3 \leq 50\kappa^4 D^2, \quad M_4 \leq 928\kappa^4 D^2. \quad (\text{A.111})$$

The full tensors need not be stored. At each neuron and each order $k \leq 4$, store the scalar mass obtained by putting one in every input direction. A chain-rule term is indexed by a partition of k labeled directions. Its weight is $c_{\ell, |\pi|}$ times the product of the corresponding lower-order row masses. Prefix arrays select both a partition and each predecessor. This samples an ordered chain-rule diagram with the correct normalized majorant weight. At most k paths cross any layer, and at most $k - 1$ vertices have derivative order greater than one. Its leaves supply the ordered tensor indices. The root factor $1/q$ scales the tensor and its mass together, so it does not change the diagram distribution. Throughout, M_k denotes the actual mass of the constructed tensor; the displayed inequalities are upper bounds used only in the work estimates. All stored quantities are dyadic, with $O(D[b + \log(D + n + 2)])$ bits. The four mass passes and their selectors use $O(Dn^2)$ scalar operations and storage entries.

A.21.2 A bounded coin for a chain-rule diagram

The normalized-neuron sampler used below has the following interface. Its first layer uses the arbitrary-radius majority factory of Theorem A.24, with radius C and its outer gate $\tanh C$. The resulting mean is $\tanh(Cy_i)$, which already equals $h_{1,i}/r_1$ because $r_1 = 1$. At a later row put $\lambda = \sum_j |w_{\ell, ij}|$ and $\rho = \lambda r_{\ell-1}$. If $\lambda = 0$, return a fair sign. Otherwise select signed normalized predecessor coins with probabilities $|w_{\ell, ij}|/\lambda$. Their mean z satisfies $\rho z = \sum_j w_{\ell, ij} h_{\ell-1, j}$. The scalar majority factory has mean $\tanh(\rho z)/\tanh \rho$; a final known gate $\tanh \rho/r_\ell \leq 1$ gives mean $h_{\ell, i}/r_\ell$. The inequality follows from $\rho \leq r_{\ell-1}$ and the upward radius bound. A further gate of probability r_ℓ gives an actual-activation sign of mean $h_{\ell, i}$ when required.

Lemma A.70 (A sampled derivative term). *Fix $k \leq 4$, a sampled order- k diagram, and an input point supplied by coordinate-coin access. Let α be the product of its edge signs and its scalar derivative factors divided by the majorants in (A.110). There is an exact $Z \in \{-1, 0, 1\}$ with*

$$\mathbb{E}Z = \alpha/48.$$

It makes a polynomial number of coordinate-source calls in expectation. The polynomial is uniform over the supplied point and over all diagrams.

Proof idea. The sharp bound $\tanh' \leq 1$ has an endpoint that prevents a direct positive factory. Divide each such factor by $1 + 1/(4D)$ instead. Their total loss stays below three. Every higher derivative is a fixed polynomial in a normalized activation, and there are at most three such vertices in a fourth-order diagram.

Proof. At a vertex in layer ℓ , put $h = \tanh s$, $u = h/r_\ell$. Independent normalized-neuron coins have mean u . The polynomials for derivative orders two, three, and four, after division by their majorants, are respectively

$$-u + r_\ell^2 u^3, \quad -1 + 4r_\ell^2 u^2 - 3r_\ell^4 u^4, \quad u - \frac{5}{2}r_\ell^2 u^3 + \frac{3}{2}r_\ell^4 u^5.$$

Their absolute coefficient sums are at most 2, 8, 5. Sampling a signed monomial from these coefficients, and returning a fair sign on unused mass, gives a sign with the corresponding mean divided by $K_2 = 2, K_3 = 8, K_4 = 5$.

For a first-derivative vertex put $\epsilon = 1/(4D)$ and $M = 4D + 1$. Take M independent signs of mean h , and let K be the number of plus signs. Return a bit with conditional probability

$$\frac{4K(M - K)}{M(M - 1)(1 + \epsilon)}. \quad (\text{A.112})$$

This belongs to $[0, 1]$, because $4K(M - K) \leq M^2$ and $M/(M - 1) = 1 + \epsilon$. Its expectation is $(1 - h^2)/(1 + \epsilon)$.

Use fresh neuron coins in all these constructions. There are at most $4D$ first-derivative vertices, so their lost factor is below $(1 + 1/(4D))^{4D} < 3$. The identity $\sum(\nu - 1) = k - 1 \leq 3$ bounds the product of the higher-derivative losses by 16: the largest possibility is one order-three and one order-two vertex. Multiply all sampled signs and bits and the known edge signs. Finally retain that product with probability

$$(1 + \epsilon)^{N_1} \prod_{\nu \geq 2} K_\nu / 48,$$

returning zero otherwise. This is a known rational probability at most one. Conditional independence proves the required mean. The work bound is proved quantitatively below. $\square$

Conditional on a majorant diagram with leaf indices $J_1, \dots, J_k$, the resulting bounded estimate of a directional derivative is

$$48M_k Z \prod_{a=1}^k v_{a, J_a}. \quad (\text{A.113})$$

Its expectation is $D^k f(y)[v_1, \dots, v_k]$. Diagram sampling requires only the scalar masses and prefix arrays already stored.

A.21.3 An empirical correction with sparse coordinate access

For $m \geq 2$ vector observations let z be their mean, V_m their unbiased sample covariance, and define

$$G_m = f(z) - \frac{V_m : D^2 f(z)}{2m}, \quad P_m = \mathbb{E} G_m.$$

The colon denotes contraction of the two indices. The strong law and bounded convergence give $\mathbb{E} f(z) \rightarrow f(\mathbb{E} Y)$. Every covariance entry has magnitude at most two, so the correction has magnitude at most M_2/m and $P_m \rightarrow f(\mathbb{E} Y)$. If a, b are a uniform unordered pair of observations and $u = Y_a - Y_b$, then $\mathbb{E}[uu^\top]/2 = V_m$. Thus the base coefficient has the bounded unbiased estimator

$$\widehat{G}_m = \frac{1}{2} S - \frac{48M_2}{4m} Z u_{J_1} u_{J_2}, \quad \mathbb{E} S = 2f(z).$$

The sign S is supplied by the normalized critical sampler followed by its gate $2r_D/q \leq 1$. At an empirical point z , each coordinate source call simply selects one retained observation uniformly and samples its known dyadic coordinate. The second term uses Lemma A.70. If $m \geq 384M_2$, then $|\widehat{G}_m| \leq 5/8$.

We next give an exact correction for a fixed population of $2m$ observations. Let its mean be z . A uniform half A and its complement B have means $z + \delta, z - \delta$ and covariance matrices V_A, V_B . Write $H = D^2 f(z)$ and $T = D^3 f(z)$. Define

$$\begin{aligned} R_4 &= \frac{1}{2} [f(z + \delta) + f(z - \delta)] - f(z) - \frac{1}{2} H[\delta, \delta], \\ R_H &= \frac{1}{2} V_A : [D^2 f(z + \delta) - H - T[\delta]] + \frac{1}{2} V_B : [D^2 f(z - \delta) - H + T[\delta]]. \end{aligned}$$

The exact identity is

$$G_{2m} - \mathbb{E}_A G_m(A) = c_3 + \mathbb{E}_A \left[\frac{R_H}{2m} - R_4 \right], \quad c_3 = \frac{D^3 f(z) : \sum_{a=1}^{2m} (Y_a - z)^{\otimes 3}}{4m(m-1)(2m-1)}. \quad (\text{A.114})$$

Here $T[\delta]$ leaves two indices uncontracted.

Proof of the identity. The within-population covariance decomposition and complement symmetry give

$$(m-1)(V_A + V_B) = (2m-1)V_{2m} - 2m\delta\delta^\top, \quad \mathbb{E}\delta\delta^\top = V_{2m}/(2m), \quad \mathbb{E}\delta^{\otimes 3} = 0.$$

For centered observations $v_a = Y_a - z$, subset inclusion probabilities $1/2$ and $(m-1)/(2(2m-1))$ also give

$$\mathbb{E}[V_A \otimes \delta] = \frac{\sum_a v_a^{\otimes 3}}{2(m-1)(2m-1)}.$$

Indeed the distinct-index terms sum to minus the equal-index third moment, while the third moment of δ vanishes. Substituting these expressions in the paired Taylor expansions proves (A.114).

The cubic term is estimated by a uniform observation a , the direction $u = Y_a - z$, and an order-three diagram. Its estimator is

$$\frac{48M_3Z u_{J_1} u_{J_2} u_{J_3}}{2(m-1)(2m-1)}.$$

Its magnitude is at most $48 \cdot 6M_3/m^2$.

For the fourth-order terms use integral remainders. With a fair sign σ , an unordered pair $u = Y_a - Y_b$ from the half of mean $z + \sigma\delta$, and independent beta variables, they are

$$R_4 = \frac{1}{24} \mathbb{E}_{\sigma, t \sim \text{Beta}(1,4)} D^4 f(z + \sigma t \delta) [\delta, \delta, \delta, \delta], \quad (\text{A.115})$$

$$\mathbb{E}_A \frac{R_H}{2m} = \frac{1}{8m} \mathbb{E}_{A, \sigma, u, t \sim \text{Beta}(1,2)} D^4 f(z + \sigma t \delta) [u, u, \delta, \delta]. \quad (\text{A.116})$$

The first formula is the fourth-order Taylor remainder with density $4(1-t)^3$; the second is the second-order remainder of the Hessian with density $2(1-t)$. These formulas avoid subtracting nearly equal random function estimates.

A.21.4 Exact tilting of a half-sample

Choose a uniform pairing of the $2m$ observations and select one member of each pair by independent fair signs ε_j . For any fixed coordinate i ,

$$\delta_i = \sum_{j=1}^m \varepsilon_j d_j, \quad d_j = (Y_{a_j, i} - Y_{b_j, i})/(2m), \quad |d_j| \leq 1/m.$$

For a fixed pairing put $v = \sum d_j^2$, $u = \sum d_j^4$. Its exact second and fourth moments are

$$\nu_2 = v \leq 1/m, \quad \nu_4 = 3v^2 - 2u \leq 3/m^2. \quad (\text{A.117})$$

To sample signs with density proportional to $(\sum \varepsilon_j d_j)^k$, $k = 2, 4$, expose them sequentially. At partial sum s the remaining moment polynomials are

$$P_2(s) = s^2 + v, \quad P_4(s) = s^4 + 6s^2v + 3v^2 - 2u.$$

The next plus probability is $P_k(s+d)/(2P_k(s))$, using the remaining moments after this increment in the numerator. The two probabilities add to one. A zero denominator is a null path under the tilted law. This is a finite rational Doob transform.

Accept a fresh pairing with probability $m\nu_2$ for a quadratic tilt, or $m^2\nu_4/3$ for a quartic tilt. On acceptance use its tilted signs. On rejection return zero. There is no retry loop. Consequently the accepted half-sample has unnormalized density $m\delta_i^2$ or $m^2\delta_i^4/3$ relative to a uniform half.

Here is the complete Hessian-remainder branch. Its mass is $48M_4/(2m^2)$. Sample an order-four diagram and then choose $i = J_3$ or J_4 equally. Perform the quadratic tilt in coordinate i . Put $v_\delta = (\delta_{J_3}^2 + \delta_{J_4}^2)/2$. On an accepted split, generate σ, u, t as in (A.116), and the diagram coin Z at $z + \sigma t \delta$. Return a bounded sign estimator with conditional mean

$$Z \frac{u_{J_1} u_{J_2}}{4} \frac{\delta_{J_3} \delta_{J_4}}{v_\delta}.$$

Both rational factors belong to $[-1, 1]$; accepted splits have $v_\delta > 0$. Multiplying the branch mass, the accepted density mv_δ , and the diagram mean $\alpha/48$ gives exactly the integrand coefficient $M_4/(8m)$ in (A.116).

The quartic branch has mass $48M_4/(8m^2)$. Sample an order-four diagram, choose one of its four leaves uniformly, and perform the quartic tilt in that coordinate. Put $w_\delta = \frac{1}{4} \sum_{a=1}^4 \delta_{J_a}^4$. On acceptance generate σ, t from (A.115) and return conditional mean

$$-Z \frac{\prod_{a=1}^4 \delta_{J_a}}{w_\delta}.$$

The ratio has magnitude at most one by the arithmetic-geometric mean inequality. The branch mass and accepted density $m^2 w_\delta / 3$ give precisely the negative term in (A.115). Only four coordinate sequences are ever needed to form these moments and ratios.

The beta variables require no real-number oracle. Use one Pólya urn, initially $(1, 4)$ or $(1, 2)$, for the whole integral term. A coordinate coin of mean $z_i + \sigma t \delta_i = (1 - t)z_i + t(z_i + \sigma \delta_i)$ first draws from that urn, then selects a uniform observation from the full population or the chosen half. It finally samples the known dyadic coordinate Y_i . Conditional on the beta parameter, all urn draws are independent. The same urn is shared by every neuron and every factor in that one diagram. A fresh urn per factor would change the integral and is not used. To verify this representation at finite stopping times, a specified urn word with s successes and f failures has probability

$$\frac{(a)_s (b)_f}{(a+b)_{s+f}} = \frac{1}{B(a, b)} \int_0^1 t^{s+a-1} (1-t)^{f+b-1} dt.$$

Here $(a)_s$ is a rising factorial. The first equality follows by multiplying the predictive probabilities; the second follows by integrating by parts. Hence every finite semantic transcript of source outputs and high-level choices has exactly the beta-mixture probability, even when requests are adaptive. The finite moment bounds below imply almost sure termination, so the same identity applies to the returned value.

A.21.5 Exactness, source count, and finite-bit work

Put $A = 6M_3 + M_4$ and $K_* = 48$. Let m_0 be the least power of two with

$$m_0 \geq 2, \quad m_0 \geq 8K_*M_2, \quad m_0^2 \geq 16K_*A. \quad (\text{A.118})$$

Every bounded rational estimator below is rounded to a sign of that conditional mean using a fresh known coin. Give the base mass $3/4$, and return a known sign of mean $\hat{G}_{m_0}/(3/4)$. At each level $m = 2^a m_0$, give the cubic, Hessian, and quartic branches their masses just specified. Their sum is at most K_*A/m^2 . Return a fair sign on all remaining mass, including rejected tilts. The total non-base mass is at most $4K_*A/(3m_0^2) \leq 1/12$. This countable choice is exact with finite rational arithmetic: its actual non-base mass is $4(288M_3 + 30M_4)/(3m_0^2)$. Conditional on that choice, take level a with probability $(3/4)4^{-a}$, using a geometric draw, and choose the three branches in the proportions $288M_3 : 24M_4 : 6M_4$. Zero masses are omitted. Every selected branch is bounded and its mean is its required coefficient. Therefore

$$P_{m_0} + \sum_{a \geq 0} (P_{2^{a+1}m_0} - P_{2^a m_0}) = f(\mathbb{E}Y)$$

proves exactness. Absolute convergence follows from the branch-mass sum. If a majorant mass is zero, its branch is omitted.

The base probes m_0 observations and a correction probes $2m$. All later neuron-source requests select from those retained observations. Hence

$$\mathbb{E}N \leq \frac{3}{4}m_0 + \sum_{m=2^a m_0} \frac{K_*A}{m^2} (2m) = \frac{3}{4}m_0 + \frac{4K_*A}{m_0} \leq m_0. \quad (\text{A.119})$$

For explicit constants, the bounds (A.111) give $8 \cdot 48M_2 \leq 2304\kappa^2 D$ and $16 \cdot 48(6M_3 + M_4) \leq 943104\kappa^4 D^2$. A power of two at least $2304\kappa^2 D$ therefore satisfies the three starting conditions, so the least valid power obeys $m_0 < 4608\kappa^2 D$. Opening the gate q and otherwise returning a fair sign proves $\mathbb{E}Q < 27648\kappa^2 \sqrt{D}$ for $D \geq 32$.

We finish the bit count, including the unbounded urn counters. Use full majorities in every normalized neuron call. Every monomial and every first-derivative factor executes all of its selected source calls, even if an earlier factor already makes the product zero. At a layer $\ell \geq 2$, its full arity A_ℓ has, uniformly over rows,

$$\mathbb{E}A_\ell \leq 1 + 3\rho^2, \quad \mathbb{E}A_\ell^2 \leq 1 + 18\rho^2, \quad \rho \leq r_{\ell-1}.$$

For completeness, the pgf in the majority representation has $\mathbb{E}K = \rho^2$ and $\mathbb{E}K^2 = \rho^2 + \frac{4}{3}\rho^4$ before the factor $(\rho/\tanh\rho)/C_K$. Since $C_K \geq 1$ and $\rho \coth\rho \leq 4/3$ for $\rho \leq 1$, these inequalities follow by using $\sum_k \pi_k = 1$ for the constant term:

$$\mathbb{E}_\pi(2K+1) \leq 1 + \frac{8}{3}\rho^2, \quad \mathbb{E}_\pi(2K+1)^2 = 1 + \mathbb{E}_\pi(4K^2 + 4K) \leq 1 + \frac{4}{3}(8\rho^2 + \frac{16}{3}\rho^4) \leq 1 + 18\rho^2.$$

The gate only reduces the arity. Thus the mean and second moment are bounded by $1 + 6/(\ell-1)$ and $1 + 36/(\ell-1)$.

Let A_1 be the resulting first-layer source arity, including the zero arity when the outer gate closes. The general-radius theorem gives

$$\mathbb{E}A_1 \leq 2\max\{C, C^2\} \leq 2\kappa^2, \quad \mathbb{E}[\text{local scalar work}] = O(\kappa^2 B^4). \quad (\text{A.120})$$

For its second moment use the majority-index law $\pi_k = (C/\tanh C)d_k/C_k$. The proposal has $\mathbb{E}_d K = C^2$ and $\mathbb{E}_d K^2 = C^2 + \frac{4}{3}C^4$. Since $\sum_k \pi_k = 1$ and $C_k \geq 1$,

$$\begin{aligned} \mathbb{E}A_1^2 &\leq \mathbb{E}_\pi(2K+1)^2 \\ &\leq 1 + C \coth C \left(8C^2 + \frac{16}{3}C^4 \right) \leq 28\kappa^5. \end{aligned} \quad (\text{A.121})$$

The last step uses $C \coth C \leq 1 + C \leq 2\kappa$. The semantic first-layer count $S_1 = 1 + A_1$ therefore satisfies

$$\mathbb{E}S_1 \leq 3\kappa^2, \quad \mathbb{E}S_1^2 \leq 58\kappa^5.$$

These bounds include the accepted full majority's source calls. The scalar proposal and acceptance arithmetic is paid separately by (A.120).

For completeness, we give explicit bounds for the recurrence used in the finite-bit argument. Let a_ℓ, b_ℓ bound the first two moments of the total semantic vertices and source calls below a normalized-neuron call in layer ℓ . The upper-layer arity estimates give

$$a_\ell \leq 1 + \mu_\ell a_{\ell-1}, \quad b_\ell \leq \mu_\ell b_{\ell-1} + 1 + 2\mu_\ell a_{\ell-1} + \nu_\ell a_{\ell-1}^2,$$

where $\mu_\ell = 1 + 6/(\ell-1)$ and $\nu_\ell = 1 + 36/(\ell-1)$. The latter bounds the second factorial arity moment as well. Conditional on arity and predecessor indices, fresh subcalls give this recurrence even when different predecessor neurons have different costs.

Write $p_\ell = \prod_{j=1}^{\ell-1} (1+6/j) = \binom{\ell+5}{6} \leq (\ell+5)^6$. Division of the first recurrence by p_ℓ yields $a_\ell/p_\ell \leq 3\kappa^2 + \ell - 1$ and hence

$$a_\ell \leq \kappa^2(\ell+5)^7.$$

For $\ell \geq 2$, the nonrecursive part of the second recurrence is at most $52\kappa^4(\ell+4)^{14}$, since $\mu_\ell \leq 7$ and $\nu_\ell \leq 37$. Divide by p_ℓ , bound each reciprocal by one, and sum. This gives

$$b_\ell \leq p_\ell[58\kappa^5 + 52\kappa^4\ell(\ell+5)^{14}] \leq 110\kappa^5(\ell+5)^{21}. \quad (\text{A.122})$$

Each selected empirical coefficient uses at most $24D^2$ normalized or actual neuron roots, including the base function coin when present. If their complete recursive counts are T_j , the pointwise inequality $(\sum_{j=1}^J T_j)^2 \leq J \sum_{j=1}^J T_j^2$ does not require their independence. With the linear diagram bookkeeping included, a semantic count N for a selected coefficient has

$$\mathbb{E}N^2 \leq 2 \cdot 10^5 \kappa^5 (D+5)^{25}. \quad (\text{A.123})$$

All arities, row choices, and monomial degrees use only known weights and radii. Full padding prevents source outcomes from controlling these counts. In particular the bound is uniform over the empirical observations, the accepted split, and the latent beta parameter.

We spell out the arithmetic charge to keep that last conditioning statement separate from the implementation of the urn. At the j th reached primitive visit, the actual observable history determines every rational parameter. Fresh random bits implement its next known coin. The general-radius scalar theorem, the unit-radius scalar bound at later layers, and exact rational comparison give conditional expected local work

$$O(\kappa^2[B^2 + \log(m+2) + \log(j+2)]^4).$$

This statement concerns the observable filtration. It makes no conditional precision-tail claim given the latent beta variable. Summing the nonnegative predictable charges and using $N \log^4(N+2) = O(N + N^2)$ with (A.123) bounds the neuron and urn work. The pairing, the finite Doob transform, and the four coordinate sequences take $O(m[B^2 + \log(m+2)]^4)$ further work. Consequently one selected coefficient costs at most

$$O(\kappa^2[m + \kappa^5(D+5)^{25}][B^2 + \log(m+2)]^4) \quad (\text{A.124})$$

in expectation. The constants in these bounds do not depend on C or on the point supplied to the sampler.

The correction levels have geometric ratio $1/4$ and $m = 2^a m_0$. Since $\log m_0 = O(B)$, summing (A.124) over those levels gives $O(\kappa^7(D+5)^{25}B^8)$ conditional work. Here κ^2 is the local scalar work of one visit, from (A.120), and κ^5 enters through $\mathbb{E}S_1^2$ in (A.123). The factor B^8 is the fourth power of the operand length $O(B^2)$. The amplitude gate $q \leq 6D^{-1/2}$ and $32 \leq D \leq B$ then give $O(\kappa^7\sqrt{D}(D+5)^{24}B^8) = O(\kappa^7\sqrt{D}B^{32})$ online bits, including the initial branch selection. This implies the bound with B^{40} stated in the theorem.

Finally, growing C causes no preprocessing increase beyond the original budget. Each c_j is a power of two between 2^{-b} and one; hence $C \leq n$ has $O(b + \log n)$ bits. The positive order-four mass circuits remain of size $O(Dn^2)$ with operands of $O(D[b + \log(nD+2)]) = O(B^2)$ bits. Schoolbook arithmetic uses $O(Dn^2B^4)$ additional work and no larger order of storage. At the stated precision this is $O(Dn^2\log^4 n)$, within the original $O(Dn^2\log^6 n)$ budget. The arbitrary-radius factory creates its product proposal on demand and requires no table whose size grows polynomially in C . Substituting $\kappa \leq 2K_{\text{col}}$ gives the stated online bounds. The mean-chain family has $C_{\text{col}} = 1$, so its critical lower bound proves sharpness on each fixed bounded-envelope subclass. This completes the proof of Theorem A.69.

Scope and selection of the sampler. The column envelope is independent of algebraic rank. Preprocessing may compare K_{col}^4 with $D+1$ and choose between this construction and the existing zero-bias sampler. The resulting expected probe bound is

$$O\left(\min\{D+1, K_{\text{col}}^2\sqrt{D+1}\}\right).$$

Its bit work remains within the displayed conservative bound because $D \leq B$ in the stated regime. The new source bound improves the $O(D)$ envelope when $C_{\text{col}} = o(D^{1/4})$. Its bit bound is polylogarithmic whenever C_{col} is polylogarithmic in n . The exponent 40 is an implementation bound and makes no practical prediction.

B Attention and preprocessing

This appendix proves the attention results of Sections 5 and 6 and the decoder of Section 7. Sections B.1–B.3 treat a fresh context: the exponential race behind Theorem 5.1, its positional lower bound, and the bounded-logit heads. Sections B.4–B.9 treat a cache built at prefill. They give the rank-one index, exact decisions from certified intervals, the fixed-rank index of Theorem 6.2, the implicit rank-one query with its query law, and a known query list. Section B.10 proves the decoder of Theorem 7.1, and Section B.11 proves the lower bound of Theorem 6.1.

B.1 An exact attention primitive

The arithmetic of attention can be sampled through bounded signs. Let $H_j \in [-R, R]^n$, let Q, K, V be dyadic matrices with every row having ℓ_1 norm at most s , and fix a causal position t . Write

$$q = QH_t, \quad k_j = KH_j, \quad v_j = VH_j, \quad a_j = \frac{\beta}{n} \langle q, k_j \rangle, \quad A = |\beta| s^2 R^2.$$

Here $R, s \geq 0$ are rational bounds. For the finite-bit claims, β is rational or is the standard value $\sqrt{n}$, and its finite encoding is included in B . For $sR = 0$, define the normalized output to be zero and return an independent fair sign. In the remaining case the desired coordinate is

$$y_i = \frac{\sum_{j=1}^t e^{a_j} v_{j,i}}{\sum_{j=1}^t e^{a_j}}, \quad |y_i| \leq sR. \quad (\text{B.1})$$

A child request means an independent sign of prescribed mean $H_{j,k}/R$. Its own cost is excluded from the local arithmetic charge. Fresh requests remain independent conditional on the deterministic context. Repeated input addresses may be cached.

Theorem B.1 (Attention from bounded signs). *After rational row-prefix preprocessing, there is a uniform exact sign sampler of mean $y_i/(sR)$. It uses at most*

$$M_A = 2e^{2A} - 1 \quad (\text{B.2})$$

expected child requests and at most e^{2A} expected proposals of a causal position. If $B \geq 16$ bounds all rational input lengths and $\log_2(n + t + 2)$, the expected local finite-bit work is

$$O(B^4 e^{2A}). \quad (\text{B.3})$$

The constants in the O notation are absolute. Input probes, fair bits, integer addresses, arithmetic, and precision refinement are charged. There is no exact-real primitive. For $A = 0$, one uniform position and one linear-row sign request suffice.

Linear and score signs. For a row w with $\sum_k |w_k| \leq s$, choose index k with probability $|w_k|/s$, return $\text{sgn}(w_k)$ times a fresh child sign, and use an independent fair sign for the remaining probability. The returned sign has mean $wH_j/(sR)$ and uses at most one child request. Prefix sums of the rational masses implement this choice exactly.

For $A > 0$, choose I uniformly from $\{1, \dots, n\}$, independently sample signs of means $q_I/(sR)$ and $k_{j,I}/(sR)$, and multiply them. Complement the result if $\beta < 0$. The resulting sign S_j satisfies

$$\mathbb{E}S_j = a_j/A.$$

Thus $C_j = (1 + S_j)/2$ is a coin with bias

$$p_j = \frac{1 + a_j/A}{2} \in [0, 1]. \quad (\text{B.4})$$

A score-coin test uses at most two child requests. Independence of the query and key signs is needed even when their addresses agree.

The race. Propose J uniformly from $\{1, \dots, t\}$. Draw $N \sim \text{Poisson}(2A)$ and test N independent p_J -coins, stopping at the first failure. Accept the position if every test succeeds; when $N = 0$, accept without a score test. Repeat after rejection. After acceptance, return a fresh linear-row sign of mean $v_{J,i}/(sR)$.

Lemma B.2 (Acceptance and first accepted position). *A proposal of j is accepted with probability*

$$r_j = \exp(a_j - A).$$

The procedure terminates almost surely. Its accepted position has probability $e^{a_j}/\sum_k e^{a_k}$, and its returned sign has the mean in Theorem B.1.

Proof idea. The Poisson probability generating function turns repeated score-coin successes into the required exponential weight. Conditioning on the first successful proposal then cancels the common scale, and a fresh value sign supplies the attention mean.

Proof. The Poisson generating function gives

$$\mathbb{E}p_j^N = e^{-2A} \sum_{k=0}^{\infty} \frac{(2Ap_j)^k}{k!} = e^{-2A(1-p_j)} = e^{a_j-A}.$$

The series is absolutely convergent, including the endpoint coin biases. Every proposal succeeds with probability $\bar{r} = t^{-1} \sum_j r_j \geq e^{-2A} > 0$. The number of proposals is geometric with mean $1/\bar{r}$. For every j , summing over the first successful trial yields

$$\Pr(J_{\text{accepted}} = j) = \sum_{m=0}^{\infty} (1 - \bar{r})^m \frac{r_j}{t} = \frac{r_j}{\sum_k r_k} = \frac{e^{a_j}}{\sum_k e^{a_k}}.$$

The final value sign is sampled with fresh randomness, after the index is accepted. Conditioning on that index proves the mean identity. $\square$

The first failure rule removes a factor of A from the work bound.

Lemma B.3 (Stopped Poisson tests). *For a coin of bias $p \in [0, 1]$ and $N \sim \text{Poisson}(\lambda)$, let G be the number of coins inspected before the first failure or the completion of all N tests. Then*

$$\mathbb{E}G = \begin{cases} (1 - e^{-\lambda(1-p)})/(1-p), & p < 1, \\ \lambda, & p = 1. \end{cases}$$

Moreover, for $\lambda \geq 0$,

$$\mathbb{E}G \leq (e^\lambda - 1)e^{-\lambda(1-p)}. \quad (\text{B.5})$$

Proof idea. Condition on the Poisson count and sum the probabilities that each test is reached. Convexity of the exponential compares this stopped count directly with the proposal acceptance probability.

Proof. Conditional on $N = k$, the inspection count has mean $\sum_{r=0}^{k-1} p^r$. For $p < 1$, the finite geometric sum and the Poisson generating function give the formula. At $p = 1$, all N coins are read. For $q = 1 - p \in (0, 1]$, inequality (B.5) is equivalent to

$$\frac{e^{\lambda q} - 1}{q} \leq e^\lambda - 1.$$

Convexity of the exponential function gives $e^{\lambda q} \leq (1 - q) + qe^\lambda$. At $q = 0$, the required inequality is $\lambda \leq e^\lambda - 1$. $\square$

Proof idea. Charge a score test only when its proposal is reached. The preceding ratio bound survives averaging over proposed positions, so dividing by the success probability bounds the whole race.

Child-request bound in Theorem B.1. Let g_j be the mean number of inspected score coins in a proposal of j . The stopped-work identity gives

$$\mathbb{E}[\text{all score tests}] = \frac{t^{-1} \sum_j g_j}{\bar{r}} \leq e^{2A} - 1.$$

One can justify this equality without assuming the desired finite mean: truncate after m proposals, sum the predictable charges, and then apply monotone convergence. The displayed bound is uniform in the truncation. Each score test uses at most two children, and the final value uses at most one. This proves (B.2). $\square$

B.1.1 A finite-bit implementation without a large Poisson draw

Generating all of N before inspecting the coins would impose avoidable work on rejected proposals. The implementation instead partitions the Poisson time $2A$ into $m = \lfloor 2A \rfloor$ unit intervals and one final interval of length $\theta = 2A - m \in [0, 1)$. In a reached interval, generate its Poisson count, test the corresponding score coins, and stop the entire proposal on the first failure. Unvisited intervals are never generated. Independent Poisson increments give exactly the same acceptance law and inspection count as above. The loop counter has $O(B)$ bits, even when A is large. For $\beta = \sqrt{n}$, the integer part of $2A$ is obtained by an integer square-root computation after clearing rational denominators; the final interval length is a rational plus a rational multiple of $\sqrt{n}$.

Lemma B.4 (Small Poisson draws). *For $\theta \in [0, 1]$ that is rational, or a rational plus a rational multiple of $\sqrt{n}$, with total encoding length and $\log_2(n + 2)$ at most B , a $\text{Poisson}(\theta)$ variable can be generated exactly with $O(B^4)$ expected bit operations. Its auxiliary proposal lengths and precision refinements have uniformly bounded fourth moments after adding B .*

Proof idea. A geometric proposal dominates the small-parameter Poisson probabilities by a constant factor. Both the proposed count and the precision needed for its acceptance decision have geometric tails, which pay for finite arithmetic.

Proof. For $\theta = 0$, return zero. Otherwise propose $k \geq 0$ with probability 2^{-k-1} and accept with probability

$$a_k = e^{-\theta} \frac{(2\theta)^k}{4k!}.$$

Since $(2\theta)^k/k! \leq e^{2\theta}$, $a_k \leq e^\theta/4 < 1$. The product of proposal and acceptance masses is $\frac{1}{8}e^{-\theta}\theta^k/k!$, so acceptance has probability $1/8$ and the accepted count has exactly the Poisson law. The proposal length has a geometric tail. All polynomial moments of the total work over the mean eight proposals are bounded by the corresponding one-proposal moments.

In the quadratic case, exact comparisons in the quadratic field detect $\theta = 0$, and rational bisection of $\sqrt{n}$ supplies certified intervals for θ to any required absolute precision. The numerator magnitude requires only $O(B)$ extra guard bits. Here are sufficient elementary precision bounds. At requested absolute error 2^{-r} , use

$$V = 32(B + k + r + 1)$$

working bits. Evaluate $e^{-\theta}$ by its alternating Taylor series through degree $2V$. Its omitted tail is at most $1/(2V + 1)! < 2^{-V}$. Compute $(2\theta)^k/k!$ by k rational multiplications and divisions, retaining V fractional bits and propagating interval endpoints. Its exact magnitude is at most $e^2 < 8$, while each partial product is bounded by e^2 . Enlarging every intermediate interval by $2^{-V+4k+\lceil \log_2(2V+2) \rceil + 8}$ covers all accumulated rounding error, including the sum of the Taylor terms. With the displayed choice of V , the final acceptance interval has width less than 2^{-r-4} . There are $O(V)$ operations on $O(V)$ -bit numbers; schoolbook arithmetic therefore gives an $O(V^3)$ enclosure cost, and $O(V^4)$ is a common bound for both this calculation and address bookkeeping.

Lemma B.14, with $d = 4$ and $B + k + 1$ in place of B , compares the acceptance probability exactly with one lazy uniform variable. Its expected work is $O((B + k + 1)^4)$ for every value of a_k , including values arbitrarily close to zero or one, and its precision tail is geometric. Averaging over the geometric tail of k proves the result. $\square$

Lemma B.5 (Expected number of reached Poisson intervals). *Across the complete attention race, the expected number of reached unit intervals and nonempty final intervals is at most $3e^{2A}$.*

Proof idea. Every reached unit interval has a fixed positive chance of requesting at least one score coin. Charging intervals to these requests bounds their total number; the final partial interval contributes at most one per proposal.

Proof. Conditional on reaching a unit interval and on its chosen position, its inspection count is at least the indicator that its Poisson count is positive. Its conditional mean is therefore at least $1 - e^{-1}$, regardless of the coin bias. Predictable charging and Lemma B.3 bound the expected number of unit intervals by

$$\frac{e^{2A} - 1}{1 - e^{-1}}.$$

At most one final interval is visited in each position proposal, giving at most e^{2A} additional intervals in expectation. Since $(1 - e^{-1})^{-1} < 2$, the sum is less than $3e^{2A}$. $\square$

Proof idea. All local choices have polynomial bit cost. Summing those costs over reached score tests and reached Poisson intervals gives the total, including the work of unsuccessful proposals.

Bit bound in Theorem B.1. Uniform integer sampling in $\{1, \dots, t\}$ and $\{1, \dots, n\}$ uses binary rejection and $O(B)$ expected fair bits. A rational row choice uses a random integer in the common denominator, followed by binary search through a prefix table. It requires $O(B^3)$ expected bit work with the conservative representation above. Each score test consequently costs $O(B^4)$ local work in addition to its child calls. Small Poisson generation has the bound in Lemma B.4. The counter and arithmetic cost per reached interval is also $O(B^4)$, and Lemma B.5 bounds the expected number of reached intervals. Combining the expected number of proposals, score tests, and intervals proves (B.3). All choices have countable discrete outputs; the comparisons terminate almost surely. The geometric race then terminates almost surely. $\square$

The race is the exponential Bernoulli race of Dughmi, Hartline, Kleinberg, and Niazadeh [13] applied to a random coordinate product. Their fast race gives another dependence on the number of alternatives and the temperature. The bound here uses the known score range and records first-failure and bit costs because these quantities control recursive composition.

B.1.2 A bounded categorical output

Theorem B.6 (Bounded softmax interface). *Suppose independent signs of means $z_1, \dots, z_K \in [-1, 1]$ are available. An exact softmax sample uses at most $e^2 - 1$ expected sign requests, e^2 expected proposals, and $O(B^4)$ local expected bit work, where B also includes $\log_2(K + 2)$. Its probability of category k is $e^{z_k} / \sum_j e^{z_j}$.*

Proof idea. The same Poisson test applies directly to a logit sign. Its success probability is an exponential of the logit with a common shift, so the accepted category has the softmax law.

Proof. Propose k uniformly, test a Poisson number of mean two of the coin with bias $(1 + z_k)/2$, and accept if all tests succeed. A test is exactly a request for a logit sign followed by checking whether it is positive. Its acceptance probability is $e^{z_k - 1}$. Apply Lemmas B.2, B.3, and B.4 with $A = 1$. $\square$

This theorem is an interface condition: the logits must come with sign samplers, as in Section 2. Coordinate logits and appropriately normalized bounded-row linear heads supply them.

B.2 Positional search when the context is fresh

The following bounds apply when preprocessing is independent of the context. They therefore concern fresh context probes; an index built from that context is handled by the separate static-cache theorems. We give a lower bound for the final categorical output of one complete block. All weights used below are shared across positions. Let $n \geq 2$, $s \geq 1$, $\beta \geq 0$, and $\alpha \in [0, 1]$, with the chosen weights legal on the parameter grid. Fix a coordinate to one at every position and write another coordinate as $x_j \in \{-1, 1\}$. Every query row selects the fixed coordinate with coefficient s ; every key row selects x_j with coefficient s . The value row of interest selects x_j with coefficient one. Other value rows may be zero. Then, at position T ,

$$A = \beta s^2, \quad E = e^{2A}, \quad Y(x) = \frac{\sum_{j=1}^T e^{Ax_j} x_j}{\sum_{j=1}^T e^{Ax_j}}, \quad U(x) = (1 - \alpha)x_T + \alpha Y(x). \quad (\text{B.6})$$

Apply a tanh row of weight one and bias zero through the second residual sublayer, giving

$$Z(x) = (1 - \alpha)U(x) + \alpha \tanh U(x).$$

The two output logits are $Z(x)$ and $-Z(x)$. Both lie in $[-1, 1]$, as do the streams in the relevant coordinate. The token-one probability is

$$P(x) = \frac{1 + \tanh Z(x)}{2}. \quad (\text{B.7})$$

The fixed coordinates define an input subcube. A lower bound on this subcube also lower-bounds a sampler required to handle every context.

Lemma B.7 (A transcript coupling bound). *Let x and $x^{(j)}$ differ at one input coordinate. If an algorithm terminates almost surely and samples the required output distribution exactly on both inputs, then*

$$\Pr_x(\text{coordinate } j \text{ is queried}) \geq \text{TV}(\mathcal{L}_x(\text{output}), \mathcal{L}_{x^{(j)}}(\text{output})).$$

Arbitrary weight-only preprocessing and adaptive arithmetic are allowed.

Proof idea. Run both inputs with the same random tape. Their executions can separate only after reading the changed coordinate, so the probability of that read must cover the difference between the output laws.

Proof. Use the same preprocessing randomness and the same online random tape on both inputs. The transcripts agree until coordinate j is queried. If the run on x terminates without that query, the other run follows the same transcript and returns the same output. The probability of unequal outputs in this coupling is therefore at most the query probability on x . Total variation is at most the disagreement probability of any coupling. Almost-sure termination handles all finite transcripts; a null set of nonterminating tapes does not affect the inequality. $\square$

Theorem B.8 (A positional lower bound for the final token). *On the all-negative context, every exact sampler for (B.7) has expected input-query cost at least*

$$\frac{1}{16} \left[(1 - \alpha) + \alpha \frac{TE}{T - 1 + E} \right] \geq \frac{1 + \alpha \min\{T, E\}}{64}. \quad (\text{B.8})$$

If x_T is fixed in advance and only positions $j < T$ are counted, the bound is

$$\frac{\alpha}{32} \min\{T - 1, E\}. \quad (\text{B.9})$$

In particular, $\beta s^2 \geq \frac{1}{2} \log T$ forces $\Omega(\alpha T)$ expected probes of context positions in this class.

Proof idea. A single positive position changes the attention mean by an explicitly known amount. The two subsequent scalar maps have derivatives bounded below on the relevant interval. Summing the resulting coupling bounds over one common baseline input gives the search cost.

Proof. Let x^- have every $x_j = -1$, and let $x^{(j)}$ flip only coordinate j . Put

$$p = \frac{E}{T - 1 + E}.$$

Then $Y(x^-) = -1$ and $Y(x^{(j)}) = -1 + 2p$. Consequently

$$U(x^{(j)}) - U(x^-) = 2\alpha p + 2(1 - \alpha)\mathbf{1}_{\{j=T\}}.$$

Both endpoints lie in $[-1, 1]$. On that interval,

$$\frac{d}{du} [(1 - \alpha)u + \alpha \tanh u] = (1 - \alpha) + \alpha \text{sech}^2 u \geq \frac{1}{4},$$

and, on $[-1, 1]$,

$$\frac{d}{dz} \frac{1 + \tanh z}{2} = \frac{1}{2} \operatorname{sech}^2 z \geq \frac{1}{8}.$$

The inequalities follow from $\cosh 1 < 2$. Integrating the derivatives gives

$$P(x^{(j)}) - P(x^-) \geq \frac{1}{16} [\alpha p + (1 - \alpha) \mathbf{1}_{\{j=T\}}].$$

Lemma B.7 lower-bounds the chance of querying each coordinate on the same baseline context. Summing these probabilities proves the first expression in (B.8). For positive a, b , $ab/(a+b) \geq \frac{1}{2} \min\{a, b\}$, whence $TE/(T-1+E) \geq \frac{1}{2} \min\{T, E\}$. Writing $M = \min\{T, E\} \geq 1$, the inequality

$$\frac{1 - \alpha}{16} + \frac{\alpha M}{32} \geq \frac{1 + \alpha M}{64}$$

follows from $3 - 4\alpha + \alpha M \geq 0$. If position T is fixed, sum only over $T-1$ possible flips and use $(T-1)E/(T-1+E) \geq \frac{1}{2} \min\{T-1, E\}$. $\square$

Theorem B.9 (A matching law for the binary attention family). *Let Q_{bin}^* be the optimum worst-context expected number of distinct x_j probes for the family (B.6), with the fixed coordinate known. Then*

$$\frac{1 + \alpha \min\{T, e^{2\beta s^2}\}}{64} \leq Q_{\text{bin}}^* \leq 1 + 32\alpha \min\{T, e^{2\beta s^2}\}. \quad (\text{B.10})$$

There is a uniform finite-bit sampler with expected work

$$O\left(B^4 [1 + \alpha \min\{T, e^{2\beta s^2}\}]\right).$$

Thus the law is sharp up to the displayed constants in queries and up to polylogarithmic encoding and address factors in bit work.

Proof idea. The residual choice makes an attention request only with probability proportional to its coefficient. Answer such a request either by the exponential race or by reading and counting the entire context. Selecting the cheaper method matches the lower bound.

Proof. The lower bound is Theorem B.8. For the upper bound, cache x_T when it is first needed. A sign of mean U is obtained by choosing x_T with probability $1 - \alpha$ and a sign of mean Y with probability α . A sign of mean Z chooses a U sign with probability $1 - \alpha$ and a sign of mean $\tanh U$ with probability α . Lemma A.26 gives at most four expected U -sign requests, including this residual choice. Applying the same lemma once more produces the sign whose positive probability is $P(x) = (1 + \tanh Z(x))/2$. The construction uses at most sixteen expected U -sign requests and at most 16α expected requests for Y . This is a predictable charge: each request is reached before its fresh residual-choice coin is drawn.

There are two ways to answer a Y request. Theorem B.1 uses at most $2E - 1$ probes, with repetitions counted. Alternatively, on the first such request read all T bits, cache their number k of positive values, and use

$$Y = \frac{k - (T - k)e^{-2A}}{k + (T - k)e^{-2A}}. \quad (\text{B.11})$$

When $k = 0$, the answer is exactly -1 . The chance of ever making this first request is at most the expected number of requests. Thus the second implementation has expected probe cost at most $16\alpha T$ beyond the cached x_T . Choosing the better implementation proves the upper query bound.

For a uniform bit implementation, the choice need only be within a factor two: compare $2A$ with an approximation to $\log T$ of absolute error at most $1/4$. Either choice has cost within a factor two of the smaller alternative. In (B.11), $k \geq 1$ gives a denominator at least one, so Y has derivative at most $2T$ in e^{-2A} . To compute Y to error 2^{-r} , Lemma B.12 encloses e^{-2A} to width $2^{-r - \lceil \log_2(T+2) \rceil - 2}$ in $O((B+r)^4)$ bit work. The sign comparison has a geometric refinement tail. The cached count requires $O(TB)$ bit work, incurred only if a Y request is reached. The remaining scalar and categorical factories have constant expected arity and polynomial moments at their fixed argument bounds. Predictable charging yields the stated bit bound. $\square$

If the coordinate used by the query is also supplied as an unknown input, its sign can be cached on the first attention request. Complementing all key signs when this bit is negative reduces the calculation to the same family. This changes the upper bound by an absolute constant and leaves the lower bound unchanged.

B.3 A quadratic interface for bounded logits

The direct scaled Poisson race is sufficient for context independence. The bounded-logit promise also permits a polynomial dependence on the normalization radius of a logit source. This bounds the cost of passing from a normalized stream sign to the final categorical token.

Theorem B.10 (Bounded logits from attenuated signs). *Let $R \geq 1$ be rational. Suppose a request for category i returns an independent sign of mean z_i/R , where the promised logits obey $|z_i| \leq 1$. There is an exact softmax sampler using $O(R^2)$ expected source-sign requests and $O(R^2 B^4)$ expected auxiliary bit work. The constants are absolute and do not depend on the number of categories, whose address length is included in B .*

Proof idea. A bounded tanh factory first produces a logistic coin with fixed slack. Linear amplification and a geometric product turn it into an exponential acceptance coin, whose common scale cancels in the final categorical race.

Proof. From a sign of mean z_i/R , the biased row sampler in Corollary A.25, with bias -1 and coefficient $R/2$, produces the coin

$$q_i = \frac{1 + \tanh(z_i/2 - 1)}{2} = \frac{e^{z_i-2}}{1 + e^{z_i-2}}.$$

Its expected source count is at most $R(1 + R/2) \leq 3R^2/2$, and its expected auxiliary work is $O(R^2 B^4)$. Since $z_i \leq 1$ and $e > 2$, one has $q_i \leq 1/(1 + e) < 1/3$. Huber’s linear factory with multiplier two and slack $1/3$ therefore returns a $2q_i$ -coin using at most $57 q_i$ -coins in expectation. Lemma C.3 supplies a constant expected auxiliary bit bound for this fixed multiplier and slack.

Draw $K \geq 1$ with $\Pr(K = k) = 2^{-k}$ and test K independent $2q_i$ -coins, stopping at the first failure. The probability of success is

$$\sum_{k=1}^{\infty} 2^{-k} (2q_i)^k = \frac{q_i}{1 - q_i} = e^{z_i-2}.$$

The expected number of tested amplified coins is at most $\mathbb{E}K = 2$, so the expected number of q_i -coins is at most 114 . This calculation charges each fresh call before its own randomness and remains valid when its cost and output are correlated.

Propose a category uniformly and accept it using this success coin. The first accepted category has the softmax law, since the common factor e^{-2} cancels. A trial succeeds with probability at least e^{-3} , so the mean number of trials is at most e^3 . The total expected number of source requests is bounded by

$$114e^3 \frac{3R^2}{2} = 171e^3 R^2 < 4096R^2.$$

For example $e < 11/4$ implies $e^3 < 21$ and $171 \cdot 21 < 4096$. All geometric and fixed-parameter amplification costs have finite moments. The row sampler gives its stated uniform finite-bit bound conditional on the requested category. Predictable charging over the race proves $O(R^2 B^4)$ auxiliary work. Almost-sure termination follows from the component factories and the positive trial acceptance probability. $\square$

B.4 A cache on one affine line: every temperature

A cache on one affine line permits exact sampling whose cost depends only on the encoding length of the temperature, even when the numerical temperature is very large. The proof uses the geometric rejection envelope for decreasing weights studied by Chewi, Gerber, Lu, Le Gouic, and Rigollet [11]. We include the short envelope argument and account separately for finite arithmetic.

Lemma B.11 (A geometric partition of decreasing weights). *Let $w_1 \geq \dots \geq w_T > 0$ and $Z = \sum_j w_j$. Partition $[T]$ into $G_0 = \{1\}$ and the nonempty blocks*

$$G_\ell = \{2^{\ell-1} + 1, \dots, \min(2^\ell, T)\}, \quad \ell \geq 1.$$

Write $m_\ell = \min G_\ell$, $N_\ell = |G_\ell|$, and $W = \sum_\ell N_\ell w_{m_\ell}$. Then

$$Z \leq W \leq 2Z.$$

Thus sampling a block proportionally to $N_\ell w_{m_\ell}$, sampling a uniform member j , and accepting it with probability w_j/w_{m_ℓ} gives an exact sample from w_j/Z with at most two proposals in expectation.

Proof idea. Each geometric block borrows its envelope mass from the preceding block. Every true weight is charged at most twice, giving a constant proposal count for any decreasing sequence.

Proof. The lower bound follows from monotonicity within each block. If $T \leq 2$ the envelope is exact. Otherwise, for every $\ell \geq 2$ with a nonempty block, its preceding block is full, has size $2^{\ell-2}$, and all its weights are at least w_{m_ℓ} . Consequently

$$N_\ell w_{m_\ell} \leq 2 \sum_{j \in G_{\ell-1}} w_j.$$

If L is the last block, summation gives

$$W \leq w_1 + w_2 + 2 \sum_{\ell=1}^{L-1} \sum_{j \in G_\ell} w_j = 2Z - w_1 + w_2 - 2 \sum_{j \in G_L} w_j \leq 2Z.$$

The accepted mass in a trial is w_j/W , and the success probability is Z/W . Summing the geometric series proves both exactness and the expected proposal count. $\square$

Lemma B.12 (Exponential evaluation on the negative half-line). *Suppose $x \leq 0$ is either a rational with encoding length b , or a rational plus a rational divided by $\sqrt{n}$, with total encoding length b . For every integer $p \geq 1$, a deterministic finite algorithm returns a dyadic interval containing e^x of length at most 2^{-p} in $O((b+p+\log(n+2))^4)$ bit operations. The bound has no dependence on the numerical magnitude $|x|$ beyond its encoding length.*

Proof idea. An exponential that is already smaller than the requested error can be enclosed immediately. Every remaining argument lies in a precision-dependent bounded interval, where a finite Taylor calculation and outward rounding suffice.

Proof. First obtain an interval of length at most $1/4$ for x . Rational arithmetic, and rational bisection for $\sqrt{n}$ when needed, have the asserted polynomial cost. If the upper endpoint is at most $-(p+2)$, return $[0, 2^{-(p+2)}]$; this is valid because $e > 2$. Otherwise the upper endpoint is greater than $-(p+2)$ and the interval has width at most $1/4$, so the known sign gives $|x| \leq p+3$. Refine x to absolute error $2^{-(p+3)}$ and evaluate its exponential on $[-p-3, 0]$. One explicit procedure divides the argument by $m = p+4$, uses the alternating exponential series on $[-1, 0]$, and raises the resulting enclosure to the power m . Take $K = 4(p + \lceil \log_2(m+1) \rceil + 10)$ terms. The alternating remainder is at most $1/(K+1)! \leq 2^{-K}$. Perform interval arithmetic with outward rounding to $P = p+2\lceil \log_2(K+1) \rceil + \lceil \log_2(m+1) \rceil + 20$ fractional bits. In the term recurrence, multiplication by a number of absolute value at most one and division by a positive integer do not increase the preceding absolute error. Summing the K terms therefore adds at most $4(K+1)2^{2-P}$ of rounding error. Clip the resulting exponential enclosure to $[0, 1]$ and multiply it by itself m times, again clipping to $[0, 1]$. These products increase interval width by at most the sum of their input widths plus 2^{1-P} per product. The chosen K, P make the final error at most $2^{-(p+2)}$. Monotonicity and $|\exp'(x)| \leq 1$ on the negative half-line control the input enclosure error. There are $O(p + \log(p+2))$ fixed-point operations on $O(b + p + \log n)$ bits; schoolbook arithmetic is bounded by the fourth power displayed above. In the algebraic case, an absolute interval for the rational numerator divided by $\sqrt{n}$ needs only a linear number of extra precision bits in the numerator's encoding length, even if the numerator has large magnitude. $\square$

Theorem B.13 (Exact indexed attention in affine dimension one). *Let $T \geq 2$. Let cached keys $k_j \in \mathbb{Q}^n$ lie on one affine line, with arbitrary cached scalar values $v_j \in [-1, 1]$. All keys, values, and the fresh query $q \in \mathbb{Q}^n$ have dyadic coordinates with at most b bits per coordinate. Let $\beta \geq 0$ be dyadic, also with at most b bits, or let $\beta = \sqrt{n}$. There is a uniform algorithm with the following bounds:*

$$\begin{aligned} \text{prefill work} &= O((Tn + T \log T)(b + \log(T + n))^2), \\ \text{stored bits} &= O(Tnb + T \log T), \\ \text{expected online work} &= O((n + \log T)(b + \log(T + n))^4). \end{aligned}$$

It samples an exact attention index for scores $\beta \langle q, k_j \rangle / n$, and hence an exact sign of mean $\sum_j e^{a_j} v_j / \sum_j e^{a_j}$. Its expected number of proposed cache members is at most $17/8$. These bounds hold at every numerical temperature covered by the stated finite encoding. No upper bound on the scores is required.

Proof idea. Project the query onto the cache line once. The projection chooses one of two stored orders; a geometric envelope then samples from that order. A small dyadic floor makes every acceptance comparison efficient even at very large temperature.

Proof. If all keys coincide, attention is uniform and the claim follows immediately. Otherwise choose two distinct keys, set $c = k_1$, and let u be their nonzero difference. Choose a pivot coordinate r with $u_r \neq 0$. Prefill can verify the affine-line promise by checking

$$u_r(k_{j,i} - c_i) = u_i(k_{j,r} - c_r) \quad \text{for every } i, j.$$

Sort the keys by their pivot coordinate, and store that permutation. Also store the rational affine coordinates $\xi_j = (k_{j,r} - c_r)/u_r$, so $k_j = c + \xi_j u$. All these tests and comparisons use exact rational arithmetic.

At query time, compute $v = \langle q, u \rangle$ once and the sign of v/u_r . It determines the order of all projected keys; use the stored permutation in the appropriate direction. If the sign is zero, all scores coincide. Otherwise relabel the keys in descending score order and shift by the first score. Set

$$w_j = \exp(a_j - a_1), \quad 1 = w_1 \geq \dots \geq w_T > 0.$$

Each shift is computed as $\beta v(\xi_j - \xi_1)/n$. It is rational, or a rational multiple of $1/\sqrt{n}$ at standard scaling. Its encoding has $O(b + \log n)$ bits. Thus one initial projection suffices for every subsequent score evaluation, and no subtraction of numerically approximated scores is required. Use the blocks of Lemma B.11; their number is at most $1 + \lceil \log_2 T \rceil$.

Put $P = \lceil \log_2(16T) \rceil$. For each first member of a block, Lemma B.12 gives a dyadic interval of length at most 2^{-P} for w_{m_ℓ} . Add 2^{-P} to its upper endpoint, and call the result u_ℓ . Then

$$2^{-P} \leq u_\ell, \quad w_{m_\ell} \leq u_\ell \leq w_{m_\ell} + 2^{1-P}.$$

The actual dyadic envelope satisfies

$$Z \leq W' := \sum_\ell N_\ell u_\ell \leq 2Z + T2^{1-P} \leq 2Z + \frac{1}{8} \leq \frac{17}{8}Z,$$

where $Z \geq w_1 = 1$. Its weights have $O(P)$ bits after a common dyadic scaling, and exact integer prefix sums sample a block in $O((P + \log T)^2)$ expected bit operations. Select a uniform member j of that block and accept it with probability w_j/u_ℓ . This lies in $[0, 1]$, and the trial's accepted mass is w_j/W' . Thus the index law is exact and the mean number of trials is $W'/Z \leq 17/8$.

To implement the last coin to precision t , it suffices to enclose w_j to precision $P+t+3$, since $u_\ell \geq 2^{-P}$. The negative-exponential lemma does this in $O((b + P + t + \log(n+2))^4)$ bit work, regardless of the temperature's numerical size. Lemma B.14 therefore makes the coin exact with expected work $O((b + P + \log(n+2))^4)$. There is one n -coordinate projection, followed by $O(\log T)$ representative score evaluations and a constant expected number of proposed-member score evaluations. A fresh known coin of mean v_j completes the attention sign.

The preprocessing bounds follow from the collinearity tests and sorting. Rational bit lengths remain $O(b + \log n)$ up to constant factors for each comparison and inner product. The stored permutation, block positions, and exact dyadic coordinates have the stated total size. $\square$

What is being sampled. The theorem gives a sample from the attention distribution and a sign with the exact attention-coordinate mean. A nonlinearity applied to one sampled value vector does not in general give the required law. An additional nonlinear layer instead requests fresh independent attention signs through its Bernoulli factory. For example, a tanh update followed by a two-logit head $(z, -z)$ with $z = \tanh Y$ needs only a constant expected number of these signs, using the bounded scalar backend. Thus the same asymptotic query bound applies to that exact final token.

B.5 Finite cached data and exact scalar decisions

In the static-cache problem, the cache is built before the query is revealed, and cached coordinates are finite dyadic numbers. This model already contains the hard instances below. Section 2 describes what a cache of transcendental neural values must provide in addition.

Lemma B.14 (A precision stopping bound). *Let $p \in [0, 1]$. Suppose a deterministic routine returns an interval $I_t \ni p$ of length at most 2^{-t} at cost at most $C(B+t)^d$, for every integer $t \geq 1$, where $B \geq 1$ and $d \geq 1$. Compare these intervals with the first t bits of one uniform $U \in [0, 1]$, refining until the intervals separate. The resulting bit is exactly $\mathbf{1}_{\{U < p\}}$. It stops almost surely and has expected work at most*

$$2^{d+5} d! C(B+1)^d + O(B+1)$$

after increasing C by an absolute constant to include interval arithmetic. The probability of reaching beyond precision t is at most $\min\{1, 2^{2-t}\}$.

Proof idea. An undecided comparison places the uniform random number close to one fixed probability. This event shrinks geometrically with precision, and its tail can be summed against every reached arithmetic cost.

Proof. Replace each interval by its intersection with previous intervals. The uniform's prefix interval has length 2^{-t} . If the two intervals overlap, then $|U - p| \leq 2^{1-t}$, so the overlap event has probability at most 2^{2-t} . Their intersection can persist forever only when $U = p$, an event of probability zero. Every decision agrees with the comparison against p .

The expected sum of stage costs is at most

$$C(B+1)^d + 8C \sum_{t \geq 2} 2^{-t} (B+t)^d.$$

Since $B \geq 1$, $(B+t)^d \leq B^d(t+1)^d$. The inequalities $(t+1)^d \leq d! \binom{t+d}{d}$ and the negative-binomial generating function give

$$\sum_{t \geq 0} 2^{-t} (t+1)^d \leq d! 2^{d+1}.$$

The displayed constant is larger than the resulting total. The random prefix itself uses one additional bit per stage and has bounded expected length. No derivative under a random stopping time is used in this argument. $\square$

Lemma B.15 (Exactification from certified intervals). *Suppose a fast query routine supplies an interval for a desired Bernoulli probability p of width at most 2^{-m} , at cost F_m . Suppose a fallback routine supplies intervals of width 2^{-t} at cost at most $H(B+t)^d$ for every $t > m$. There is an exact sampler of expected cost*

$$F_m + O(m) + 2^{d+5} d! 2^{-m} H(B+m+1)^d.$$

In particular, $m = \max\{1, \lceil \log_2 \max\{1, H\} \rceil\}$ makes the fallback contribution polynomial in $B + \log(H+2)$.

Proof idea. The fast enclosure settles most uniform comparisons. A complete evaluation is needed only near its boundary, and choosing the initial precision from the fallback cost makes that rare work affordable.

Proof. Use the fast interval and m uniform bits first. If the intervals overlap, continue with the same uniform and the fallback routine at precisions $m+1, m+2, \dots$. At stage $m+r$, the probability of being reached is at most 2^{3-m-r} . Thus the expected fallback cost is at most

$$82^{-m}H \sum_{r \geq 1} 2^{-r}(B+m+r)^d.$$

The same negative-binomial estimate proves the stated bound. Every interval is certified, so the stopping decision has no error. $\square$

This lemma does not apply merely because an approximation is correct with high probability. An invalid interval, however rare, can change the output law. A randomized approximation procedure can be used when its returned interval is always valid and its expected cost has the required bound, with that conditional cost accounted for.

Lemma B.16 (A finite-bit scan for cached attention). *Let $q, k_1, \dots, k_T$ be dyadic vectors of width n , with b bits per coordinate, and let $v_j \in [-1, 1]$ be dyadic. Let $\beta \geq 0$ be dyadic of b bits or let $\beta = \sqrt{n}$. For $Y = \sum_j e^{a_j} v_j / \sum_j e^{a_j}$, with $a_j = \beta \langle q, k_j \rangle / n$, the Bernoulli probability*

$$p = \frac{1 + \tanh(\tanh Y)}{2}$$

can be sampled exactly with $O(Tn(b + \log(T+n))^4)$ expected bit operations. The same asymptotic bound samples an attention index exactly.

Proof idea. Shift all scores by their maximum, so a denominator term equals one. Certified absolute errors then control the normalized mean or cumulative probabilities, and interval refinement turns those enclosures into an exact sample.

Proof. Find a largest inner product using exact dyadic arithmetic. As $\beta \geq 0$, it also maximizes the score. Shift by that score and put $w_j = e^{a_j - a_{\max}} \leq 1$. At least one w_j is one, so $Z = \sum_j w_j \geq 1$.

At requested output precision t , enclose every w_j to width 2^{-P} with $P = t + \lceil \log_2(T+1) \rceil + 8$. The negative exponential algorithm in Lemma B.12 applies. Both the numerator and denominator sums then have error at most $T2^{-P}$. Since the exact numerator has absolute value at most Z and $Z \geq 1$, interval division gives an interval for Y of width at most 2^{-t-3} after clipping to $[-1, 1]$. The two $\tanh$ maps are 1-Lipschitz and have bounded arguments. Their certified series evaluations supply a probability interval of width at most 2^{-t} . Its cost is $O(Tn(b+t+\log(T+n))^4)$, and Lemma B.14 proves the expected bound.

For an attention index, form intervals for all normalized prefix weights and compare one lazy uniform with them. There are at most $T-1$ interior boundaries. Use precision $P = t + 2\lceil \log_2(T+1) \rceil + 8$ per weight and read the same P bits of the uniform. Each boundary enclosure has width at most $8T2^{-P}$, and the uniform-prefix interval adds at most 2^{-P} on either side. The union over the boundaries consequently has measure at most $10T^22^{-P} \leq 2^{2-t}$. The same stopping and work calculation applies. This also proves almost-sure termination if several neighboring weights are extremely small. $\square$

B.6 An append-only index at fixed affine rank

This subsection proves the fixed-rank index of Theorem 6.2. A grid gives the index when the temperature and coordinate bounds are polynomial in the width. The grid controls variation of scores inside a cell. It need not recover their complete order. A new key changes one cell, while a query combines the cell envelopes into one exact rejection sampler. Keeping an earlier coordinate chart when the affine span grows avoids any migration of stored records.

Theorem B.17 (Dynamic fixed-rank attention). *Fix $r \geq 1$ and positive dyadic coordinate bounds K, Q . Let $\beta \geq 0$ be dyadic, or $\beta = \sqrt{n}$, and choose a fixed dyadic bound $\beta \leq \bar{\beta} \leq 2\beta$ when $\beta > 0$, and set $\bar{\beta} = 0$ when $\beta = 0$. A stream appends dyadic key-value records (k_j, v_j) with $k_j \in [-K, K]^n$ and $v_j \in [-1, 1]$. Every prefix of keys has affine dimension at most r . Queries $q \in [-Q, Q]^n$ are explicitly supplied dyadic vectors and are made only at nonempty prefixes $T \geq 1$. All scalar encodings and fixed parameters have at most b bits, with dyadics represented by binary integer numerators and denominators. Put $B = 16 + b + \lceil \log_2(T + n + r + 2) \rceil$ at cache length T .*

There is a uniform exact attention-index sampler with

$$\begin{aligned} \text{worst-case append work} &= O_r((n + \log(T + 2))B^3), \\ \text{stored bits} &= O_r((T + 1)nB), \\ \text{expected query work} &= O_r((n + M_T + \log(T + 2))B^4), \end{aligned}$$

where

$$M_T \leq \min \{T, (r + 1)(5 + 32r\bar{\beta}QK)^r\}.$$

The expected number of proposed members is at most $17/8$. The output index has probabilities proportional to $\exp(\beta \langle q, k_j \rangle / n)$. A fresh value sign then has the exact attention-coordinate mean. The claim holds conditionally on every adaptive history of legal records and queries. At $\beta = 0$, uniform index sampling suffices.

For fixed r, K, Q and $\beta = \sqrt{n}$, expected query work is

$$O_{r,K,Q}((n + n^{r/2} + \log(T + 2))B^4).$$

The construction never scans earlier records after an append, including an append that enlarges the span of a large degenerate refill.

The theorem charges operations on the supplied finite records. Section B.10 also charges the construction of neural keys and values.

Lemma B.18 (A bounded rational coordinate chart). *Let U be an n by d dyadic matrix of full column rank, $1 \leq d \leq r$, with entries of encoding length $O(b)$. In $O_r(nB^3)$ bit operations one can find d distinct row indices I such that*

$$A = U(U_I)^{-1}, \quad A_I = I_d, \quad |A_{ij}| \leq 2.$$

The coefficients have $O_r(B)$ bits and a common determinant denominator.

Proof idea. A large reconstruction coefficient identifies a row replacement that more than doubles the determinant. Integer determinant bounds limit the number of such replacements. This is the classical maximum-volume argument; the coefficient and row-replacement identities appear, for example, in Mikhalev and Oseledets [31], Section 3.

Proof. Find any nonsingular row minor by Gaussian elimination. Clear the common dyadic denominator of U , obtaining an integer matrix E . Suppose $C = E(E_I)^{-1}$ has an entry $|C_{ij}| > 2$. Replace selected row j by row i . This is a distinct row, because the coefficients of selected rows form the identity. Multilinearity of the determinant gives

$$|\det(E_{I,\text{new}})| = |C_{ij}| |\det(E_I)| > 2 |\det(E_I)|.$$

Every nonzero integer determinant has magnitude at least one. If entries of E have magnitude at most H_E , every minor is at most $d!H_E^d$. Its logarithm is $O_r(B)$. Therefore only $O_r(B)$ replacements occur. At each stage Cramer's rule or fixed-dimensional elimination computes all nd coefficients in $O_r(nB^2)$ bits. Every intermediate determinant and coefficient has $O_r(B)$ bits. When no offending entry remains, return the current indices. The adjugate formula gives the common denominator, and multiplication by the selected minor gives $A_I = I_d$. $\square$

Lemma B.19 (A finite cell envelope). *Partition T members into nonempty cells. Suppose cell C has a proxy score a_C and each actual score satisfies $|a_j - a_C| \leq 1/4$ for $j \in C$. Suppose proxy comparisons are exact and shifted negative exponentials admit the finite enclosures of Lemma B.12. Then all cell weights can be prepared using one exponential enclosure per cell, and exact sampling needs at most $17/8$ proposed members in expectation. Final acceptance comparisons have a geometric precision tail.*

Proof idea. One score represents a cell to constant multiplicative accuracy. A dyadic floor prevents a very small envelope weight from requiring unbounded initial precision in the acceptance coin. Adding an approximation allowance to construct a rejection envelope, then refining only ambiguous comparisons, is an established exact-sampling technique; see Brassard, Devroye, and Gravel [6], Sections 2 and 4.

Proof. Let $a_0 = \max_C a_C$, and write

$$w_j = e^{a_j - a_0 - 1/4}, \quad Z = \sum_j w_j, \quad e_C = e^{a_C - a_0}, \quad W_0 = \sum_C |C| e_C.$$

Every defining exponent is nonpositive. For $j \in C$, $w_j \leq e_C \leq e^{1/2} w_j < 2w_j$, so $Z \leq W_0 < 2Z$. A member of a cell whose proxy equals a_0 has $w_j \geq e^{-1/2} > 1/2$, giving $Z > 1/2$.

Set $p = \lceil \log_2(64T) \rceil$. Enclose each e_C in a dyadic interval of width 2^{-p-2} , round the upper endpoint up to the grid 2^{-p} , and add 2^{-p} . The resulting dyadic u_C satisfies

$$u_C \geq 2^{-p}, \quad e_C \leq u_C \leq e_C + 3 \cdot 2^{-p}.$$

Thus the actual envelope obeys

$$Z \leq W := \sum_C |C| u_C < 2Z + \frac{3}{64} \leq \frac{67}{32} Z < \frac{17}{8} Z.$$

Draw a cell proportionally to $|C| u_C$, then a uniform member j in it, and accept with probability w_j / u_C . Its accepted mass is

$$\frac{|C| u_C}{W} \cdot \frac{1}{|C|} \cdot \frac{w_j}{u_C} = \frac{w_j}{W}.$$

The success probability is Z/W . Independent trials therefore return w_j/Z and use $W/Z < 17/8$ proposals in expectation.

The dyadic cell weights have a common denominator and $O(B)$ bits; integer cumulative weights sample a cell exactly. Since $u_C \geq 2^{-p}$, an enclosure of width 2^{-p-t-3} for w_j suffices at acceptance precision t . Such an enclosure costs $O((B+t)^4)$, so Lemma B.14 decides the acceptance coin exactly in expected $O(B^4)$ bit work per proposal, with a geometric precision tail. This stopping argument charges the actual work of every reached precision stage. $\square$

Proof idea for Theorem B.17. Selected ambient coordinates describe every key in a bounded chart. Their grid controls the score variation for every legal query. When the span grows, a new chart receives only future records. There are at most $r+1$ charts, so querying all of them has a bounded rank-dependent overhead.

Proof of Theorem B.17. Fix the first key c . Maintain a basis U of all differences $k_j - c$ encountered so far. An epoch starts at rank zero and whenever this rank increases. An epoch of dimension $d > 0$ uses the chart of Lemma B.18, with

$$c_0 = c - A c_I, \quad k = c_0 + A k_I.$$

The selected coordinates $y = k_I$ lie in $[-K, K]^d$. For a query let $v = q^T A$. Since $|A_{ij}| \leq 2$, $|v_i| \leq 2nQ$. Choose

$$m = \lceil \log_2 \max\{1, 8r\bar{\beta}QK\} \rceil, \quad \Delta = K 2^{-m}.$$

The integer cell tuple is $\lfloor y_i/\Delta \rfloor$. Two members in one cell have coordinate differences less than Δ , so their scores differ by at most

$$\frac{\beta}{n} \sum_{i=1}^d |v_i| \Delta \leq 2\beta r Q \Delta \leq \frac{1}{4}.$$

In one coordinate there are at most $2^{m+1} + 1$ possible cell indices. This is at most $5 + 32r\bar{\beta}QK$. The rank-zero epoch has one cell.

Appends. At an insertion test membership in the current affine span by the exact equality $k - c = A(k_I - c_I)$. This costs $O_r(nB^2)$. If it holds, insert in the current epoch. If it fails, append $k - c$ to U , build its new bounded chart, and create a new epoch containing only the new record. Earlier epochs and their records remain untouched. At most r increases are possible, but even an individual increase costs only $O_r(nB^3)$.

Each epoch stores its occupied cells in a balanced search tree keyed by their integer tuples. A cell stores one representative identifier and an appendable array of member identifiers. For completeness, the latter can be a binary tree whose leaves have local indices $0, \dots, N - 1$. At a power-of-two capacity boundary attach a new root and insert along the binary path for the new local index. There are $O(N)$ stored nodes and $O(\log(N + 2))$ pointer operations per append or random access. All pointers, counters, and cell coordinates have $O_r(B)$ bits. Drawing a uniform integer in $[N]$ by rejection from binary bits takes expected $O(B^2)$ bit work. These structures give the stated worst-case append cost.

Queries. At a query compute $\langle q, c_0 \rangle$ and $q^T A$ for each epoch. The number of epochs is at most $r + 1$. The entries of A have one common determinant denominator. Combining that denominator with the common dyadic query denominator shows that an n -term projection adds only $O(\log n)$ numerator bits. Each projection therefore has $O_r(B)$ encoding length and costs $O_r(nB^2)$ bit work. Evaluate each cell representative's score from its d selected coordinates, including the affine offset so that scores in different epochs are comparable.

Apply Lemma B.19 to the union of these cells. Their total number is at most

$$\min \left\{ T, \sum_{d=0}^r (5 + 32r\bar{\beta}QK)^d \right\},$$

which gives the displayed M_T . At positive temperature, comparisons of representative scores reduce to rational comparisons before multiplication by β/n . At standard scaling, all shifted exponents are a rational plus a rational multiple of $1/\sqrt{n}$, within Lemma B.12. Thus one length- n projection per epoch, one enclosure per occupied cell, and a constant expected number of member enclosures give the query bit bound. Full records occupy $O(Tnb)$ bits, cell structures $O_r(TB)$, and the bounded number of charts $O_r(nB)$.

Adaptive histories. Finally condition on any realized history of preceding legal records and queries. The current partition and its score formulas are fixed. Fresh random bits give exactly the law proved by the envelope lemma. Induction gives the claim for adaptive histories. At $\beta = \sqrt{n}$, a power-of-two upper bound $\bar{\beta} < 2\sqrt{n}$ gives $M_T = O_{r,K,Q}(n^{r/2})$. $\square$

Corollary B.20 (Dynamic rank one at every temperature). *For an append-only cache of affine rank at most one, numerical temperature can be removed from the preceding complexity bound. Worst-case append work is $O((n + \log(T + 2))B^3)$ and expected query work is $O((n + \log(T + 2))B^4)$ at every allowed temperature. The expected proposal count is at most $17/8$.*

Proof idea. Maintain the scalar coordinate order with subtree member counts. Rank selection replaces access to a stored permutation in the static monotone envelope.

Proof. When the first nonconstant key arrives, choose a nonzero pivot coordinate on its line. Every preceding record has the same key, so its existing member array becomes a single duplicate group without scanning that

array. The rank promise prevents any later change of line. Maintain the ordered distinct pivot coordinates in a balanced tree with subtree member counts. Verify line membership and insert each new member exactly.

A query's one length- n projection chooses the forward or reverse order. Retrieve the geometric-block endpoints of Lemma B.11 by rank selection and apply the dyadic envelope of Theorem B.13. One rank selection costs $O(\log(T+2)B^2) = O(B^3)$ bits. It is absorbed by that endpoint's $O(B^4)$ exponential enclosure. There are $O(\log(T+2))$ endpoints and a constant expected number of proposed ranks. Hence the static query bound is retained, including all exact arithmetic and precision refinement. $\square$

Corollary B.21 (A certified neighborhood of a known key space). *Suppose a fixed rational chart $c_0 + Ay$ is known, $A_I = I_r$, $(c_0)_I = 0$, and $|A_{ij}| \leq 2$. The actual keys may have arbitrary rank, provided each satisfies*

$$\|k - c_0 - Ak_I\|_\infty \leq \eta, \quad \bar{\beta}Q\eta \leq \frac{1}{8}.$$

The same append and query bounds hold with $M_T \leq \min\{T, (5 + 64r\bar{\beta}QK)^r\}$. Only a constant expected number of proposed members require length- n actual-score evaluations.

Proof idea. Split the cell's score allowance between variation inside the projected grid and the certified residual outside the key space. The acceptance coin evaluates the actual score.

Proof. Use $m = \lceil \log_2 \max\{1, 16r\bar{\beta}QK\} \rceil$. Projected scores in a cell vary by at most $1/8$. The score error from a residual of norm at most η is at most $\beta Q\eta \leq 1/8$. Use the projected score of a stored representative as the proxy. Every actual member lies within $1/4$ of it, so Lemma B.19 applies. Proxies use the fixed low-dimensional projection. Actual candidate scores use all n coordinates, but there are fewer than $17/8$ candidates in expectation. The residual certificate is checked by rational arithmetic at insertion.

If the promise is instead ordinary ℓ_∞ distance at most η_0 from a known affine space, its bounded coordinate chart has projection $P = AR_I$ with $\|P\|_{\infty \rightarrow \infty} \leq 2r$. Applying $I - P$ to an approximating point gives residual at most $(1 + 2r)\eta_0$. Thus $\bar{\beta}Q(1 + 2r)\eta_0 \leq 1/8$ suffices. $\square$

B.7 A rank-one cache with an implicit query

An explicit query can locate its score order. When the query is supplied only through fresh signs of an unknown scalar mean, that comparison is unavailable. A prefilled scalar factory still gives a polynomial source bound at standard scaling. We use a bias-corrected Bernstein expansion. Bernstein approximation and smoothness-based simulation are established Bernoulli-factory methods [21, 34]; the role of prefill here is to store their cache-dependent coefficient computations. The correction is needed because the ordinary Bernstein approximation has an error of order m^{-1} , whose direct mixture has infinite mean source work.

Lemma B.22 (A summable Bernstein correction). *Let $f \in C^4([0, 1])$ satisfy $\|f\|_\infty \leq 4/5$. Suppose f and f'' have algorithms returning certified absolute enclosures at every rational argument and requested precision. Let known nonnegative rational numbers $M_j \geq \|f^{(j)}\|_\infty$ be supplied for $j = 2, 3, 4$. Put*

$$A = M_2 + M_3 + M_4, \quad h(p) = p(1-p)f''(p), \quad g_m(p) = f(p) - \frac{h(p)}{2m}.$$

For $K \in \{0, \dots, 2m\}$, let J be hypergeometric, counting the successes in a uniformly selected half of a population of $2m$ bits containing K successes. Write $p = K/(2m)$ and

$$d_{m,K} = g_{2m}(p) - \mathbb{E}g_m(J/m).$$

Then

$$|d_{m,K}| \leq \frac{A}{4m^2}. \tag{B.12}$$

If m_0 is a power of two satisfying $m_0 \geq \max\{1, 2M_2, 2\sqrt{A}\}$, there is an exact factory returning a sign of mean $f(p)$ from independent source bits of unknown mean p , using at most

$$\frac{7}{8}m_0 + \frac{A}{m_0} \tag{B.13}$$

source bits in expectation. This count does not include the work needed to evaluate the known coefficient coins.

Proof idea. The second-derivative correction cancels the leading Bernstein approximation error. Degree elevation is described by a symmetric hypergeometric sample, so a fourth-moment estimate leaves summable correction coefficients. A mixture of these corrections yields the exact scalar mean.

Proof. Put $Y = J/m$ and $\Delta = Y - p$. A sample and its complement have the same distribution, so Δ is symmetric. In particular $\mathbb{E}\Delta = \mathbb{E}\Delta^3 = 0$, and the usual two-indicator calculation gives

$$\nu := \mathbb{E}\Delta^2 = \frac{p(1-p)}{2m-1}.$$

We also need the bound $\mathbb{E}\Delta^4 \leq m^{-2}$. Here is an elementary derivation. Reveal the m sampled bits in order, and take the Doob martingale of their final mean. At reveal i , its increment is

$$\frac{X_i - \mathbb{E}[X_i \mid X_1, \dots, X_{i-1}]}{2m - i}.$$

Its conditional range has length at most $1/m$. The elementary Hoeffding moment bound for a centered variable in an interval of length c , $\mathbb{E}e^{tX} \leq e^{t^2 c^2/8}$, follows by convexity of the exponential and one-variable differentiation. Indeed convexity bounds the moment by that of the two endpoints with the same mean. After scaling by c , the logarithm of this endpoint moment has value and derivative zero at the origin, and its second derivative is the variance of a tilted Bernoulli bit, at most $1/4$. Integrating twice proves the bound for either sign of t . Iterating this bound gives $\Pr(|\Delta| > x) \leq 2e^{-2mx^2}$. Integrating the tail yields $\mathbb{E}\Delta^4 \leq 8 \int_0^\infty x^3 e^{-2mx^2} dx = m^{-2}$.

The coefficient bound. The second derivative of h satisfies

$$H_2 := \|h''\|_\infty \leq 2M_2 + 2M_3 + M_4/4.$$

Taylor expansion of f through degree three, symmetry, and the fourth-moment bound give

$$\mathbb{E}f(Y) = f(p) + \frac{1}{2}f''(p)\nu + R, \quad |R| \leq \frac{M_4}{24m^2}.$$

The first-order expansion of h gives $|\mathbb{E}h(Y) - h(p)| \leq H_2\nu/2$. The leading terms cancel as follows:

$$\frac{h(p)}{4m} - \frac{1}{2}f''(p)\nu = -\frac{h(p)}{4m(2m-1)}.$$

Since $|h| \leq M_2/4$ and $\nu \leq 1/(4m)$, it follows that

$$|d_{m,K}| \leq \frac{1}{m^2} \left(\frac{M_2 + H_2}{16} + \frac{M_4}{24} \right) \leq \frac{M_2 + M_3 + M_4}{4m^2}.$$

The factory. Let $B_m g$ denote the degree- m Bernstein polynomial of g . Conditional on the total number K of successes among $2m$ independent source bits, the number in their first half is the hypergeometric variable above. Therefore the coefficient of degree $2m$ in $B_{2m}g_{2m} - B_m g_m$ is exactly $d_{m,K}$. Also $B_m g_m \rightarrow f$ uniformly: ordinary Bernstein approximation converges uniformly for continuous functions, and $\|g_m - f\|_\infty \leq M_2/(8m)$. For the former claim, if $X \sim \text{Bin}(m, p)$, split $\mathbb{E}|f(X/m) - f(p)|$ at $|X/m - p| \leq \delta$. Uniform continuity bounds the first part, and Chebyshev's inequality bounds the other part by $2\|f\|_\infty/(4m\delta^2)$, uniformly in p . Thus, for $m_\ell = 2^\ell m_0$,

$$f = B_{m_0}g_{m_0} + \sum_{\ell \geq 0} (B_{2m_\ell}g_{2m_\ell} - B_{m_\ell}g_{m_\ell}).$$

Set $b_0 = 7/8$ and $c_\ell = A/(4m_\ell^2)$. We have $\|g_{m_0}\|_\infty \leq 4/5 + 1/16 < 7/8$, and

$$\sum_{\ell \geq 0} c_\ell = \frac{A}{3m_0^2} \leq \frac{1}{12}.$$

Select the base with probability b_0 , select correction ℓ with probability c_ℓ , and otherwise return a fair sign. On the base draw m_0 source bits and, conditional on their count K , return a known sign of mean $g_{m_0}(K/m_0)/b_0$. On correction ℓ draw $2m_\ell$ bits and return a known sign of mean $d_{m_\ell, K}/c_\ell$. All these means are in $[-1, 1]$. The absolutely convergent displayed identity proves exactness. The mean source count is

$$b_0 m_0 + \sum_{\ell \geq 0} 2m_\ell c_\ell = \frac{7}{8} m_0 + \frac{A}{m_0}.$$

If $A = 0$, omit all corrections. Otherwise the conditional level law is $(3/4)4^{-\ell}$, sampled by pairs of fair random bits. The level, source count, and every known coefficient comparison are finite almost surely. $\square$

Lemma B.23 (Uniform scalar softmax derivatives). *Let $|v_j| \leq 1$ and $|a_j| \leq \Lambda$, and define*

$$u(p) = \frac{\sum_{j=1}^T v_j e^{a_j p + c_j}}{\sum_{j=1}^T e^{a_j p + c_j}}, \quad f(p) = u(p)/2.$$

For arbitrary real offsets c_j and $p \in [0, 1]$,

$$\|f\|_\infty \leq 1/2, \quad \|f''\|_\infty \leq 3\Lambda^2, \quad \|f'''\|_\infty \leq 13\Lambda^3, \quad \|f''''\|_\infty \leq 75\Lambda^4.$$

The bounds are independent of T and of the score offsets.

Proof idea. Differentiate the numerator identity repeatedly, dividing throughout by the positive softmax denominator. The bounded slopes control each normalized derivative independently of the number of keys and their offsets.

Proof. Write the numerator as N and denominator as Z . Then $|N^{(k)}|/Z \leq \Lambda^k$ and $|Z^{(k)}|/Z \leq \Lambda^k$. Differentiating $uZ = N$ inductively gives

$$|u'| \leq 2\Lambda, \quad |u''| \leq 6\Lambda^2, \quad |u'''| \leq 26\Lambda^3, \quad |u''''| \leq 150\Lambda^4.$$

Dividing these inequalities by two proves the claim. The radius bound follows from $|u| \leq 1$. $\square$

Theorem B.24 (A prefilled scalar softmax factory). *Consider the family in Lemma B.23. Assume a_j, c_j, v_j and a rational bound Λ have at most b bits, or that the slopes and offsets are rational multiples of a common $1/\sqrt{n}$ with rational encodings of at most b bits. The online input consists of independent source bits of an unknown mean $p \in [0, 1]$. Put*

$$M_2 = 3\Lambda^2, \quad M_3 = 13\Lambda^3, \quad M_4 = 75\Lambda^4, \quad A = M_2 + M_3 + M_4,$$

and let m_0 be the least power of two at least $\max\{1, 2M_2, 2\sqrt{A}\}$. Then a finite-bit data structure samples a sign of mean $u(p)/2$ exactly, using at most $(7/8)m_0 + A/m_0 = O(1 + \Lambda^2)$ source requests and expected auxiliary bit work

$$O((1 + \Lambda^2)B^4), \quad B = 16 + b + \lceil \log_2(T + n + \Lambda + 2) \rceil.$$

The preprocessing work is $O(T^3(1 + \Lambda^4)B^4)$ and stored bits are $O(T(1 + \Lambda^2)B)$. All constants are absolute. No exact-real operations are required. The output is an attention-coordinate sign at radius two. It need not reveal an attention index. More explicitly, the source count is less than $45 \max\{1, \Lambda^2\}$: one has $A \leq 91 \max\{\Lambda^2, \Lambda^4\}$, $m_0 < 40 \max\{1, \Lambda^2\}$, and $A/m_0 \leq m_0/4$.

Proof idea. Prefill stores the corrected Bernstein coefficients through a context-dependent cutoff. The geometric correction law makes higher levels rare enough to evaluate when requested, and certified intervals resolve coefficient coins without altering their biases.

Proof. If $A = 0$, omit every correction table and correction branch below; the base branch alone suffices. Assume $A > 0$ whenever a correction is normalized by its mixture mass. We first specify a certified coefficient routine. For rational $x \in [0, 1]$, a precision- p enclosure for $f(x)$ and $f''(x)$ costs $O(T(B + \log m + p)^4)$ bit operations when x has $O(\log m)$ bits. To see this, subtract the largest score before evaluating exponentials. Its weight is one, so the denominator is at least one. Weighted moments of the slopes up to order two give u, u', u'' by the quotient rule, and division by two gives f, f'' . The slope bounds require only $O(B)$ extra guard bits. Certified negative-exponential enclosures have fourth-power bit cost by the preceding arithmetic lemma. In the common quadratic-field case, score comparisons reduce to rational comparisons, and certified square-root enclosures suffice for all evaluations.

Correction coefficients. For a correction coefficient, the hypergeometric mass is proportional to

$$w_j = \binom{m}{j} \binom{m}{K-j}, \quad \max(0, K-m) \leq j \leq \min(m, K).$$

Its mode is $j_0 = \lfloor K/2 \rfloor$, possibly one of two tied modes. Set the mode weight to one and generate weights outwards. The forward ratio is

$$\frac{w_{j+1}}{w_j} = \frac{(m-j)(K-j)}{(j+1)(m-K+j+1)};$$

on the upper side it is at most one, and the reverse ratio is at most one on the lower side. Every ratio has $O(\log(m+2))$ bits. Using fixed-point intervals at r fractional bits, each outward path accumulates at most $O(m2^{-r})$ error. The total mass error is $O(m^2 2^{-r})$. The exact total is at least one because the mode weight is one, so division is stable. Taking $r = p + 2\lceil \log_2(m+2) \rceil + O(1)$ therefore gives the weighted average to error $O(2^{-p})$. Evaluate the $O(m)$ required $g_m(j/m)$ values by the preceding routine with $p + \lceil \log_2(m+2) \rceil + O(1)$ precision bits. Thus a correction coefficient has a precision- p enclosure in

$$O(mT(B + \log(m+2) + p)^4)$$

bit operations. Normalizing by $c_\ell = A/(4m^2)$ adds only $O(B + \log(m+2))$ precision bits. No binomial integer with $\Theta(m)$ bits is ever formed.

The prefilled table. Let M be the smallest level m_ℓ at least Tm_0 ; then $Tm_0 \leq M < 2Tm_0$. During prefill, store certified enclosures for all base and correction coefficient coin probabilities through level M . For degree $2m$ use width at most 2^{-P_m} , where

$$P_m = 8 + \lceil \log_2(Tm + 2) \rceil.$$

These dyadic interval data are used for exact comparisons with the desired probabilities. The sum of the table sizes is $O(MB)$. Evaluating all entries costs $O(TM^2B^4)$, giving the stated bounds.

Online sampling. Online, sample the exact mixture of Lemma B.22 and draw its required source bits. Lemma B.15 decides each stored coefficient coin exactly. Its fast enclosure is the stored interval, of width 2^{-P_m} , and its fallback is the certified routine above, of cost $O(mT(B + \log(m+2) + t)^4)$ at precision t . Since $2^{P_m} \geq 2^8(Tm + 2)$ and $\log m = O(B)$ for stored levels, the lemma bounds the expected work by $O(B^4)$.

Levels above the table. For a level $m > M$, compute the coefficient directly. Its expected cost is $O(mT(B + \log m)^4)$, and its mixture probability is $A/(4m^2)$. Summing over dyadic levels gives

$$\sum_{m>M} \frac{A}{4m^2} mT(B + \log m)^4 = O\left(\frac{AT}{M}(B + \log M)^4\right) = O\left(\frac{A}{m_0}B^4\right).$$

Source counting, source addresses, and mixture selection have the same bound after logarithmic factors: the weighted level tail is geometric. Together with the base work and the displayed source count, this proves the online bound. Uniform comparisons stop almost surely, and the infinite mixture assigns probability one to finite levels. The coefficient intervals only decide exact coins, so the uniform Bernstein limit proves exactness for every source bias, including zero and one. $\square$

For a rank-one cache $k_j = c + t_j u$, choose a known R with $|\langle q, u \rangle| \leq R$ on the allowed query domain, and suppose fresh signs of mean $z = \langle q, u \rangle / R$ are available. Writing $p = (1 + z)/2$, the common cache offset cancels and the scores have slopes $a_j = 2\beta R t_j / n$ and offsets $c_j = -\beta R t_j / n$. The theorem therefore applies with $\Lambda \geq 2\beta R \max_j |t_j| / n$. For instance, if $k_j = c + t_j u$ with $|t_j| \leq 1$ and $\|q\|_\infty, \|u\|_\infty \leq 1$, take $R = n$ and any rational $\Lambda \geq 2\beta$. For $\beta \geq 1$ one may choose $\Lambda \leq 4\beta$. One source sign of mean z uses at most one query-coordinate sign by a signed mixture. At standard scaling $\beta = \sqrt{n}$, the expected source and auxiliary costs are $O(nB^4)$, independently of T after the stated polynomial prefill. As after Theorem B.13, a unit tanh row and a bounded two-logit head consume the radius-two output through constant-cost scalar factories, so the exact final token has the same asymptotic bound. Tables can also be built for every cached value coordinate; this multiplies the prefill and storage by the value dimension. The two-key lower bound of Theorem B.25 has order β^2 before the finite-input cap. It is a worst-case statement over caches.

B.8 A sharp query law for a cache with two keys

A fast cache index can remove the length of the cached prefix from the query cost. It cannot generally remove the width of a fresh query. The following family makes this distinction quantitative.

Theorem B.25 (Two-key attention). *Let $n \geq 1$, and let $\beta \geq 0$ be dyadic or equal to $\sqrt{n}$. A cache contains equal numbers of the keys $\mathbf{1}$ and $-\mathbf{1}$, with respective scalar values $+1$ and -1 . The fresh query is $x \in \{-1, +1\}^n$, accessed by coordinate probes. Scores are $\beta \langle x, k \rangle / n$. Apply one tanh row to the attention value and use the two logits given by that row and its negative. Let $Q^*(n, \beta)$ be the infimum, over exact samplers, of the worst-case expected number of fresh-query probes. Arbitrary preprocessing independent of x is allowed. Then, with absolute implicit constants,*

$$Q^*(n, \beta) = \Theta(\min\{n, \beta + \beta^2\}).$$

There is a finite-bit implementation with expected online work

$$O(B^4 [1 + \min\{n, \beta + \beta^2\}]),$$

where B contains the parameter encoding lengths and $\log(n + 2)$. The cache can be stored by its two key types and multiplicities. At standard scaling, the fresh-query probe complexity is $\Theta(n)$, independently of the number of cached copies.

Proof idea. The two-key cache turns the query mean into a scalar tanh composition. Factories give the upper bound. For the lower bound, a product-distributed query and the transcript score relate the output derivative to the number of coordinates read.

Proof. Write $M = n^{-1} \sum_i x_i$ and $F_3 = \tanh \circ \tanh \circ \tanh$. The attention value is $\tanh(\beta M)$, so the final token has mean $F_3(\beta M)$ when written as a sign.

For the upper bound, obtain a source sign of mean M by querying one uniform coordinate. The arbitrary-radius tanh factory uses at most $2 \max\{\beta, \beta^2\}$ such requests, and each of the next two unit-radius tanh factories uses at most two requests to its predecessor. Fresh randomness gives the factor eight. Repeated input coordinates may be cached, which can only reduce the number of probes. Alternatively read all n input coordinates and make a certified scalar comparison. Choose the cheaper construction from the known parameters. The quadratic-radius bit bound and the precision stopping lemma give the stated finite-bit work.

For $0 \leq \beta \leq 1$, the output laws on $x = \mathbf{1}$ and $x = -\mathbf{1}$ have total variation distance $F_3(\beta)$. Couple all preprocessing and online random bits. Until the first probe the executions are identical, so the probability

of making at least one probe is at least that distance. The inequalities $\tanh u \geq u/2$ for $u \in [0, 1]$ give $F_3(\beta) \geq \beta/8$. This proves the required bound in this range, since $\beta + \beta^2 \leq 2\beta$.

For $\beta \geq 1$, draw the query coordinates independently with common mean t , and let $H(t)$ be the mean of the output sign. At $t = 0$, put $S = \sum_i x_i$ and $Z = S/\sqrt{n}$. Differentiating the finite product distribution gives

$$H'(0) = \mathbb{E}_0[F_3(\beta S/n)S] = \sqrt{n} \mathbb{E}_0 \left[Z F_3 \left(\frac{\beta}{\sqrt{n}} Z \right) \right].$$

Here $\mathbb{E}Z^2 = 1$ and $\mathbb{E}Z^4 = 3 - 2/n \leq 3$. Paley–Zygmund’s inequality, or its elementary Cauchy–Schwarz proof, therefore gives

$$\Pr(|Z| \geq 1/2) \geq \frac{(1 - 1/4)^2}{3} = \frac{3}{16}.$$

Let $d = \min\{\beta/\sqrt{n}, 1\}$. On this event, oddness and monotonicity imply

$$Z F_3 \left(\frac{\beta}{\sqrt{n}} Z \right) \geq \frac{1}{2} F_3(d/2) \geq \frac{d}{32}.$$

The integrand is nonnegative everywhere, hence $H'(0) \geq 3 \min\{\beta, \sqrt{n}\}/512$. Lemma A.17 and Cauchy–Schwarz give

$$\sup_x \mathbb{E}_x Q \geq \mathbb{E}_0 Q \geq H'(0)^2 \geq \frac{1}{32768} \min\{\beta^2, n\}.$$

Finally $\beta + \beta^2 \leq 2\beta^2$ in this range. Combining both ranges gives the universal lower constant $1/65536$. The case $\beta = 0$ is a fair output coin and uses no input probes. $\square$

Theorem B.26 (Worst-case query cost for a rank-one cache). *Let $T, n \geq 2$, and let $\beta \geq 1$ be dyadic or equal to $\sqrt{n}$. Consider all length- T caches with finitely encoded dyadic keys and values, whose keys lie in $[-1, 1]^n$ on one affine line, whose scalar values lie in $[-1, 1]$, and whose fresh query is $x \in \{-1, 1\}^n$ accessed by coordinate probes. The required output sign has mean $\tanh(\tanh Y(x))$, where Y is the attention-coordinate mean with scores $\beta \langle x, k_j \rangle / n$. Allow preprocessing independent of x and polynomial in the cache length, width, and parameter encoding size.*

The optimal worst-cache, worst-query expected probe cost is

$$\Theta(\min\{n, \beta^2\}),$$

with absolute implicit constants. The upper bound has expected online bit work

$$O([1 + \min\{n, \beta^2\} + \log T]B^4)$$

after polynomial (T, n, B) prefill, where $B \geq 16$ includes all parameter encoding lengths and $\log_2(T + n + 2)$. At standard scaling the sharp fresh-query probe order is n , independently of the prefix length. The theorem is a worst-case law over the stated class of caches.

Proof idea. The two-key construction supplies the lower bound for every cache length. For the upper bound, express an arbitrary line cache using its two extreme keys and expose the resulting one-dimensional query through source signs; use the explicit index when reading the whole query is cheaper.

Proof. For even T , the equal-multiplicity two-key cache in Theorem B.25 supplies the lower bound even with arbitrary preprocessing independent of x . For odd T , use $T - 1$ equally split keys and one zero key with value zero. Its attention output has the same sign as $\tanh(\beta M)$ and magnitude at least $(2/3)|\tanh(\beta M)|$. The map $\phi(u) = \tanh(\tanh u)$ is increasing and concave for $u \geq 0$, so the absolute output mean is at least $(2/3)|F_3(\beta M)|$. The first-score calculation in that theorem therefore gives $H'(0) \geq \min\{\beta, \sqrt{n}\}/256$. Lemma A.17 proves the same lower constant $1/65536$ in this case as well.

For the upper bound, find the two extreme keys along the affine line. Their midpoint is c and their half-difference is u . Then

$$k_j = c + t_j u, \quad |t_j| \leq 1, \quad \|u\|_\infty \leq 1.$$

If all keys agree, the attention value is independent of the query. Otherwise each t_j is a rational ratio of two dyadic coordinate differences, so its bit length is $O(B)$. A fresh sign of mean $z = \langle x, u \rangle / n$ needs at most one input probe: select coordinate i with mass $|u_i|/n$, return its signed input, and complete the unused mass with a fair sign. Independent selections give independent source signs for every fixed query.

The rank-one reduction after Theorem B.24, with $R = n$ and $p = (1 + z)/2$, gives slopes $a_j = 2\beta t_j$ and a rational slope bound $2\beta \leq \Lambda \leq 4\beta$. Its radius-two attention sign needs fewer than $45\Lambda^2 \leq 720\beta^2$ source requests. The arbitrary-radius tanh factory in Theorem A.24 needs at most eight such signs to generate a sign of mean $\tanh Y$, and at most two further calls produce a sign of mean $\tanh(\tanh Y)$. Predictable charging gives at most $11520\beta^2$ input requests. Its expected bit work is $O(\beta^2 B^4)$.

Alternatively, read all n input signs and use the explicit rank-one index in Theorem B.13. Its work is $O((n + \log T)B^4)$ and its probe count is n . Use the implicit construction only when $12000\beta^2 < n$. Then its prefill factor Λ^4 is $O(n^2)$, so its $O(T^3(1 + \Lambda^4)B^4)$ preprocessing is polynomial in T, n, B . In the other case the near-linear explicit index suffices. Thus no preprocessing exponential in the encoding of a numerically huge temperature is being claimed. This proves the two upper bounds and almost-sure termination. $\square$

Adding the current position. The affine-line index also tolerates a fixed number of explicitly available exceptional keys, such as the current position's own key. Use the monotone geometric blocks for the line and one singleton block per exception. Shift by the largest weight among the line and the exceptions. The ideal envelope is at most twice the total mass; the same dyadic floor argument gives at most $17/8$ proposals. The additional query work is proportional to the number of exceptional inner products. Thus including the current position does not require its key to lie on the cached affine line.

B.9 A finite known query list

Theorem B.27 (A finite known query list can move evaluation to prefill). *Fix a cache and a list of M permitted queries, known before preprocessing. Query q has a categorical output law $p_{q,1}, \dots, p_{q,V}$. Suppose a deterministic certified algorithm, given q and an integer $t \geq 1$, returns dyadic intervals of width at most 2^{-t} containing all V probabilities, with endpoints of length $O(B + t)$, in at most*

$$H(B + t)^d$$

bit operations. Here $H \geq V$, $B \geq 2$, and the fixed integer $d \geq 2$ are known bounds; B includes addresses and parameter lengths. No exact real probabilities need be stored.

There is a preprocessor using

$$O(MH[B + \log(VH + 2)]^d)$$

bit operations and $O(MV[B + \log(VH + 2)])$ additional bits of storage, such that, when the online query is supplied by its list identifier, one exact output sample takes

$$O([B + \log(VH + 2)]^d)$$

expected bit operations. It halts almost surely, and the bound is uniform over the query list. The cache and the certified evaluation procedure remain available on a rare fallback branch. If H is polynomial in T, n, D, V and the model encoding size, and B is polylogarithmic in those quantities, this online bound is polylogarithmic in those quantities, regardless of the attention temperature. Identifying a query supplied as an arbitrary vector is a separate cost and is not covered by the list-identifier interface.

Proof idea. Prefill stores dyadic masses lying below the desired probabilities. Their missing mass is positive but small. Most samples finish from the stored table, while the residual branch recomputes enough certified digits to sample its exact conditional law.

Proof. The case $V = 1$ is immediate. Choose a dyadic δ with $1/(8H) < \delta \leq 1/(4H)$. For each query, compute dyadic lower bounds $\ell_k \leq p_k$ with $0 \leq p_k - \ell_k \leq \delta/V$. Such bounds follow by evaluating at

precision $t_0 = \lceil \log_2(V/\delta) \rceil + 2$ and rounding downward to a common dyadic grid, clamping negative lower bounds to zero. Put

$$a_k = (1 - \delta)\ell_k, \quad r = 1 - \sum_{k=1}^V a_k.$$

All these numbers are stored dyadics of length $O(B + \log(VH + 2))$. They satisfy $0 \leq a_k \leq p_k$ and

$$\delta \leq r \leq 1 - (1 - \delta)^2 \leq 2\delta.$$

The left inequality uses $\sum \ell_k \leq 1$; the right uses $\sum \ell_k \geq 1 - \delta$.

Store the prefix sums of $a_1, \dots, a_V, r$ on their common dyadic grid. Draw from these $V + 1$ known masses using uniform grid bits and binary search of the prefix sums. This costs a polynomial in $B + \log(VH + 2)$, covered by its d th power. On outcome $k \leq V$, return token k . On the residual outcome, use fresh randomness to sample the categorical law

$$b_k = \frac{p_k - a_k}{r}.$$

It is a probability distribution because its entries are nonnegative and sum to one. The final probability of token k is exactly $a_k + rb_k = p_k$.

We give the precision and stopping bound for the residual sampler. Let U be a fresh uniform number on $[0, 1]$, exposed through its binary digits, and use the cumulative masses of b . At stage $j \geq 0$, evaluate all p_k with width at most 2^{-t_j} , where

$$t_j = \lceil \log_2(16V^2/\delta) \rceil + j.$$

Each resulting residual cumulative interval has width at most $V2^{-t_j}/r \leq 2^{-j}/(16V)$. Expose enough bits of U that its dyadic interval has width at most $2^{-j}/(16V)$. If this interval is certified to lie between two adjacent cumulative boundaries, return the corresponding token; otherwise refine. Division by the known positive dyadic r can be implemented by cross-multiplication in every comparison, using finite integers.

Failure to decide at stage j places U within distance at most $2^{-j}/(8V)$ of one of the $V - 1$ true boundaries. A union bound therefore gives probability at most $2^{-j}/4$. The probability of never deciding is zero. The same estimate gives conditional expected fallback work

$$O\left(H \sum_{j \geq 0} 2^{-j} [B + \log(V/\delta + 2) + j]^d\right) = O(H[B + \log(VH + 2)]^d).$$

The sum also covers the fresh uniform bits and the V cumulative interval additions, since $H \geq V$ and $d \geq 2$. Boundary equalities have probability zero and do not require an equality oracle.

The residual branch has probability $r \leq 2\delta \leq 1/(2H)$. Multiplying its conditional expected work by this known probability gives the stated online bound. The initial dyadic draw contributes the same order or less. Preprocessing performs one precision- t_0 evaluation and stores $V + 1$ prefix masses for each of the M known queries, giving the displayed preprocessing and storage bounds. Every arithmetic operation uses finite integers or certified dyadic intervals. $\square$

For decoding, this theorem distinguishes a finite advertised set of next queries from an unrestricted fresh query. A finite vocabulary can sometimes determine such a list once the previous state is fixed; constructing that list and all its output laws is then part of prefill. The lower bound of Section B.11 concerns arbitrary fresh queries and does not apply to such a list. The tables are specific to one cache, and a sampled token that changes the cache requires new tables. For fixed-rank key maps, Theorem B.28 provides a decoder construction that pays for those updates.

B.10 Maintaining fixed-rank attention through exact decoding

This subsection proves the decoder of Theorem 7.1. An index that accepts already computed keys leaves their construction outside its cost. For a fixed key-map rank, polynomial moments also permit the construction of the new keys and values. We first give a deterministic finite-precision evaluator. We then use certified

probability intervals to sample exactly, with rare precision increases paid for by their probability. The maintained Taylor feature sums [20, 24] and exact sampling by interval refinement [6, 18] are established; Section 10 says what we take from them. The claims here concern finite arithmetic, a uniform depth error bound, maintenance while precision increases, and the exact original autoregressive law.

Model. The vocabulary has size V , the residual width is n , and there are D causal attention blocks. Each block first applies

$$N_\varepsilon(h) = \frac{h}{\sqrt{\varepsilon + \|h\|_2^2/n}}, \quad \varepsilon > 0,$$

then affine query, key, and value maps. Its attention scores are $\beta \langle q_t, k_j \rangle / n$, $j \leq t$. The attention update is added to the residual stream. A second normalization followed by an affine map and coordinatewise tanh gives the second additive update. Either additive update may be multiplied by a fixed dyadic coefficient $\alpha \in [0, 1]$. Convex residual mixtures with coefficients in $[0, 1]$ are also allowed. An additional fixed affine attention output map can be included. The output consists of V affine logits followed by softmax; the logits are promised to lie in $[-1, 1]$. Input tokens use a fixed dyadic embedding table with entries of magnitude at most s .

All model parameters are dyadic with encoding length at most b , using binary integer numerators and denominators; in particular, their denominators divide 2^b . The row sums of absolute values and the bias bounds are at most $s \geq 1$. Every positive normalization parameter is at least 2^{-b} . The temperature $\beta \geq 0$ is dyadic of at most b bits, or $\beta = \sqrt{n}$. Each layer uses one fixed affine key map at every position, and its linear part has rank at most the fixed integer r . Thus there is one common known affine key space per layer containing the keys at all positions. Separate rank bounds for position-dependent maps, such as positional rotations that enlarge the joint key span, do not imply this joint-space condition. The affine key space is computed from the fixed weights, independently of the token history.

The statement also applies to a bounded number of heads, with a separate rank promise for each head. More generally, the number of heads can be included as another polynomial parameter. We state the formulas for one head to keep notation short.

Theorem B.28 (Exact decoding with a fixed-rank key map). *For each fixed r , the model above has a uniform exact autoregressive sampler. Let $T_0 \geq 1$ be the prompt length and $T \geq T_0$ the current context length. There is a constructive polynomial F_r such that prefill costs at most*

$$T_0 F_r(n, D, V, b, s, \beta, \log(T_0 + 2)),$$

and every subsequent token has expected bit cost at most

$$F_r(n, D, V, b, s, \beta, \log(T + 2)).$$

The expectation is conditional on every previous token history. It includes the construction and insertion of the new keys and values, random bits, and all precision refinement. The sampler uses no exact-real oracle. At standard scaling $\beta = \sqrt{n}$, if s, D, V, b are polynomially bounded in n , the expected work is $\text{poly}_r(n, \log T)$. The context dependence outside the stored token history is polylogarithmic.

The same argument applies to a specified family of other feedforward updates when fixed polynomial bounds for its range, certified evaluation cost, and logarithm of its Lipschitz bound are supplied as part of the theorem's assumptions. The resulting polynomial also depends on those fixed bounds. The degree of F_r depends on r ; the uniform polynomial dependence on width requires a rank bound independent of width.

We prove the theorem after three lemmas: a finite moment index (Lemma B.29), uniform precision for a causal replay (Lemma B.30), and exact sampling from certified replays (Lemma B.31).

Lemma B.29 (A finite moment index). *Fix r . Let $z_j \in [-1, 1]^r$ and $u_j \in [-1, 1]^a$ be dyadic records on a common grid 2^{-P} , and let $t \in \mathbb{R}^r$ satisfy $\|t\|_1 \leq H$, where $H \geq 1$ is an integer. The coordinates of t are either rational with a common denominator, or have the form $c_i/\sqrt{n}$ with rational c_i having a common denominator. Write*

$$A(t) = \frac{\sum_{j=1}^T e^{\langle t, z_j \rangle} u_j}{\sum_{j=1}^T e^{\langle t, z_j \rangle}}.$$

For each $p \geq 1$, an append-only deterministic index returns certified intervals of width at most 2^{-p} for all coordinates of $A(t)$. Its update and query costs are polynomial, at fixed r , in a, H, P, p , the rational coefficient lengths, $\log(T+2)$, and, in the second case, $\log(n+2)$. Its arithmetic has no error that accumulates with the number of insertions.

Proof idea. Expand the exponential in the fixed number of key coordinates. The index stores exact sums of monomials and value-weighted monomials. Their numerator lengths grow by $\log T$. A uniform Taylor error controls the normalized attention mean because its average denominator is at least e^{-H} .

Proof. Put

$$q = p + 2H + 6, \quad m = \lceil 8(H + q + 1) \rceil, \quad \mathcal{I}_m = \{\nu \in \mathbb{N}^r : |\nu| \leq m\}.$$

Let $N_m = |\mathcal{I}_m| = \binom{m+r}{r}$. The Taylor polynomial $E_m(x) = \sum_{d=0}^m x^d/d!$ satisfies

$$\sup_{|x| \leq H} |E_m(x) - e^x| \leq 2^{-q}. \quad (\text{B.14})$$

Indeed, the Lagrange remainder is at most $e^H H^{m+1}/(m+1)!$. The elementary bound $N! \geq (N/e)^N$ follows by integrating $\log x$ below its sum. Since $m+1 \geq 8H$ and $e < 4$, the remainder is at most $e^H 2^{-(m+1)} \leq 2^{2H-m-1} \leq 2^{-q}$.

Store the exact rational sums

$$S_\nu = \sum_{j=1}^T z_j^\nu, \quad M_{\nu,i} = \sum_{j=1}^T z_j^\nu u_{j,i}, \quad \nu \in \mathcal{I}_m, \quad i \leq a.$$

They can be stored as integer numerators over the one denominator $2^{P(m+1)}$. An insertion computes its monomials exactly and adds the corresponding integer increments. The absolute value of every stored sum is at most T , so its numerator has $O(P(m+1) + \log(T+2))$ bits. No rounding occurs during these additions. Different dyadic coordinate grids can be put over one common power-of-two denominator before forming the moments.

The multinomial theorem gives

$$E_m(\langle t, z \rangle) = \sum_{\nu \in \mathcal{I}_m} \frac{t^\nu z^\nu}{\nu!}.$$

Consequently the truncated numerator and denominator are computed from the stored moments by $O(aN_m)$ exact rational operations. If the common denominator of the rational coordinates of t is d_t , all Taylor coefficients have a common denominator dividing $d_t^m m!$. This follows because $\nu!$ divides $|\nu|!$, which divides $m!$. Integer numerator lengths are polynomial in m, P , the coefficient encoding lengths, and $\log T$.

Divide the true and truncated sums by T . Their denominators are Z and $\tilde{Z}$, and their coordinate numerators are N_i and $\tilde{N}_i$. Then

$$Z \geq e^{-H}, \quad |N_i| \leq Z, \quad |Z - \tilde{Z}| \leq \delta, \quad |N_i - \tilde{N}_i| \leq \delta, \quad \delta = 2^{-q}.$$

The choice of q makes $\delta \leq Z/2$. Thus

$$\left| \frac{\tilde{N}_i}{\tilde{Z}} - \frac{N_i}{Z} \right| \leq \frac{2\delta}{Z - \delta} \leq 4\delta e^H \leq 2^{-p-4}.$$

In particular the truncated denominator is positive. A rational division enclosure of width 2^{-p-2} , enlarged by the displayed Taylor error on both sides, has width less than 2^{-p} .

At standard scaling the coordinates are $t_i = c_i/\sqrt{n}$. Separate multiindices by the parity of $|\nu|$. Every truncated sum is then of the form $A + B/\sqrt{n}$ with rational A, B . A common denominator is $d_c^m n^{\lfloor m/2 \rfloor} m!$, where d_c is a common denominator for the c_i . Its encoding length is $O(m \log n + m \log(m+2))$ plus the rational input lengths. The ratio of two such expressions is evaluated by certified bisection for $\sqrt{n}$. The positive lower bound $\tilde{Z} \geq e^{-H}/2$ bounds the required guard precision. If the rational coefficients have at most L bits, precision $O(L + H + p + \log(T+2))$ in the square root suffices. This remains polynomial. If n is a square, ordinary rational arithmetic suffices. Hence the temperature is exactly $\sqrt{n}$ throughout; it has not been replaced by a floating-point constant. $\square$

An explicit polynomial bound. The preceding construction can deliberately use schoolbook integer arithmetic and recompute monomials. With

$$L = C_r \{m[P + b + \log(n + m + D + V + 2)] + \log(T + 2) + 1\},$$

one may bound an entire token's deterministic finite-precision evaluation by

$$C_r(D + 1)(n + V + 1)^2(m + 1)^{2r+4}L^4. \quad (\text{B.15})$$

Here b also bounds the fixed rational chart lengths up to a factor depending on r , and P includes the guard precision described below. To check (B.15), note that every integer formed for one token has $O(L)$ bits, so a schoolbook product costs $O(L^2)$. The layers and the output head use $O((D + 1)(n + V + 1)^2)$ products in affine maps and $O((D + 1)(n + V))$ certified scalar evaluations, each of cost $O(L^4)$ by Lemma B.12 and rational bisection. The moment updates and queries use $O((D + 1)(n + m + 1)N_m)$ products, where $N_m = \binom{m+r}{r} \leq (m + 1)^r$: each monomial and each Taylor coefficient takes $O(m)$ products and meets n value coordinates. The total is $O_r((D + 1)(n + V + 1)^2(m + 1)^{r+1}L^4)$. The displayed bound is intentionally loose; it counts bit operations.

Lemma B.30 (Uniform stability over a causal prefix). *For the model in Theorem B.28, a token history of any length has a deterministic certified replay algorithm. To return all output probabilities and their cumulative sums within absolute interval width 2^{-p} , it uses working precision*

$$P = p + O((D + 1)[b + \log(n + s + \beta + 2)] + \log(D + V + 2)).$$

Its work per replayed token is bounded by a fixed polynomial in $n, D, V, b, s, \beta, p, \log(T + 2)$ at fixed r . The stability guard does not grow linearly with the context length.

Proof idea. An attention row is stable in the largest coordinate error, with a bound independent of the number of keys. Each layer reads only the preceding layer's prefix. Errors therefore propagate through network depth. Exact moment counters introduce no additional recurrence over token positions.

Proof. For scores a and values v , write $p(a)$ for softmax and $F(a, v) = \sum_j p_j(a)v_j$. Along a differentiable score path,

$$\frac{d}{dt}p_j = p_j \left(\dot{a}_j - \sum_i p_i \dot{a}_i \right).$$

Summing absolute values shows that softmax is Lipschitz from ℓ_∞ scores to ℓ_1 probabilities with constant two, for every number of positions. If the value coordinates are bounded by K , this gives

$$\|F(a, v) - F(a', v')\|_\infty \leq \|v - v'\|_\infty + 2K\|a - a'\|_\infty. \quad (\text{B.16})$$

For $|q_i|, |k_{j,i}| \leq K$, perturbations bounded by δ_q, δ_k change any score by at most $\beta K(\delta_q + \delta_k)$. The same estimate follows by interpolating between two bounded endpoint arrays, so it has no extra product-error term.

Normalization has a global bound independent of the residual radius. Put $v = \varepsilon + \|h\|_2^2/n$. Its Jacobian is

$$\frac{\partial N_i}{\partial h_j} = \frac{\mathbf{1}_{i=j}}{\sqrt{v}} - \frac{h_i h_j}{nv^{3/2}}.$$

Using $|h_i| \leq \sqrt{n(v - \varepsilon)}$ and $\|h\|_1 \leq n\sqrt{v - \varepsilon}$, every absolute row sum is at most $(1 + \sqrt{n})/\sqrt{\varepsilon}$. Hence the ℓ_∞ Lipschitz constant is at most $(1 + \sqrt{n})2^{b/2}$. Also $|N_i(h)| \leq \sqrt{n}$. Choose a known power of two

$$K \geq s(\lceil \sqrt{n} \rceil + 1), \quad K < 2s(\lceil \sqrt{n} \rceil + 1).$$

Queries, keys, and values obtained from normalized inputs are bounded by K . Affine maps have Lipschitz bound s , tanh has bound one, and residual addition has bound two on a product of streams. A fixed

affine output map only introduces another factor polynomial in s . Thus the network is a composition of $J = O(D + 1)$ maps on prefix arrays, each with a common Lipschitz bound $L_0 \geq 2$ satisfying

$$\log_2 L_0 = O(b + \log(n + s + \beta + 2)).$$

These maps act on all positions with the maximum norm. The bound is independent of the number of positions by (B.16).

Induction over depth. If each primitive’s computed output has local error at most δ , induction bounds the final error by $JL_0^J\delta$. Choose

$$\delta = 2^{-P}, \quad P \geq p + \lceil \log_2(J + 1) \rceil + J\lceil \log_2 L_0 \rceil + 10.$$

Known interval error bounds can be propagated with these same constants, so this argument returns certified enclosures. The output head and its cumulative probabilities are evaluated with the same margin. Exponentials on bounded logit intervals and a positive softmax denominator have elementary polynomial certified evaluation cost. The V sums and output coordinates are charged explicitly.

Rank-preserving rounding. A deterministic rounding operation first computes a certified interval of width at most 2^{-P-3} , then rounds its rational midpoint to the grid 2^{-P} . Its local error is less than 2^{-P} after the fixed guard margin is included. It does not decide the exact nearest grid point of an unknown real value. Apply this rule to the output of normalization, clipping the rounded value to the rational interval $[-\lceil \sqrt{n} \rceil, \lceil \sqrt{n} \rceil]$, and then compute

$$\tilde{k} = W_K \tilde{N} + b_K$$

exactly. The result is dyadic on the grid $2^{-(P+b)}$ and lies exactly in $b_K + \text{image}(W_K)$. Independent rounding of the coordinates of $\tilde{k}$ is not used. Query and value affine products can be computed exactly in the same way. The completed residual updates are rounded to 2^{-P} . All these local rounding errors are included in δ by requesting a fixed number of guard bits inside each primitive.

The moment chart. Use a bounded rational coordinate chart for the key image, $k = c_0 + Ak_I$, with $|A_{ij}| \leq 2$. The chart is computed once from the weights by Lemma B.18, applied to a column basis of W_K . Put $z_j = (k_j)_I/K$. Then $z_j \in [-1, 1]^r$, and the common score offset $\beta\langle q, c_0 \rangle/n$ cancels. The remaining coefficient vector is

$$t = \frac{\beta K}{n} q^T A, \quad \|t\|_1 \leq 2r\beta K^2.$$

Take $H = \lceil \max\{1, 2r\beta K^2\} \rceil$, using a rational upper bound for $\sqrt{n}$ if necessary. Scale values by K and apply Lemma B.29; restoring that scale needs only $O(\log(K + 2))$ additional requested precision bits. The key and value grids are common throughout a precision phase, so the moment additions remain exact. The common determinant denominator in A gives rational coefficient lengths $O_r(P + b + \log(n + 2))$; an n -term projection adds $O(\log n)$ numerator bits.

Causal order. Process a prefix in increasing position order. At a layer and position t , first form its query, key and value from the current preceding-layer residual. Append the current key and value to that layer’s moment index *before* querying it, so the attention range is exactly $j \leq t$. Complete the residual and feedforward operations, then move to the next layer. This is the same deterministic computation whether performed during prefill, an online append, or a replay.

To see why no token-count error factor is hidden, regard each layer’s exact and approximate states as arrays over all positions in the current prefix. Every attention input belongs to the preceding-layer array. Its maximum error is already bounded by the preceding induction step. A previous token’s error is never fed back into the same layer’s residual state. The only cross-position operation is the normalized weighted average, whose uniform Lipschitz bound was proved above. Exact integer moment sums evaluate the attention of the approximated arrays without a further time-accumulation error. The induction therefore runs over the J network stages only. The explicit polynomial bound (B.15) completes the claim. $\square$

Lemma B.31 (Exact sampling from certified replay). *Suppose a deterministic causal evaluator can maintain a token history at any fixed accuracy 2^{-p} , spending at most $G(p + \log(T + 2))$ bits per appended token, where G is a fixed polynomial with nonnegative coefficients. Suppose a replay of all T tokens costs at most $TG(p + \log(T + 2))$ and returns certified intervals of width at most 2^{-p} for all categorical cumulative probabilities. Then exact autoregressive sampling can be maintained with expected polynomial work per token, including precision changes, after polynomial prefill.*

Proof idea. Routine precision is a sufficiently large multiple of $\log T$. Only a uniform random number close to a cumulative boundary requires a complete finer replay. Background processing prepares the next routine index before the current precision expires. The interval decision is classical. The additional point here is paying for a complete causal replay only on its small ambiguity event, while preparing routine precision increases with bounded work at each token.

Proof. For a power-of-two horizon R , choose

$$p_R = \lceil \log_2(64V(R + 2)^3) \rceil.$$

At a query draw the first p_R bits of a fresh uniform binary expansion U . Let $I_U = [a, b]$ be its dyadic interval, of width 2^{-p_R} . Denote the certified cumulative intervals by $[\underline{F}_i, \bar{F}_i]$, and use the exact endpoints $F_0 = 0$ and $F_V = 1$. If $\bar{F}_{i-1} \leq a$ and $b \leq \underline{F}_i$, output token i . Equality at a dyadic cell endpoint is permitted; the rule can then differ from the inverse CDF only when U equals a true boundary, an event of probability zero. Otherwise replay the current history at accuracy $p_R + 1$, reveal one more uniform bit, and try again. Continue with $p_R + j$ as needed. Intervals from successive replays need not be nested; every one contains the same true cumulative probabilities.

If the precision- p comparison is unresolved, U is within 2^{1-p} of one of the at most $V - 1$ cumulative boundaries. The conditional probability of this event, given the token history, is at most

$$4V2^{-p}. \tag{B.17}$$

Each realized uniform number outside the finite set of true boundaries eventually lies in a unique certified categorical interval. The algorithm therefore stops almost surely, and its output is exactly the inverse-CDF sample. It does not substitute an approximate probability law.

Cost of temporary replays. The expected work of temporary replays is bounded by

$$8VT2^{-p_R} \sum_{j \geq 0} 2^{-j} G(p_R + j + \log(T + 2)).$$

As long as $T \leq R$, its prefactor is at most a constant times $(R + 2)^{-2}$. If G has degree k , the sum is polynomial in $p_R + \log(T + 2)$. One explicit inequality, valid for $P \geq 1$, is

$$\sum_{j \geq 0} 2^{-j} (P + j)^k \leq 2^{k+1} k! P^k.$$

It follows from $(P + j)^k \leq P^k(j + 1)^k$ and $(j + 1)^k \leq k! \binom{j+k}{k}$, followed by the generating function for these binomial coefficients. This proves the expected work bound without assuming finite expected stopping time in advance. The same tail bounds all additional random bits. After the token is returned, discard every temporary fine-precision index and retain the deterministic routine index.

The background schedule. It remains to maintain routine precision without a large online rebuild. Let N be the least power of two at least T_0 . During prefill build the routine index at precision p_{2N} . This horizon is at least $2T_0$, so the first append is not forced to pay for a prefix rebuild. Before context length N , use this index alone. At length N , begin an empty future index at precision p_{4N} . During each of the next N appends, perform the current index's one-token update and replay two successive history tokens into

the future index. After j such appends, the history has length $N + j$ and the future index has replayed $2j$ tokens. Since $2j \leq N + j$ for $j \leq N$, the next replay tokens always exist. At length $2N$ the future index has replayed exactly the entire current history. Switch to it, replace N by $2N$, and begin the next future index.

Every online append therefore performs at most three deterministic token evaluations. The future precision differs from the current precision by only a constant number of bits in p_R ; its other model-dependent guards are unchanged. No operation scans the context as part of the deterministic maintenance schedule. The only complete scans are temporary random refinements, whose expected cost was bounded above.

The exact law. Store the actual token history for all replays. A replay is driven only by this history and the fixed model. It never uses the approximate hidden states of a different precision index. Conditional on every realized token history, its intervals therefore enclose the same exact next-token law. Fresh uniform bits give the correct next conditional distribution, and induction establishes the entire autoregressive joint law. Background replays and temporary refinements do not change any previously sampled token. $\square$

Proof idea for Theorem B.28. The moment index supplies a maintained evaluator at each fixed precision. Uniform stability sets the required precision from network depth, and the replay lemma handles its increase with context length. Combining their bit bounds charges both normal updates and the rare refinements needed for the exact law.

Proof of Theorem B.28. Apply Lemma B.30 to obtain the deterministic fixed-precision evaluator, with the exact moment counters of Lemma B.29. Its one-token work is bounded by (B.15), with $m = O_r(\beta K^2 + P + 1)$ and the working precision stated in the stability lemma. Apply Lemma B.31. The explicit geometric-moment inequality converts the temporary replay cost into another polynomial in the same parameters. All storage consists of the fixed model, the token history, at most two routine moment indices, and temporary replay storage when required. The routine indices have polynomial size in the displayed parameters and $\log T$. Temporary precision has no deterministic upper bound, but the same geometric moment estimate bounds its expected space by such a polynomial. No infinite expansion or real arithmetic oracle is used.

For the normalized model, $K = O(s\sqrt{n})$, so $H = O_r(1 + \beta s^2 n)$. At $\beta = \sqrt{n}$ this is polynomial in n and s . The normalization guard depends on b only through $\log L_0$, and the degree of the polynomial approximation has no factor 2^b . Thus a very small positive normalization floor changes precision polynomially. The stipulated bounds on the remaining parameters give the claimed $\text{poly}_r(n, \log T)$ work. Substituting other polynomially evaluable updates only changes the range and Lipschitz constants in the same finite-depth induction. $\square$

Corollary B.32 (The affine output head needs no fixed logit range). *The conclusions of Theorem B.28 remain valid when the promise that logits lie in $[-1, 1]$ is removed, keeping the stated affine-head row and bias bounds. The cost is bounded by another constructive polynomial in the same parameters. The fixed-model corollaries below also hold for this extension.*

Proof idea. Pre-normalization bounds every attention update, so additive residuals give a polynomial logit range. Subtracting an upper bound for the largest logit makes all exponentials nonpositive and leaves a constant lower bound on their total. Certified softmax evaluation therefore has no cost exponential in the logit range.

Proof. With the power-of-two value bound K used above, the residual range after D blocks is at most

$$R_D = s + D(sK + s + 1).$$

Indeed the embedding range is at most s ; an attention mean has range at most K , its optional affine output map has range at most $sK + s$, and the tanh update has range at most one. This bound also covers convex residual coefficients. Hence each output logit has absolute value at most $Z = sR_D + s$, which is polynomial in n, D, s .

For completeness, consider certified intervals of width at most $1/4$ for the real logits $z_1, \dots, z_V$. Let A be the largest of their rational upper endpoints. Then every $z_i - A$ is nonpositive and at least one is at least $-1/4$. Consequently

$$S = \sum_i e^{z_i - A} \geq e^{-1/4} > 1/2.$$

Enclose each exponential to width $\eta = 2^{-p}/(64V)$. On the negative half-line the exponential has Lipschitz constant at most one, so this only requires $p + O(\log(V + 2))$ certified logit bits, followed by the negative-exponential routine. Intersect any refined logit interval with its original enclosure, so its upper endpoint remains at most A . It does not require an exact comparison of two unknown logits.

Every numerator partial sum and the denominator has error at most $V\eta$. Put $\delta = V\eta \leq S/2$. Outward interval division, after clipping each numerator interval to nonnegative values, has width at most

$$\frac{N + \delta}{S - \delta} - \frac{N - \delta}{S + \delta} \leq \frac{4S\delta}{S^2 - \delta^2} \leq \frac{16\delta}{3S} < 11\delta < 2^{-p-2}.$$

Here $0 \leq N \leq S$ is the true partial sum. A final outward dyadic rounding leaves width below 2^{-p} . All exponents have polynomial encoding length and are nonpositive; no factor e^Z occurs in their bit cost. The global score-to-probability Lipschitz bound for softmax used in Lemma B.30 is valid at every logit range. Thus both its stability guard and the replay argument apply with the stated polynomial evaluation cost. $\square$

Corollary B.33 (Specified GELU and SwiGLU updates). *Theorem B.28, with the unrestricted affine head of Corollary B.32, also holds when the tanh feedforward update is replaced by either exact GELU or SwiGLU applied to affine projections of the normalized stream, followed by an affine output map. The hidden width is included as a polynomial parameter, or is bounded by a fixed polynomial in n . All affine row and bias bounds remain at most s . The update can be multiplied by any fixed dyadic coefficient $\alpha \in [0, 1]$.*

Proof idea. Normalization bounds each affine input independently of the current residual magnitude. GELU and SwiGLU consequently have polynomial range and derivative bounds. Their scalar functions have certified finite series evaluations, so they satisfy the explicit hypotheses of the decoder theorem.

Proof. Write $K = s(\sqrt{n} + 1)$, and increase it to an integer when forming series bounds. Every coordinate of every affine projection of a normalized vector has magnitude at most K . For exact GELU, put $g(u) = u\Phi(u)$, where Φ is the standard normal distribution function and ϕ its density. Since $0 \leq \Phi \leq 1$ and $0 \leq \phi \leq 1$,

$$|g(u)| \leq K, \quad |g'(u)| = |\Phi(u) + u\phi(u)| \leq 1 + K \quad (|u| \leq K).$$

For a SwiGLU coordinate put $w(u, v) = uv\sigma(v)$, where $\sigma(v) = (1 + e^{-v})^{-1}$. The identities $\sigma' = \sigma(1 - \sigma)$ and $0 \leq \sigma' \leq 1/4$ give

$$|w(u, v)| \leq K^2, \quad |\partial_u w| \leq K, \quad |\partial_v w| \leq K(1 + K/4).$$

Thus its Lipschitz constant from the two-coordinate maximum norm is at most $2K + K^2/4$. Composing the input projections and the output map multiplies these bounds by at most s^2 . The output ranges are at most $sK + s$ and $sK^2 + s$, respectively. These estimates do not depend on hidden width because they use row sums. Multiplication by $\alpha \in [0, 1]$ only decreases both range and derivative bounds. Adding D such bounded updates leaves a residual and affine head range polynomial in n, D, s , as required by Corollary B.32.

Here is a direct certified evaluation procedure. The sigmoid uses a negative exponential and a denominator at least one: for $v \geq 0$ use $1/(1 + e^{-v})$, and for $v < 0$ use $e^v/(1 + e^v)$. Evaluation at a dyadic argument therefore has polynomial cost in K and the requested precision, by Lemma B.12. For GELU use

$$\Phi(u) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^u e^{-t^2/2} dt.$$

Expand the integrand in powers of $-t^2/2$ and integrate its finite polynomial exactly. Put $H = \max(1, K^2/2)$. The same factorial remainder bound as in (B.14) shows that $O(H + p + \log(K + 2))$ terms suffice for integral error $2^{-p-O(1)}$ uniformly on $[-K, K]$. The intermediate rational coefficients and arithmetic have polynomial

bit length in K, p, b . Certified values of π follow from the arctangent identity used below in Corollary B.35, and rational bisection gives its square root and reciprocal. Guard precision $O(K^2 + p + \log(K + 2))$ controls every product and final division. Request an additional $\lceil \log_2(K + 2) \rceil$ bits for Φ before multiplying it by u . Thus both updates have polynomial certified evaluation cost. For a real input already enclosed by a certified interval, evaluate at its dyadic midpoint and add the displayed derivative bound times its radius. No exact comparison with a transcendental value is required.

The range bounds, evaluation algorithms, and derivative bounds verify the specified-family assumptions of Theorem B.28. Its depth stability and replay proofs therefore apply without further changes. $\square$

Corollary B.34 (A completely fixed model). *Fix every parameter of one finite model covered by Theorem B.28, including its width, depth, vocabulary and weights. Its key ranks are then fixed integers, even if its key maps have full rank. After prefill, exact autoregressive decoding takes $\log^{O_{\text{model}}(1)}(T + 2)$ expected bit work per token. For the basic tanh model, including the affine head extension, the explicit implementation above permits the bound $O_{\text{model}}(\log^{2r+12}(T + 2))$. Other specified feedforward families retain the $\log^{O_{\text{model}}(1)}(T + 2)$ bound, whose exponent includes their evaluation polynomial.*

Proof idea. Only the requested precision increases with context length. The number of monomials is a polynomial in that precision when the key dimension is fixed.

Proof. For a fixed model, H and all stability constants are constant. Routine accuracy and working precision are $O_{\text{model}}(\log(T + 2))$, as is m . The integer length L in (B.15) is $O_{\text{model}}(\log^2(T + 2))$. For the basic tanh model, substitution gives exponent $2r + 12$, and the temporary-replay geometric moment bound preserves that degree. The background schedule has constant overhead. Taking the largest key rank across layers proves the claim. For another specified feedforward family, replace this explicit polynomial by the fixed polynomial that also charges its certified scalar evaluations; the same argument preserves a model-dependent polylogarithmic exponent. $\square$

Corollary B.35 (Fixed positional rotations). *In the completely fixed model of Corollary B.34, allow query and key vectors at position t to undergo coordinate-pair rotations with angles $t\omega_1, \dots, t\omega_{\lfloor n/2 \rfloor}$. Each frequency is fixed and comes with a specified finite algorithm that returns a certified interval of width 2^{-p} in time polynomial in p . Fixed rational and real algebraic frequencies satisfy this assumption. Exact autoregressive decoding still has $\log^{O_{\text{model}}(1)}(T + 2)$ expected bit work at each token after prefill. The same conclusion holds for a fixed finite collection of such rotations and fixed unrotated coordinates.*

Proof idea. Rotation does not increase Euclidean length and has a bounded coordinate Lipschitz factor. Evaluating an angle at a late position needs additional frequency bits proportional to $\log T$. The full ambient key space avoids a rank condition on the rotating subspaces.

Proof. A pair rotation has matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Its maximum absolute row sum is at most $\sqrt{2}$, uniformly in θ . Consequently it introduces only a fixed factor in the key and query coordinate bounds and in their Lipschitz bounds with respect to preceding-layer states. The position is a known exact integer and is never perturbed.

We give a finite computation of its coefficients. Fix a frequency bound $C_\omega \geq 1 + |\omega|$. At position $1 \leq t \leq T$, form a dyadic approximation θ_0 to $t\omega$, with error at most 2^{-p-10} , by requesting $p + \lceil \log_2(T + 1) \rceil + O(1)$ frequency bits. Compute a dyadic $\pi_0 \in [3, 4]$ with error at most $2^{-p-10}/[C_\omega(T + 1)]$. This has polynomial bit cost in $p + \log(T + 2)$: for example, the identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the alternating arctangent series give certified intervals using a linear number of terms in the requested precision, with polynomial-length integer arithmetic. Set $k = \lfloor \theta_0/(2\pi_0) \rfloor$ by exact rational division and put

$u_0 = \theta_0 - 2k\pi_0$. Then $0 \leq u_0 < 8$ and $|k| \leq C_\omega(T+1)$. The true reduced argument $u = t\omega - 2k\pi$ satisfies

$$|u - u_0| \leq |t\omega - \theta_0| + 2|k||\pi - \pi_0| \leq 3 \cdot 2^{-p-10}.$$

No test of whether a true angle lies exactly on a period boundary is made. Sine and cosine have Lipschitz constant one. Their rational Taylor polynomials through degree $32(p+8)$ on $[0, 8]$ have remainder at most $8^{32(p+8)+1}/[32(p+8)+1]! < 2^{-p-6}$. Evaluating these finite polynomials exactly and rounding outward gives certified error 2^{-p-4} in polynomial bit work. Thus both coefficients have the required certified evaluation bound. For a fixed algebraic frequency, an isolating interval for its minimal polynomial and rational bisection give the assumed polynomial-time frequency approximations.

Use the common ambient chart $A = I_n$, $c_0 = 0$, of dimension $r = n$. After a rotation, round each approximate query and key coordinate to the common working grid by the certified midpoint rule. The keys and the coefficient vector t then have the finite form that Lemma B.29 requires. At full dimension, rounding the keys cannot violate the key-space condition. Request an additional constant number of coefficient bits to account for the fixed coordinate range. The rotation error and the final grid rounding are then included in the same local error budget δ . The resulting moment counters remain exact integers. All score and state stability bounds are independent of position, while coefficient evaluation adds only a fixed polynomial in $p + \log(T+2)$ to per-token work. Lemma B.31 applies with this enlarged polynomial and proves the claimed conditional exact law and expected cost. $\square$

Scope. For a completely fixed full-rank model, Corollaries B.34 and B.35 are formal asymptotic statements with a potentially enormous model-dependent exponent and constant. They do not assert a practical improvement at present model dimensions. The moment dimension grows as $\binom{O_r(\beta s^2 n + P) + r}{r}$. The source of the positive result is a structural restriction on the key maps. The arbitrary-fresh-query attention hardness results of Section B.11 permit ranks that grow with the context, so they do not exclude this theorem. Conversely, the theorem does not remove their barrier for unrestricted caches.

B.11 A lower bound after preprocessing the keys and values

The context-probe bounds proved earlier concern preprocessing that is independent of the context. An index built during prefill is a different model. The index can already contain every input sign, so changing one sign can change the index before a query begins. The earlier coupling argument gives no query lower bound in this model.

We therefore use randomized OVH (Section 6) in the following precise form: for each $\eta > 0$ there is a constant $c > 0$ such that no randomized algorithm, with error at most $1/3$, decides whether two sets of N vectors in $\{0, 1\}^{\lceil c \log_2 N \rceil}$ contain an orthogonal pair in $O(N^{2-\eta})$ bit operations. The randomized form is needed because sampling a bounded-logit token gives a randomized decision algorithm. The standard split-and-list reduction from randomized SETH gives randomized OVH. The constant-gap attention obstruction with polynomial prefill is also established by Haverbeck et al. [19, Theorem 4.1 and Lemma A.5]; the direct reduction below keeps sign-valued coordinates and the finite-bit stopping conditions explicit.

Our indexed attention problem has a static cache $\mathcal{C} = ((k_i, v_i))_{i=0}^{T-1}$ and an online query q . It must sample the binary token with probability

$$\pi_{\mathcal{C}}(q) = \frac{1 + \tanh(\tanh u_{\mathcal{C}}(q))}{2}, \quad u_{\mathcal{C}}(q) = \frac{\sum_{i=0}^{T-1} e^{\langle q, k_i \rangle / \sqrt{n}} v_i}{\sum_{i=0}^{T-1} e^{\langle q, k_i \rangle / \sqrt{n}}}. \quad (\text{B.18})$$

This is a scalar attention coordinate, a unit tanh row, and a softmax over the two logits $\pm \tanh u_{\mathcal{C}}(q)$. Both logits lie in $[-1, 1]$ whenever $|v_i| \leq 1$.

Theorem B.36 (Polynomial prefill does not give uniformly sublinear queries). *Assume randomized OVH. For every fixed $a \geq 1$, $b \geq 0$, and $\epsilon \in (0, 1)$, there is a constant $C > 0$ such that no uniform randomized data structure has both of the following guarantees on all caches of width $n \leq C \log^4(T+2)$:*

1. *it preprocesses the complete cache in expected $O(T^a(n + \log T)^b)$ bit operations;*
2. *for every query it samples (B.18) in expected $O(T^{1-\epsilon}(n + \log T)^b)$ additional bit operations.*

The conclusion already holds when n is a perfect square, $q \in \{0, 1\}^n$, $k_i \in \{-1, 0\}^n$, and $v_i \in \{-1, 1\}$. In particular $\|q\|_2, \|k_i\|_2 \leq \sqrt{n}$ and all input numbers have constant bit length. The only score scaling is the standard $1/\sqrt{n}$.

The same conclusion holds if the marginal law of each sample may have total variation distance at most $1/64$ from the required token law. It suffices to require the query guarantee for a fresh query to a fresh index. Correctness conditional on a random prefill state, or on an adaptive query history, is not needed for this lower bound.

The constant C is allowed to depend on the proposed preprocessing degree and the claimed query saving. This is the usual uniform quantification in indexed fine-grained lower bounds. The theorem excludes a uniform $\text{poly}(n, \log T)$ query algorithm after any fixed polynomial prefill budget. It does not give an unconditional query lower bound, nor a lower bound for each individual cache.

Remark B.37 (Translating entry bounds into width). The parameter regimes can be compared directly. Suppose old vectors have dimension d , coordinate magnitudes at most B_0 , and scores $\langle q, k \rangle / d$. Choose an integer $H \geq \max\{1, B_0^2/d\}$. Divide each coordinate by $\sqrt{dH}$ and repeat it dH^2 times. The new width is $n = (dH)^2$, the coordinate bound is one, and

$$\frac{\langle q_{\text{new}}, k_{\text{new}} \rangle}{\sqrt{n}} = \frac{dH^2}{dH\sqrt{n}} \langle q, k \rangle = \frac{\langle q, k \rangle}{d}.$$

Thus $d = O(\log T)$ and $B_0 = O(\sqrt{\log T})$, with $B_0^2/d = O(1)$, translate to $n = O(\log^2 T)$ at standard scaling. Rounding the new coordinates within $\eta \leq 1/(100\sqrt{n})$, while keeping their magnitudes at most one, changes every score by at most $\sqrt{n}(2\eta + \eta^2) \leq 0.03$. The dummy keys can remain exactly zero. These bounded perturbations preserve a scalar separation tending to zero and one, since they multiply all non-dummy exponential weights by factors in $[e^{-0.03}, e^{0.03}]$. Replacing scalar values $v \in \{0, 1\}$ by $2v - 1$ then gives attention mean $u = 2y - 1$. Once the two cases have $u \geq 1/2$ and $u \leq -1/2$, the token map in (B.18) separates them by $9/16$ and $7/16$, using the inequality $\tanh(\tanh x) \geq x/4$ on $[0, 1]$ shown in the proof of Lemma B.38.

B.11.1 A constant token gap from orthogonality

Lemma B.38 (Dummy-key gap). Given $m \geq 1$ vectors $a_1, \dots, a_m \in \{0, 1\}^d$, $d \geq 1$, define

$$L = \lceil \log_2(16m) \rceil, \quad H = dL, \quad n = H^2, \quad r = dL^2.$$

Repeat every coordinate of $-a_j$ exactly r times to form k_j , and repeat every coordinate of a query $b \in \{0, 1\}^d$ exactly r times to form q . Let $k_0 = 0$, $v_0 = -1$, and $v_j = 1$ for $j \geq 1$. Then

$$\langle q, k_j \rangle / \sqrt{n} = -L \langle a_j, b \rangle.$$

If some a_j is orthogonal to b , then $\pi_C(q) \geq 1/2$. If none is orthogonal, then $\pi_C(q) \leq 1/2 - 15/136 < 7/16$.

Proof idea. A zero dummy key gives a known reference weight. An orthogonal vector matches that weight; without one, all real-key weights are exponentially suppressed. Coordinate repetition realizes the needed penalty at standard scaling with sign entries.

Proof. There are $dr = d^2 L^2 = n$ coordinates, and $r/H = L$, proving the score identity. Put

$$W = \sum_{j=1}^m e^{-L \langle a_j, b \rangle}.$$

The dummy has weight one, so the attention output is $u = (W - 1)/(W + 1)$. An orthogonal vector contributes one and gives $W \geq 1$, hence $u \geq 0$. Otherwise every dot product is at least one, and

$$0 \leq W \leq m e^{-L} \leq m 2^{-L} \leq 1/16.$$

It follows that $u \leq -15/17$.

For $0 \leq x \leq 1$, concavity of $\tanh$ on this interval and $\tanh 1 \geq 1/2$ give $\tanh x \geq x/2$. The endpoint inequality follows from $e^2 \geq 3$. Applying this inequality twice gives $\tanh(\tanh x) \geq x/4$. Oddness and monotonicity therefore imply

$$\frac{1 + \tanh(\tanh u)}{2} \leq \frac{1}{2} - \frac{15}{136} < \frac{7}{16}.$$

All computations specifying the cache and query use integers. The reduction never evaluates an exponential, $\tanh$, or an exact real number. $\square$

B.11.2 Absorbing every fixed polynomial preprocessing degree

Proof idea. Partition the hard instance into small buckets so any fixed preprocessing degree is affordable. Independent index copies amplify the token gap, and a global runtime cutoff converts the expected query guarantee into a bounded-time randomized decision algorithm.

Proof of Theorem B.36. Suppose the proposed data structure exists. We may enlarge a and b to integers and decrease ϵ to a positive rational if necessary. Set

$$\gamma = \frac{1}{2(a+1)}, \quad m = \lceil N^\gamma \rceil.$$

Take a balanced randomized-OVH instance A, B , each of size N , in dimension $d = \lceil c \log_2 N \rceil$. Partition A into at most $\lceil N/m \rceil$ buckets of size at most m . Apply Lemma B.38 to each bucket. Empty buckets are omitted. For a smaller final bucket one may duplicate any of its vectors until its size is m ; orthogonality existence is unchanged.

Each index has $T = m + 1$ entries and width $n = (d \lceil \log_2(16m) \rceil)^2 = O(\log^4 N) = O(\log^4 T)$. The constant in this relation may depend on c and γ . For each $b \in B$, query every bucket. There is an orthogonal pair in $A \times B$ precisely when at least one bucket-query pair is in the yes case of Lemma B.38.

Let

$$K = 2048 \lceil \log_2(24N^2) \rceil.$$

Build K mutually independent copies of each bucket index, and obtain one sample for a fixed bucket-query pair from each independent copy. For this fixed pair, the K outputs are independent even when prefill is randomized. If the data structure can alter its index during a query, record every changed cell and restore its old contents afterwards. Every query to a given copy then starts in its original prefill state. We bound the logging overhead after imposing a global time cutoff below; this avoids any assumption on higher moments of a query's runtime. No correctness assumption conditional on earlier outputs is being introduced.

In the yes case each output has success probability at least $31/64$, even with total variation error $1/64$. In the no case it has success probability at most $29/64$. Classify a pair as yes if its empirical mean is at least $15/32$. Hoeffding's inequality bounds its error by

$$\exp(-2K(1/64)^2) \leq \frac{1}{24N^2}.$$

There are at most N^2 bucket-query pairs. A union bound gives total decision error at most $1/24$ before runtime truncation. Outputs from different pairs may be correlated; the union bound needs no independence between pairs.

The expected preprocessing cost, including the independent copies, is

$$\tilde{O}\left(\frac{N}{m} m^a\right) = \tilde{O}\left(N^{1+\gamma(a-1)}\right),$$

and the total expected query cost is

$$\tilde{O}\left(\frac{N^2}{m} m^{1-\epsilon}\right) = \tilde{O}\left(N^{2-\gamma\epsilon}\right).$$

Here the tilde hides only fixed powers of $\log N$. Construction of the explicit caches, formation of the explicit query vectors, sampling, counting, and comparison are included. In particular the query vectors cost $O(n)$ bits each; this is absorbed by the polylogarithmic factors.

Since $\gamma(a-1) < 1/2$ and $\gamma\epsilon < 1/4$, the total expected original charged work is bounded by $F(N) = O(N^{2-3\gamma\epsilon/4})$ for all sufficiently large N . Simulate at most $24F(N)$ original charged steps, returning an arbitrary answer on timeout. Logging and undoing each memory write then has at most an $O(\log(F(N) + 2)) = O(\log N)$ overhead in the charged-address bit model: the log contains at most $24F(N)$ entries, and the original address and changed bits were already charged. The complete bounded simulation therefore takes $O(F(N) \log N) = O(N^{2-\gamma\epsilon/2})$ bit operations. Markov's inequality, applied to the original work, adds at most $1/24$ error. No bound on $\mathbb{E}[Q \log Q]$ is used. We have obtained a worst-case $O(N^{2-\gamma\epsilon/2})$ -time randomized OV algorithm with error at most $1/12$, contradicting randomized OVH for the corresponding constant c . $\square$

B.11.3 Realization by row-norm-one transformer maps

Corollary B.39 (The hard cache is realized by one legal block). *The caches and queries in Theorem B.36 arise from a one-block causal transformer with sign inputs, query, key, value, and tanh row norms at most one, residual parameter $\alpha = 1$, and standard scaling $\beta = \sqrt{n}$. The current position's own key and value supply the dummy. Consequently the lower bound applies to an index over the already computed prefix keys and values.*

Proof idea. Separate the raw coordinates used by cached keys from those used by the fresh query. Shared half-sum rows select the appropriate role, and the current position contributes the same dummy key for every query.

Proof. Reserve two groups of d input coordinates, a constant coordinate, and a value marker. There is room because $n = d^2 L^2 \geq 2d + 2$. Fill all unused input coordinates with $+1$. At cache position a_j , put $2a_j - 1$ in the first group, -1 in the second, $+1$ in the constant coordinate, and $+1$ in the marker. At the current query position b , put -1 in the first group, $2b - 1$ in the second, $+1$ in the constant coordinate, and -1 in the marker.

For every one of the r output rows assigned to original coordinate i , let the query map be one half of the second-group coordinate plus one half of the constant coordinate. Let the key map be minus one half of the first-group coordinate minus one half of the constant coordinate. These rows have ℓ_1 norm one. They produce exactly the repeated query b and repeated keys $-a_j$ in the lemma. The current position has key zero. Let the value map send the marker to a chosen output coordinate and send zero to all other coordinates. The selected attention output is u . With $\alpha = 1$, a unit tanh row gives $\tanh u$, and the signed-coordinate head gives (B.18). All coefficients lie in $\{0, \pm 1/2, \pm 1\}$.

Only the first m positions depend on the prefill input. The key zero and value -1 at the query position are the same for every query, so an index may include this dummy during prefill. No future query information is used to build it. $\square$

B.11.4 A well-conditioned pre-norm additive realization

Corollary B.40 (Normalization and additive residuals preserve the obstruction). *The conclusion of Theorem B.36 also holds for a single pre-norm additive attention block*

$$h \mapsto h + \text{Att}(\text{RMS}_0(h)),$$

followed by a zero feed-forward update and a two-logit affine head whose rows have ℓ_1 norm at most one. The RMS denominator before attention equals one on every constructed input, and its squared denominator after attention is at least $(n-1)/n$. Thus the lower bound does not rely on a vanishing normalization denominator.

Proof idea. All constructed raw streams are sign vectors, so normalization initially does nothing. Store the attention output in the marker coordinate; an affine readout cancels the known residual offset and preserves a constant token gap.

Proof. Use the sign inputs and maps of Corollary B.39. Each sign vector has mean squared coordinate one, so $\text{RMS}_0(h) = h$. Choose the marker itself as the output coordinate of the value map. The current marker is initially -1 , so after the additive attention update it is $-1 + u$. The constant coordinate stays equal to one. Their quarter-sum is $z = u/4 \in [-1/4, 1/4]$. Use logits $\pm z$; the token probability is $(1 + \tanh(u/4))/2$. This dyadic readout also satisfies the bounded-logit promise on the entire sign cube: the updated marker has magnitude at most two and the unchanged constant coordinate has magnitude one, so the quarter-sum has magnitude at most $3/4$ even outside the restricted hard family.

For $0 \leq x \leq 1/4$, one has $\tanh x \geq x - x^3/3 \geq (7/8)x$. Indeed $0 \leq \tanh x \leq x$ gives $\tanh'(x) = 1 - \tanh^2 x \geq 1 - x^2$, and integration proves the first inequality. The second follows from $x^2 \leq 1/16$. The no case has $u \leq -15/17$, hence

$$(1 + \tanh(u/4))/2 \leq 1/2 - 105/1088 < 7/16;$$

the yes case is at least $1/2$. The same amplification proof therefore applies. All other coordinates are unchanged signs. At every position after attention, their contribution to the mean square is $(n-1)/n$. Set the subsequent $\tanh$ feed-forward map to zero. Its normalization is well-defined with the displayed floor, and the update is zero. $\square$

For a fixed positive dyadic stabilizer ϵ_0 , the same construction rescales query and key vectors by $1/\sqrt{1+\epsilon_0}$. Replacing L by an integer at least $(1+\epsilon_0)\lceil \log_2(16m) \rceil$ restores the score gap. The value amplitude is reduced by the fixed factor $1/\sqrt{1+\epsilon_0}$, leaving a positive constant token gap. The number of independent copies changes by a constant depending on ϵ_0 . All matrix coefficients remain rational constants. The reduction does no normalization arithmetic; the exact sampler performs it. For exact sampling this proves the conclusion for every fixed $\epsilon_0 > 0$. A fixed total-variation tolerance must be chosen smaller than a constant multiple of $(1+\epsilon_0)^{-1/2}$; the particular tolerance $1/64$ continues to work, with the displayed thresholds, whenever $0 \leq \epsilon_0 \leq 1$.

B.11.5 Width $O(\log^2 T)$ from Hamming-distance gaps

The elementary orthogonality construction used a penalty for a single violated coordinate. A known gap reduction gives a stronger dimension bound. We state the imported hardness fact precisely before using it.

Lemma B.41 (Gap Hamming hardness, extracted from Rubinfeld). *Assume randomized OVH. For every $\rho > 0$, there are fixed positive constants ζ, c_0, C_0 such that the following promise problem has no randomized $O(N^{2-\rho})$ -time algorithm with error at most $1/3$. Given two lists A, B of N vectors in $\{0, 1\}^d$, with $c_0 \log_2 N \leq d \leq C_0 \log_2 N$, and an integer $d/4 \leq t \leq d/2$, distinguish*

$$\min_{a \in A, b \in B} \text{dist}_H(a, b) \leq t \quad \text{from} \quad \min_{a \in A, b \in B} \text{dist}_H(a, b) \geq (1 + \zeta)t.$$

The gap constant may be chosen rational.

This is the Hamming part of Rubinfeld's deterministic reduction in the proof of Theorem 4.1 of [38]. In its notation, the output dimension is $d = 2T'm_0$, the yes distance is $t = (T' - 1)m_0$, and the no distance is at least $T'm_0$, where $T' \geq 2$ is constant. These formulas give $d/4 \leq t < d/2$ and a fixed rational gap. The verifier's random tapes become vector coordinates; they are enumerated by the reduction. Its deterministic construction therefore also transfers hardness against randomized algorithms. Duplicating vectors balances the two output sets without changing the distances.

Theorem B.42 (Indexed lower bound at quadratic logarithmic width). *Under randomized OVH, Theorem B.36 remains true with $n \leq C \log^2(T+2)$, allowing $q, k_i \in \{-1, 0, 1\}^n$. The queries and all non-dummy keys can in fact have every coordinate in $\{-1, 1\}$, so their Euclidean norms equal $\sqrt{n}$. The total-variation tolerance $1/64$ is unchanged.*

Proof idea. A Hamming-distance gap turns the required score penalty into a constant number of coordinate repetitions. The same dummy-key separation and bucket amplification then give the sharper width bound.

Proof. Fix proposed preprocessing degree a and query saving ϵ , and set $\gamma = 1/\lceil 2(a+1) \rceil$ as before. Use Lemma B.41 with $\rho = \gamma\epsilon/2$; let its constants be ζ, c_0, C_0 . Take one of its instances of size N , and partition A into buckets of size $m = \lceil N^\gamma \rceil$, duplicating the final bucket's elements if necessary.

For the known threshold t , fix a vector $c_t \in \{-1, 1\}^d$ with exactly t positive entries. Given $a, b \in \{0, 1\}^d$, form

$$\hat{k}_a = (2a - 1, c_t), \quad \hat{q}_b = (2b - 1, \mathbf{1}) \quad \text{in } \{-1, 1\}^{2d}.$$

Their dot product is exactly

$$\langle \hat{q}_b, \hat{k}_a \rangle = d - 2 \operatorname{dist}_H(a, b) + (2t - d) = 2(t - \operatorname{dist}_H(a, b)). \quad (\text{B.19})$$

Let $L = \lceil \log_2(16m) \rceil$ and choose

$$H = \max\{1, \lceil L/(2\zeta t) \rceil\}, \quad r = 2dH^2, \quad n = (2dH)^2.$$

The gap constant is a fixed rational number, so this choice uses only rational arithmetic and an integer ceiling. Repeat every coordinate of $\hat{k}_a$ and $\hat{q}_b$ exactly r times. Since $2dr = n$ and $r/\sqrt{n} = H$, the standard-scaled score is

$$\frac{\langle q_b, k_a \rangle}{\sqrt{n}} = 2H(t - \operatorname{dist}_H(a, b)).$$

Append a zero dummy key with value -1 and give all real keys value $+1$.

In a yes bucket-query pair some real key has score at least zero, so its total real-key weight W is at least one. In a no instance, every real score is at most $-2H\zeta t \leq -L$, and every bucket has $W \leq me^{-L} \leq 1/16$. The dummy-key argument gives exactly the same constant token gap as Lemma B.38.

Here $t \geq d/4 \geq (c_0/4) \log_2 N$, whereas $L \leq \gamma \log_2 N + O(1)$. Therefore H is bounded by a constant depending only on γ, ζ, c_0 . Consequently

$$n = O(d^2) = O(\log^2 N) = O(\log^2 T).$$

All input coordinates are 0 or signs; the scaling denominator $\sqrt{n} = 2dH$ is an integer.

Use the independent prefill copies, sample threshold, union bound, and runtime truncation from Theorem B.36. A yes instance has at least one yes bucket-query pair. A no instance has only no pairs. Under the promise, these are the only cases needed; intermediate distances in a yes instance can produce arbitrary additional yes classifications without affecting the decision. The expected time is

$$\tilde{O}\left(N^{1+\gamma(a-1)} + N^{2-\gamma\epsilon}\right),$$

which after truncation is $O(N^{2-\gamma\epsilon/2})$. This contradicts Lemma B.41. $\square$

This matches the exponent of the straightforward scan. At width $n = O(\log^2 T)$ and input bit lengths polylogarithmic in T , Lemma B.16 samples (B.18) exactly in T polylog T expected bit operations. Thus the lower and upper bounds agree in the power of T , under the stated hypothesis.

Corollary B.43 (Quadratic-logarithmic hard blocks). *The row-norm-one block realization and the well-conditioned pre-norm additive realization in Corollaries B.39 and B.40 also hold for the $n = O(\log^2 T)$ lower bound of Theorem B.42.*

Proof idea. Opposite sign pairs encode a coordinate at its intended position and give zero at the other role. Fixed padding and the marker coordinate complete the Hamming-gap gadget using only rows of absolute sum one.

Proof. For $d \geq 2$, reserve $4d + 2$ source coordinates, which fit in $n = (2dH)^2$. For each $i \leq d$, use two key-source coordinates and two query-source coordinates. At a cache position a , the key pair is $(2a_i - 1, 1 - 2a_i)$ and the query pair is $(1, 1)$. At the current position b , the key pair is $(1, 1)$ and the query pair is $(2b_i - 1, 1 - 2b_i)$. Half the difference of each pair gives the required bipolar key or query coordinate; it is zero on the other role. Reserve a constant coordinate $c = 1$ and a marker v that is $+1$ on cache positions and -1 on the query position.

The i th fixed padding coordinate of a key is $(c_t)_i(c+v)/2$, and the corresponding query padding coordinate is $(c - v)/2$. Thus cache keys have padding c_t , the query has padding 1 , and its own key is zero. Replicate every one of these $2d$ output rows r times. Each row has absolute sum one. The marker supplies the scalar value. All raw source coordinates are signs, including the unused coordinates, which are set to one.

The remainder of Corollary B.39, or respectively Corollary B.40, now applies verbatim. The asymptotic hardness instances have $d \geq 2$; finitely many smaller instances have no effect on the contradiction. $\square$

C Normalization and additive residuals

This appendix proves the results of Section 8. Sections C.1 and C.3 prove the two interfaces of Theorem 8.1. Section C.2 shows that a denominator floor is needed, and Section C.4 proves the scalar law (8.1). Section C.5 proves Theorem 8.2 (as Theorem C.15) and the relative-floor bound of Theorem C.16. Sections C.6 and C.7 prove the tanh case of (8.5), Sections C.8 and C.9 prove Theorem 8.3, and Section C.10 treats GELU and SwiGLU.

C.1 Normalization with a denominator floor

We use the explicit convention

$$\mathcal{N}_\epsilon(a)_i = \frac{a_i}{\sqrt{\epsilon + n^{-1} \sum_{j=1}^n a_j^2}}, \quad |a_j| \leq R. \quad (\text{C.1})$$

The original RMSNorm definition has $\epsilon = 0$ in this formula [46, Eq. (4)]. The authors' released implementation instead adds its numerical stabilizer outside the square root [45]. Llama's reference implementation uses the inside-root convention in (C.1) [30]. These are different functions. Every theorem below refers to (C.1). A gain of absolute value at most G simply multiplies the output radius by G .

Theorem C.1 (A normalization factory with a denominator floor). *Suppose that $R > 0$, $\epsilon \geq 0$, and $\lambda > 0$ are known rational numbers, and that*

$$v = \epsilon + \frac{1}{n} \sum_{j=1}^n a_j^2 \geq \lambda.$$

Replace λ by $\max\{\lambda, \epsilon\}$, and put

$$\kappa = \max\{1, R^2/\lambda\}, \quad R_{\text{out}} = \frac{2R}{\sqrt{\lambda}}.$$

Assume that a request at coordinate j returns an independent sign of mean a_j/R . There is an exact sign sampler of mean $\mathcal{N}_\epsilon(a)_i/R_{\text{out}}$ whose expected number of such requests is at most

$$1 + 228\kappa^2. \quad (\text{C.2})$$

If B_N includes the binary lengths of the rational parameters and $\lceil \log_2(n+2) \rceil$, its expected auxiliary bit work is $O(\kappa^2 B_N^4)$. The constant in this bound is absolute. No numerical evaluation of the vector a is required.

Proof idea. Two independent signs give a disagreement coin whose bias records the average squared coordinate. A linear factory moves this bias away from its endpoint, and a negative-binomial generating function supplies the reciprocal square root. Multiplying by a fresh numerator sign completes the normalized coordinate.

Proof. An independent product of two signs with mean a_j/R has mean $(a_j/R)^2$. It does not give a coin of that probability. The latter function vanishes when the source bias is zero, an interior point of the sign-coin parameter interval. A finite Bernoulli factory for that square coin would therefore be impossible. We instead use the disagreement coin

$$q = \frac{1}{2} \left(1 - \frac{1}{n} \sum_j (a_j/R)^2 \right).$$

It costs two source requests at one uniform coordinate. This exact coin is available even when all coordinates vanish.

Set

$$U = \epsilon + R^2, \quad d = \frac{\lambda}{2U}, \quad C = \frac{2R^2}{U - \lambda/2}, \quad \zeta = \frac{\lambda}{2U - \lambda}. \quad (\text{C.3})$$

The promise implies $0 < \lambda \leq U$. In particular $0 < d \leq 1/2$, $C \leq 4$, and

$$r = Cq \leq 1 - \zeta, \quad \frac{C}{\zeta} = \frac{4R^2}{\lambda}. \quad (\text{C.4})$$

If $C \leq 1$, obtain an r -coin by thinning the q -coin. If $C > 1$, use the linear Bernoulli factory with supplied slack $\min\{\zeta, 1/2\}$.

We use the following published input-count result: for $Cp \leq 1 - e$, Huber's linear factory is exact and uses at most $9.5C/e$ source coins in expectation [22, Theorem 1]. Its finite-bit implementation in the parameter range used here is recorded below. Thus an r -coin costs at most 76κ q -coins in expectation. This bound also covers thinning.

Let

$$c_k = \binom{2k}{k} 4^{-k}, \quad \Pr(K = k) = \sqrt{d} c_k (1 - d)^k, \quad k \geq 0.$$

Independently draw a known coin of probability $1/\sqrt{2}$. If it fails, return $A = 0$. Otherwise return $A = 1$ exactly when K independent r -coins are all one. For $K = 0$, the empty conjunction is one. The binomial generating function gives

$$\Pr(A = 1) = \frac{1}{\sqrt{2}} \sqrt{\frac{d}{1 - (1 - d)r}} = \frac{\sqrt{\lambda}}{2\sqrt{v}}, \quad (\text{C.5})$$

$$\mathbb{E}K = \frac{1 - d}{2d} = \frac{U}{\lambda} - \frac{1}{2} \leq \frac{3}{2}\kappa. \quad (\text{C.6})$$

The algebra in the first line uses $(1 - d)C = 2R^2/U$ and $1 - (1 - d)Cq = v/U$.

On $A = 1$, return a fresh sign of mean a_i/R . On $A = 0$, return a fair sign. Its mean is

$$\frac{a_i}{R} \frac{\sqrt{\lambda}}{2\sqrt{v}} = \frac{\mathcal{N}_\epsilon(a)_i}{R_{\text{out}}}.$$

The fair branch completes the output law to a sign on every run.

The number of r -coins is at most K . Each q -coin uses two child signs. The uniform conditional input bound for the amplifier therefore gives

$$1 + 2(76\kappa)(3\kappa/2) = 1 + 228\kappa^2.$$

This argument charges a fresh factory invocation before its own randomness is exposed. It does not require the cost of that invocation to be independent of its output. All component procedures terminate almost surely, and K is finite almost surely. This proves exactness, termination, and (C.2).

The arithmetic bound is proved below. □

For recursive composition, the possibly irrational radius R_{out} can be replaced by a rational one without changing the child bound. Choose a power of two $\bar{R}$ with $R_{\text{out}} \leq \bar{R} < 2R_{\text{out}}$. Its exponent is found by comparisons of the rational quantities $\bar{R}^2 \lambda$ and $4R^2$. With probability $R_{\text{out}}/\bar{R}$, invoke the normalization sampler; otherwise return a fair sign. The resulting mean is $\mathcal{N}_\epsilon(a)_i/\bar{R}$. This known algebraic coin has a certified square-root implementation using $O(B_N^4)$ expected bit work and introduces no child requests. All later attention and row samplers may therefore use rational radius bounds.

C.1.1 Finite-bit implementation

Lemma C.2 (Half-order negative-binomial sampling). *For rational $0 < d \leq 1/2$, the law $\Pr(K = k) = \sqrt{d} c_k (1 - d)^k$ can be sampled exactly with $O(d^{-1}(B + \log_2(d^{-1} + 2))^4)$ expected bit work, where B bounds the input length of d .*

Proof idea. A geometric proposal acquires the required central-binomial factor through a product of rational acceptance coins. Most unsuccessful products stop early. Their square-root mean length compensates for the square-root acceptance probability.

Proof. Propose $G \geq 0$ with law $d(1 - d)^k$. Accept it with probability

$$c_G = \prod_{j=1}^G \frac{2j-1}{2j},$$

stopping this product of rational coins at its first failure. The mean acceptance probability is $\sqrt{d}$. Thus the accepted law is the claimed law. The elementary bound $c_j \leq 1/\sqrt{j+1}$ follows by induction from $c_{j+1}/c_j = (2j+1)/(2j+2)$. Conditional on $G = k$, the expected number of product tests is at most

$$\sum_{j=0}^{k-1} c_j \leq 2\sqrt{k}.$$

An unconditional trial uses at most $2/\sqrt{d}$ tests in expectation. Dividing by its success probability bounds the total expected number of tests by $2/d$.

Geometric inversion generates G from

$$G = \left\lfloor \frac{-\log V}{-\log(1-d)} \right\rfloor, \quad V \sim \text{Unif}(0, 1).$$

The proof of Lemma A.23 performs this inversion with certified logarithms. There the rate is $-\log(1 - p_j)$; here it is $-\log(1 - d) \in [d, \log 2]$. The leading-zero count of V has a geometric tail, and the bound $6e$ there on the chance of lying within e of a positive integer needs only $e \leq 1/2$ and a rate times e at most one. Another $\lceil \log_2(d^{-1} + 2) \rceil$ working bits absorb the division by the rate. Consequently extra refinement has a geometric tail, and a certified m -bit logarithm costs a cubic number of bit operations in its input and output lengths. Rational product coins use the usual uniform-bit comparison.

Both the proposal law and the accepted law have geometric tails on scale d^{-1} . Their logarithmic moments, together with the same bound for the stopped product tests, show that the expected cost per test is absorbed by the fourth power in the stated bound. This reasoning counts uniform bits, counters, rational arithmetic, and precision refinement. $\square$

Lemma C.3 (Finite-bit linear factories with arbitrary known multiplier). *Let $C > 1$ and $0 < e \leq 1/2$ be known rationals, and suppose $Cp \leq 1 - e$. Huber's linear factory has an implementation with $O(Ce^{-1}B^4)$ expected auxiliary bit work, where $B \geq 16$ includes the parameter lengths, source-address length, and $\log_2(1/(C - 1) + 2)$. Source invocations are counted separately.*

Proof idea. Huber’s counter has a harmonic function equal to the desired output bias. Its global source bound controls the early stages, while the rapidly decreasing probability of reaching a late stage pays for that stage’s larger counters and finer arithmetic.

Proof. Use Huber’s stage algorithm [22, Theorem 1]. At stage j , let $e_j = 2^{-j}e$, $C_j = C \prod_{r < j} (1 + e_r/2) < 2C$, and $k_j = (23/5)/e_j$. Initially the integer counter is $i = 1$. While $0 < i < k_j$, draw a source bit X and an independent positive geometric variable G of parameter $(C_j - 1)/C_j$, and replace i by $i - 1 + (1 - X)G$. At $i = 0$, return one. At an upper exit, continue with probability $(1 + e_j/2)^{-i}$; on continuation replace C_j by $C_j(1 + e_j/2)$ and e_j by $e_j/2$. If continuation fails, return zero. The source bound for this algorithm is at most $9.5C/e$.

Its exactness also follows from the counter identity. At a within-stage step the conditional expectation of $(C_j p)^i$ is unchanged, because $\mathbb{E}(C_j p)^G = (C_j - 1)p/(1 - p)$. At an upper exit the continuation probability times $(C_{j+1} p)^i$ equals $(C_j p)^i$. The slack promise is preserved. The finite stopping bounds below imply almost-sure termination, at which the harmonic function is the returned bit; bounded convergence therefore gives bias Cp . At an upper exit the probability of continuing is at most $\exp(-46/25) < 1/6$, so the probability of reaching stage j is at most 6^{-j} . The stopped-drift calculation, including the geometric overshoot, bounds the expected update count from any entry state by

$$\frac{(C_j - 1)[k_j] + C_j}{e_j} \leq 14C e_j^{-2}.$$

If the entry counter already exceeds the threshold, no updates are needed before the exit test. A positive geometric variable is sampled by rational-parameter inversion, so its numerical magnitude is not charged as a unary loop. All arithmetic and certified logarithm comparisons use dyadic fixed-point intervals with outward rounding and guarded summation. The required numerical estimates hold uniformly over the stages. A geometric draw has parameter at least $(C - 1)/(2C)$, so its logarithm has moments $O((B + j)^r)$ for each fixed r . At an upper exit, memorylessness bounds the overshoot by a fresh geometric excess. Conditioning on continuation multiplies its mass by a decreasing exponential, which improves this tail. This remains true at subsequent stage entries.

At requested precision m , the product defining C_j is recomputed with $O(B + j + m)$ working bits and j guarded multiplications. Subtracting one needs only $O(B)$ additional guard bits, since $C_j - 1 \geq C - 1$. Geometric inversion uses the logarithm procedure in Lemma C.2. For a continuation coin, if $i e_j > 3(m + 1)$ its probability is below 2^{-m} ; otherwise its exponent has $O(B + j + m)$ relevant bits. Binary range reduction and certified logarithm and exponential series cost a cubic number of operations in those lengths. Fresh uniform comparison bits give a geometric refinement tail. These bounds give stage work $O(C e_j^{-2} (B + j)^4)$. To sharpen the total, let N_j count updates at stage j . Each update requests exactly one source coin, so Huber’s global source bound gives $\sum_j \mathbb{E} N_j \leq 9.5C/e$. At the start of an update its counter is below k_j , hence has length $O(B + j)$. Its fresh geometric draw and subsequent comparison have conditional expected auxiliary work $O((B + j)^4)$ uniformly over the past. Predictable charging bounds the update work by $O(\sum_j (B + j)^4 \mathbb{E} N_j)$.

Choose $J = \lceil \log(1/e)/\log(3/2) \rceil = O(B)$. The stages $j \leq J$ have total work $O(C e^{-1} B^4)$ by the global source bound. For the remaining stages, the reach and stopped-drift estimates give

$$\sum_{j > J} (B + j)^4 \mathbb{E} N_j \leq 14C e^{-2} \sum_{j > J} (2/3)^j (B + j)^4 = O(C e^{-1} B^4).$$

Exit tests have one invocation per reached stage. The conditional overshoot log-moment estimate and reach probability 6^{-j} give a further $O(B^4)$ term, which is absorbed by the claimed bound. The construction and its exactness are those of Huber’s factory; the calculation supplies the bit bound at arbitrary known multiplier. $\square$

For the floor parameters, $C/\zeta = 4R^2/\lambda \leq 4\kappa$. With supplied slack $e = \min\{\zeta, 1/2\}$ and $C \leq 4$, one has $C/e = O(\kappa)$. Lemma C.3 therefore costs $O(\kappa B_N^4)$ auxiliary work per amplified coin. There are at most $3\kappa/2$ such coins in expectation. Predictable charging bounds their local work by $O(\kappa^2 B_N^4)$; source invocations remain counted separately. Lemma C.2 costs $O(\kappa B_N^4)$ for the negative-binomial draw. Known gates, coordinate choices, and the output sign contribute $O(B_N^4)$. This proves the bit bound in Theorem C.1.

C.2 A sharp normalization obstruction

Theorem 8.1 assumes a certified denominator floor. This subsection shows that the floor matters: after uniform attention, an unstabilized normalization can force a linear number of context probes, and a stabilizer ϵ caps the cost at order ϵ^{-1} . Corollary C.6 keeps every weight on one fixed grid.

Theorem C.4 (A sharp normalization family). *For every $T \geq 2$ and rational $0 \leq \epsilon \leq 1$, there is a width-two causal block consisting of uniform attention, RMSNorm, and a tanh map, each replacing its input stream, with row norms at most one and two logits in $[-1, 1]$, on the two-token embedding alphabet $\{(-1, 1), (1, 1)\}$, such that its exact-sampling query optimum satisfies*

$$\frac{1}{384} \min\{T, \epsilon^{-1}\} \leq Q^* \leq 64 \min\{T, \epsilon^{-1}\}. \quad (\text{C.7})$$

Here $\epsilon^{-1} = +\infty$ at zero. The lower bound applies to all exact samplers, with arbitrary weight-only preprocessing. The upper bound has an implementation using $O(B_N^4 \min\{T, \epsilon^{-1}\})$ expected bit operations. All nonzero network coefficients are dyadic with $O(\log T)$ bits. The denominator is nonzero on every context, including when $\epsilon = 0$.

The score matrices in this construction are zero. Consequently the lower bound holds at every inverse temperature, including $\beta = 0$.

The matching sampler uses the following identity.

Lemma C.5 (A geometric majority identity). *Let $\theta > 0$. Draw $K \geq 0$ with*

$$\Pr(K = k) = \frac{\theta}{1 + \theta} (1 + \theta)^{-k}.$$

The majority of $2K + 1$ independent signs of mean z has mean

$$G_\theta(z) = \frac{z\sqrt{1+\theta}}{\sqrt{\theta+z^2}}. \quad (\text{C.8})$$

Its expected number of source signs is $1 + 2/\theta$.

Proof idea. Differentiate the majority polynomials before summing the geometric mixture. Their derivatives form a binomial series with exponent $-3/2$, which is also the derivative of the required normalized mean.

Proof. Write $M_k(z)$ for the mean of majority of $2k + 1$ signs. The proof of Lemma A.20 gives $M'_k(z) = C_k(1 - z^2)^k$ with $C_k = (2k + 1)\binom{2k}{k}4^{-k}$, and $C_k = (3/2)_k/k!$. The geometric mixture can be differentiated term by term uniformly on $[-1, 1]$. The binomial series gives its derivative as

$$\frac{\theta}{1 + \theta} \left(1 - \frac{1 - z^2}{1 + \theta}\right)^{-3/2} = \frac{\theta\sqrt{1+\theta}}{(\theta + z^2)^{3/2}} = G'_\theta(z).$$

Both functions vanish at zero, proving the identity. The arity expectation follows from $\mathbb{E}K = 1/\theta$. $\square$

Proof idea for Theorem C.4. Uniform attention writes the context mean M next to a small dyadic sentinel, and normalization makes the output a steep function of M near zero. The transcript score turns this steepness into order $\min\{T, \epsilon^{-1}\}$ probes. A geometric mixture of majorities realizes the same function at that cost, and a full scan is cheaper when $\epsilon < 1/T$.

Proof of Theorem C.4. Take width two and contexts

$$h_{0,j} = (x_j, 1), \quad x_j \in \{-1, +1\}.$$

Let $c = 2^{-\lceil \log_2 T \rceil}$, so $0 < c \leq 1/T$. The query and key maps are zero, and the value map is $\text{diag}(1, c)$. At the final position, uniform causal attention gives

$$a = (M, c), \quad M = T^{-1} \sum_{j=1}^T x_j.$$

Apply (C.1), then a $\tanh$ row with coefficient $1/2$ on the first coordinate. Give the two output categories opposite logits. If the category is encoded by $Y \in \{-1, +1\}$, its target mean is

$$f_\epsilon(M) = \tanh\left(\tanh\left(\frac{M}{\sqrt{2\sqrt{2\epsilon + c^2 + M^2}}}\right)\right). \quad (\text{C.9})$$

Every parameter except ϵ is dyadic. The value and $\tanh$ row norms are at most one, and the logits lie in $[-1, 1]$. The sentinel c prevents a zero denominator.

The lower bound. Lemma A.17 gives $H'(0)^2 \leq \mathbb{E}_0 Q$ by Cauchy–Schwarz. This is the finite-input form of the sequential information bound; the first-score Bernoulli-factory lower bound is standard [28].

For $0 \leq u \leq 1$, $\tanh u \geq u/2$. The argument of the inner $\tanh$ in (C.9) has absolute value at most $1/\sqrt{2}$. Oddness consequently gives

$$M f_\epsilon(M) \geq \frac{M^2}{8\sqrt{2\epsilon + c^2 + M^2}}.$$

For independent fair signs,

$$\mathbb{E} M^2 = \frac{1}{T}, \quad \mathbb{E} M^4 = \frac{3T - 2}{T^3} \leq \frac{3}{T^2}.$$

For any nonnegative Z with positive mean and any $a > 0$, weighted Jensen gives

$$\mathbb{E} \frac{Z}{\sqrt{a + Z}} \geq \frac{\mathbb{E} Z}{\sqrt{a + \mathbb{E} Z^2 / \mathbb{E} Z}}. \quad (\text{C.10})$$

To see this, tilt the probability law by $Z/\mathbb{E} Z$ and apply convexity of $u \mapsto (a + u)^{-1/2}$. Taking $Z = M^2$ yields

$$H'(0) = T \mathbb{E}[M f_\epsilon(M)] \geq \frac{1}{8\sqrt{2\epsilon + c^2 + 3/T}}.$$

Lemma A.17 now implies

$$\mathbb{E}_0 Q \geq \frac{1}{64(2\epsilon + c^2 + 3/T)} \geq \frac{1}{384} \min\{T, \epsilon^{-1}\}. \quad (\text{C.11})$$

The last inequality uses $c^2 \leq 1/T^2 \leq 1/T$ and considers $\epsilon \leq 1/T$ and $\epsilon \geq 1/T$ separately. An average-case lower bound is also a worst-context lower bound. This proves the lower part of Theorem C.4.

A matching sampler. For (C.9), take $\theta = 2\epsilon + c^2$. A known gate of probability $[2(1 + \theta)]^{-1/2}$ turns the majority sign into a sign of mean $M/[\sqrt{2}\sqrt{\theta + M^2}]$, using a fair sign on the other branch. A source sign of mean M is obtained by querying one uniform context position.

Lemma A.26 supplies a sign of mean $\tanh z$ from signs of mean $z \in [-1, 1]$. Its expected source count is at most two, which is in particular at most $2e$, and its local bit work is $O(B_N^4)$.

Applying this $\tanh$ factory twice realizes (C.9). Its expected number of context requests is at most

$$4e^2(1 + 2/\theta) \leq 32(1 + \epsilon^{-1}) \leq 64\epsilon^{-1}, \quad 0 < \epsilon \leq 1.$$

Alternatively, read all T inputs, form the exact rational M , and use certified square roots and exponentials to sample the two-category probability. Choose this alternative when $\epsilon < 1/T$. The first construction is used when $\epsilon \geq 1/T$. Every geometric parameter then has inverse at most $1 + T$, and its arity and arithmetic tails give $O(B_N^4/\epsilon)$ expected bit work. In the scan case, the sentinel is at least $1/(2T)$, so the additional precision needed by the square root is $O(\log T)$ and the total work is $O(TB_N^4)$. This proves both upper bounds in Theorem C.4. $\square$

C.2.1 A fixed coefficient grid

The width-two construction uses a weight with $O(\log T)$ bits. That precision is permitted by the finite-encoding model above, but a fixed grid such as $b = 20\lceil \log_2 n \rceil$ would not allow this weight at fixed width and unbounded T . The next construction generates its sentinel through averaging. Its weights are independent of the context length.

Corollary C.6 (A fixed-grid normalization law). *There is a width-four block consisting of uniform attention, RMSNorm, and tanh, each replacing its input stream, with row norms at most one and nonzero matrix coefficients belong to $\{1/2, 1\}$, such that*

$$\frac{1}{128} \min\{T, \epsilon^{-1}\} \leq Q^* \leq 32 \min\{T, \epsilon^{-1}\}, \quad 0 \leq \epsilon \leq 1. \quad (\text{C.12})$$

The upper bound has $O(B_N^4 \min\{T, \epsilon^{-1}\})$ expected bit work. The input has one unknown sign per position and a public first-position feature. Every normalization denominator is positive at every causal position on every allowed context. In particular the unstabilized linear lower bound holds with one fixed coefficient grid as T tends to infinity.

Proof idea. A public first-position feature produces the sentinel $1/t$ through uniform averaging, so no context-dependent matrix coefficient is needed. The score lower bound and geometric-majority sampler then apply to the resulting scalar normalization, with a full scan available at small stabilization.

Proof. Use

$$h_{0,j} = (x_j, 1, \eta_j, 1), \quad \eta_1 = 1, \quad \eta_j = -1 \ (j > 1).$$

The first-position feature η_j is public and requires no probe of x_j . Both query and key maps vanish. The first value row selects x_j , the second takes the average of coordinates two and three, and the other rows vanish. Consequently, at every causal position t ,

$$y_t = (M_t, 1/t, 0, 0), \quad M_t = t^{-1} \sum_{j \leq t} x_j.$$

The sentinel is strictly positive for every t . The tanh output row is one half of the first normalized coordinate. Opposite output logits give target sign mean

$$\tilde{f}_\epsilon(M) = \tanh\left(\tanh\left(\frac{M}{\sqrt{4\epsilon + T^{-2} + M^2}}\right)\right).$$

All matrix rows have norm at most one and all biases are zero. The inner argument belongs to $[-1, 1]$. The same two tanh bounds give

$$M \tilde{f}_\epsilon(M) \geq \frac{M^2}{4\sqrt{4\epsilon + T^{-2} + M^2}}.$$

Weighted Jensen and the finite-input score lemma imply

$$\mathbb{E}_0 Q \geq \frac{1}{16(4\epsilon + T^{-2} + 3/T)} \geq \frac{1}{128} \min\{T, \epsilon^{-1}\}.$$

For the last step, use $T^{-2} \leq 1/T$ and $\epsilon + 1/T \leq 2 \max\{\epsilon, 1/T\}$.

For the upper bound, put $\theta = 4\epsilon + T^{-2}$ in Lemma C.5. A known gate of probability $(1 + \theta)^{-1/2}$ gives the inner argument. Applying the unit-interval tanh factory from Lemma A.26 twice costs fewer than sixteen requests of that source sign. Thus for $0 < \epsilon \leq 1$ the expected number of context requests is at most

$$16(1 + 2/\theta) \leq 16(1 + (2\epsilon)^{-1}) \leq 24\epsilon^{-1}.$$

When $\epsilon < 1/T$, scan the context instead. This gives the constant 32 in (C.12) with slack. The bit implementation and its precision bound are identical to those in Theorem C.4; now the rational sentinel $1/T$ is computed from the position index, without being a network coefficient. $\square$

C.3 A Gaussian construction for regularized normalization

A positive stabilizer permits a factory with output radius $4\sqrt{n}$ and source bound $1 + nR^2/\epsilon$. The construction uses the Gaussian sign identity, an error-function mixture of majority rules, and a sine series. Its denominator parameter is the stabilizer itself.

Write

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt, \quad A_k = \frac{(2k+1)\binom{2k}{k}}{4^k}.$$

For a sign of mean z , let $M_k(z)$ be the mean of the majority of $2k+1$ independent copies. The derivative identity $M'_k(z) = A_k(1-z^2)^k$ from the proof of Lemma A.20 and $M_k(0) = 0$ imply

$$\operatorname{erf}(cz) = \sum_{k \geq 0} \frac{2c}{\sqrt{\pi}} e^{-c^2} \frac{c^{2k}}{k! A_k} M_k(z), \quad c \geq 0, \quad |z| \leq 1. \quad (\text{C.13})$$

All coefficients are nonnegative, and their sum is $\operatorname{erf}(c)$.

Lemma C.7 (An error-function factory). *For every known computable $c \geq 0$, there is an exact factory with output mean $\operatorname{erf}(cz)$. Its expected number of input signs, using complete majorities, is exactly*

$$q(c) = 2c^2 \operatorname{erf}(c) + \frac{2c}{\sqrt{\pi}} e^{-c^2} \leq 1 + 2c^2.$$

One implementation draws $G \sim N(0, 1)$. If $G^2 > 2c^2$, it returns a fair sign. Otherwise it draws $K \sim \operatorname{Pois}(c^2 - G^2/2)$ and returns the majority of $2K+1$ fresh input signs.

Proof idea. The derivative of an odd-majority mean is a power of $1 - z^2$. Poisson coefficients turn their sum into the Gaussian density, and a truncated Gaussian draw realizes those coefficients as probabilities. The same integral computes the expected majority size.

Proof. Differentiating the right side of (C.13), and using nonnegative convergence on $[0, 1]$, gives

$$\frac{2c}{\sqrt{\pi}} e^{-c^2} \sum_{k \geq 0} \frac{[c^2(1-z^2)]^k}{k!} = \frac{2c}{\sqrt{\pi}} e^{-c^2 z^2}.$$

Both sides vanish at zero; oddness handles negative z . At $z = 1$ every majority has mean one, proving the coefficient sum.

The variable $Y = |G|/\sqrt{2}$ has density $(2/\sqrt{\pi})e^{-y^2}$ for $y \geq 0$. For each k , the probability that the algorithm enters the majority branch and selects k is

$$\frac{2}{\sqrt{\pi}} \int_0^c e^{-y^2} e^{-(c^2-y^2)} \frac{(c^2-y^2)^k}{k!} dy = \frac{2c}{\sqrt{\pi}} e^{-c^2} \frac{c^{2k}}{k!} \int_0^1 (1-u^2)^k du.$$

Integration by parts, starting from $k = 0$, gives $\int_0^1 (1-u^2)^k du = 1/A_k$. The algorithm therefore uses the mixture in (C.13) exactly. It never forms a square of an unknown mean.

The number of input signs has conditional expectation $1 + 2(c^2 - G^2/2)$ on the majority branch. Consequently its mean is

$$\frac{2}{\sqrt{\pi}} \int_0^c e^{-y^2} (1 + 2c^2 - 2y^2) dy = 2c^2 \operatorname{erf}(c) + \frac{2c}{\sqrt{\pi}} e^{-c^2},$$

where the last equality uses $(ye^{-y^2})' = e^{-y^2} - 2y^2 e^{-y^2}$. The bound $1 + 2c^2$ also follows directly from the conditional expectation. A sequential majority may stop earlier. $\square$

Lemma C.8 (A sine factory). *Given independent signs of unknown mean $z \in [-1, 1]$, an exact factory of mean $\frac{1}{4} \sin(\pi z/2)$ uses*

$$C_{\sin} = \frac{\pi}{8} \cosh(\pi/2) < 1$$

input signs in expectation.

Proof idea. The sine series has a summable total absolute coefficient mass below one at the chosen scale. Products of fresh signs realize its monomials; a fair-sign remainder supplies the unused probability mass.

Proof. Put $c_0 = \pi/2$. For each $k \geq 0$, choose degree $2k + 1$ with probability $c_0^{2k+1}/[4(2k + 1)!]$. Return $(-1)^k$ times the product of that many independent input signs. On the remaining probability return a fair sign. The total specified mass is $\sinh(c_0)/4 < 1$, and the Taylor series gives the stated mean. The weighted degree sum is $c_0 \cosh(c_0)/4 = C_{\sin}$.

For an explicit rational check on the strict inequalities, use $\pi < 22/7$ and

$$\cosh(11/7) \leq 1 + \frac{(11/7)^2}{2!} + \frac{(11/7)^4}{4!} + \frac{(11/7)^6}{6!} + \frac{(11/7)^8}{8![1 - (11/7)^2/90]} < \frac{101}{40}.$$

Thus $C_{\sin} < 2222/2240 < 1$, and also $\sinh(c_0)/4 < \cosh(c_0)/4 < 101/160 < 1$. The series can be sampled by an odd Poisson variable with parameter c_0 , preceded by the known gate $\sinh(c_0)/4$. All parameters are computable constants, and the degree has an exponential tail. $\square$

Theorem C.9 (An intrinsic-radius RMS factory). *Let $n \geq 1$, $R > 0$, and $\epsilon > 0$. Suppose fresh independent signs of means a_j/R are available, where $|a_j| \leq R$. For every coordinate i , an exact factory returns a sign of mean*

$$\frac{a_i}{4\sqrt{n}\sqrt{\epsilon + n^{-1}\sum_j a_j^2}}.$$

Its expected number of source-sign requests is at most

$$C_{\sin} \left(1 + \frac{R^2}{\epsilon} \left[1 + \frac{2(n-1)}{\pi} \right] \right) \leq 1 + \frac{nR^2}{\epsilon}. \quad (\text{C.14})$$

All Gaussian and Poisson variables can be represented by successively refined rational intervals. For finite rational R, ϵ , and $B \geq 16$ containing their encoding lengths and $\log(n+2)$, the auxiliary expected bit work is

$$O(n(1 + R^2/\epsilon)B^4).$$

This counts Gaussian draws, random bits, arithmetic, and precision refinement; it uses no exact-real oracle.

Proof idea. A Gaussian projection converts the normalized coordinate into the correlation of two Gaussian signs. That correlation is an arcsine, which the sine factory inverts. Fresh Gaussian vectors for successive calls let the source costs multiply by conditional expectation.

Proof. Let $G_1, \dots, G_n$ be independent standard Gaussian variables. Conditional on this vector, put

$$A = R \sum_j |G_j|, \quad c = \frac{A}{\sqrt{2n\epsilon}}, \quad z = \frac{\sum_j G_j a_j}{A}.$$

The event $A = 0$ has probability zero. A source sign of mean z is obtained by selecting j proportionally to $|G_j|$, requesting a fresh sign of mean a_j/R , and multiplying it by $\text{sgn } G_j$. Use Lemma C.7 and multiply its output by $\text{sgn } G_i$. This produces a sign whose conditional mean is

$$\text{sgn } G_i \operatorname{erf} \left(\frac{\sum_j G_j a_j}{\sqrt{2n\epsilon}} \right).$$

Introduce an independent $G_0 \sim N(0, 1)$ only for the analysis. Integrating it out gives the error function above as the conditional mean of $\text{sgn}(\sum_j G_j a_j + \sqrt{n\epsilon}G_0)$. The two centered Gaussian variables

$$G_i, \quad H = \sum_j G_j a_j + \sqrt{n\epsilon}G_0$$

have correlation $u = a_i / \sqrt{\sum_j a_j^2 + n\epsilon}$. The Gaussian sign identity gives

$$\mathbb{E}[\text{sgn } G_i \text{sgn } H] = \frac{2}{\pi} \arcsin u.$$

For completeness, write a pair with correlation $u = \cos \theta$ as $(X, \cos \theta X + \sin \theta Y)$ with independent standard Gaussians X, Y . Their angle is uniform on the circle; the two signs disagree on sectors of total angle 2θ . Their product therefore has mean $1 - 2\theta/\pi = (2/\pi) \arcsin u$.

Request independent copies of this sign as inputs to Lemma C.8. The returned mean is

$$\frac{1}{4} \sin(\arcsin u) = \frac{u}{4},$$

which is the desired output mean. Each request draws its own Gaussian vector and all subsequent randomness afresh.

By Lemma C.7, a single intermediate sign uses at most $1 + 2c^2$ source requests. Since $\mathbb{E}G_j^2 = 1$ and $\mathbb{E}|G_j| = \sqrt{2/\pi}$,

$$2\mathbb{E}c^2 = \frac{R^2}{n\epsilon} \mathbb{E} \left(\sum_j |G_j| \right)^2 = \frac{R^2}{\epsilon} \left[1 + \frac{2(n-1)}{\pi} \right].$$

Multiplication by the mean input count of the independent sine factory proves (C.14). Its degree is drawn before any of its input signs, so this multiplication needs only Tonelli's theorem and conditional expectation. The finite-bit implementation is proved in the accompanying precision lemma. $\square$

The vector result provides a second option when $\epsilon > 0$. The floor factory of Theorem C.1 has radius $2R/\sqrt{\lambda}$ and cost $1 + 228 \max\{1, R^2/\lambda\}^2$. The Gaussian factory has radius $4\sqrt{n}$ and cost at most $1 + nR^2/\epsilon$. A large promise floor $\lambda \gg \epsilon$ can favor the floor factory.

Corollary C.10 (Centered normalization). *For $\epsilon > 0$, $\mu = n^{-1} \sum_j a_j$, and $|a_j| \leq R$, regularized centered normalization*

$$\frac{a_i - \mu}{\sqrt{\epsilon + n^{-1} \sum_j (a_j - \mu)^2}}$$

has an exact sign factory at output radius $4\sqrt{n}$, using at most $1 + 4nR^2/\epsilon$ original source signs in expectation.

Proof idea. Centering is a signed mixture of one coordinate and the uniform coordinate average. It enlarges the coordinate radius by a factor of two, after which the Gaussian normalization theorem applies directly.

Proof. A sign of mean $(a_i - \mu)/(2R)$ is a fair choice between an a_i/R sign and the negative of a sign from a uniformly chosen coordinate. It costs one original request. Apply Theorem C.9 to the centered vector with coordinate bound $2R$. $\square$

Thinning trades output radius for source cost. The next corollary opens a known gate before any Gaussian work and reaches radius $2R/\sqrt{\epsilon}$, which is the radius of the floor factory at $\lambda = \epsilon$.

Corollary C.11 (Thinning the vector Gaussian interface). *Under Theorem C.9, put $\kappa = R^2/\epsilon$ and suppose $\kappa \geq 4n$. There is an exact factory returning the normalized coordinate at radius $2\sqrt{\kappa}$ with at most*

$$2\sqrt{n/\kappa} (1 + n\kappa)$$

expected source requests and

$$O\left(B^4 [1 + n^{3/2} \sqrt{\kappa}]\right)$$

expected auxiliary bit work.

Proof idea. Open the radius-adjustment gate before drawing any Gaussians. Both the source count and the expensive local work are then charged only on the gate's open branch.

Proof. Before doing any Gaussian work, open a known gate with probability $p = 2\sqrt{n/\kappa} \leq 1$. On its open branch run the Gaussian factory at radius $4\sqrt{n}$, and otherwise return a fair sign. The output radius becomes $(4\sqrt{n})/p = 2\sqrt{\kappa}$. The source bound follows by multiplication. Lemma C.12 gives conditional auxiliary cost $O(n(1 + \kappa)B^4)$. Multiplying by p , and using $\kappa \geq 4n \geq 4$, gives the stated bound after adding the $O(B^4)$ initial gate. As before, every gate is sampled by certified interval comparisons. $\square$

C.3.1 Finite-bit implementation

Lemma C.12 (Linear stabilizer dependence in bit work). *Let $n \geq 1$ and let $R, \epsilon > 0$ be rational. Put $\kappa = R^2/\epsilon$, and let $B \geq 16$ include their binary encoding lengths and $\lceil \log_2(n+2) \rceil$. The factory in Theorem C.9, including all Gaussian draws, weighted coordinate selections, Poisson variables, random bits, counters, and precision refinement, has expected auxiliary bit work*

$$O(n(1 + \kappa)B^4). \quad (\text{C.15})$$

Source-sign invocations are charged separately. The implicit constant is absolute. The output radius $4\sqrt{n}$ may be replaced by the integer radius $4\lceil \sqrt{n} \rceil$ without increasing the expected source count or changing this bit bound.

Proof idea. Two decompositions control precision costs. Split the Poisson mean into a rational bulk and a bounded residual, and select Gaussian coordinates from an integer envelope. Only a selected coordinate needs further refinement; Gaussian fourth moments then keep the stabilizer dependence linear.

Proof. We describe rational interval algorithms and bound their refinement costs. Elementary series and their arithmetic are evaluated in dyadic fixed-point intervals with outward rounding; logarithmically many guard bits in the number of summed terms bound the accumulated rounding error. Thus denominators from separate series terms are never multiplied together. Each comparison uses an independent uniform bit stream. Almost sure decisions suffice; equality with a random continuously distributed comparison variable has probability zero.

Gaussian intervals. Two independent uniform variables $U, V \in (0, 1)$ give independent standard Gaussians by

$$G_1 = \sqrt{-2 \log U} \cos(2\pi V), \quad G_2 = \sqrt{-2 \log U} \sin(2\pi V).$$

The change to polar coordinates proves this distribution. The first nonzero bit of U locates it in $[2^{-L}, 2^{1-L}]$. The integer L has a geometric tail. The proof of Lemma A.23 gives certified intervals for π and, after binary range reduction, for $\log U$, at a bit cost cubic in the precision. The usual sine and cosine series have factorial tail bounds on their bounded arguments. Square-root intervals can be found by rational bisection or integer square roots. Using twice the desired absolute precision before the square-root step handles arbitrarily small nonnegative arguments. These procedures produce an interval of width at most 2^{-q} for a Gaussian in $O((q + L + 1)^3)$ bit operations. They never evaluate a Gaussian as an exact real.

Generate $G_1, \dots, G_n$ and the additional Gaussian Z used by the erf factory in this way. Set $S = \sum_j |G_j|$. A dyadic interval locating S within a factor of two is obtained by refinement. Its precision has an exponential tail after $O(\log(n+2))$ initial bits: if this search has not resolved at absolute precision 2^{-q} , then $S \leq Cn2^{-q}$ for an absolute constant C , whereas

$$\Pr(S \leq u) \leq \Pr(|G_1| \leq u) \leq u \quad (u \geq 0).$$

The last bound uses the standard normal density and may be weakened to $2u$ without changing any estimate. The signs of the finitely many Gaussians are found by the same refinement; the union bound gives $\Pr(\min_j |G_j| \leq u) \leq 2nu$.

To simplify the remaining accounting, let H be one plus the largest leading-zero count in the Gaussian uniforms, the largest required logarithmic precision for these signs and the dyadic scale of S , and the logarithmic magnitude of the Gaussian variables. It can be increased by an absolute multiple of $\log(n+2)$. The preceding bounds, the normal density bound, and the Gaussian tail give, for every fixed integer $r \geq 1$,

$$\mathbb{E}H^r \leq C_r[1 + \log(n+2)]^r. \quad (\text{C.16})$$

The constants C_r are absolute. A union bound over $n+1$ variables followed by integration of an exponential tail proves this display.

The erf gate. The gate compares $|Z|$ with $\sqrt{2}c$, where $c = RS/\sqrt{2n\epsilon}$. At refinement level q , evaluate both sides in intervals of absolute width 2^{-q-3} . The arithmetic cost is $O(n(B+H+q)^3)$. Conditional on $G_1, \dots, G_n$, the variable Z is independent and its density is bounded. Therefore the probability of an unresolved gate at level q is at most $C2^{-q}$, uniformly in the threshold c . Include the gate's extra refinement depth in H . This leaves (C.16) valid with different absolute constants. The event of exact equality has probability zero.

A Poisson variable with a random computable mean. On the open gate branch, $\mu = c^2 - Z^2/2 \geq 0$. First compute a rational interval $[\ell_0, u_0]$ for μ of width at most $1/4$. Set $\ell = \max\{0, \ell_0\}$ and $r = \mu - \ell$. Then

$$0 \leq \ell \leq \mu \leq c^2, \quad 0 \leq r \leq 1/4.$$

No decision about the fractional part of μ is needed. The lower approximation ℓ is an explicitly represented rational. Generate $P_\ell \sim \text{Pois}(\ell)$ from $\lfloor \ell \rfloor$ independent Poisson variables of mean one and one with the known rational mean $\ell - \lfloor \ell \rfloor$. This part uses no further Gaussian refinement. Independently generate $P_r \sim \text{Pois}(r)$ by thinning a Poisson variable of mean one with success probability r . Each thinning test compares a fresh uniform with successive intervals for $r = \mu - \ell$. The retained count is exactly Poisson with mean r , so $P_\ell + P_r$ has the required Poisson law.

Lemma B.4 generates a Poisson variable whose mean is a rational in $[0, 1]$ of bit length b with $O(b^4)$ expected bit work and finite polynomial work moments; at mean one this work is constant. The bulk rational part therefore costs $O((1+c^2)(B+H)^4)$ conditionally on the Gaussian variables. The number of Gaussian-dependent thinning comparisons has mean one. Each comparison has a geometric refinement tail, because its uniform is fresh, and evaluating r to q bits costs $O(n(B+H+q)^3)$. This part costs $O(n(B+H)^4)$ in conditional expectation. The denominator, scale, and polynomial coefficients involved have logarithmic size $O(B+H)$; in particular B includes $\log(2+\kappa)$.

Unbounded Poisson counters do not change these estimates. For every fixed p , a Poisson variable P of mean $\lambda \geq 1$ satisfies

$$\mathbb{E}[(1+P)(1+\log(P+2))^p] \leq C_p(1+\lambda)(1+\log(\lambda+2))^p.$$

One proof divides the values into $P \leq 4\lambda$ and the intervals $2^j\lambda < P \leq 2^{j+1}\lambda$, and uses $\Pr(P \geq v) \leq (e\lambda/v)^v$ for $v \geq \lambda$. Summing the resulting convergent series proves the display. It accounts for counters, majority arities, and source-address bookkeeping. The case $\lambda \leq 1$ has an absolute bound by the same argument.

Choosing a Gaussian-weighted coordinate. The Gaussian vector is fixed throughout one erf call. Find a positive rational σ for which $\sigma \leq S \leq 2\sigma$, using the interval scale search described above. Choose a dyadic number h with

$$\frac{\sigma}{32n} \leq h \leq \frac{\sigma}{16n}.$$

For each j , compute an interval for $|G_j|$ of width at most $h/4$, and let b_j be its rational upper endpoint. Define

$$u_j = h \max\{1, \lceil b_j/h \rceil\}, \quad U = \sum_j u_j.$$

Only rational rounding is used. The inequalities

$$|G_j| \leq u_j \leq |G_j| + 2h, \quad u_j \geq h, \quad S \leq U \leq S + \sigma/8 \leq 9S/8$$

follow directly. All weights u_j/h are positive integers, and their sum is at most $66n$, hence also at most $72n$. Prepare their integer prefix table once. This costs $O(n(B+H)^3)$ bits of work.

For every required coordinate draw, propose j with probability u_j/U using the integer table and retain it with probability $|G_j|/u_j$. At acceptance, its law is exactly proportional to $|G_j|$. The success probability is $S/U \geq 8/9$, so the mean number of proposals is at most $9/8$. A rejection comparison refines only the selected Gaussian. Since $u_j \geq h$, obtaining absolute error 2^{-q} in this ratio costs $O((B+H+q)^3)$, without a factor n . Fresh comparison uniforms give a geometric refinement tail. The conditional expected work per accepted coordinate is therefore $O((B+H)^4)$, including its address and sign. The majority arity is drawn before its input signs are requested, and has conditional mean at most $1+2c^2$. Predictable charging bounds the expected total coordinate-selection work by $O((1+c^2)(B+H)^4)$ in addition to the one-time table preparation.

Averaging the random precision cost. The preceding bounds, including the initial Gaussian interval work, are controlled by $C\mathbb{E}[(n+1+c^2)(B+H)^4]$. This expectation remains linear in κ . In fact, $S^4 \leq n^3 \sum_j G_j^4$, so

$$\mathbb{E}c^4 = \frac{\kappa^2}{4n^2}\mathbb{E}S^4 \leq \frac{3}{4}n^2\kappa^2, \quad 2\mathbb{E}c^2 \leq n\kappa.$$

Consequently $\mathbb{E}(1+c^2)^2 \leq (1+n\kappa)^2$. By the triangle inequality in L^2 , $\{\mathbb{E}(n+1+c^2)^2\}^{1/2} \leq n+1+n\kappa$. Cauchy–Schwarz and (C.16), with $r=8$, give

$$\mathbb{E}[(n+1+c^2)(B+H)^4] \leq (n+1+n\kappa)\{\mathbb{E}(B+H)^8\}^{1/2} \leq Cn(1+\kappa)B^4.$$

This also accounts for correlations between large Gaussian values, large Poisson means, and difficult comparisons. The constants do not depend on R, ϵ, n , or the unknown source means.

Finally, the sine factory draws its odd degree before any Gaussian factory calls. Its degree has an exponential tail and mean less than one; all its scalar parameters are fixed computable constants. A known constant gate and a bounded-parameter Poisson generator give $O(B^4)$ auxiliary work. Conditional charging proves (C.15). Every comparison terminates almost surely, every Poisson variable is finite almost surely, and every selected majority is finite. The entire factory therefore terminates almost surely.

Let $m = \lceil \sqrt{n} \rceil$, found by integer arithmetic. Thin the output by the known algebraic probability $\sqrt{n}/m$, returning a fair sign otherwise. Squaring rational interval endpoints gives an exact implementation of this comparison with $O(B^4)$ expected work. The new mean is the required normalized coordinate at radius $4m$. $\square$

C.4 Scalar normalization at a specified output radius

We retain the output radius as a parameter. A larger radius reduces the requested sign mean and permits an initial thinning gate. The geometric-majority identity in Lemma C.5 gives the upper bound in the scalar law below.

Lemma C.13 (A second-derivative bound from uniformly finite mean work). *Let $f: [-1, 1] \rightarrow [-1, 1]$ be twice continuously differentiable. Suppose a single exact factory, valid for every source-sign mean $z \in [-1, 1]$, returns a sign of mean $f(z)$ and uses at most K source requests in expectation uniformly over z . Then*

$$K \geq \frac{1}{2} \sup_{-1 < t < 1} (1-t^2)|f''(t)|.$$

Only the first moment of the source count is assumed.

Proof idea. Run the factory on uniformly sampled entries of a finite hidden vector. The finite transcript curvature bound applies to this experiment without a second moment for the stopping time. Second derivatives of its Bernstein polynomial converge to the requested scalar derivative.

Proof. For an integer $m \geq 2$, give a query algorithm a hidden sign vector $x \in \{-1, 1\}^m$. Simulate each requested source sign by drawing an independent uniform index in $[m]$ and returning its coordinate. Repeated coordinates can be cached. Conditional on x , these source signs are independent and have mean $M(x) = m^{-1} \sum_i x_i$. Thus the output has exact conditional mean $f(M(x))$, and its number Q of distinct input probes satisfies $\mathbb{E}[Q \mid x] \leq K$.

Give the hidden coordinates independent signs of common mean $t \in (-1, 1)$ and put $H_m(t) = \mathbb{E}_t f(M)$. The finite-input transcript bound of Lemma A.17 gives

$$|H_m''(t)| \leq \frac{2\mathbb{E}_t Q}{1-t^2} \leq \frac{2K}{1-t^2}.$$

It requires no moment assumption on the stopping time beyond the stated mean-work bound.

It remains to pass to the desired scalar function. With $p = (1+t)/2$, the function H_m is the Bernstein polynomial

$$H_m(t) = \sum_{j=0}^m \binom{m}{j} p^j (1-p)^{m-j} f(2j/m - 1).$$

Let J have distribution $\text{Bin}(m-2, p)$, and let U, V be independent uniforms on $[0, 1]$, independent of J . Twice differentiating the polynomial, then expressing its second finite difference by two integrals, gives

$$H_m''(t) = \left(1 - \frac{1}{m}\right) \mathbb{E} f''\left(\frac{2J}{m} - 1 + \frac{2(U+V)}{m}\right).$$

The argument lies in $[-1, 1]$ and converges to t in probability, since its variance is $O(1/m)$ and its mean differs from t by $-2t/m$. Uniform continuity and boundedness of f'' imply $H_m''(t) \rightarrow f''(t)$. Pass to the limit in the preceding inequality and then take the supremum over t . $\square$

Theorem C.14 (Scalar RMS cost at a specified amplitude). *Let $\kappa > 0$ and $A \geq 1$. Consider factories valid for every unknown sign mean $z \in [-1, 1]$ whose output mean is*

$$f_{\kappa, A}(z) = \frac{z}{A\sqrt{\kappa^{-1} + z^2}}.$$

Let $Q^(\kappa, A)$ be the infimum, over all such exact factories, of their worst-case expected number of source requests. Then*

$$\frac{\sqrt{\kappa} + \kappa}{8A} \leq Q^*(\kappa, A) \leq \frac{1 + 2\kappa}{A\sqrt{1 + \kappa^{-1}}} \leq \frac{3(\sqrt{\kappa} + \kappa)}{A}. \quad (\text{C.17})$$

For rational κ, A , the upper construction has expected auxiliary bit work

$$O\left(B^4 \left[1 + \frac{\sqrt{\kappa} + \kappa}{A}\right]\right),$$

where $B \geq 16$ includes their encoding lengths and source-address lengths. Its expected source count has the explicit middle bound in (C.17).

Proof idea. A known thinning gate rescales the geometric-majority identity to the requested amplitude. Curvature gives the lower bound when the normalization is steep; coupling the two deterministic source biases handles the remaining parameter range.

Proof. Put $\theta = \kappa^{-1}$ and $q = 1/[A\sqrt{1 + \theta}] \leq 1$. Lemma C.5 gives

$$\frac{z\sqrt{1 + \theta}}{\sqrt{\theta + z^2}} = \mathbb{E} M_K(z), \quad \Pr(K = k) = \frac{\theta}{1 + \theta} \left(\frac{1}{1 + \theta}\right)^k, \quad k \geq 0.$$

First open a known gate of probability q . On its closed branch return a fair sign. On its open branch draw K and return the majority of $2K + 1$ fresh source signs. This yields exactly $f_{\kappa, A}$. Since $\mathbb{E}K = \kappa$, its

complete-majority source cost is $q(1 + 2\kappa)$. If $\kappa \leq 1$, use $q \leq \sqrt{\kappa}/A$ and $1 + 2\kappa \leq 3$. If $\kappa \geq 1$, use $q \leq 1/A$ and $1 + 2\kappa \leq 3\kappa$. These prove the upper bounds.

For $\kappa \geq 1$, direct differentiation gives

$$|f''_{\kappa,A}(t)| = \frac{3\kappa^{-1}|t|}{A(\kappa^{-1} + t^2)^{5/2}}.$$

Take $t = 1/(2\sqrt{\kappa})$. Then

$$|f''_{\kappa,A}(t)| = \frac{48\kappa}{25\sqrt{5}A}, \quad 1 - t^2 \geq \frac{3}{4}.$$

Lemma C.13 gives

$$Q^*(\kappa, A) \geq \frac{18\kappa}{25\sqrt{5}A} > \frac{\kappa}{4A} \geq \frac{\sqrt{\kappa} + \kappa}{8A}.$$

The strict comparison follows by squaring $72 > 25\sqrt{5}$.

For $0 < \kappa \leq 1$, run the factory at source means $+1$ and -1 . Every source request then returns the same deterministic sign, so the factory reads one input coordinate in $\{-1, 1\}$. Lemma B.7 bounds the probability of any source request below by the total variation distance between the two required output laws,

$$f_{\kappa,A}(1) = \frac{\sqrt{\kappa}}{A\sqrt{1+\kappa}} \geq \frac{\sqrt{\kappa}}{\sqrt{2}A}.$$

Every uniform expected-cost bound is at least this probability, which is more than the stated lower bound.

The outer known gate uses certified square-root intervals and a lazy uniform comparison, with expected $O(B^4)$ auxiliary work. Draw K only after that gate opens. The simplest exact implementation flips the known rational coin with success probability $1/(1 + \kappa)$ until its first success and lets K count the failures. Each such coin has expected $O(B^2)$ bit work by integer rejection. Its geometric tail gives, for each fixed integer r , a bound

$$\mathbb{E}[(1 + K)(1 + \log(K + 2))^r] \leq C_r(1 + \kappa)(1 + \log(\kappa + 2))^r.$$

This follows by dividing the geometric distribution into intervals of length $\lceil 1 + \kappa \rceil$ and summing its exponential tail. Counters, source addresses, and majority evaluation therefore have conditional expected auxiliary cost $O((1 + \kappa)B^4)$. The initial gate multiplies this cost by q . The preceding inequalities finish the bit bound. Known comparisons stop almost surely, K is finite almost surely, and the finite majority terminates. $\square$

For the earlier scalar floor interface, let $\kappa = R^2/\epsilon \geq 1$ and return RMS at radius $2R/\sqrt{\epsilon} = 2\sqrt{\kappa}$. This is $A = 2\sqrt{\kappa}$ in the theorem, so its optimal uniform source cost has order $\sqrt{\kappa}$. At the fixed radius $A = 4$, the cost instead has order κ . Both statements are consistent; they request different output amplitudes.

C.5 Pre-normalization with additive residuals

The placement of a normalization changes the residual calculation. Consider

$$a_{\ell,t} = h_{\ell-1,t} + \text{Att}_{\ell,t}(\mathcal{N}_{\epsilon_{\ell,A}}(h_{\ell-1})), \tag{C.18}$$

$$h_{\ell,t} = a_{\ell,t} + \tanh(b_{\ell} + W_{\ell}\mathcal{N}_{\epsilon_{\ell,F}}(a_{\ell,t})). \tag{C.19}$$

Normalization is performed separately at each position, with the stabilizer inside the square root. All stabilizers are nonnegative rationals. Attention is causal. Its query, key, and value rows and the rows of W_{ℓ} have ℓ_1 norm at most s ; the $\tanh$ biases have magnitude at most one. Initially $|h_{0,t,i}| \leq 1$, and an input coordinate can be read at unit query cost. Sign inputs are a special case. An exactly represented bounded input coordinate can instead be converted into a sign of that mean. All quantities used by the implementation have finite rational encodings. A gain applied after RMSNorm can be absorbed into the subsequent linear maps by using their resulting row norms.

A denominator certificate is part of the hypothesis. The requirement below is a uniform statement over every allowed context and every position at which the corresponding normalization can be requested.

C.5.1 An additive composition lemma

We first separate the residual bookkeeping from the normalized updates. Index the sublayers by $j = 1, \dots, J$ and write

$$H_j = H_{j-1} + F_j(H_{j-1}), \quad R_0 = 1, \quad R_j = 1 + \sum_{k=1}^j U_k.$$

The coordinate and position sets can be arbitrary finite sets. Each U_j is a nonnegative rational number, and $|F_j(H)_{t,i}| \leq U_j$ on every admissible predecessor stream. Zero-radius updates are omitted. Suppose one exact sign of mean $F_j(H)_{t,i}/U_j$ uses at most m_j expected predecessor signs of means $H_{t',i'}/R_{j-1}$ and at most $C_0 B^4 L_j$ expected auxiliary bit operations. These are uniform conditional bounds for every requested address and fixed predecessor stream. The fresh predecessor calls may have costs correlated with their returned signs. Assume $L_j \geq 0$ and enlarge the common absolute constant C_0 to cover a source probe and one prefix-table draw. Put

$$P_j = \prod_{k=1}^j \left(1 + \frac{U_k m_k}{R_{k-1}}\right), \quad P_0 = 1.$$

Here B includes the bit lengths of the radius prefixes and all addresses.

Theorem C.15 (Exact additive composition). *There is an almost surely terminating exact sign sampler for $H_{J,t,i}/R_J$. Its expected number of input leaves is at most*

$$\frac{1}{R_J} \prod_{j=1}^J \left(1 + \frac{U_j m_j}{R_{j-1}}\right). \quad (\text{C.20})$$

Its expected bit work is at most

$$C_0 B^4 \frac{P_J}{R_J} \left[2 + \sum_{j=1}^J \frac{U_j(1 + L_j)}{P_j}\right]. \quad (\text{C.21})$$

No factor J is hidden in the cost of following identity branches. If $m_j \leq m$ for $m \geq 1$, the leaf bound is at most R_J^{m-1} . If also $L_j \leq \mathcal{L}$, the sharper bit bound is

$$C_0 B^4 (2 + \mathcal{L})(1 + \log R_J) R_J^{m-1}. \quad (\text{C.22})$$

Thus the uniform source and bit bounds have the same polynomial exponent in the cumulative residual radius.

Proof idea. Normalize each additive update by its contribution to the total radius. The resulting mixture selects either an earlier state or an update. Sampling the first selected update directly skips all intermediate identity choices, and the remaining expected-cost recurrence telescopes to a product.

Proof. At a normalized request at level j , take the identity branch with probability R_{j-1}/R_j and the update branch with probability U_j/R_j . The corresponding means sum to H_j/R_j . The mean input-leaf recurrence is bounded by

$$C_j \leq \frac{R_{j-1} + U_j m_j}{R_j} C_{j-1}, \quad C_0 = 1,$$

which proves (C.20) by telescoping.

The implementation does not perform those identity trials individually. Starting at level j , the first selected update is $k \leq j$ with probability

$$\left(\prod_{r=k+1}^j \frac{R_{r-1}}{R_r} \right) \frac{U_k}{R_k} = \frac{U_k}{R_j}.$$

The probability of reaching the input without an update is $1/R_j$. Choose directly from these $j+1$ alternatives using the preprocessed prefix weights $1, U_1, \dots, U_j$. At an update k , invoke its factory, whose predecessor

requests start at level $k - 1$. This constructs exactly the same probability law and the same recursive calls as the uncompressed mixture. For dyadic weights, an exact uniform integer draw on the common integer denominator followed by binary search implements this choice. General rational prefixes can also be used, with their full common representation length included in B . Rejection for the uniform integer has expected fewer than two attempts. The draw and search cost $O(B^3)$, which is included in $C_0 B^4$.

For completeness, let T_j be the maximum expected bit work of a compressed request at level j . Conditional charging at the start of each fresh update call gives

$$R_j T_j \leq C_0 B^4 (R_j + 1) + \sum_{k=1}^j U_k (C_0 B^4 L_k + m_k T_{k-1}).$$

This includes one prefix draw and the terminal probe. Set $S_j = R_j T_j$, and take the equality majorant W_j of the displayed recurrence, starting at $W_0 = 2C_0 B^4$. Subtracting successive equalities gives

$$W_j = \left(1 + \frac{U_j m_j}{R_{j-1}}\right) W_{j-1} + C_0 B^4 U_j (1 + L_j).$$

Division by P_j and summation give $W_j/P_j = C_0 B^4 [2 + \sum_{k \leq j} U_k (1 + L_k)/P_k]$. The majorant is finite and bounds S_j by induction, proving (C.21). This calculation does not require $m_j \geq 1$. Tonelli's theorem applies to the nonnegative charges, so the recurrence itself proves that the expected work is finite. Each component makes finitely many requests almost surely, and every recursive call lowers the level, so finite-depth induction proves termination and exactness.

For uniform bounds, replace each source budget by m and each local budget by $\mathcal{L}$ in the established estimates. Put $\hat{P}_j = \prod_{k \leq j} (1 + m U_k / R_{k-1})$. Bernoulli's inequality $1 + mx \leq (1 + x)^m$ for $m \geq 1$ gives $\hat{P}_j / R_j \leq R_j^{m-1}$, proving the leaf bound. Also $\hat{P}_j \geq R_j$ and

$$\sum_{k=1}^j \frac{U_k}{\hat{P}_k} \leq \sum_{k=1}^j \frac{U_k}{R_k} \leq \log R_j.$$

The last inequality integrates $1/r$ over each interval $[R_{k-1}, R_k]$. Substitution in (C.21) gives

$$T_j \leq C_0 B^4 R_j^{m-1} [2 + (1 + \mathcal{L}) \log R_j] \leq C_0 B^4 (2 + \mathcal{L}) (1 + \log R_j) R_j^{m-1}.$$

This proves (C.22). Local arithmetic enters its prefactor without increasing the polynomial depth exponent. $\square$

C.5.2 A relative energy floor gives polynomial depth

Fix a rational $\gamma > 0$. Let G be the least power of two with $G \geq 2/\sqrt{\gamma}$; rational comparisons determine it, and

$$2/\sqrt{\gamma} \leq G < 4/\sqrt{\gamma}.$$

Define the radii

$$R_{2\ell-2} = 1 + (\ell - 1)(sG + 1), \quad R_{2\ell-1} = R_{2\ell-2} + sG, \quad R_{2\ell} = R_{2\ell-1} + 1.$$

The j th normalization is assumed to satisfy, at every admissible position and context,

$$\epsilon_j + \frac{1}{n} \sum_i H_{j-1,t,i}^2 \geq \gamma R_{j-1}^2. \quad (\text{C.23})$$

Let

$$\begin{aligned} k &= \max\{1, \gamma^{-1}\}, & K &= 1 + 228k^2, & \rho &= sG, \\ E &= \exp(2|\beta|s^2G^2), \\ m_A &= (2E - 1)K, & L_A &= 4E(1 + k^2), \\ m_F &= 2\rho(1 + \rho)K, & L_F &= 4(1 + \rho)^2(1 + k^2), \\ m &= \max\{1, m_A, m_F\}, & \mathcal{L} &= \max\{L_A, L_F\}. \end{aligned}$$

Write $\Psi(R) = (2 + \mathcal{L})(1 + \log R)$. These numbers are analysis bounds. The implementation never needs to compute the exponential or the product defining the work bound.

Theorem C.16 (Pre-norm additive sampling under an energy certificate). *Under (C.23), every coordinate of the architecture (C.18)–(C.19), normalized by $R_{2D} = 1 + D(sG + 1)$, has an exact sampler with at most R_{2D}^{m-1} expected input leaves and*

$$O(B^4 \Psi(R_{2D}) R_{2D}^{m-1}) \quad (\text{C.24})$$

expected bit work. The implicit constant is absolute. Weight preparation takes $O(Dn^2 B^6)$ time and $O(Dn^2 B)$ stored bits. Neither bound has a linear factor in T .

If the promised bounded logits are signed coordinates of the final unnormalized stream, the complete softmax sample costs $O(B^4 \Psi(R_{2D}) R_{2D}^{m+1})$. If the model instead applies a final RMSNorm with the same relative energy certificate and an affine readout of row norm at most u and bias magnitude at most b , put $H = \max\{1, b + uG\}$. For promised logits in $[-1, 1]$, the complete token sample costs

$$O(B^4 H^2 (K + 1 + k^2) \Psi(R_{2D}) R_{2D}^{m-1}). \quad (\text{C.25})$$

For fixed γ, s, β, u, b , these are polynomial bounds in depth. In particular, logarithmic depth gives polylogarithmic work, including position-address costs when T is polynomially bounded in n .

Proof idea. The relative energy certificate makes every normalization factory have the same bounded output radius and source budget. Composing one such factory with attention or a tanh row supplies the local costs in the additive product. The final readout is charged through its own interface.

Proof. At level j , supply the floor $\lambda_j = \gamma R_{j-1}^2$ to the normalization factory, replacing it by $\max\{\lambda_j, \epsilon_j\}$ when appropriate. Its condition number is at most k . Its natural output radius is at most $2/\sqrt{\gamma} \leq G$. An algebraic thinning coin makes an exact sign of mean $\mathcal{N}_{\epsilon_j}(H)_i/G$. By Theorem C.1, this uses at most K predecessor requests and $O(k^2 B^4)$ auxiliary work, including the known thinning coin.

Theorem B.1 gives an attention update of radius sG . Its sampler makes at most $2E - 1$ requests for signs from the normalized stream. Its own auxiliary work is $O(EB^4)$. Composing these requests gives m_A and a local bound $C_0 B^4 L_A$ after enlarging the common absolute scalar constant. The raw tanh update has radius one. Corollary A.25 makes at most $2\rho(1 + \rho)$ normalized requests, with auxiliary work $O((1 + \rho)^2 B^4)$. This gives m_F, L_F . Uniformity of these bounds permits charging each normalization call before its fresh randomness.

The specified radii bound the true streams by induction. Apply Theorem C.15 with alternating update radii sG and one. All radii are rational, so the exact origin tables have the claimed representation and preparation costs. A uniform causal-position draw uses $O(\log(T + 2))$ expected random bits, included in B .

For an unnormalized coordinate head, use Theorem B.10 with radius R_{2D} . It makes at most $4096 R_{2D}^2$ expected stream requests. Its auxiliary work is absorbed since $m \geq 1$ and $\Psi(R) \geq 1$. For the final-normalization head, a normalized logit sign at radius H uses at most one sign at radius G , hence at most K stream requests and $O((1 + k^2) B^4)$ local work. The bounded-logit interface makes at most $4096 H^2$ such calls. Conditional charging proves (C.25) and almost-sure termination. $\square$

Corollary C.17 (Step sizes and cumulative residual scale). *Give sublayer j any known nonnegative rational step size a_j , so that its update is $H_j = H_{j-1} + a_j F_j(H_{j-1})$. Under the relative floor $\epsilon_j + n^{-1} \sum_i H_{j-1,t,i}^2 \geq \gamma R_{j-1}^2$, replace U_j by $a_j U_j$ and use*

$$R_j = 1 + \sum_{k \leq j} a_k U_k.$$

The exact product, the direct origin draw, and the bounds $O(B^4 \Psi(R_j) R_j^{m-1})$ for a normalized stream and (C.25) for a final-normalization head remain valid, with the same local constants m_j, L_j . A zero step is omitted. For a common step $\alpha \in [0, 1]$ on both sublayers of each block,

$$R_{2D} = 1 + \alpha D(sG + 1).$$

At fixed local geometry γ, s, β and fixed readout bounds, the cost is polynomial in $1 + \alpha D$. Consequently every $\alpha D = \log^{O(1)} n$ gives polylogarithmic work when the address and encoding factor B is polylogarithmic. This conclusion treats those local parameters as constants; standard scaling $\beta = \sqrt{n}$ requires retaining their displayed dependence or supplying a better indexed-attention budget.

Proof idea. Scaling an update also scales its radius, leaving its normalized sign law unchanged. Only the update-selection probabilities and radius prefixes change, so the same product formula handles the entire schedule.

Proof. For $a_j > 0$, a sign of mean $a_j F_j / (a_j U_j)$ is exactly a sign of mean F_j / U_j , so neither its predecessor count nor its auxiliary cost changes. Only the probability of choosing that update and the radius prefix change. Apply Theorem C.15. The direct origin distribution has masses $a_k U_k / R_j$ and $1/R_j$, and can be prepared on the same rational prefix grid. $\square$

What the certificate supplies. The relative floor is a quantitative restriction on the residual stream. A fixed positive stabilizer alone supplies the weaker floor $\lambda_j = \epsilon_j$. It does not imply (C.23) with a constant γ , since the radius R_j can grow with depth while a particular stream remains near zero. For a general certified floor sequence, choose a rational power of two $G_j \geq 2R_{j-1}/\sqrt{\lambda_j}$ and use the exact product in Theorem C.15, with

$$\kappa_j = \max\{1, R_{j-1}^2/\lambda_j\}, \quad K_j = 1 + 228\kappa_j^2.$$

Attention then has radius sG_j , score bound $|\beta|s^2G_j^2$, and the corresponding local normalization charge $O(\kappa_j^2 B^4)$. This remains a finite exact algorithm, but the resulting product need not be polynomial in depth.

An indexed attention sampler can be substituted in the abstract theorem. It must supply uniform conditional bounds for the normalized update's predecessor requests and auxiliary bit work, including all searches, retries, and precision refinement. Its prefill construction and stored cache are charged separately. The proof uses only those two bounds and exactness, so it does not require a Poisson race or forbid an index. This substitution is especially relevant under standard scaling, where $|\beta| = \sqrt{n}$ makes the unindexed constant E large.

C.6 Lower bounds for additive depth

C.6.1 A fixed-stabilizer lower bound

Theorem C.16 needs its relative floor. With a fixed stabilizer alone, logarithmic depth can force a cost linear in the context length, even at zero attention scores. The following construction concerns weight-only preprocessing, as in the input-probe model. It also applies when the unknown signs arrive as a fresh query. It is not a lower bound against an index that has already read and summarized these particular signs during prefill.

Theorem C.18 (Additive pre-norm blocks can reveal a majority). *For every $T \geq 2$, there is a width-three instance of (C.18)–(C.19), with $s = 4$, stabilizer $\epsilon = 3$ at every normalization, zero attention scores, and $O(\log T)$ blocks, whose bounded two-logit exact-token query complexity is at least $T/2^{17}$ in the worst context. All nonzero weights are in $\{1, 2, -2\}$ except a dyadic final readout coefficient. The input alphabet is $\{(-1, 1, -1), (1, 1, -1)\}$, and every normalization denominator is at least $\sqrt{3}$.*

Proof idea. Uniform attention writes a hidden input average into one coordinate. Subsequent normalized residual updates amplify small averages until their sign becomes visible at a bounded readout. A fourth-moment estimate and the transcript score convert this sensitivity into a linear query bound.

Proof. Write the unknown first coordinates as $x_1, \dots, x_T$. Initially every coordinate has magnitude one, so the first normalization denominator is exactly two. Use zero queries and keys. The sole nonzero value row is

row two, with coefficients 2 on coordinate one and -2 on coordinate two. The attention residual therefore changes the stream at position t to

$$(x_t, M_t, -1), \quad M_t = \frac{1}{t} \sum_{j \leq t} x_j.$$

Take the tanh update of this first block to be zero. All later attention updates are zero. In each later feed-forward sublayer, the sole nonzero row is row two, selecting its own normalized coordinate with weight one. At the last position this produces the scalar recurrence

$$F_0(z) = z, \quad F_{r+1}(z) = F_r(z) + \tanh\left(\frac{F_r(z)}{\sqrt{11/3 + F_r(z)^2/3}}\right), \quad z = M_T.$$

Every recurrence is odd and increasing. If $0 \leq y \leq 1$, its normalized argument is at least $y/2$ and at most one. The inequality $\tanh v \geq v/2$ for $0 \leq v \leq 1$ therefore gives an increment at least $y/4$. If $y \geq 1$, the argument is at least $1/2$, so the increment is at least $1/4$.

Set

$$k = \left\lceil \log_{5/4}(2\sqrt{T}) \right\rceil, \quad d = 2^{\lceil \log_2(2k) \rceil}.$$

Thus $2k \leq d < 4k$. Starting at $z \geq 1/(2\sqrt{T})$, the recurrence reaches one within k steps. During its remaining $d - k$ steps it increases by at least $1/4$ per step. Consequently $F_d(z) \geq d/8$. The trivial upper bound $|F_d(z)| \leq 1 + d \leq 2d$ holds for all $|z| \leq 1$. Take logits $\pm F_d(M_T)/(2d)$, a legal dyadic readout. Their output-sign mean is

$$g(M_T) = \tanh(F_d(M_T)/(2d)).$$

For $z \geq 1/(2\sqrt{T})$, one has $g(z) \geq 1/32$. Oddness gives the corresponding negative inequality.

Let the input signs be independent with common mean t , and put $H(t) = \mathbb{E}_t g(M_T)$. Differentiating the finite input sum at zero gives

$$H'(0) = T\mathbb{E}_0[M_T g(M_T)].$$

Now $\mathbb{E}_0 M_T^2 = 1/T$ and $\mathbb{E}_0 M_T^4 = (3T^2 - 2T)/T^4 \leq 3/T^2$. The elementary second-moment inequality gives

$$\Pr_0\{|M_T| \geq 1/(2\sqrt{T})\} \geq 3/16.$$

Indeed, Cauchy–Schwarz applied to that event gives

$$\Pr\{M_T^2 \geq (\mathbb{E} M_T^2)/4\} \geq (3/4)^2 \frac{(\mathbb{E} M_T^2)^2}{\mathbb{E} M_T^4}.$$

Hence

$$H'(0) \geq T \frac{1}{32} \frac{1}{2\sqrt{T}} \frac{3}{16} = \frac{3\sqrt{T}}{1024}.$$

For any exact sampler, expose only its distinct unknown input probes. At $t = 0$, the transcript score Z has $\mathbb{E} Z^2 = \mathbb{E} Q$ and $H'(0) = \mathbb{E}[YZ]$, where $Y \in \{-1, 1\}$ is the output sign. Cauchy–Schwarz gives $\mathbb{E} Q \geq H'(0)^2 \geq 9T/2^{20} \geq T/2^{17}$. The average lower bound implies a worst-context lower bound. The construction has $d + 1 = O(\log T)$ blocks and row norms at most four. $\square$

The public constant coordinates make the denominator floor explicit. The lower bound does not rely on normalization being undefined on any input, on growing row norms, or on nonzero attention temperature. An index built after reading the context can store all prefix means in this example. It consequently evades this particular probe lower bound, which is why cache preprocessing must be stated separately.

Corollary C.19 (A fresh-query obstruction survives a cache). *For every power-of-two width $n \geq 2$, a pre-norm additive network with one position, row norms at most four, stabilizer three, and $O(\log n)$ blocks has a bounded binary output whose exact sampler requires at least $n/2^{17}$ expected fresh-query coordinate probes in the worst query. Any preprocessing independent of that query is allowed.*

Proof idea. At one position, an affine attention value map averages the unknown query coordinates. This places the same hard scalar recurrence inside a single fresh query, so preprocessing independent of that query cannot remove the information bound.

Proof. The query is $x \in \{-1, 1\}^n$. Its first RMSNorm denominator is two. At the sole position, attention has weight one. Give its value map the matrix

$$V = 2(J_n - I_n), \quad (J_n)_{ij} = 1/n.$$

The first residual becomes $J_n x = M \mathbf{1}$, where $M = n^{-1} \sum_i x_i$. Every row has norm $4(1 - 1/n) < 4$, and the coefficients are dyadic. Use a zero first tanh update, zero attention updates thereafter, and identity tanh rows. All coordinates now obey

$$F_{r+1}(z) = F_r(z) + \tanh(F_r(z)/\sqrt{3 + F_r(z)^2}).$$

For $0 \leq y \leq 1$ this adds at least $y/4$, and for $y \geq 1$ it adds at least $1/4$. Choose d exactly as in the preceding theorem with n in place of T , and use logits $\pm F_d(M)/(2d)$. The same moment and finite-transcript argument gives $n/2^{17}$. An index built before x arrives is part of the fixed preprocessed state in that argument. $\square$

C.6.2 Relative-floor lower bounds

The next constructions keep a relative energy floor on the entire input cube. They show that the polynomial dependence on the cumulative residual radius in Theorem C.16 cannot be removed.

Theorem C.20 (A relative-floor depth lower bound, uniform in the step size). *Fix a dyadic $s > 1$ and put*

$$\delta = s - 1, \quad C = \frac{s(s^2 + 1/2)}{2(s - 1)}.$$

There is a positive constant c_s , depending only on s , with the following property. For every power-of-two $N \geq 2$, every dyadic $0 < \alpha \leq 1$, and every integer D with $\alpha D \geq 4$, there is a one-position, width- $2N$ pre-norm additive network with D blocks, row norms at most s , stabilizer $\epsilon = 3$ in every normalization, and the same step size α on every residual sublayer, whose exact binary output requires

$$\mathbb{E}Q \geq c_s \min\{(1 + \alpha D)^{2\delta}, N\} \tag{C.26}$$

expected fresh-query coordinate probes in the worst input. Arbitrary preprocessing independent of that query is allowed. On the entire input sign cube, every actual normalization denominator squared satisfies

$$v_j \geq \frac{1}{12}(1 + \alpha j)^2$$

at the corresponding residual radius $R_j = 1 + \alpha j$. Attention updates are zero, so the temperature is arbitrary. The output uses one coordinate and a dyadic readout, and its logits lie in $[-1, 1]$ on every permitted input. The constant in (C.26) is given explicitly in the proof.

Proof idea. Separate feature coordinates from deterministic carriers that control the normalization scale. Dividing by the residual radius gives a scalar gain whose product grows polynomially. A reciprocal-square comparison keeps that gain valid through nonlinear saturation, and the transcript score turns it into a probe lower bound.

Proof. The network accepts every input in $\{-1, 1\}^{2N}$. For its lower-bound distribution, let the first N coordinates be unknown signs $x_1, \dots, x_N$ and fix the remaining N coordinates to -1 . All query, key, and value maps are zero. Every feature tanh row places weight s/N on each of the first N normalized coordinates, with zero bias. Every carrier row has zero weights and bias -1 . These are legal dyadic parameters with the asserted row norms.

Let $M = N^{-1} \sum_i x_i$ and $t_0 = \tanh 1$. After j blocks the feature coordinates are

$$x_i - M + F_j(M),$$

and all carrier coordinates are $C_j = -(1 + \alpha j t_0)$. This follows by induction, with $F_0(z) = z$ and

$$F_{j+1}(z) = F_j(z) + \alpha \tanh \left(\frac{s F_j(z)}{\sqrt{3 + [1 - z^2 + F_j(z)^2 + C_j^2]/2}} \right). \quad (\text{C.27})$$

Indeed, the mean of the feature coordinates is $F_j(M)$, and their mean square is $1 - M^2 + F_j(M)^2$. Every coordinate has magnitude at most $R_j = 1 + \alpha j$. Since $t_0 > 1/2$, $|C_j| \geq R_j/2$, so the denominator squared is at least $R_j^2/8$. The attention residual has zero update and leaves these statements unchanged. The network also has a relative floor on the entire sign cube. For arbitrary initial carrier signs $e_i \in \{-1, 1\}$, their values are $e_i - \tau t_0$, with $\tau = \alpha j$. If $\tau \leq 2$, the stabilizer alone gives $v_j \geq 3 \geq (1 + \tau)^2/12$. If $\tau \geq 2$, every carrier has magnitude at least $\tau/2 - 1$, so

$$12v_j - (1 + \tau)^2 \geq 12 \left[3 + \frac{1}{2}(\tau/2 - 1)^2 \right] - (1 + \tau)^2 = \frac{1}{2}(\tau - 8)^2 + 9 > 0.$$

Thus the claimed floor $R_j^2/12$ holds on every sign input; the stronger $R_j^2/8$ remains available on the lower-bound subdomain.

For $z \geq 0$, all $F_j(z)$ are nonnegative and $F_j(z) \geq z$; the recurrences are odd. Set $u_j = F_j/R_j$ and $a_j = \alpha/R_{j+1}$. Their normalized update is

$$u_{j+1} = (1 - a_j)u_j + a_j \tanh \left(\frac{s u_j}{\sqrt{b_j(z) + u_j^2/2}} \right),$$

where

$$b_j(z) = \frac{3 + (1 - z^2)/2 + C_j^2/2}{R_j^2} \leq \frac{7}{2R_j^2} + \frac{1}{2}.$$

Let $j_0 = \lceil 2/\alpha \rceil$. Then $3 \leq R_{j_0} \leq 4$, hence $u_{j_0}(z) \geq z/4$, and for every $j \geq j_0$ one has $b_j(z) \leq 8/9 < 1$.

For $u \geq 0$, the bound $\tanh y \geq y/\sqrt{1 + y^2}$ gives

$$\tanh \left(\frac{s u}{\sqrt{b_j(z) + u^2/2}} \right) \geq \frac{s u}{\sqrt{1 + (s^2 + 1/2)u^2}}.$$

Put $g_j = 1 + \delta a_j$ and $d_j = a_j s(s^2 + 1/2)/g_j$. Convexity of $v \mapsto (1 + v)^{-1/2}$ gives $u_{j+1} \geq g_j u_j / \sqrt{1 + d_j u_j^2}$. Define

$$G = \prod_{j=j_0}^{D-1} g_j.$$

For a positive z , write $G_j = \prod_{r=j_0}^{j-1} g_r$. The reciprocal-square inequality gives

$$\frac{G_{j+1}^2}{u_{j+1}(z)^2} \leq \frac{G_j^2}{u_j(z)^2} + d_j G_j^2.$$

The initial reciprocal term is at most $16/z^2$. Moreover, $d_j \leq a_j s(s^2 + 1/2)$ and $g_j^2 - 1 \geq 2\delta a_j$, so

$$\sum_{j=j_0}^{D-1} d_j G_j^2 \leq C \sum_{j=j_0}^{D-1} (g_j^2 - 1) G_j^2 = C(G^2 - 1).$$

Summing and inverting therefore gives

$$u_D(z) \geq \frac{G(z/4)}{\sqrt{1 + C(G^2 - 1)(z/4)^2}}, \quad z \geq 0. \quad (\text{C.28})$$

Continuity covers $z = 0$. The calculation uses the pointwise comparison at each step and does not assume that $b_j(z)$ is independent of z .

Choose the least power of two $L \geq R_D$. Then $R_D \leq L < 2R_D$. The logits are plus and minus the first feature coordinate divided by L . They lie in $[-1, 1]$, and $1/L$ is dyadic. Conditional on $M = z$, the selected sign x_1 has mean z . Put

$$k = 1/L, \quad A = (F_D(z) - z)/L \geq 0.$$

The conditional output-sign mean is

$$\bar{g}(z) = \frac{1+z}{2} \tanh(A+k) + \frac{1-z}{2} \tanh(A-k).$$

At $A = 0$, this equals $z \tanh k \geq zk/4$. Increasing A from zero to its actual value keeps both arguments in $[-1, 1]$, since $F_D(z) - z \leq \alpha D = R_D - 1$. Throughout that interval the derivative of the displayed conditional mean with respect to A is at least $\text{sech}^2(1) > 1/4$. Consequently

$$\bar{g}(z) \geq \frac{zk + A}{4} = \frac{F_D(z)}{4L} \geq \frac{u_D(z)}{8}, \quad z \geq 0. \quad (\text{C.29})$$

Oddness gives the signed counterpart for negative z .

Give the unknown query signs independent common mean t , and let $H(t)$ be the output-sign mean. Differentiating the finite input sum at zero and conditioning on M gives $H'(0) = N\mathbb{E}_0[M\bar{g}(M)]$. Apply (C.28)–(C.29) and Jensen's inequality weighted by M^2 . Since $\mathbb{E}M^2 = 1/N$ and $\mathbb{E}M^4/\mathbb{E}M^2 \leq 3/N$,

$$H'(0) \geq \frac{G}{32\sqrt{1 + 3CG^2/(16N)}}.$$

The finite-transcript information inequality therefore gives

$$\mathbb{E}Q \geq \frac{G^2}{1024(1 + 3CG^2/(16N))} \geq \frac{\min\{G^2, N\}}{1024(1 + 3C/16)}.$$

It holds on average over unbiased inputs, and hence for a worst input.

It remains to bound G uniformly in α . Monotonicity of $x \mapsto (1+x)^{-1}$ and its square implies

$$\sum_{j=j_0}^{D-1} a_j \geq \log \frac{R_{D+1}}{R_{j_0+1}}, \quad \sum_{j=j_0}^{D-1} a_j^2 \leq \frac{\alpha}{R_{j_0}} \leq \frac{1}{3}.$$

Also $R_{j_0+1} \leq 5$ and $R_{D+1} \geq 1 + \alpha D$. The inequality $\log(1+x) \geq x - x^2/2$ gives

$$G \geq e^{-\delta^2/6} \left(\frac{1 + \alpha D}{5} \right)^\delta.$$

Since $e^{-\delta^2/3} 5^{-2\delta} \leq 1$, substitution into the minimum bound proves (C.26) with

$$c_s = \frac{e^{-\delta^2/3} 5^{-2\delta}}{1024(1 + 3C/16)}.$$

The assumption $\alpha D \geq 4$ ensures $D \geq j_0$; all sums and comparisons above are then valid. $\square$

Corollary C.21 (A fixed final RMSNorm readout). *In Theorem C.20, replace the depth-dependent readout by a final RMSNorm₃. Use the fixed logits $\pm N_3(h_D)_1/4$. On the entire sign cube their absolute values are at most $\sqrt{12}/4 < 1$. The lower bound (C.26) remains valid with $c_s/16$ in place of c_s . Every normalization, including the final one, has the same fixed stabilizer and the relative floor $R_D^2/12$.*

Proof idea. The final normalization denominator stays between two fixed multiples of the residual radius. Replacing the scaled readout by this denominator therefore preserves the mean sensitivity up to a constant.

Proof. Conditional on $M = z$ in the lower-bound distribution, write $L(z) = 4\sqrt{v_D(z)}$. The full-domain floor gives $L(z) \geq R_D$. The coordinate-radius bound gives $v_D(z) \leq 3 + R_D^2 \leq 4R_D^2$, hence $L(z) \leq 8R_D$. The conditional shift calculation in (C.29) now gives

$$\bar{g}(z) \geq \frac{F_D(z)}{4L(z)} \geq \frac{u_D(z)}{32} \quad (z \geq 0).$$

The denominator is even in z , so oddness remains valid. The score lower bound loses a factor four, and its square loses a factor sixteen. This gives the asserted constant. The full-domain logit bound is $|h_{D,1}|/(4\sqrt{v_D}) \leq \sqrt{12}/4$, and the readout coefficient $1/4$ is fixed and dyadic. $\square$

Corollary C.22 (Padding to every width). *There is a constant $c_s^{\text{pad}} > 0$, depending only on s , such that for every integer $n \geq 4$, the fixed final-normalization construction can be made at width n , with all the assumptions of Corollary C.21, and with*

$$\mathbb{E}Q \geq c_s^{\text{pad}} \min\{(1 + \alpha D)^{2\delta}, n\}.$$

An explicit value for the constant is given in the proof. The relative floor $1/12$, fixed stabilizer 3, and fixed readout coefficient $1/4$ hold on the entire input sign cube.

Proof idea. Use a power-of-two feature group so its averaging weights remain dyadic, and fill the remaining coordinates with carriers. Their positive fraction preserves the energy floor, while the feature group still contains a constant fraction of the unknown input.

Proof. Choose the largest power of two $N \leq n/2$. Use N feature coordinates and $n - N$ carrier coordinates. Put $\eta = N/n$, so $1/4 < \eta \leq 1/2$. Each feature row still has the dyadic weights s/N . On the lower-bound subdomain the squared denominator is

$$3 + \eta(1 - z^2 + F_j(z)^2) + (1 - \eta)C_j^2.$$

The carrier fraction is at least $1/2$, so the full-cube floor proof above is unchanged. Use $j_0 = \lceil 3/\alpha \rceil$. Then $4 \leq R_{j_0} \leq 5$, $u_{j_0} \geq z/5$, and the normalized nonsignal part of the denominator satisfies

$$b_j(z) \leq \frac{7}{2R_j^2} + \frac{3}{4} \leq \frac{31}{32} < 1.$$

Because $\eta \leq 1/2$, the same Jensen comparison applies with the same C , replacing $z/4$ by $z/5$. The fixed final-normalization head still gives $\bar{g}(z) \geq u_D(z)/32$. Thus the score and query inequalities become

$$H'(0) \geq \frac{G}{160\sqrt{1 + 3CG^2/(25N)}}, \quad \mathbb{E}Q \geq \frac{\min\{G^2, N\}}{25600(1 + 3C/25)}.$$

Now $\sum a_j^2 \leq 1/4$ and $R_{j_0+1} \leq 6$, whence $G \geq e^{-\delta^2/8}((1 + \alpha D)/6)^\delta$. Finally, $N > n/4$ implies $\min\{x, N\} \geq \min\{x, n\}/4$ for $x \geq 0$. These estimates give the explicit constant

$$c_s^{\text{pad}} = \frac{e^{-\delta^2/4} 6^{-2\delta}}{102400(1 + 3C/25)}.$$

Since $\alpha D \geq 4$, $D \geq j_0$ as required. $\square$

Corollary C.23 (Prescribed bounded step schedules). *The arbitrary-width construction in Corollary C.22 also permits prescribed dyadic block steps $a_j \in [0, 1]$ for $0 \leq j < D$, used on both residual sublayers of their respective blocks. Put $\tau_j = \sum_{k < j} a_k$, and suppose $\tau_D \geq 4$. For a positive constant c_s^{sched} depending only on s , its fixed final readout requires*

$$\mathbb{E}Q \geq c_s^{\text{sched}} \min\{(1 + \tau_D)^{2\delta}, n\}.$$

The constant is specified in the proof. All full-cube floors hold at radius $r_j = 1 + \tau_j$ with the same constant $1/12$.

Proof idea. The carrier displacement depends on the cumulative step size. Reciprocal increments telescope even for an irregular schedule, and their squares have a bounded sum. These two facts control the product of normalized gains without a regularity assumption on the steps.

Proof. Replace every occurrence of αj in the carrier and feature recurrences by τ_j , and every increment α by a_j . Let j_0 be the first index with $\tau_{j_0} \geq 3$. Since $a_j \leq 1$, $4 \leq r_{j_0} \leq 5$, so the same initial value $u_{j_0} \geq z/5$ and nonsignal denominator bound apply. The normalized branch size is $q_j = a_j/r_{j+1}$. Let $G = \prod_{j=j_0}^{D-1} (1 + \delta q_j)$. The reciprocal-square comparison and the score lower bound are unchanged.

For completeness, the needed product estimate follows without any regularity assumption on the schedule. Since $r_{j+1} - r_j = a_j$,

$$q_j^2 \leq \frac{a_j^2}{r_j r_{j+1}} \leq \frac{1}{r_j} - \frac{1}{r_{j+1}}.$$

Also, integrating $1/r - 1/r_{j+1}$ over $[r_j, r_{j+1}]$ gives

$$0 \leq \log \frac{r_{j+1}}{r_j} - q_j \leq \frac{a_j^2}{r_j r_{j+1}}.$$

Therefore $\sum q_j^2 \leq 1/4$ and $\sum q_j \geq \log(r_D/r_{j_0}) - 1/4$. The inequality $\log(1+x) \geq x - x^2/2$ yields

$$G \geq e^{-\delta/4 - \delta^2/8} \left(\frac{1 + \tau_D}{5} \right)^\delta.$$

Substitution into the score lower bound proves the claim with

$$c_s^{\text{sched}} = \frac{e^{-\delta/2 - \delta^2/4} 5^{-2\delta}}{102400(1 + 3C/25)}.$$

Zero steps leave the state unchanged and cause no exception in these identities. $\square$

Corollary C.24 (An effective-depth threshold for the certified pre-norm class). *Fix $s > 1$, a nonnegative temperature β , a rational $0 < \gamma \leq 1/12$, an output row-norm bound $u \geq 1/4$, and an output bias bound $c_{\text{out}} \geq 0$, all independent of width and depth. Consider widths tending to infinity, polynomially bounded position counts and depths, and encoding/address length $B = \log^{O(1)} n$. Use a common dyadic residual step $\alpha \in [0, 1]$. For each attention update take the valid radius $U_j = G \max_i \sum_k |V_{j,ik}|$, and for a tanh update take $U_j = 1$, omitting updates certified to be zero. Let $R_j = 1 + \alpha \sum_{k \leq j} U_k$. Restrict to networks whose pre-norm and final-normalization denominators satisfy the relative certificate $v_j \geq \gamma R_{j-1}^2$ of (C.23), with $j = 2D + 1$ for the final normalization. The affine output logits lie in $[-1, 1]$; their row norms are at most u and their biases have magnitude at most c_{out} .*

This class has a uniform polylogarithmic worst-case exact-sampling bound if and only if

$$\alpha D = \log^{O(1)} n.$$

The equivalence concerns asymptotic families. It determines the scale of effective depth; it does not identify the optimal polynomial exponent.

Proof idea. The composition theorem gives the upper bound from the cumulative residual radius. For the converse, one fixed gain above one yields a positive lower exponent on every width, so a radius larger than every polylogarithm cannot have a uniform polylogarithmic sampling bound.

Proof. The positive direction follows from Corollary C.17, since $U_j \leq sG$ at attention and $U_j \leq 1$ at tanh, while all local factory constants are fixed. The final-normalization interface has fixed radius and cost.

For necessity, choose a fixed dyadic σ with $1 < \sigma \leq s$ (if s is dyadic, take $\sigma = s$). For every width $n \geq 4$, use Corollary C.22 with gain σ . Its attention update radii are zero and its tanh update radii are one, so the certificate radii are exactly $1 + \alpha j$. Its full-domain floor $1/12$ meets any stated $\gamma \leq 1/12$. Its

fixed coefficient $1/4$ gives bounded affine logits. The lower bound is a constant depending only on s times $\min\{(1 + \alpha D)^{2(\sigma-1)}, n\}$. If αD is not bounded by a fixed power of $\log n$, this lower bound is not uniformly polylogarithmic. The regime $\alpha D < 4$ cannot cause that failure, so the lower theorem's cutoff has no effect on the equivalence. $\square$

C.7 The gain created by pre-normalization

The row norm alone understates the amplification in an additive pre-norm block. Deterministic carrier coordinates can keep the normalization denominator proportional to the residual radius while leaving its proportionality constant below one. The following lower bound retains that constant.

Theorem C.25 (An additive lower bound from normalized gain). *Put $c = \tanh 1$. Fix a legal dyadic row bound $s > c/\sqrt{2}$, and define*

$$\nu = \frac{\sqrt{2}s}{c} - 1 > 0.$$

For every width $n \geq 4$, every depth $D \geq 0$, and every prescribed dyadic block-step schedule $a_j \in [0, 1]$, there is a one-position pre-norm additive tanh network with row norms at most s , biases at most one, zero attention, and fixed stabilizer 3 whose exact binary output requires

$$\Omega_s(\min\{(1 + \tau_D)^{2\nu}, n\}), \quad \tau_D = \sum_{j < D} a_j,$$

expected fresh-query probes in the worst input. Preprocessing independent of the query is arbitrary. On the entire input sign cube, every pre-norm and final-normalization denominator satisfies

$$v_j \geq \gamma_0(1 + \tau_j)^2, \quad \gamma_0 = \frac{3c^2}{12 + (1 + c)^2} > \frac{1}{12}.$$

The fixed output logits are $\pm \mathcal{N}_3(h_D)_1/4$ and have magnitude below one on the entire sign cube. The constant in the lower bound is positive and depends only on s ; an explicit choice is given in the proof.

Proof idea. Half of a dyadic-sized group carries the unknown mean, and an equally large group moves deterministically at speed $\tanh 1$. Unused coordinates remain fixed. After dividing the state by its residual radius, the scalar gain approaches $s/(\sqrt{\eta} \tanh 1) \geq \sqrt{2}s/\tanh 1$, where $\eta \leq 1/2$ is the fraction of carrier coordinates. The proof bounds the gain below by $\sqrt{2}s/\tanh 1 - O(1/R)$, and the error term has a summable contribution along the residual step schedule. A reciprocal-square comparison controls the nonlinear saturation. The transcript score turns the resulting mean sensitivity into a query lower bound.

Proof. Let N be the largest power of two at most $n/2$, and put $\eta = N/n \in (1/4, 1/2]$. Use N feature coordinates and N carrier coordinates. All remaining coordinates are inert. A feature row has weight s/N on each normalized feature coordinate and zero bias. A carrier row has zero weights and bias -1 . An inert row is zero. All attention maps are zero. These parameters are legal, and every coordinate has magnitude at most $r_j = 1 + \tau_j$, where $\tau_j = \sum_{k < j} a_k$.

For arbitrary initial carrier signs e_i , their values are $e_i - c\tau_j$. Thus

$$v_j \geq 3 + \eta(c\tau_j - 1)^2 \geq 3 + \frac{1}{4}(c\tau_j - 1)^2.$$

For $\theta > 0$ and every real τ , completing a square gives

$$3 + \theta(c\tau - 1)^2 \geq \frac{3\theta c^2}{3 + \theta(1 + c)^2}(1 + \tau)^2. \quad (\text{C.30})$$

Indeed, after multiplication by $A = 3 + \theta(1 + c)^2$, the difference is $[A - \theta c(1 + c)(1 + \tau)]^2$. Taking $\theta = 1/4$ proves the stated full-cube floor. The inequality $c > 3/4$ follows from $e^2 > 7$, for example by summing the first five terms of its series. It gives $35c^2 - 2c - 13 > 0$, which implies $\gamma_0 > 1/12$.

On the lower-bound subdomain, let the features be unknown signs $x_1, \dots, x_N$, set every carrier to -1 , and fix the inert coordinates to signs. Write $M = N^{-1} \sum_i x_i$. After j blocks, the feature coordinates are $x_i - M + F_j(M)$, where $F_0(z) = z$ and

$$F_{j+1}(z) = F_j(z) + a_j \tanh\left(\frac{sF_j(z)}{\sqrt{4 - \eta - \eta z^2 + \eta F_j(z)^2 + \eta(1 + c\tau_j)^2}}\right).$$

The functions are odd, and $F_j(z) \geq z$ for $z \geq 0$. Put $u_j = F_j/r_j$ and $q_j = a_j/r_{j+1}$. The squared normalized denominator is $b_j(z) + \eta u_j^2$, where

$$b_j(z) \leq \frac{c^2}{2} + \frac{c(1-c)}{r_j} + \frac{4}{r_j^2} \leq A_0 + \frac{A_1}{r_j}, \quad A_0 = c^2/2, \quad A_1 = 4 + c(1-c).$$

Set $\kappa = s/\sqrt{A_0} = 1 + \nu$ and $K = \kappa A_1/(2A_0)$. The gains

$$t_j = \frac{s}{\sqrt{A_0 + A_1/r_j}}$$

satisfy $0 \leq \kappa - t_j \leq K/r_j$. This follows from $1 - (1+x)^{-1/2} \leq x/2$ for $x \geq 0$.

Choose

$$R_* = \lceil \max\{4, 2K/\nu\} \rceil, \quad M_* = R_* + 1, \quad C = \frac{\kappa(s^2 + 1/2)}{A_0\nu}.$$

First suppose $r_D \geq R_*$. Let j_0 be the first index with $r_{j_0} \geq R_*$. Since $a_j \leq 1$, $R_* \leq r_{j_0} \leq M_*$ and $u_{j_0}(z) \geq z/M_*$. For $j \geq j_0$, put $\nu_j = t_j - 1$, so $\nu/2 \leq \nu_j \leq \nu$, and $g_j = 1 + \nu_j q_j$. Run the comparison in the proof of Theorem C.20 with the gain t_j in place of s and $B_j = (s^2 + 1/2)/(A_0 + A_1/r_j)$ in place of $s^2 + 1/2$. Its tanh bound and its convexity step, now with weights $(1 - q_j)/g_j$ and $q_j t_j/g_j$, give

$$u_{j+1} \geq \frac{g_j u_j}{\sqrt{1 + d_j u_j^2}}, \quad d_j = \frac{q_j t_j B_j}{g_j} \leq C(g_j^2 - 1),$$

since $d_j \leq q_j \kappa(s^2 + 1/2)/A_0$ and $g_j^2 - 1 \geq \nu q_j$. Its reciprocal-square summation, started from $u_{j_0}(z) \geq z/M_*$, gives for $G = \prod_{j=j_0}^{D-1} g_j$

$$u_D(z) \geq \frac{G(z/M_*)}{\sqrt{1 + C(G^2 - 1)(z/M_*)^2}}, \quad z \geq 0. \quad (\text{C.31})$$

The bound holds by continuity at $z = 0$; the comparison also allows the empty product when $j_0 = D$.

The schedule needs no regularity beyond $0 \leq a_j \leq 1$. Telescoping and integrating $1/r$ give

$$\sum_{j=j_0}^{D-1} q_j^2 \leq \frac{1}{r_{j_0}}, \quad \sum_{j=j_0}^{D-1} q_j \geq \log \frac{r_D}{r_{j_0}} - \frac{1}{r_{j_0}}, \quad \sum_{j=j_0}^{D-1} \frac{q_j}{r_j} \leq \frac{1}{r_{j_0}}.$$

The last sum is exactly $1/r_{j_0} - 1/r_D$. Using $\log(1+x) \geq x - x^2/2$ and $0 \leq \nu - \nu_j \leq K/r_j$, one obtains

$$G \geq e^{-E_*} (r_D/M_*)^\nu, \quad E_* = (\nu + K + \nu^2/2)/R_*.$$

We verify the readout on the entire sign cube. Feature coordinates always have the form $x_i - M + F$, even when the initial carrier and inert signs vary. Hence $|h_i| \leq |F| + 2$ and $v \geq 3 + \eta F^2 \geq 3 + F^2/4$. The identity

$$\frac{16}{3}(3 + F^2/4) - (|F| + 2)^2 = (|F| - 6)^2/3$$

shows $|h_i|/(4\sqrt{v}) \leq 1/\sqrt{3} < 1$.

On the lower-bound subdomain, conditional on $M = z$, set $L(z) = 4\sqrt{v_D(z)}$. Since $v_D(z) \leq 3 + r_D^2 \leq 4r_D^2$, one has $L(z) \leq 8r_D$. The conditional shift argument in the proof of Corollary C.21 applies, because the

readout bound above keeps every intermediate head argument in $[-1, 1]$. It shows that the conditional output-sign mean $\bar{g}$ is odd and satisfies

$$\bar{g}(z) \geq \frac{F_D(z)}{4L(z)} \geq \frac{u_D(z)}{32}, \quad z \geq 0.$$

Let $H(t)$ be the output-sign mean when the unknown signs have common mean t . The score calculation in the proof of Theorem C.20, with (C.31) and this head bound in place of (C.28) and (C.29), gives

$$H'(0) \geq \frac{G}{32M_*\sqrt{1+3CG^2/(M_*^2N)}}, \quad \mathbb{E}Q \geq \frac{\min\{G^2, N\}}{1024M_*^2 + 3072C}.$$

Since $N > n/4$, this proves the theorem in the present case with constant

$$c_1 = \frac{e^{-2E_*}M_*^{-2\nu}}{4096M_*^2 + 12288C}.$$

If $r_D < R_*$, the same head calculation and $F_D(z) \geq z$ give $H'(0) \geq 1/(32r_D)$. Thus $\mathbb{E}Q \geq 1/(1024R_*^2)$, which proves the required lower bound with constant $c_2 = 1/(1024R_*^{2\nu+2})$. Taking $c_s = \min\{c_1, c_2\}$ handles every schedule and depth. The average lower bound yields a worst-query lower bound. $\square$

Dependence on the normalization floor. The same construction gives a parameterized form. At compatible power-of-two widths, use half the coordinates for features, a fixed dyadic fraction $\theta \in (0, 1/2]$ for carriers, and leave the rest inert. Give carriers a fixed negative dyadic bias $-b$, where $0 < b \leq 1$, and put $c_b = \tanh b$. The full-cube certificate and normalized gain are

$$\gamma(\theta, b) = \frac{3\theta c_b^2}{3 + \theta(1 + c_b)^2}, \quad \kappa(\theta, b) = \frac{s}{c_b\sqrt{\theta}}.$$

Whenever $\kappa(\theta, b) > 1$, this argument gives a lower bound with exponent $2(\kappa(\theta, b) - 1)$. The proof above applies with $A_0 = \theta c_b^2$, $A_1 = 4 + 2\theta c_b(1 - c_b)$, and the same reciprocal-square calculation. In particular,

$$\kappa(\theta, b) = \frac{s}{\sqrt{\gamma(\theta, b)}} \sqrt{\frac{3}{3 + \theta(1 + c_b)^2}}.$$

This identifies the source of the amplification: the relevant gain in these lower examples is the row gain divided by the asymptotic normalization scale. The next two subsections prove a matching law for this explicit carrier family.

C.8 A matching query law for the carrier family

The preceding lower construction also admits a matching upper bound on input probes. This result concerns that explicit family on its entire input sign cube. It does not identify the optimal exponent for arbitrary pre-norm networks.

Theorem C.26 (The carrier family's query exponent). *Fix a legal dyadic $s > \tanh(1)/\sqrt{2}$. Consider the networks constructed in Theorem C.25, allowing every initial feature, carrier, and inert coordinate to be an arbitrary sign. Let N be the largest power of two at most $n/2$, put $\eta = N/n$, $r_D = 1 + \sum_{j < D} a_j$, and define*

$$\nu_\eta = \frac{s}{\tanh(1)\sqrt{\eta}} - 1.$$

Their optimal worst-input expected query cost is

$$\Theta_s\left(\min\{r_D^{2\nu_\eta}, n\}\right).$$

An exact sampler attaining the upper bound has finite expected bit work. A direct certified implementation uses

$$O_s\left(\min\{(D+1)r_D^{2\nu_\eta}, n+D+1\}B^4\right),$$

where $B \geq 16$ includes all parameter and radius-prefix encoding lengths and $\log(n+D+2)$. Theorem C.27 removes the depth factor by preprocessing the two-mean function.

Proof idea. On the full cube, the network depends on two unknown averages: the feature mean and the carrier mean. Its normalized scalar state has derivative growth $r_D^{\nu_\eta}$. The discrepancy from that limiting gain has summable contribution along the step schedule. Derivatives through order six then fit the two-variable corrected Bernstein factory with a base sample size of order $r_D^{2\nu_\eta}$. One observed feature sign determines which of two bounded head functions the factory must simulate.

Proof. Write $c = \tanh 1$, and let z and w denote the initial feature and carrier means. Inert coordinates always have squared value one. Every feature coordinate is $x_i - z + F_j(z, w)$. The squared RMS denominator is

$$v_j = 4 - \eta z^2 + \eta F_j^2 - 2\eta c \tau_j w + \eta c^2 \tau_j^2, \quad \tau_j = r_j - 1.$$

Set $u_j = F_j/r_j$ and $q_j = a_j/r_{j+1}$. Then

$$u_{j+1} = (1 - q_j)u_j + q_j T_j(z, w, u_j), \quad T_j(z, w, u) = \tanh\left(\frac{su}{\sqrt{b_j(z, w) + \eta u^2}}\right),$$

where

$$b_j(z, w) = \frac{4 - \eta z^2 - 2\eta c \tau_j w + \eta c^2 \tau_j^2}{r_j^2}.$$

For all $(z, w) \in [-1, 1]^2$ and j ,

$$r_j^2 b_j \geq 4 - 2\eta + \eta(c\tau_j - 1)^2 \geq 3 + \eta(c\tau_j - 1)^2.$$

The carrier-floor identity in (C.30) therefore gives $b_j \geq 1/12$. Moreover

$$|b_j - \eta c^2| \leq 8/r_j.$$

For example, expansion in $1/r_j$ gives first-order coefficient $-2\eta c(c + w)$ of magnitude at most two, and a second-order coefficient of magnitude at most six. The state satisfies $|u_j| \leq 1$, since its update is a convex combination of numbers in $[-1, 1]$.

Put $\nu = \nu_\eta$, $C_u = 192s$, and $M = e^{C_u}$. Direct differentiation and the mean-value theorem give

$$0 \leq \partial_u T_j = \operatorname{sech}^2\left(\frac{su}{\sqrt{b_j + \eta u^2}}\right) \frac{sb_j}{(b_j + \eta u^2)^{3/2}} \leq 1 + \nu + C_u/r_j.$$

Here the derivative of $x^{-1/2}$ on $x \geq 1/12$ is at most $12^{3/2}/2$, and $4 \cdot 12^{3/2} < 192$. Consequently a derivative of the scalar state propagated from level k to level j has multiplier at most

$$\prod_{i=k}^{j-1} (1 + \nu q_i + C_u q_i/r_i) \leq M(r_j/r_k)^\nu. \quad (\text{C.32})$$

This uses $\sum q_i \leq \log(r_j/r_k)$ and $\sum q_i/r_i = 1/r_k - 1/r_j \leq 1$.

We give explicit uniform local derivative bounds. Set

$$h = \frac{1}{10000(1+s)}, \quad C_0 = 1440(3/h)^6.$$

For real $z, w, u \in [-1, 1]$, extend T_j to the complex polydisk of coordinate radius h . The squared denominator changes by at most $4h$, while its real value is at least $1/12$. The principal inverse square root is therefore holomorphic there. Its argument inside $\tanh$ changes by at most $6sh \cdot 12^{3/2} < 1/8$. $\tanh$ has no pole there and has modulus at most two. Cauchy's estimates show that the entrywise ℓ_1 norm of every local derivative tensor of order at most six is at most C_0 ; the factor 3^6 covers the sum over coordinates. The same bound holds for the head function, after enlarging C_0 by an absolute factor if necessary; the argument below permits the choice $2C_0$ instead.

Let $E_k(j)$ be the maximum entrywise ℓ_1 norm of the k th derivative tensor of u_j on $[-1, 1]^2$. For $k = 1$, the direct parameter derivatives have total magnitude at most C_0 . Equations (C.32) and

$$\sum_i q_i r_{i+1}^{-\nu} \leq 1/\nu$$

give $E_1(j) \leq C_1 r_j^\nu$, where $C_1 = M(1 + C_0/\nu)$. For $k = 2, \dots, 6$, define recursively

$$H_k = \max\{2 + C_1, C_2, \dots, C_{k-1}\}, \quad C_k = \frac{MC_0 k^k H_k^k}{(k-1)\nu}.$$

Then

$$E_k(j) \leq C_k r_j^{k\nu}. \quad (\text{C.33})$$

To verify this induction, use the multivariate chain rule in partition form. The single-block term is $\partial_u T_j D^k u_j$. Every other partition has at least two blocks, all of size below k . There are at most k^k partitions. The first derivative of the map $(z, w) \mapsto (z, w, u_j)$ has total norm at most $(2 + C_1)r_j^\nu$; each higher derivative is bounded by the induction hypothesis. Thus the remainder has norm at most $C_0 k^k H_k^k r_j^{k\nu}$. After propagation by (C.32), its sum is bounded using

$$\sum_i q_i r_i^p \leq (r_j^p - 1)/p, \quad p > 0.$$

For this inequality, integrate t^{p-1} on $[r_i, r_{i+1}]$ and use $t^{p-1} \geq r_i^p/r_{i+1}$. This proves (C.33) with the displayed constants.

Read $x_1 = e \in \{-1, 1\}$ once. The required output-sign mean is the known function

$$f_e(z, w) = \tanh\left(\frac{u_D(z, w) + (e - z)/r_D}{4\sqrt{b_D(z, w) + \eta u_D(z, w)^2}}\right).$$

The full-domain head estimate in Theorem C.25 gives $|f_e| \leq \tanh(1/\sqrt{3}) < 4/5$, even at parameter pairs inconsistent with the observed sign. The same local complex argument just used applies to the head: its numerator has absolute value at most three, its squared denominator is at least $1/12$, and changing each variable by h changes its tanh argument by less than $1/8$. The partition chain rule and (C.33) therefore bound its derivatives of orders two through six by $A_{k,s,\eta} r_D^{k\nu}$, with explicit constants obtained from the displayed C_k and $2C_0 k^k \max\{2 + C_1, C_2, \dots, C_k\}^k$.

Convert the means to Bernoulli biases by $p_1 = (1 + z)/2$, $p_2 = (1 + w)/2$; this multiplies the order- k derivative bounds by 2^k . The two-variable corrected Bernstein lemma, Lemma A.62, requires known rational derivative budgets. Enclose $k\nu \log_2 r_D$ by certified logarithm and exponential arithmetic, and round each displayed derivative bound upward to a dyadic rational within a fixed factor. This costs $O_s(B^4)$ work and gives budgets of $O_s(B)$ encoding length. The lemma applies with base size $m_0 = O_{s,\eta}(r_D^{2\nu})$. Each source bit draws a uniformly random feature or carrier coordinate, using at most one input probe. The two streams use independent coordinate draws. They are independent conditional on every fixed input, including the already observed x_1 . The returned sign has exactly the desired mean. Alternatively, read all $2N$ relevant coordinates and evaluate the two known means directly. This proves the upper query bound.

For clarity, the coefficient computations can be implemented directly without an uncharged function oracle. Evaluate the scalar recurrence and its bivariate derivatives through order four by the chain rule, using certified dyadic interval arithmetic. There are $O(D + 1)$ fixed-size derivative operations. Local derivative coefficients are bounded by the constants above. Here is an explicit form of the error estimate used for guards. Let $e_k(j)$ bound the error in the order- k derivative tensor, for $0 \leq k \leq 4$, and suppose every local update has absolute rounding error at most ϵ . Clip only the approximate state value to $[-1, 1]$, which cannot increase its error. On the known derivative ranges, the product rule gives

$$e_k(j+1) \leq (1 + \nu q_j + C_u q_j / r_j) e_k(j) + C_s q_j \sum_{l < k} r_j^{(k-l)\nu} e_l(j) + \epsilon.$$

For $k = 0$ the sum is empty. The constants are obtained by differentiating the finite partition formulas; they use local derivatives through order five, already bounded above. Induction on k , followed by (C.32) and the integral estimate with $p = k\nu$, gives

$$e_k(j) \leq A_{k,s}(D+1)\epsilon r_j^{\nu(k+1)}.$$

The rounding-error sum contributes at most $M(D+1)\epsilon r_j^\nu$; each lower-order forcing term has the displayed exponent $(k+1)\nu$, which proves this claim. The same estimate applies to the head derivatives. Choosing $\epsilon \leq 2^{-t}/[C_s(D+1)r_D^{5\nu}]$ both gives the requested final accuracy and closes the assumption that the computed derivatives lie within one of their known bounds. Evaluating the fixed number of local operations to an additional $O_s(\log(r_D+2))$ guard bits makes their aggregate error at most ϵ . The resulting guard length is $O_s(\log(D+2) + \log(r_D+2))$, already included in $O_s(B)$. The known constant c is enclosed to the same accuracy. This gives $O_s((D+1)(B+t+\text{bits}(p_1, p_2))^4)$ bit work per value or derivative evaluation.

A Bernstein correction at scale m requires one scalar hypergeometric sum of $O(m)$ such evaluations. Its normalized coefficient needs $O_s(B + \log(m+2))$ initial precision; further precision has a geometric tail under the fresh uniform comparison. The correction masses are $O_s(m_0^2/m^2)$ on dyadic levels, so summing the resulting expected costs gives $O_s((D+1)m_0B^4)$. Reading all inputs instead costs $O_s((n+D+1)B^4)$. These two implementations give the displayed finite bit bound, including all scalar decisions and refinement.

For the lower bound, repeat Theorem C.25 on the subdomain with all carrier signs equal to -1 , now retaining $A_0 = \eta c^2$ and $A_1 = 4 + 2\eta c(1-c)$ in its normalized-gain comparison. This gives exponent $2\nu_\eta$ with the same fixed readout and a full-cube floor at least $1/12$. Finally $\eta \in [1/4, 1/2]$ and $\nu_\eta \geq \sqrt{2}s/c - 1 > 0$. All constants in both proofs are bounded uniformly on this compact parameter interval. For example, use $\eta c^2 \geq c^2/4$, $\nu_\eta \geq \sqrt{2}s/c - 1$, and $\nu_\eta \leq 2s/c - 1$ in every prefactor. Thus the constants may be chosen to depend only on s , as claimed. $\square$

C.9 A finite-bit oracle for the carrier family

The direct implementation in Theorem C.26 has a factor $D+1$ in its bit work. This subsection removes that factor by preprocessing the two-mean function, which gives the bit bound in Theorem 8.3.

Theorem C.27 (The carrier family's bit cost). *Under the hypotheses of Theorem C.26, weight-only preprocessing of time and storage $O_s((D+1)n^2B^6)$ gives an exact sampler with expected online bit work*

$$O_s\left(\min\{r_D^{2\nu_\eta}, n\}B^4\right).$$

The lower query bound of that theorem therefore matches this bit bound up to the displayed factor.

Proof idea. The normalized recurrence has a complex neighborhood whose radius shrinks as $r_D^{-\nu_\eta}$. Preprocessing covers the two unknown means with Taylor patches at that scale. A patch contains only polynomially many precision bits, since the recurrence propagates rounding error through factors $1 + O_s(q_j)$. Cached values answer almost every factory comparison. An unusually close comparison recomputes the recurrence; its probability pays for the depth.

Proof. Write $r = r_D$, $\nu = \nu_\eta$, and retain h, C_0, C_u, M from the preceding proof. We first establish a uniform complex neighborhood. Fix a real parameter pair (z_0, w_0) and its real trajectory. Perturb each parameter by at most δ in the complex plane, and let e_j be the absolute perturbation in u_j . The local derivative bounds and Taylor's formula give, as long as $e_j + 2\delta \leq h/2$,

$$e_{j+1} \leq L_j e_j + C_0 q_j \delta + C_2 q_j (e_j + 2\delta)^2, \quad L_j = 1 + \nu q_j + C_u q_j / r_j,$$

where a fixed $C_2 = 100C_0$ suffices. These bounds follow either by integrating the second derivative along the complex segment or by its convergent local Taylor series. Enlarge C_0 by two if needed to cover the sum of the direct parameter derivatives.

Set $K = \max\{2, 2C_0/\nu\}$ and $d_j = e_j + K\delta$. Since $L_j - 1 \geq \nu q_j$,

$$d_{j+1} \leq L_j d_j + C_2 q_j d_j^2.$$

Let $P_j = \prod_{i < j} L_i$. The preceding product estimate gives $P_j \leq M r_j^\nu$. For $v_j = d_j/P_j$ this implies

$$v_{j+1} \leq v_j + C_2 q_j P_j v_j^2, \quad \sum_{j < D} q_j P_j \leq M r^\nu / \nu.$$

The last inequality follows from $\sum q_j r_j^\nu \leq (r^\nu - 1)/\nu$. The elementary bootstrap for $v_{j+1} \leq v_j + a_j v_j^2$ says $v_j \leq 2v_0$ whenever $4v_0 \sum a_j \leq 1$: under this bound the total increment is at most $4v_0^2 \sum a_j \leq v_0$. Taking a strict inequality first and then continuity handles the boundary.

Choose a positive rational constant c_s below

$$\min \left\{ \frac{\nu}{16C_2 M(K+1)}, \frac{h}{16M(K+1)}, \frac{h}{16} \right\}, \quad \delta = c_s r^{-\nu}.$$

Uniform choices in $\eta \in [1/4, 1/2]$ are possible because $\nu \geq \sqrt{2}s/c - 1 > 0$. Then $e_j \leq 2M(K+1)\delta r_j^\nu \leq h/8$. The local analytic extensions remain valid at every step. This bootstrap, first on a slightly smaller polydisk and then by continuation, proves that every intermediate state and the head are holomorphic on the parameter polydisk of radius δ . Their moduli there are at most two.

Choose a dyadic mesh spacing a with $\delta/32 < a \leq \delta/16$. Cover $[-1, 1]^2$ by grid squares with a real corner as center; truncate boundary squares at one. There are $O_s(r^{2\nu})$ squares. At a center (z_*, w_*) , use scaled coordinates $z = z_* + 2aU$, $w = w_* + 2aV$. A point in its square has $|U|, |V| \leq 1/2$, while all trajectories are holomorphic and bounded by two on $|U|, |V| \leq 2$. If b_{kl} are Taylor coefficients of any such state or head, Cauchy's estimate gives

$$|b_{kl}| \leq 2^{1-k-l}.$$

Retain the rectangle $0 \leq k, l \leq J$. Its value error and its derivatives of total order at most four, with respect to U, V , are bounded by $C(J+2)^4 4^{-J}$. Conversion to the original parameters costs at most $(2a)^{-4}$. Thus degree $J = O_s(P + \log(r+2))$ and coefficient precision $S = P + 4\lceil \log_2(1/a) \rceil + O(\log(J+2))$ suffice for absolute error 2^{-P} in the value and the derivatives f_{11}, f_{22}, f_{1122} needed by the two-variable correction. Changing from means to Bernoulli biases changes these estimates only by fixed factors.

We justify preprocessing these coefficient arrays without an exponential-in-depth precision allowance. For polynomials truncated to the rectangle, use the sum of absolute coefficients as norm. Truncation preserves submultiplicativity. The exact state jets have norm at most eight. All local analytic functions and their required inverses have bounded coefficient norm $O_s(1)$, by the same radius-two Cauchy estimate. In particular, the squared denominator, its inverse, and its inverse square root have such bounds; the argument remains in the right half-plane bounded away from zero. The exponential and its denominator in $\tanh$ also have bounded norms depending on s . Neumann series, or the convergent analytic power series about the exact jet, consequently show that each local operation is $O_s(1)$ -Lipschitz in this coefficient norm whenever the input error is sufficiently small. For inverse square roots, this follows explicitly by writing $(V + E)^{-1/2} = V^{-1/2}(1 + V^{-1}E)^{-1/2}$.

For explicit error budgets, one may take

$$K_s = 2^{100(\lceil s \rceil + 2)}, \quad C_s = K_s^{100}, \quad d_{\text{err}} = 12.$$

Indeed, on the radius-two parameter polydisk the squared denominator has modulus at most ten and real part at least $1/24$. Its square root, inverse, and inverse square root have modulus at most 24. The argument of $\tanh$ has modulus at most $10s$ and imaginary part at most $1/4$. Thus its exponentials, their reciprocals, and the inverse denominator of $\tanh$ have modulus bounded by $2e^{20s} + 24$. Cauchy's coefficient bound multiplies these supremum bounds by at most four. The displayed K_s bounds all these coefficient norms; the elementary product and inverse identities then give the stated larger C_s as a common local Lipschitz and residual-error constant.

At M_* working bits, the polynomial procedures below contribute at most $C_s(J+1)^{12}2^{-M_*}$ error. The normalized update therefore satisfies

$$E_{j+1} \leq (1 + C_s q_j) E_j + C_s(J+1)^{12}2^{-M_*}.$$

Its product is at most r^{C_s} , because $\sum q_j \leq \log r$. It follows that

$$E_D \leq C_s(D+1)r^{C_s}(J+1)^{12}2^{-M_*}.$$

For example, take an integer M_* at least

$$S + \lceil \log_2(C_s(D+1)) \rceil + \lceil C_s \log_2 r \rceil + 12\lceil \log_2(J+1) \rceil + \lceil \log_2(64K_s) \rceil + 16.$$

Certified dyadic upper enclosures of these logarithms determine such an integer without testing transcendental equality. This is $S + O_s(\log(D+2) + \log(r+2) + \log(J+2))$. It ensures all promised final errors and, by induction, keeps the intermediate errors inside the neighborhoods just used. The known constant $c = \tanh 1$ is enclosed to the same accuracy. Stable evaluation of the known scalar coefficient b_j uses

$$b_j = \eta c^2 - \frac{2\eta c(c+w)}{r_j} + \frac{4 - \eta z^2 + 2\eta cw + \eta c^2}{r_j^2},$$

so its coefficient magnitudes remain bounded.

For completeness, the polynomial computations take $O(J^4)$ coefficient operations per layer. Multiplication and inversion use the elementary coefficient recurrences. Let $\mathcal{E} = U\partial_U + V\partial_V$ be the Euler operator, which multiplies a coefficient by its total degree. It satisfies the product rule in the rectangular quotient algebra and its inverse on zero-constant polynomials has norm at most one. For $Y = \exp P$, solve $\mathcal{E}Y = Y\mathcal{E}P$ with $Y_{00} = e^{P_{00}}$; explicitly, for nonzero multi-index λ ,

$$|\lambda|Y_\lambda = \sum_{0 < \mu \leq \lambda} |\mu|P_\mu Y_{\lambda-\mu}.$$

For $Y = V^{-1/2}$, compute V^{-1} and $H = -V^{-1}\mathcal{E}V/2$, then solve $\mathcal{E}Y = YH$ with $Y_{00} = V_{00}^{-1/2}$. Here $H_{00} = 0$. Both recurrences divide by total degrees between one and $2J$, and use $O(J^4)$ operations.

These recurrences have polynomial residual error at fixed precision. For the exponential, a residual R obeys $\mathcal{E}(e^{-P}Y) = e^{-P}R$; for the inverse square root it obeys $\mathcal{E}(V^{1/2}Y) = V^{1/2}R$. Apply $\mathcal{E}^{-1}$ after removing the constant term, then multiply by the exact solution. The coefficient norms bounded above show that the output error is at most a fixed C_s times the residual norm and the constant-term error. Each residual coefficient sums $O(J^2)$ rounded products, of magnitude at most $O_s(J)$, and there are $O(J^2)$ coefficients. The exponent $d_{\text{err}} = 12$ therefore dominates all residual accumulation, including the fixed number of intermediate multiplications and inversions. Inversion itself uses the identity $VY - 1 = R$ and the bound on V^{-1} . A simultaneous induction over total coefficient degree keeps computed coefficient norms within one of their stated bounds: apply the same residual identity in the further quotient that discards all larger total degrees, and use the precision margin above to make the error smaller than $1/(64K_s)$. This closes the bounded-intermediate assumption used in charging products. Direct fixed-precision arithmetic and certified scalar exponential and square-root routines therefore take $O_s((D+1)B^6)$ bit work per patch when $P = O(B)$. The Taylor degree, coefficient precision, and all guard lengths are $O_s(B)$. Nested evaluation of the stored polynomial and its required derivatives at rational arguments uses $O_s((B + \text{bits}(z, w))^4)$ bit work.

Compute a rational Λ with $r^{2\nu} \leq \Lambda \leq 2r^{2\nu}$ by certified logarithm and exponential arithmetic. If $\Lambda \leq n$, use the Taylor table; otherwise use the finite lattice table described below. This decision involves only rational comparisons.

In the first case $r^{2\nu} \leq n$. Precompute both head patch tables at precision

$$\epsilon = 2^{-P}, \quad P = 32 + 2\lceil \log_2(D+2) \rceil.$$

This costs $O_s((D+1)nB^6)$. A corrected Bernstein coefficient is a fixed linear combination of the value and the three derivatives above, followed by at most one hypergeometric average. The polynomial derivative

prefactors have bounded magnitude and the averaging weights are nonnegative and sum to one. Hence cached evaluations determine the entire coefficient to error $C\epsilon$, independently of its scale. Rounding the rational prefactors and the averaging computation to the same final tolerance uses the displayed polynomial bit budget. For the hypergeometric average, use the mode-centered recurrence in the proof of Theorem B.24: set the mode weight to one and generate adjacent weights by rational ratios. All weights are at most one, their sum is at least one, and $O(\log(m+2))$ guard bits suffice. This performs $O(m)$ arithmetic steps without constructing binomial integers of length proportional to m .

Let $m_0 = O_s(r^{2\nu})$ be the factory base size and let $c_m = \bar{A}/(4m^2)$ be a correction mass, on dyadic levels $m = 2^k m_0$. The known constant $\bar{A} = O_s(m_0^2)$ is the sum of the derivative budgets from the two-variable factory. Use the cached enclosure first when comparing the normalized coefficient with its fresh uniform random number. Conditional on selecting this correction, a direct recurrence evaluation is necessary only with probability at most $C \min\{1, \epsilon/c_m\}$. If $c_m < \epsilon$, simply use the direct method immediately. Thus its unconditional probability is at most $C \min\{c_m, \epsilon\}$. We charge its refinements by unconditional interval measures. If $c_m \geq \epsilon$, after t additional bits the unresolved set of uniform values has measure at most $C(\epsilon/c_m)2^{-t}$. Computing that enclosure costs $O_s((D+1)m(B+\log(m+2)+P+t)^4)$. Multiply by the branch mass c_m and sum over $t \geq 0$. The resulting unconditional contribution is $O_s((D+1)m\epsilon(B+\log(m+2)+P)^4)$. If $c_m < \epsilon$, start the direct computation at absolute precision proportional to c_m ; the same argument gives c_m in place of ϵ . This argument does not assume that the work and the cache-miss event are independent. Since $P = O(B)$, splitting the geometric sum at $m \asymp \sqrt{\bar{A}/\epsilon}$ gives

$$\sum_{m=2^k m_0} \min\{\bar{A}/m^2, \epsilon\} m(B+\log(m+2))^4 = O_s(m_0 \sqrt{\epsilon} B^4).$$

Here $\log m_0$ and $\log(1/\epsilon)$ are $O_s(B)$; the remaining polynomial factors have finite geometric sums. Since $(D+1)\sqrt{\epsilon} \leq 1$, all direct fallbacks cost $O_s(m_0 B^4)$ in expectation. The base comparison costs less: its fallback probability is $O(\epsilon)$. The cached work and input probes themselves have expected cost $O_s(m_0 B^4)$ by the source-cost accounting of the factory. Every comparison refines certified enclosures until it decides, so this modification preserves the exact output law.

In the second case $r^{2\nu} > n/2$. Both unknown means belong to the known lattice $\{-1, -1 + 2/N, \dots, 1\}$. Precompute the two head values at all $(N+1)^2$ pairs, each to error ϵ . Direct scalar recurrence evaluation costs $O_s((D+1)n^2 B^4)$ for this table. Read the $2N$ relevant input signs, retrieve the appropriate enclosure, and compare with a fresh uniform. With probability $O(\epsilon)$ refine by direct recurrence. Its expected extra cost is $O_s((D+1)\epsilon B^4) = O_s(B^4)$, and the total online cost is $O_s(nB^4) = O_s(\min\{r^{2\nu}, n\}B^4)$. Both preprocessing alternatives meet the claimed budget. Upward rational enclosures decide all mesh sizes and derivative budgets, as in the preceding theorem. $\square$

C.10 GELU and sigmoid-gated rows

The additive composition argument applies to updates other than $\tanh$. The following elementary constructions record the constants for two common feed-forward choices. They use fresh randomness for each factor in a product. Every closed gate returns a fresh fair sign.

Suppose normalized-stream coordinates satisfy $|a_i| \leq G$, and fresh signs of means a_i/G are available. Let u and v be affine rows with biases of magnitude at most one and row norms at most s . Put $A = 1 + sG$ and $\rho = sG/2$. A sign of mean u/A or v/A uses at most one normalized-stream request, by a signed row mixture. Write

$$\sigma(v) = \frac{1}{1 + e^{-v}}, \quad \text{GELU}(v) = \frac{v}{2} \left(1 + \text{erf}(v/\sqrt{2}) \right).$$

Lemma C.28 (Exact GELU and gated-product interfaces). *The following updates have exact normalized sign factories with the stated output radii and expected normalized-stream request counts:*

update	radius	expected requests, at most
$\text{GELU}(v)$	A	$3/2 + A^2/2$
$u \text{GELU}(v)$	A^2	$5/2 + A^2/2$
$uv\sigma(v)$	A^2	$2 + 2\rho(1 + \rho)$

Their expected auxiliary bit costs are respectively $O((1 + A^2)B^4)$, $O((1 + A^2)B^4)$, and $O((1 + \rho)^2 B^4)$. All constants are absolute. A following affine projection with bias bound b_0 and row norm s_0 changes a radius U to $b_0 + s_0 U$ and needs at most one such update request per normalized projected sign.

Proof idea. Write GELU as a linear term plus a linear term times an error function. Independent signs realize these products exactly. For SwiGLU, a logistic gate selects a product of two fresh affine-row signs; the final projection is a signed mixture.

Proof. Let $z = v/A$. Lemma C.7, with $c = A/\sqrt{2}$, gives an erf sign of mean $\text{erf}(v/\sqrt{2})$ and source count at most $1 + A^2$. Each of those sources is one fresh v/A row sign. With probability one half, return a v/A sign. With probability one half, return the product of an independent v/A sign and the erf sign. Its mean is

$$\frac{z}{2} + \frac{z}{2} \text{erf}(v/\sqrt{2}) = \frac{\text{GELU}(v)}{A}.$$

The expected source count is $1 + (1 + A^2)/2$. The finite-bit erf implementation uses one Gaussian and a Poisson mean at most $A^2/2$; the rational-plus-residual Poisson construction in Lemma C.12 gives the displayed auxiliary bound. Multiply this output by a fresh u/A sign for the gated GELU row.

For the last row, the biased tanh factory produces a sign of mean $\tanh(v/2)$ with at most $2\rho(1 + \rho)$ normalized-stream requests. Its positive outcome is a gate of probability $(1 + \tanh(v/2))/2 = \sigma(v)$. On an open gate, multiply independent signs of means u/A and v/A . On a closed gate, return a fair sign. The mean is $uv\sigma(v)/A^2$. Its two additional source requests and the tanh factory's local bound prove the claimed costs.

Finally, a signed mixture over projection-row coefficients and a known bias sign gives the normalized affine projection, with a fair-sign completion if its supplied radius exceeds the exact row bound. Every mixture is finite, every component factory terminates almost surely, and each product uses independent fresh calls. Conditional expectations prove exactness for every fixed input stream. $\square$

Under a relative energy certificate, G is a fixed normalization radius, so all entries of the table are constants at fixed row norms. Theorem C.15 therefore still yields a polynomial depth bound, with the corresponding update radius and source count. The presence of these activations does not itself certify the energy floor or bound the learned matrix norms.

C.10.1 Reducing a supplied radius

Lemma C.29 (Amplifying a sign mean with a fixed margin). *Let $C \geq 1$ be rational. From independent signs of unknown mean z with the promise $|z| \leq 1/(2C)$, one can generate an exact sign of mean Cz using at most $722C(C + 1) \leq 1444C^2$ source requests in expectation and $O(C^2 B^4)$ expected auxiliary bit work. The factory terminates almost surely.*

Proof idea. Two affine changes of Bernoulli bias turn sign amplification into two linear factories with explicit slack. Their exact affine identities determine the output bias, and conditional charging multiplies their source budgets.

Proof. The case $C = 1$ returns a source sign. Otherwise let

$$p = (1 + z)/2, \quad a = (C - 1)/(2C), \quad D = \frac{1}{1 - a} = \frac{2C}{C + 1}, \quad E = \frac{C + 1}{2}.$$

Apply the linear factory to a $(1 - p)$ -coin with multiplier D . The promise gives

$$D(1 - p) \leq 1 - \frac{1}{2(C + 1)}.$$

Complement the result. Its bias is $r = 1 - D(1 - p) = (p - a)/(1 - a)$. A second linear factory with multiplier E gives bias

$$Er = C(p - a) = \frac{1 + Cz}{2} \in [1/4, 3/4].$$

Read this bit as a sign. Huber’s 9.5 bound gives at most $38C$ source requests for an r -coin, and at most $19(C + 1)$ r -coins for the second factory. Predictable charging yields their product, $722C(C + 1)$. The first factory has expected auxiliary work $O(CB^4)$; the second has $O(CB^4)$. Charging the expected $O(C)$ first-stage invocations proves the total $O(C^2B^4)$ bound. Exactness follows from the displayed affine identities. Almost-sure termination follows from the two linear factories and their almost surely finite call counts. $\square$

Corollary C.30 (Reducing a supplied normalization radius). *Suppose a source returns signs of mean F/R , where a separately certified bound gives $|F| \leq U$. There is an exact sign sampler for $F/(2U)$ with at most*

$$\max\{1, 361(R/U)^2\}$$

expected source invocations and $O((1 + (R/U)^2)B^4)$ auxiliary bit work. Here $R, U > 0$ are finitely encoded rationals.

Proof idea. A supplied radius can be increased by thinning. When it must be decreased, the separate magnitude certificate provides the fixed margin required by the sign-amplification lemma.

Proof. If $R \leq 2U$, use a known thinning coin of probability $R/(2U)$ and return a fair sign on the closed branch. If $R > 2U$, apply Lemma C.29 with $C = R/(2U) > 1$ and $z = F/R$. Its premise holds since $|F|/R \leq U/R = 1/(2C)$. The input-count bound $1444C^2 = 361(R/U)^2$ proves the claim. $\square$

C.10.2 Depth lower bounds for GELU and SwiGLU

We now carry the relative-floor lower construction of Theorem C.20 over to GELU and SwiGLU blocks. Lemma C.28 and Corollary C.30 supply exact update factories at the declared radii. Corollary C.32 gives the GELU and SwiGLU case of (8.5) for every dyadic step schedule.

Theorem C.31 (A depth obstruction with GELU or SwiGLU). *Fix a dyadic $s \geq 2$, and set*

$$\delta = s - 1, \quad C_* = \frac{s}{4(s - 1)}, \quad \gamma_* = \frac{1}{192s^2}.$$

There is a positive constant c_s^{ff} , depending only on s , with the following property. For every $n \geq 4$, dyadic $0 < \alpha \leq 1$, and D with $\alpha D \geq 4$, there is a one-position pre-norm additive network of width n and depth D whose feed-forward activation is either exact GELU or SwiGLU, with all affine row norms at most s , biases at most one, zero attention, fixed stabilizer 3, and step α on every sublayer, for which

$$\mathbb{E}Q \geq c_s^{\text{ff}} \min\{(1 + \alpha D)^{2\delta}, n\}.$$

An explicit value for the constant is given in the proof. Preprocessing independent of the fresh query is arbitrary. The output is a fixed final RMSNorm₃ followed by logits $\pm \mathcal{N}_3(h_D)_1/4$. These logits are bounded by one on the entire input sign cube. Every normalization has the relative floor $v_j \geq \gamma_ R_j^2$ for the valid supplied radii $R_j = 1 + 4s\alpha j$. Exact update factories at these radii have costs depending only on s .*

Proof idea. Both activations satisfy an exact antisymmetric identity that recovers a linear signal. A constant hidden unit drives the carriers. This embeds the normalized-gain lower construction into each feed-forward architecture while preserving its full-cube energy floor.

Proof. Let N be the largest power of two at most $n/2$, and put $\eta = N/n \in (1/4, 1/2]$. The first N coordinates are features and the remaining coordinates are carriers. Write $\phi(v) = \text{GELU}(v)$ or $\phi(v) = v\sigma(v)$. Both choices satisfy

$$\phi(v) - \phi(-v) = v, \quad \frac{1}{2} < \phi(1) < 1.$$

The first identity follows from the oddness of the error function for GELU, and from $\sigma(v) + \sigma(-v) = 1$ for SiLU. Use three hidden units with preactivations

$$\frac{s}{N} \sum_{i \leq N} \mathcal{N}_3(h)_i, \quad -\frac{s}{N} \sum_{i \leq N} \mathcal{N}_3(h)_i, \quad 1.$$

Every feature projection row is $(1, -1, 0)$ and every carrier row is $(0, 0, -1)$. Extra hidden coordinates may be zero. In the SwiGLU case $uv\sigma(v)$, the displayed preactivations are the gate arguments v , and the ungated factor u of each of these three hidden units is the constant one. Thus the SwiGLU units equal ϕ exactly. The input rows have norms at most s , and the projection rows have norms at most two, which is allowed because $s \geq 2$.

For the lower-bound distribution let the features be independent unknown signs x_i and set every initial carrier to -1 . Put $M = N^{-1} \sum_i x_i$ and $c = \phi(1)$. After j blocks the feature coordinates are $x_i - M + F_j(M)$ and all carriers are $C_j = -(1 + \alpha jc)$, where $F_0(z) = z$ and

$$F_{j+1}(z) = F_j(z) + \alpha \frac{sF_j(z)}{\sqrt{3 + \eta(1 - z^2 + F_j(z)^2) + (1 - \eta)C_j^2}}.$$

This follows from the exact difference identity above. These recurrences are odd in z , and $F_j(z) \geq z \geq 0$ when $z \geq 0$.

For every input, the mean of the feature coordinates has magnitude at most $\sqrt{v_j/\eta}$, by Cauchy–Schwarz and the definition of the RMS denominator. The feature update therefore has magnitude at most $s/\sqrt{\eta} \leq 2s$, and the carrier update has magnitude below one. In particular, $|h_{j,i}| \leq 1 + 2s\alpha j$. The coordinate update bound $2s$ is valid on the entire sign cube.

Put $r_j = 1 + \alpha j$. For arbitrary initial carrier signs their values are $e_i - \alpha jc$, with $c > 1/2$. The full-cube calculation in Corollary C.22, using the carrier fraction $1 - \eta \geq 1/2$ and the stabilizer 3, gives

$$v_j \geq r_j^2/12.$$

Because $R_j = 1 + 4s\alpha j \leq 4sr_j$, this implies the stated $v_j \geq R_j^2/(192s^2)$. To obtain the exact update factory at radius $4s$, first use Lemma C.28 and its affine projection to obtain a normalized sign with some fixed radius depending only on s and the supplied floor γ_* . The actual update bound is $2s$. Corollary C.30 supplies an exact sign at radius $4s$, still with costs depending only on s . This accounts for the declared radii without requiring the generic activation bound to equal the true update range.

For the growth comparison, normalize the feature mean by r_j , writing $u_j = F_j/r_j$ and $a_j = \alpha/r_{j+1}$. After the burn-in $j_0 = \lceil 3/\alpha \rceil$, one has $4 \leq r_{j_0} \leq 5$ and $u_{j_0}(z) \geq z/5$. The nonsignal part of the normalized denominator satisfies

$$b_j(z) \leq \frac{7}{2r_j^2} + \frac{3}{4} \leq \frac{31}{32} < 1.$$

Consequently, for $z \geq 0$,

$$u_{j+1} \geq (1 - a_j)u_j + a_j \frac{su_j}{\sqrt{1 + u_j^2/2}}.$$

Apply the convexity comparison used in Theorem C.20, now with $g_j = 1 + \delta a_j$ and $d_j = a_j s/(2g_j)$. Its reciprocal-square sum has coefficient $C_* = s/(4\delta)$. Thus, for $G = \prod_{j=j_0}^{D-1} g_j$,

$$u_D(z) \geq \frac{G(z/5)}{\sqrt{1 + C_*(G^2 - 1)(z/5)^2}}, \quad G \geq e^{-\delta^2/8} \left(\frac{1 + \alpha D}{6} \right)^\delta.$$

The latter product bound is uniform in α .

It remains to control the fixed output readout. On every sign input the feature decomposition $h_i = x_i - M + F$ still holds, even when the initial carrier signs vary. Hence $|h_i| \leq |F| + 2$ and $v \geq 3 + \eta F^2 \geq 3 + F^2/4$.

The readout calculation in the proof of Theorem C.25 then shows that the absolute logit is at most $1/\sqrt{3} < 1$. On the lower-bound subdomain, conditional on $M = z$, set $L(z) = 4\sqrt{v_D(z)}$. The coordinate bound gives

$$L(z) \leq 4\sqrt{3 + (2sr_D)^2} \leq 8(s+1)r_D.$$

The conditional shift proof in Corollary C.21 applies verbatim: the actual and intermediate head arguments lie in $[-1, 1]$, $\text{sech}^2(1) > 1/4$, and

$$\bar{g}(z) \geq \frac{F_D(z)}{4L(z)} \geq \frac{u_D(z)}{32(s+1)} \quad (z \geq 0).$$

Weighted Jensen and the Rademacher fourth-moment bound, as in the proof of Theorem C.20, now give

$$\begin{aligned} H'(0) &\geq \frac{G}{160(s+1)\sqrt{1 + 3C_*G^2/(25N)}}, \\ \mathbb{E}Q &\geq \frac{\min\{G^2, N\}}{25600(s+1)^2(1 + 3C_*/25)}. \end{aligned}$$

Substitute the bound for G and use $N > n/4$. This proves the theorem with

$$c_s^{\text{ff}} = \frac{e^{-\delta^2/4} 6^{-2\delta}}{102400(s+1)^2(1 + 3C_*/25)}.$$

□

Corollary C.32 (GELU and SwiGLU with a prescribed step schedule). *Theorem C.31 also holds for every depth $D \geq 0$ and every prescribed dyadic schedule $a_j \in [0, 1]$, with αD replaced by $\tau_D = \sum_{j < D} a_j$ and the condition $\alpha D \geq 4$ dropped, the supplied radius replaced by $R_j = 1 + 4s \sum_{k < j} a_k$, and a positive lower-bound prefactor depending only on s . Its value is given in the proof.*

Proof idea. The activation identities and output-radius bounds hold at every step size. Replacing elapsed depth by cumulative carrier displacement therefore transfers the scheduled normalized-gain argument directly. A small total step needs only the head bound.

Proof. The states and relative floors depend on the step schedule through the cumulative carrier displacement and the feature recurrence. Put $\tau_j = \sum_{k < j} a_k$ and $r_j = 1 + \tau_j$. If $\tau_D \geq 4$, the proof of Corollary C.23 applies with the same linear-update comparison constant C_* . The head and source radius arguments in Theorem C.31 are unchanged. The resulting prefactor is

$$\frac{e^{-\delta/2 - \delta^2/4} 5^{-2\delta}}{102400(s+1)^2(1 + 3C_*/25)}.$$

If $\tau_D < 4$, no growth comparison is needed. The head bound in Theorem C.31, with $L(z) \leq 8(s+1)r_D$ and $F_D(z) \geq z$, gives $\bar{g}(z) \geq z/(32(s+1)r_D)$ for $z \geq 0$. Hence $H'(0) \geq 1/(32(s+1)r_D)$, and since $r_D < 5$,

$$\mathbb{E}Q \geq H'(0)^2 > \frac{1}{25600(s+1)^2}.$$

Because $\min\{(1 + \tau_D)^{2\delta}, n\} < 5^{2\delta}$, the displayed prefactor also covers this case. □

D Instance complexity

This appendix studies the optimal query cost of one fixed network and proves the obstruction quoted in Section 11. Section D.1 shows that absolute weights and one-parameter profiles do not determine this cost, and gives a multivariate information certificate. Section D.2 shows that, at every fixed row-norm bound

above one, a polynomial-time approximation of the optimum within a polylogarithmic factor would imply $P = NP$. Section D.3 characterizes the optimum exactly by a flow program of exponential size.

For a network W , let $Q^*(W)$ denote the infimum of the worst-input expected number of distinct probes among all almost surely terminating exact samplers for W . This definition permits unlimited weight-only preprocessing and ignores internal arithmetic. In particular, a network with output identically zero has $Q^*(W) = 0$.

None of these results gives an efficiently computable instance bound. The examples of Section D.1 exclude bounds computed from absolute weights or from a single profile; a bound computed from the complete signed weights remains possible. The certificate and the flow program are defined from the complete output table, which has 2^m entries for m inputs. We give no efficient way to evaluate either, and no matching upper bound in terms of either. The hardness results concern approximation of $Q^*(W)$ for each individual network and are conditional on $P \neq NP$, or on $NP \not\subseteq BPP$ for randomized algorithms. These results leave open a useful efficiently computed upper bound, the sharp critical depth exponent, and sharper instance bounds within the unit-norm class.

D.1 Instance query complexity and input influence

The norm threshold concerns the worst network in a class. A useful bound for a particular network would also recognize cancellation and the direction in which its inputs affect the output. The following examples show why both are necessary.

Proposition D.1 (Equal weight magnitudes, different sampling costs). *Let $D \geq 2$, let $m = 2^k \leq n$ with $m \geq 2$, and let $a = A2^{-r} > 0$, where A, r are nonnegative integers. Suppose $b \geq r + k + 1$. There are two width- n , depth- D networks with zero biases, row norms at most a , and identical matrices $(|w_{\ell,ij}|)$ at every layer, whose outputs are respectively*

$$0 \quad \text{and} \quad F_{a,D} \left(\frac{1}{m} \sum_{i=1}^m x_i \right).$$

The first output can be sampled with no input probes and one fair bit. Every exact sampler for the second satisfies the lower bound (A.5) and, if $a > 1$, the lower bound (A.6), with this m .

Proof idea. Duplicate one signal along two paths and subtract them at the output. Negating one path changes cancellation into addition while preserving every absolute weight. The amplified mean-chain lower bound separates the two sampling costs.

Proof. Only neurons 1 and 2 can be nonzero before the output layer. In both networks the first row of the first layer has weights a/m on the first m input coordinates. In the first network, the second row is identical. In the second network, every weight in that row is negated. Every other first-layer entry is zero. For $2 \leq \ell \leq D-1$, set $w_{\ell,11} = w_{\ell,22} = a$, and set all other weights to zero. At layer D , set

$$w_{D,11} = a/2, \quad w_{D,12} = -a/2,$$

and again set every other entry to zero. When $D = 2$, the intermediate range of layers is empty.

Write $u = m^{-1} \sum_{i=1}^m x_i$. In the first network the two active neurons at layer $D-1$ both equal $F_{a,D-1}(u)$, so the last preactivation is zero. In the second network their values are $F_{a,D-1}(u)$ and $-F_{a,D-1}(u)$, because $\tanh$ is odd. The last preactivation is therefore $aF_{a,D-1}(u)$. This proves the two output identities.

The construction changes signs only. Its nonzero row norms are a , and its parameters belong to the grid $2^{-b}\mathbb{Z}$: the only divisions of a are by $m = 2^k$ and by 2. The mean-chain lower bound in Theorem A.2, which holds for every power of two $m \leq n$ by Remark A.3, applies unchanged to the second network, regardless of the preprocessing. $\square$

For $D = \lceil \log_2 n \rceil$ and fixed dyadic $a > 1$, this is a polynomial separation. Thus row norms, absolute path products, and even the complete sequence of entrywise absolute weight matrices cannot determine instance

sampling cost within polylogarithmic factors. The output of the first network is a fair coin; a uniform algorithm can detect this displayed cancellation during preprocessing.

Proposition D.2 (A zero product profile can conceal a hard network). *Let $m \leq n$ be even, and define*

$$f(x) = F_{a,D} \left(\frac{1}{m} \sum_{i=1}^{m/2} x_i - \frac{1}{m} \sum_{i=m/2+1}^m x_i \right).$$

Whenever a and a/m belong to the parameter grid, this is the output of a legal width- n , depth- D network with row norms at most a and zero biases. Its optimal worst-input expected query cost equals that of the unsigned mean chain. Nevertheless, under independent input signs of common mean t ,

$$H_f(t) = \mathbb{E}_t f(X) = 0 \quad (-1 \leq t \leq 1).$$

Consequently all derivatives of this one-parameter profile vanish.

Proof idea. Flipping half the input signs preserves the query problem but changes the direction of its hard average. Exchanging the two halves leaves the common-mean input distribution invariant and negates the output, forcing that one-dimensional profile to vanish.

Proof. Use weights a/m on the first half of the first row and $-a/m$ on its second half. Subsequent distinguished rows have their sole nonzero weight equal to a . All other weights and all biases are zero. This realizes the displayed function.

Replacing x_i by $-x_i$ in the second half maps this function to the unsigned mean chain. A query to either input can be answered by one query to its image and a deterministic sign flip. The map is an involution, so it gives reductions in both directions without changing the number of probes.

To compute the product profile, exchange the first and second input halves. The common-mean product law is invariant under this permutation. The argument of $F_{a,D}$ is negated, and the function $F_{a,D}$ is odd. Hence $\mathbb{E}_t f(X) = -\mathbb{E}_t f(X)$. The profile is zero. It is a polynomial in t , so every derivative is also zero. $\square$

This example rules out every instance statistic that uses only the single profile $t \mapsto \mathbb{E}_t f(X)$, including all of its derivatives. It leaves the multivariate product extension available.

D.1.1 A multivariate information certificate

For $\mathbf{t} \in (-1, 1)^m$, let

$$\mathcal{H}(\mathbf{t}) = \sum_{x \in \{-1, 1\}^m} f(x) \prod_{i=1}^m \frac{1 + t_i x_i}{2}.$$

Let $q_i(\mathbf{t})$ be the probability that an arbitrary exact sampler probes coordinate i under this product law. Repeated queries can be answered from a cache. For a direction $v \in \mathbb{R}^m$, put

$$I_v(\mathbf{t}) = \sum_{i=1}^m \frac{v_i^2}{1 - t_i^2} q_i(\mathbf{t}).$$

Lemma D.3 (Directional transcript inequalities). *Every almost surely terminating exact sign sampler satisfies*

$$(v \cdot \nabla \mathcal{H}(\mathbf{t}))^2 \leq (1 - \mathcal{H}(\mathbf{t})^2) I_v(\mathbf{t}), \quad (\text{D.1})$$

$$|v^\top \nabla^2 \mathcal{H}(\mathbf{t}) v| \leq 2 I_v(\mathbf{t}). \quad (\text{D.2})$$

The assertions allow arbitrary weight-only preprocessing and require no moment bound on arithmetic time.

Proof idea. Perturb all input biases in a chosen direction and differentiate the transcript likelihood. Fresh-coordinate scores are orthogonal martingale increments. Their variance controls the first derivative, and the second likelihood derivative controls curvature.

Proof. Follow the finite decision-tree reduction used in Lemma A.17. On a neighborhood of zero, perturb the input mean vector to $\mathbf{t} + uv$. At $u = 0$, the first and negative second log-likelihood derivatives of a leaf are

$$Z = \sum_{i \text{ queried}} \frac{v_i X_i}{1 + t_i X_i}, \quad A = \sum_{i \text{ queried}} \frac{v_i^2}{(1 + t_i X_i)^2}.$$

A new coordinate retains its product law conditional on the past. Its score has conditional mean zero and conditional second moment $v_i^2/(1 - t_i^2)$. The same expression is the conditional mean of its contribution to A . There are at most m distinct probes, and therefore

$$\mathbb{E}Z = 0, \quad \mathbb{E}Z^2 = \mathbb{E}A = I_v(\mathbf{t}).$$

Finite likelihood differentiation gives

$$v \cdot \nabla \mathcal{H} = \mathbb{E}[YZ], \quad v^\top \nabla^2 \mathcal{H} v = \mathbb{E}[Y(Z^2 - A)].$$

Since $\mathbb{E}Z = 0$, the first expectation equals $\mathbb{E}[(Y - \mathcal{H})Z]$. Cauchy-Schwarz and $\text{Var}(Y) = 1 - \mathcal{H}^2$ prove (D.1). The inequalities $|Y| = 1$ and $|Z^2 - A| \leq Z^2 + A$ prove (D.2). $\square$

For a nonconstant f , the product extension lies in $(-1, 1)$ at every interior point. Taking directions with $v_i^2 \leq 1 - t_i^2$ gives the instance lower bound

$$\sup_x \mathbb{E}[Q \mid x] \geq \sup_{\mathbf{t} \in (-1, 1)^m} \max \left\{ \frac{\left(\sum_i \sqrt{1 - t_i^2} |\partial_i \mathcal{H}(\mathbf{t})| \right)^2}{1 - \mathcal{H}(\mathbf{t})^2}, \frac{1}{2} \sup_{v_i^2 \leq 1 - t_i^2} |v^\top \nabla^2 \mathcal{H}(\mathbf{t}) v| \right\}. \quad (\text{D.3})$$

The first term follows by choosing each v_i with the sign of $\partial_i \mathcal{H}$. The second term follows directly from (D.2). If f is constant, both terms are defined to be zero.

This certificate recognizes the signed mean chain: take $t_i = t$ in its positive half and $t_i = -t$ in its negative half, then differentiate in the corresponding signed direction. Computing the displayed extension by its definition entails 2^m network evaluations.

D.2 The complexity of approximating an instance optimum

The query optimum has a finite flow characterization, proved at the end of this appendix. Computing a close approximation within polynomial time is harder. The following reduction starts at the fixed row-norm bound eight, then extends to every fixed bound above one. The optimum $Q^*(W)$ is defined at the start of this appendix.

Proposition D.4 (The norm-eight reduction). *Suppose a deterministic algorithm, polynomial in the network description length, outputs a nonnegative rational number $\Lambda(W)$. Suppose there are fixed constants $K \geq 1$ and $c \geq 0$ such that, for every width- n network with*

$$D = \lceil \log_2 n \rceil, \quad b = 20D, \quad s \leq 8,$$

it satisfies

$$\frac{1 + Q^*(W)}{K(\log(n+2))^c} \leq \Lambda(W) \leq K(\log(n+2))^c(1 + Q^*(W)).$$

Then $P = NP$. The same approximation guarantee with probability at least $2/3$ from a polynomial-time randomized algorithm implies $NP \subseteq BPP$.

Lemma D.5 (Exact Boolean gates in a fixed amplitude range). *Encode false by zero and true by any number in $[1/2, 1)$. The functions*

$$\text{Or}(u_1, \dots, u_j) = \tanh\left(2 \sum_{i=1}^j u_i\right), \quad 1 \leq j \leq 3,$$

preserve this encoding and compute the Boolean disjunction. Define

$$\begin{aligned} B(u, v) &= -\tanh(1/2 + 4u + 4v) + \tanh(1/2 + 4u) \\ &\quad + \tanh(1/2 + 4v) - \tanh(1/2), \\ \text{And}(u, v) &= \tanh(2B(u, v)). \end{aligned}$$

For encoded inputs, And preserves the encoding and computes conjunction. The OR gate uses one layer and row norm at most six. The AND gate uses two layers, each with row norms at most eight. All biases are at most one in absolute value, and all parameters are dyadic half-integers.

Proof idea. Encode false by exact zero and true by an interval separated from zero. Cancellation makes the AND gate vanish exactly when an input is false, while elementary tanh bounds keep every true output inside the encoding interval.

Proof. For OR, the preactivation is zero when every input is zero. If one input is at least $1/2$, the preactivation is at least one, and $\tanh 1 \geq 1/2$.

If $u = 0$, the first and third terms of B cancel, as do the second and fourth. The case $v = 0$ is symmetric. Thus a false input makes the AND output exactly zero. If $u, v \geq 1/2$, monotonicity and $\tanh x \leq 1$ give

$$B(u, v) \geq 2 \tanh(5/2) - 1 - \tanh(1/2) \geq \frac{1}{3}.$$

Here $\tanh(1/2) \leq 1/2$, and $\tanh(5/2) \geq 11/12$ follows from $e^5 \geq 23$. The exponential series gives, for example, $e^5 \geq 1 + 5 + 25/2 + 125/6 > 23$. Also $\tanh(2/3) \geq 1/2$, since $e^{4/3} \geq 1 + 4/3 + 8/9 > 3$. Therefore $\text{And}(u, v) \geq 1/2$. Every finite tanh output is strictly smaller than one.

The first AND layer computes the four displayed tanh terms. Its largest row has weights $(4, 4)$, and all four biases are $1/2$. The second layer has weights $(-2, 2, 2, -2)$ and zero bias. The stated parameter bounds follow. $\square$

Proof idea. A formula circuit gates a balanced parity circuit. Unsatisfiable formulas produce a constant output, while one satisfying assignment leaves a parity signal of fixed amplitude. Any transcript missing a parity input has zero correlation with that signal, which creates a polynomial query gap.

Proof of Proposition D.4. Start from a formula φ in conjunctive normal form with at most three literals per clause, v variables, and k clauses. Satisfiability for these formulas is NP-complete [12, 23]. Let $m = 2^r$ be the least power of two with $m \geq \max\{v, k, 2\}$; the network will have width $n = 16m^3$. Inputs consist of its v assignment signs x , another m signs y , and unused coordinates. Formulas with an empty clause or with no clauses can be decided directly. Delete repeated literals within each clause, and replace a clause containing complementary literals by a constant true clause. Pad the list of clauses to length m with constant true clauses.

The formula circuit. At layer one, the two literal values for an assignment sign are

$$\tanh((1 + x_i)/2), \quad \tanh((1 - x_i)/2).$$

Exactly one is zero and the other equals $\tanh 1 \geq 1/2$. At layer two, apply the OR gate to each clause's literal values. A padded true clause is the constant $\tanh 1$, obtained from bias one and zero weights. Apply a balanced binary tree of AND gates to these $m = 2^r$ clause outputs. The final value $V_\varphi(x)$, at depth $2 + 2r$, is zero if the assignment does not satisfy the formula and belongs to $[1/2, 1)$ otherwise.

The parity circuit. Use two outputs to encode a Boolean value: one belongs to $[1/2, 1)$ and the other is zero. The two literal values at layer one give such a pair for each y_i . Given encoded pairs (u_+, u_-) and (v_+, v_-) , their product sign is encoded by

$$\begin{aligned} w_+ &= \text{Or}(\text{And}(u_+, v_+), \text{And}(u_-, v_-)), \\ w_- &= \text{Or}(\text{And}(u_+, v_-), \text{And}(u_-, v_+)). \end{aligned}$$

Only the pair matching the product sign is nonzero, and its value is at least $1/2$. This gate uses three layers. A balanced binary tree produces $(P_+(y), P_-(y))$, encoding $\chi(y) = \prod_{i=1}^m y_i$, at depth $1 + 3r$.

The formula circuit finishes no later than this, since $2 + 2r \leq 1 + 3r$ for $r \geq 1$. At each remaining layer, relay its value through $u \mapsto \tanh(2u)$. This relay preserves the encoding and uses row norm two. Denote the resulting formula value at depth $1 + 3r$ again by $V_\varphi(x)$.

The output. Apply the AND gadget to (V_φ, P_+) and (V_φ, P_-) in parallel, obtaining G_+, G_- after two further layers. The final neuron is

$$f_\varphi(x, y) = \tanh(2(G_+(x, y) - G_-(x, y))).$$

The total depth is $D = 4 + 3r = \log_2(16m^3) = \log_2 n$. If the assignment is unsatisfying, both gated values are exactly zero. If it is satisfying, exactly one gated value is at least $1/2$, and it has the sign selected by parity. Therefore

$$\varphi(x) = 0 \implies f_\varphi(x, y) = 0, \quad \varphi(x) = 1 \implies \chi(y)f_\varphi(x, y) \geq \tanh 1 \geq \frac{1}{2}. \quad (\text{D.4})$$

At layer one there are at most $4m$ active neurons. A first layer of parallel parity-combination gates has at most $8m$ neurons: each pair uses four AND gates, each initially using four neurons. All subsequent parity layers use fewer. The formula circuit uses at most $4m$ neurons in any layer, including padding and relays. Thus at most $12m$ neurons are active in a layer. The two final AND gadgets use eight neurons in their first layer. Since $12m \leq 16m^3 = n$, zero padding gives the required width at every layer. All weights belong to $\{0, \pm 1/2, \pm 2, \pm 4\}$, and all biases belong to $\{0, 1/2, 1\}$. They lie on the prescribed 2^{-20D} grid. Lemma D.5 verifies all row norms. The dense network description has polynomial size in $v + k$, and its construction takes polynomial time.

The cost gap. If φ is unsatisfiable, (D.4) gives $f_\varphi \equiv 0$. One independent fair bit samples the output, so $Q^*(W_\varphi) = 0$.

If φ is satisfiable, fix one satisfying assignment x^* . Giving this assignment to the sampler for free can only help. Its output sign Y satisfies, for every y ,

$$\Pr(Y = \chi(y) \mid x^*, y) = \frac{1 + \chi(y)f_\varphi(x^*, y)}{2} \geq \frac{3}{4}.$$

Now draw the m parity signs independently and uniformly. Conditional on any transcript that leaves at least one of them unqueried, their parity is uniform. This remains true after including all preprocessing and online randomness in the transcript's independent tape. Thus, if Q_y counts the distinct parity coordinates queried,

$$\Pr(Y = \chi(y)) \leq \frac{1}{2} + \frac{1}{2} \Pr(Q_y = m).$$

It follows that $\Pr(Q_y = m) \geq 1/2$ and $\mathbb{E}Q_y \geq m/2$. Averaging supplies a fixed y with this expected cost. This argument applies to every exact sampler with unlimited preprocessing, so $Q^*(W_\varphi) \geq m/2$.

Choose fixed positive integers K', c' such that the rational quantity $g(n) = K'(\lceil \log_2 n \rceil + 2)^{c'}$ bounds $K(\log(n + 2))^c$ from above. In the unsatisfiable case, $\Lambda(W_\varphi) \leq g(n)$. In the satisfiable case, $\Lambda(W_\varphi) \geq (1 + m/2)/g(n)$. For all sufficiently large m , $g(16m^3)^2 < 1 + m/2$; the two ranges are disjoint. Comparing the output with $g(n)$ decides satisfiability in polynomial time. The finitely many smaller formula sizes can be handled by exhaustive search. This proves $P = NP$. With a randomized approximation, the same reduction decides satisfiability with probability at least $2/3$, giving $NP \subseteq BPP$. $\square$

Theorem D.6 (Conditional threshold for efficient instance optima). *Fix $s > 1$. Suppose a deterministic polynomial-time algorithm computes a rational value $\Lambda(W)$ satisfying*

$$\frac{1 + Q^*(W)}{K(\log(n+2))^c} \leq \Lambda(W) \leq K(\log(n+2))^c(1 + Q^*(W))$$

for fixed constants $K \geq 1, c \geq 0$, on every width- n , depth- $\lceil \log_2 n \rceil$ network with row norms at most s and parameter precision $b = 20\lceil \log_2 n \rceil$. Then $P = NP$. A polynomial-time randomized approximation with probability at least $2/3$ implies $NP \subseteq BPP$. For every fixed $s \leq 1$, the constant value one approximates $1 + Q^(W)$ within a polylogarithmic factor on the same class.*

Proof idea. For an arbitrary fixed row bound above one, insert relay layers that restore the amplitude lost by smaller gates. Their number is a constant depending on the row bound. Padding preserves logarithmic depth and the polynomial gap used by the satisfiability reduction.

Proof. Fix an integer $R \geq 0$ with $\delta = 2^{-R} \leq \min\{s-1, 1\}$, and put

$$a = 1 + \delta, \quad c = \delta/16, \quad K = 3 \cdot 2^R(R+18), \quad C = 2K + 3.$$

All these quantities are fixed as the formula size increases. We will encode false by zero and true by any value in $[c, 1)$.

First consider the relay $T(u) = \tanh(au)$. The inequality $\tanh v \geq v - v^3/3$ for $v \geq 0$ follows by integrating $\tanh'(v) = 1 - \tanh^2 v \geq 1 - v^2$. For $0 \leq u \leq c$, it gives

$$T(u) \geq \left(a - \frac{a^3 c^2}{3}\right)u \geq (1 + \delta/2)u,$$

because $a^3 c^2/3 \leq \delta^2/96 \leq \delta/2$. Monotonicity also implies $T(u) \geq c$ whenever $u \geq c$. Consequently, if $d = c^2/2^{14}$, then K relay layers take every input in $[d, 1)$ into $[c, 1)$, and preserve zero exactly. Indeed, until the value reaches c , each relay multiplies it by at least $1 + \delta/2$. Moreover,

$$K \log(1 + \delta/2) \geq K\delta/3 = R + 18 \geq (R + 18) \log 2 = \log(c/d).$$

The inequality $\log(1+x) \geq x/(1+x)$ gives the first bound.

Use first-layer coefficient $\gamma = a/2$ and second-layer coefficient $\kappa = a/4$ in the AND construction:

$$\begin{aligned} B_a(u, v) &= -\tanh(1/2 + \gamma u + \gamma v) + \tanh(1/2 + \gamma u) \\ &\quad + \tanh(1/2 + \gamma v) - \tanh(1/2), \\ A_a(u, v) &= \tanh(\kappa B_a(u, v)). \end{aligned}$$

The largest first-layer row norm is $2\gamma = a$, and the second-layer row norm is $4\kappa = a$. A zero input still causes exact cancellation. If $u, v \geq c$, the integral identity

$$B_a(u, v) = \int_0^{\gamma u} \int_0^{\gamma v} 2 \tanh(1/2 + y + z) \operatorname{sech}^2(1/2 + y + z) dz dy$$

gives an explicit positive lower bound. Every argument belongs to $[1/2, 5/2]$. On this interval, $\tanh \geq 1/3$, and $\operatorname{sech}^2 w \geq e^{-2w} \geq e^{-5} > 1/256$. For the last inequality, the exponential series gives $e < 3$, so $e^5 < 243 < 256$. Therefore

$$B_a(u, v) \geq \frac{\gamma^2 uv}{384} \geq \frac{c^2}{1536}.$$

Also $0 \leq B_a(u, v) \leq 2$ and $\kappa \leq 1/2$. Using $\tanh z \geq z/2$ on $[0, 1]$ gives

$$A_a(u, v) \geq \frac{\kappa B_a(u, v)}{2} \geq \frac{c^2}{12288} \geq d.$$

Append K relay layers. The resulting AND gate has depth $K + 2$, preserves the new encoding, and has row norms at most a .

For OR with at most three inputs, use $\tanh(\kappa \sum_i u_i)$ followed by K relays. Its initial row norm is at most $3a/4$. A true input implies an initial output at least $\tanh(\kappa c) \geq \kappa c/2 \geq c/8 \geq d$. Thus this OR gate has depth $K + 1$, computes disjunction in the new encoding, and respects the norm bound. The original literal outputs are at least $1/2$, hence at least c , and their row norms are at most a .

Repeat the formula and parity construction with these gates. Each parity-combination level now has depth $(K + 2) + (K + 1) = C$, so the parity pair is ready at depth $1 + Cr$. The formula value is ready at depth $(r + 1)(K + 2)$, which is no larger since the difference is $(K + 1)(r - 1) \geq 0$. Relay the formula value to synchronize the two circuits. Gating the two parity outputs by the formula value uses $K + 2$ further layers. Take the final output to be

$$f_\varphi(x, y) = \tanh\left(\frac{a}{2}(G_+(x, y) - G_-(x, y))\right).$$

Its row norm is a . On a satisfying assignment its correlation with parity is at least $\tanh(ac/2) \geq c/4$; on an unsatisfying assignment it is exactly zero.

Set

$$n = 2^{K+4}m^C, \quad D = K + 4 + Cr = \log_2 n, \quad b = 20D.$$

The same active-width bound $12m$ applies, since relay layers do not increase the number of signals. Zero padding yields the required width n . All parameter denominators divide 2^{R+2} . Since $R + 2 \leq 20(K + 4) \leq b$, they lie on the prescribed grid. All biases remain at most one. For fixed s , K and C are constants, so the reduction has polynomial size.

The parity transcript argument now yields $Q^*(W_\varphi) \geq cm/4$ in the satisfiable case and zero in the unsatisfiable case. This polynomial gap again exceeds the square of any fixed polylogarithmic factor. The approximation consequences follow exactly as in Proposition D.4.

When $s \leq 1$, Theorem A.1 gives $Q^*(W) \leq \log^{O(1)} n$. Since $Q^*(W) \geq 0$, the constant value one satisfies a two-sided polylogarithmic approximation guarantee for $1 + Q^*(W)$. $\square$

Corollary D.7 (Computing the multivariate score certificate). *Let $l(W)$ be the first term in the supremum in (D.3), with value zero on constant networks. For every fixed $s > 1$, a deterministic polynomial-time approximation to $1 + l(W)$ within a fixed polylogarithmic factor on the depth- $\lceil \log_2 n \rceil$, $b = 20\lceil \log_2 n \rceil$, row-norm- s class would imply $P = NP$. The corresponding randomized approximation implies $NP \subseteq BPP$.*

Proof idea. The balanced parity construction has the same nonzero amplitude for every parity pattern. Fixing a satisfying assignment therefore exposes an exact parity factor. A product distribution near the boundary makes its multivariate score certificate polynomially large.

Proof. In the reduction, the true amplitude of a parity pair is the same for every input pattern at a given level. This follows by induction: at the leaves it equals $\tanh 1$; at each balanced combination, the correct branch applies the same AND gate to two copies of that amplitude and the same OR gate to the result and zero. Relays preserve the equality. The two final gated branches are also identical functions of the formula value and the parity amplitude. Thus the network output factorizes exactly as

$$f_\varphi(x, y) = A_\varphi(x) \prod_{i=1}^m y_i,$$

where $A_\varphi(x) = 0$ on an unsatisfying assignment and $A_\varphi(x) \geq c/4$ on a satisfying assignment, with c as in Theorem D.6.

If the formula is unsatisfiable then $l(W_\varphi) = 0$. Otherwise, take product means for the assignment coordinates that approach a satisfying assignment x^* . Give every parity coordinate mean $t = \sqrt{1 - 1/m}$, and use direction $v_i = 1/\sqrt{m}$ on parity coordinates and zero on the others. The direction is admissible in Lemma D.3. In the limit the output mean is αt^m , where $\alpha = A_\varphi(x^*) \geq c/4$, and its directional derivative is

$\alpha\sqrt{m}t^{m-1}$. The denominator in the centered information bound is at most one. Since the product extension and its derivatives are continuous,

$$I(W_\varphi) \geq \alpha^2 m \left(1 - \frac{1}{m}\right)^{m-1} \geq \frac{\alpha^2 m}{3} \geq \frac{c^2 m}{48}.$$

The middle inequality follows from $(1 + 1/(m-1))^{m-1} \leq e < 3$. For the fixed norm s , this is a polynomial gap in the network width. The same comparison used for Theorem D.6 decides satisfiability. $\square$

D.3 An exact query characterization with exponential size

For completeness, an instance has an exact finite optimization description if exponential preprocessing is allowed.

Let $p : \{-1, 1\}^m \rightarrow [0, 1]$ be the required probability of returning one. Write $\Sigma = \{-1, *, 1\}^m$ for the partial assignments, and $\sigma \preceq x$ if x extends σ . At a partial assignment σ , introduce nonnegative variables u_σ, v_σ for stopping with one or zero and $a_{\sigma i}$ for querying an unassigned coordinate i . Define the incoming flow by

$$w_{*^m} = 1, \quad w_\sigma = \sum_{i: \sigma_i \neq *} a_{\sigma(i \leftarrow *), i} \quad (\sigma \neq *^m).$$

Minimize C , subject to

$$\begin{aligned} w_\sigma &= u_\sigma + v_\sigma + \sum_{i: \sigma_i = *} a_{\sigma i} & (\sigma \in \Sigma), \\ \sum_{\sigma \preceq x} u_\sigma &= p(x) & (x \in \{-1, 1\}^m), \\ \sum_{\sigma \preceq x} \sum_{i: \sigma_i = *} a_{\sigma i} &\leq C & (x \in \{-1, 1\}^m). \end{aligned}$$

Denote the value of this program by $\text{Flow}(p)$. It has $O(m3^m)$ variables and $O(3^m + 2^m)$ constraints.

Lemma D.8 (Flow characterization). *The infimum of the worst-input expected number of distinct probes among exact samplers for p is $\text{Flow}(p)$. This statement counts queries only. It permits exponential preprocessing and storage.*

Proof idea. Merge decision-tree histories that reveal the same partial input. Their stopping and query probabilities form a conserved flow. Conversely, normalizing a feasible flow gives a sampler; a small complete-scan correction then shows that computable probabilities attain the same infimum.

Proof. First consider a randomized decision tree with no repeated query. All histories that end at the same partial assignment σ have probabilities determined by the answers recorded in σ . Their probabilities are therefore the same on every full input extending σ . Add their probabilities of stopping with one, stopping with zero, or querying i , to obtain $u_\sigma, v_\sigma, a_{\sigma i}$. The incoming-flow identity and conservation hold by construction. On a full input x , summing the one-stopping flows gives the output probability, and summing the query flows gives the expected query count. Every exact sampler supplies a feasible solution with C equal to its worst-input expected query count. The same finite-input reduction used for the transcript lemma applies to almost surely terminating algorithms with arbitrary internal work.

Conversely, at a state with $w_\sigma > 0$, stop with one, stop with zero, or query i , with respective probabilities $u_\sigma/w_\sigma, v_\sigma/w_\sigma$, and $a_{\sigma i}/w_\sigma$. A state with zero incoming flow is unreachable. Each query fixes one new coordinate, so the process stops after at most m queries. Induction on the number of fixed coordinates shows that its probability of reaching σ , on any input extending σ , is exactly w_σ . The output and cost constraints thus give the specified law and cost. This proves equality when arbitrary real transition probabilities are permitted.

For the feed-forward networks in this paper, the same infimum is obtained by computable finite-bit samplers. Their output probabilities belong to $[\varepsilon, 1 - \varepsilon]$ for an explicitly computable $\varepsilon > 0$. For example, any rational number below $(1 - \tanh(1 + s))/2$ suffices when the output row norm is at most s . All 2^m probabilities are computable to arbitrary precision by evaluating the network.

The flow value is continuous quantitatively on this interior cube. If $p, q \in [\varepsilon, 1 - \varepsilon]^{2^m}$ and $\delta = \|p - q\|_\infty \leq \varepsilon$, put $\gamma = \delta/\varepsilon$. For $\delta > 0$, the table

$$r(x) = \frac{p(x) - (1 - \gamma)q(x)}{\gamma}$$

belongs to $[0, 1]^{2^m}$. Mix a sampler for q , with probability $1 - \gamma$, and a complete input scan followed by an $r(x)$ -coin, with probability γ . It follows that

$$|\text{Flow}(p) - \text{Flow}(q)| \leq \frac{m}{\varepsilon} \|p - q\|_\infty.$$

The case $\delta = 0$ is immediate. A rational approximation q admits an exactly solved rational flow program. Using the same mixture with a certified approximation error and $\varepsilon/2$ in place of ε yields an exact finite-bit sampler whose query cost approaches $\text{Flow}(p)$. The residual scalar probability is computable; a lazy fair-bit comparison samples it exactly. Hence computability does not change the infimum. $\square$

The flow program accounts for all signed cancellations because it uses the complete output table. Its size exceeds the permitted preprocessing budget exponentially.

E Certificates and checkpoint cost formulas

This appendix specifies the measurements behind Section 9. The encoded parameters define a mathematical network with the stated nonlinearities. A deployed floating-point operation order or an additional token filter would need its own target specification. A checkpoint measurement is evidence about the measured inputs unless it yields a certificate over the theorem’s allowed input set.

E.1 Dimensions, gains, and common spaces

Let n be residual width, d_h and a_h the key and value dimensions of head h , and V the vocabulary size. A head score is $\beta_h \langle q, k \rangle / d_h$. Its standard scaling is $\beta_h = \sqrt{d_h}$. The RMS denominator continues to use n . For column-vector conventions, a learned normalization gain g is absorbed into the next map as $W^{\text{eff}} = W \text{diag}(g)$. All row norms and parameter lengths below refer to these complete effective matrices. They include a base matrix and any learned updates. Finite dyadic rank can be computed by exact elimination.

For an unrotated key map, a common affine containing space is $b_K + \text{image}(W_K^{\text{eff}})$. For position-dependent linear maps R_t , a sufficient space with fixed center c has direction

$$S = \text{span} \{ \text{image}(R_t W_K^{\text{eff}}), R_t b_K - c : t \text{ is allowed} \}.$$

Thus its rank is $\dim S$. An observed numerical rank with a threshold is a different datum; record its norm and absolute residual. A smaller rank under general rotations requires a certified common containing space. The full ambient choice $S = \mathbb{R}^{d_h}$ is always available in the fixed-rotation corollary. No claim of exact symbolic rank computation from an arbitrary computable frequency is needed.

Corollary E.1 (Rectangular heads). *The exact decoder theorem extends to residual width n , head dimensions d_h, a_h , feed-forward widths f_ℓ , and a specified finite number of heads. Suppose each head has a fixed affine key map of rank at most r , and use a dyadic temperature or $\beta_h = \sqrt{d_h}$. Include these dimensions and the head count among the size parameters. At fixed r , prefill is linear in the prompt length times a polynomial in those parameters and their encoding lengths. Every subsequent token has conditional expected work polynomial in the same parameters and $\log(T + 2)$.*

Proof idea. The key-coordinate dimension controls the feature expansion. Residual width controls normalization. The exact arithmetic uses the square root in the head’s score scaling, and never requires these dimensions to agree.

Proof. RMSNorm has coordinate bound $\sqrt{n}$ and the Lipschitz bound proved below. A bounded chart C_h for a head’s key space has d_h rows, $r_h \leq r$ columns, and $|C_{h,ij}| \leq 2$. With coordinate bound K_h , write

$$t_h = \frac{\beta_h K_h}{d_h} q^T C_h.$$

If $|q_i| \leq Q_h$, summing the d_h terms gives $\|t_h\|_1 \leq 2r_h \beta_h Q_h K_h$. The finite moment lemma already permits arbitrary value dimension a_h . At standard scaling, the even and odd total degrees produce sums of the form $A + B/\sqrt{d_h}$ with rational A, B . The same certified square-root arithmetic therefore applies. The dense products use their actual rectangular sizes. Concatenation and the attention output map are affine operations. Taking the maximum coordinate error over all heads gives the same prefix stability induction. Summing their finite work gives a polynomial at fixed r . The exact counters, causal insertion order, categorical stopping estimate, and background replay are unchanged. The replay lemma then proves the asserted conditional expected bound. $\square$

Sometimes only a projection of the keys affects the scores. The following certificate is computable from unrotated affine maps.

Corollary E.2 (Effective rank of an unrotated score). *Write an unrotated head as $q = W_Q u + b_Q$, $k_j = W_K v_j + b_K$, where $u, v_j \in \mathbb{R}^n$ are normalized streams and all matrices include learned gains. Define the dyadic matrix*

$$S = \begin{bmatrix} W_Q^T W_K \\ b_Q^T W_K \end{bmatrix}, \quad r_{\text{score}} = \text{rank } S.$$

The complete moment decoder can use r_{score} in place of the affine key-map rank for this head, with polynomial changes in its other parameters. In particular,

$$r_{\text{score}} \leq \min\{\text{rank } W_K, \text{rank } W_Q + 1\}.$$

This assertion assumes the displayed maps are independent of position. Position-dependent rotations require a certificate for the resulting common score space.

Proof idea. Remove the score terms that are common to all keys. The remaining bilinear form has a finite dyadic key map of the displayed rank. The moment construction evaluates this identical score law.

Proof. For every key,

$$\langle q, k_j \rangle = u^T W_Q^T W_K v_j + b_Q^T W_K v_j + u^T W_Q^T b_K + b_Q^T b_K.$$

The last two terms do not depend on j and cancel from softmax. Set $\tilde{q} = (u, 1)$ and $\tilde{k}_j = S v_j$. Their position-dependent score is exactly $(\beta/d) \tilde{q}^T \tilde{k}_j$. The matrix S is a fixed dyadic $(n+1)$ by n map. If every original row norm and bias magnitude is bounded by $s \geq 1$, then its row norms are at most ds^2 :

$$\sum_b |(W_Q^T W_K)_{ab}| \leq \sum_i |W_{Q,ia}| \sum_b |W_{K,ib}| \leq ds^2,$$

and the same calculation applies to the last row. Its coefficients have at most $2b + O(\log(d+2))$ encoding bits when the effective input matrices have length at most b . Computing S takes polynomial weight preprocessing.

Apply the moment construction to this new key map and its r_{score} -dimensional chart. The coefficient β/d retains its original value. At standard scaling it equals $1/\sqrt{d}$, so the moment coefficients remain in the same quadratic extension handled by the rectangular proof. The internal feature dimension $n+1$ does not change that radicand. All numerical range and stability bounds remain polynomial in the original size parameters. To evaluate the recoded cost, let G_N bound the exact and rounded normalized streams, choose

a power-of-two virtual-key scale $K_S \geq G_N \|S\|_{\infty \rightarrow \infty}$, and let C_S be its bounded chart. Then the moment coefficients and a sufficient integer bound are

$$t_S = \frac{\beta K_S}{d} (u, 1)^T C_S, \quad H_S = \left\lceil \max \left\{ 1, \frac{2r_{\text{score}} |\beta| K_S}{d} (nG_N + 1) \right\} \right\rceil.$$

Recompute this bound when substituting r_{score} in the feature count. If $r_{\text{score}} = 0$, attention is uniform and exact value sums suffice. The factorization $S = [W_Q^T; b_Q^T] W_K$ proves the rank inequalities. Thus the original head and the recoded evaluator have identical attention and final-token laws. $\square$

If W_Q and W_K both have full row rank d , this certificate still has rank d : W_Q^T is injective and W_K is surjective onto $\mathbb{R}^d$. The recoding therefore gives no hidden rank reduction in that case. The observation is a direct linear-algebra consequence of the score representation; no novelty claim is attached to it.

Proposition E.3 (An illustrative rank-one rotational example). *Let $d = 2m$ with $m \geq 1$. Let R be block diagonal with plane rotations through angles $\omega_1, \dots, \omega_m$, and suppose all d complex numbers $e^{i\omega_j}, e^{-i\omega_j}$ are distinct. Let $v \in \mathbb{R}^d$ have a nonzero component in every coordinate pair. The common containing space for the rank-one key images $\text{span}(R^t v)$, $0 \leq t < d$, has dimension d . The $d + 1$ actual keys $R^t v$, $0 \leq t \leq d$, also have affine dimension d .*

Proof idea. In complex eigen-coordinates, each rotation multiplies a component by one eigenvalue. The successive rotated vectors form a Vandermonde matrix with nonzero row factors.

Proof. Over $\mathbb{C}$, diagonalize each two-dimensional rotation. The resulting eigenvalues $\lambda_1, \dots, \lambda_d$ are the listed numbers. The two eigen-coordinates contributed by a real pair of v are nonzero because that pair is nonzero. Write these coordinates as $c_i \neq 0$. The matrix with columns $v, Rv, \dots, R^{d-1}v$ is therefore, after an invertible change of basis,

$$\text{diag}(c_i) [\lambda_i^t]_{1 \leq i \leq d, 0 \leq t < d}.$$

Its determinant is a nonzero constant times $\prod_{i < j} (\lambda_j - \lambda_i)$. The usual Vandermonde identity follows by viewing the determinant as an alternating polynomial of that degree with leading coefficient one. Thus the original real matrix has rank d . Every indicated key image contains zero, so its common affine containing space is linear and has that same dimension.

The distinct paired eigenvalues also exclude 1: otherwise the two eigenvalues of one rotation pair would coincide. Augment the columns $R^t v$, $0 \leq t \leq d$, by a coordinate one. The same argument is the Vandermonde determinant for $1, \lambda_1, \dots, \lambda_d$, with nonzero row factors. This augmented matrix has rank $d + 1$, which proves full affine rank of the $d + 1$ records. $\square$

The distinction between d position-dependent key images and $d + 1$ actual records is necessary: T points have affine dimension at most $T - 1$.

E.2 Normalization floors and a one-sided error bound

For $\varepsilon > 0$ put

$$r_\varepsilon(h) = \sqrt{\varepsilon + \|h\|_2^2/n}, \quad N_\varepsilon(h) = h/r_\varepsilon(h).$$

Then $\|N_\varepsilon(h)\|_2 \leq \sqrt{n}$. If $|h_i| \leq R$ and $r_\varepsilon(h)^2 \geq \lambda$, also

$$\|N_\varepsilon(h)\|_\infty \leq \min\{\sqrt{n}, R/\sqrt{\lambda}\}.$$

The floor λ bounds the squared denominator. The relative quantity $\gamma = \lambda/R^2$ is dimensionless.

Lemma E.4 (Error control from the exact state's floor). *For $n \geq 1$, $\varepsilon > 0$, and $h, \tilde{h} \in \mathbb{R}^n$,*

$$\|N_\varepsilon(h) - N_\varepsilon(\tilde{h})\|_\infty \leq \frac{1 + \sqrt{n}}{r_\varepsilon(h)} \|h - \tilde{h}\|_\infty.$$

Consequently a floor λ certified only for legal exact states gives the error factor $(1 + \sqrt{n})/\sqrt{\lambda}$. No floor hypothesis on $\tilde{h}$ or on the line segment between the two states is required.

Proof idea. The denominator is a Euclidean norm of an augmented vector. Its change is bounded by the coordinate error. The normalized approximate vector has a universal coordinate bound.

Proof. The reverse triangle inequality applied to $(\sqrt{\varepsilon}, h/\sqrt{n})$ gives

$$|r_\varepsilon(h) - r_\varepsilon(\tilde{h})| \leq \|h - \tilde{h}\|_2/\sqrt{n} \leq \|h - \tilde{h}\|_\infty.$$

Both denominators are positive, and $\|\tilde{h}\|_\infty/r_\varepsilon(\tilde{h}) \leq \sqrt{n}$. Use the exact identity

$$N_\varepsilon(h) - N_\varepsilon(\tilde{h}) = \frac{h - \tilde{h}}{r_\varepsilon(h)} + \frac{\tilde{h}}{r_\varepsilon(\tilde{h})} \frac{r_\varepsilon(\tilde{h}) - r_\varepsilon(h)}{r_\varepsilon(h)}.$$

Taking the maximum coordinate norm proves the result. $\square$

A fixed positive stabilizer always permits $\lambda = \varepsilon$. Lemma E.4 allows a stronger true-history certificate to improve the replay precision without extending that promise to rounded states. For maintained decoding, a floor used to set the routine precision must cover the allowed future histories as well. Discovering or weakening such a certificate online requires charging its verification and any precision-change replay. The universal $\lambda = \varepsilon$ schedule remains valid without this additional future-history certificate.

For the sign factory, use

$$\kappa = \max\{1, R^2/\lambda\}, \quad G \geq 2R/\sqrt{\lambda}, \quad K_N = 1 + 228\kappa^2,$$

after replacing λ by $\max\{\lambda, \varepsilon\}$. Choose the next power of two for G . It is less than twice the displayed natural radius. The source and local bit bounds are K_N and $O(\kappa^2 B^4)$. The Gaussian alternative has radius $4\sqrt{n}$, source count $1 + nR^2/\varepsilon$, and local work $O(n(1 + R^2/\varepsilon)B^4)$. These radii enter later row budgets, so source count alone does not determine the better composition.

E.3 Certified approximate key spaces

Proposition E.5 (A residual bound obtained from the weights). *Fix a rational chart C , selected coordinates I , and projection $P = CR_I$, with $C_I = I_r$ and $|C_{ij}| \leq 2$. For $k = b_K + W_K^{\text{eff}} N_\varepsilon(h)$ set $c_0 = (I - P)b_K$. Suppose $\|N_\varepsilon(h)\|_\infty \leq G_{\text{true}}$. Either of the following is a valid upper bound on the coordinate residual $\|k - c_0 - Pk\|_\infty$:*

$$\eta_1 = G_{\text{true}} \|(I - P)W_K^{\text{eff}}\|_{\infty \rightarrow \infty}, \quad \eta_2 = \sqrt{n} \max_i \|[(I - P)W_K^{\text{eff}}]_{i,:}\|_2.$$

Choose a rational $\eta \geq \min\{\eta_1, \eta_2\}$. If finite supplied records obey this residual certificate and $\bar{\beta}Q_{\max}\eta \leq 1/8$, the approximate-space finite-cache index applies. It has cell bound

$$M_T \leq \min\{T, (5 + 64r\bar{\beta}Q_{\max}K)^r\}.$$

Proof idea. The key residual is a fixed matrix applied to the normalized stream. Its row norms provide a uniform certificate. The index then uses the projected scores for proposals and the actual scores for acceptance.

Proof. Since $P^2 = P$ and its selected rows agree with the coordinate selector, $(c_0)_I = 0$. Direct subtraction gives

$$k - c_0 - Pk = (I - P)W_K^{\text{eff}} N_\varepsilon(h).$$

The maximum absolute row-sum inequality gives η_1 . Cauchy-Schwarz and $\|N_\varepsilon(h)\|_2 \leq \sqrt{n}$ give η_2 . Each right-hand side therefore bounds the same residual. Corollary B.21 proves the index claim. Its exact-score acceptance preserves the original attention law. $\square$

If a stored key has coordinate error at most δ relative to the mathematical key, enlarge its residual allowance by $(1 + 2r)\delta$, since $\|I - P\|_{\infty \rightarrow \infty} \leq 1 + 2r$. For already supplied dyadic cache records the residual can instead be checked directly on insertion. Positional transforms require a bound after every allowed transform. This proposition is an extension for the finite-cache index. The moment decoder still needs an exact fixed key-space certificate, or the exact score recoding above.

E.4 Three distinct score bounds

Let A_{range} denote a certified bound on the absolute centered scores. Record an empirical score maximum separately. A Poisson factory additionally needs an independent sign with mean equal to score divided by its chosen radius A_{coin} . For normalized input signs at radius G , affine mixtures give

$$Q_i^{\text{coin}} = |b_{Q,i}| + G\|W_{Q,i}^{\text{eff}}\|_1, \quad K_i^{\text{coin}} = |b_{K,i}| + G\|W_{K,i}^{\text{eff}}\|_1.$$

The weighted coordinate draw and two independent signs give

$$A_{\text{coin}} = \frac{|\beta|}{d} \sum_i Q_i^{\text{coin}} K_i^{\text{coin}}.$$

Zero coordinate budgets are omitted. If $A_{\text{coin}} = 0$, uniform attention suffices. Otherwise the attention sign uses at most $2e^{2A_{\text{coin}}} - 1$ normalized-stream requests and $O(B^4 e^{2A_{\text{coin}}})$ local bits. A request to the normalization factory is charged in addition. A common position-independent key bias may be removed from the score law. Any other score centering must retain and charge an exact sign interface.

Finite evaluation can use smaller range certificates. For example,

$$|q_i| \leq |b_{Q,i}| + \min \{G_{\text{true}}\|W_{Q,i}^{\text{eff}}\|_1, \sqrt{n}\|W_{Q,i}^{\text{eff}}\|_2\}.$$

The key and value inequalities are the same. Replacing a coin budget by this range may require an additional factory, since the direct affine mixture supplies signs only at the coin radius.

The moment construction uses a third quantity. For chart C and key scale K , choose an integer

$$H \geq \max \left\{ 1, \sup_{q \text{ allowed}} \left\| \frac{\beta K}{d} q^T C \right\|_1 \right\}.$$

A computable choice from coordinate bounds is

$$H = \left\lceil \max \left\{ 1, \frac{|\beta|K}{d} \sum_{a=1}^r \sum_{i=1}^d Q_i |C_{ia}| \right\} \right\rceil.$$

It is at most $\lceil \max\{1, 2r|\beta|Q_{\text{max}}K\} \rceil$ for a common Q_{max} . Include the finite evaluator's rounding margins in those bounds. The original proof-safe choice uses $K \in [s(\lceil \sqrt{n} \rceil + 1), 2s(\lceil \sqrt{n} \rceil + 1))$. Thus $H = O_r(1 + \beta s^2 n)$ follows from row budgets alone. Small empirical A_{range} need not make this cube bound H small.

E.5 A precision guard from individual stages

Lemma E.6 (A variable-stage precision certificate). *Suppose a prefix evaluator has J stages. For each stage j , let $\Lambda_j \geq 0$ bound the change between its legal exact input and every approximate input encountered by the evaluator. Suppose each stage can be enclosed with local error at most 2^{-P} . To make the final error less than 2^{-P} , it suffices to take*

$$P = p + \lceil \log_2(J+1) \rceil + \sum_{j=1}^J \lceil \log_2 \max\{1, \Lambda_j\} \rceil + 10.$$

Proof idea. Each local error is multiplied only by the later stage bounds. Enlarging every factor to at least one permits one common product to bound their sum.

Proof. Let e_j be the maximum error after stage j , with $e_0 = 0$. Then $e_j \leq \Lambda_j e_{j-1} + 2^{-P}$. Induction gives

$$e_J \leq 2^{-P} \sum_{i=1}^J \prod_{j=i+1}^J \Lambda_j \leq J 2^{-P} \prod_{j=1}^J \max\{1, \Lambda_j\}.$$

The prescribed ceilings make the last quantity less than 2^{-P} . Rational upper enclosures for the Λ_j and rational comparisons with powers of two implement this choice without an exact-real comparison. $\square$

For an RMS stage, Lemma E.4 supplies $\Lambda_{N,j} = (1 + \sqrt{n})/\sqrt{\lambda_j}$. An additive attention stage with step α_j , output-map row norm $s_{O,j}$, and value bound U_j has a valid bound

$$\Lambda_{A,j} \leq 1 + \alpha_j s_{O,j} \Lambda_{N,j} [s_{V,j} + 2U_j |\beta_j| (K_j s_{Q,j} + Q_j s_{K,j})].$$

Here Q_j, K_j, U_j must bound both the exact arrays and the approximate arrays used in the comparison. The universal RMS range supplies such bounds; a sharper true-state bound can be enlarged by the certified rounding error. The formula follows from $\|F(a, v) - F(a', v')\|_\infty \leq \|v - v'\|_\infty + 2U_j \|a - a'\|_\infty$ and the score difference bound $|\beta_j| (K_j \delta_q + Q_j \delta_k)$. For a single affine-tanh update, $\Lambda_{F,j} \leq 1 + \alpha_j s_{F,j} \Lambda_{N,j}$. For a convex mixture replace the leading 1 by $1 - \alpha_j$. A two-affine update uses its two row norms and a certified activation derivative bound. An affine head followed by softmax cumulative sums has Lipschitz bound at most twice its row norm. The cost of enumerating its V logits is still included.

E.6 Construction sizes and a numerical worksheet

Let R_T be a routine precision horizon with $T \leq R_T \leq 2T$, distinct from residual radius. At context length T , the comparison in Lemma B.31 permits

$$p_T = \lceil \log_2(64V(R_T + 2)^3) \rceil.$$

The maintenance schedule in its proof keeps the routine index and the future index at power-of-two horizons in $(T, 4T]$. Since $(4R_T + 2)^3 \leq 64(R_T + 2)^3$, their routine precision is at most $p_T + 6$. Use Lemma E.6 to choose P . If the value coordinate scale is U , request normalized attention precision $p_{\text{att}} = P + \lceil \log_2 \max\{1, U\} \rceil + 4$. Then set

$$m = \lceil 8(p_{\text{att}} + 3H + 7) \rceil, \quad N_m = \binom{m+r}{r}.$$

The exact number of integer counters per head and precision index is $(a+1)N_m$. Their numerator lengths are $O(P_{\text{grid}}(m+1) + \log(T+2))$, where P_{grid} includes the affine-product grid and the key and value scaling. Several heads contribute the sum of these counts.

In the original equal-width dimensions, (B.15) is a proved conservative token work bound. For rectangular heads, the sum of the actual dimensions can replace $n + V + 1$ there as a coarse size bound, with the number of stages charged as well. The constants C_r remain rank dependent.

Routine maintenance uses at most three token passes. The unresolved comparison probability is at most $4V2^{-p_T}$. For any chosen implementation whose further precision cost is a degree- k polynomial, the replay proof charges the rare complete replays using

$$\sum_{j \geq 0} 2^{-j} (P + j)^k \leq 2^{k+1} k! P^k.$$

The degree k also grows with rank in the present implementation. The included worksheet takes R_T to be the least power of two at least T . It computes the routine precision, chart bound, exact binomial count, counter count, and a coarse work proxy with C_r left unspecified. Its supplied inputs are explicitly illustrative.

E.7 An arbitrary initial residual radius

Embedding coordinates need not have radius one. The following version of Theorem C.15 lets a bound R_0 on the initial coordinates enter the additive formula directly.

Corollary E.7 (An arbitrary initial residual radius). *In Theorem C.15, suppose $R_0 > 0$ is rational, $|H_{0,t,i}| \leq R_0$, and an exact sign of mean $H_{0,t,i}/R_0$ has the stipulated input cost. Retain the uniform conditional source and local arithmetic assumptions, and put*

$$R_j = R_0 + \sum_{k \leq j} u_k, \quad P_j = \prod_{k \leq j} \left(1 + \frac{u_k m_k}{R_{k-1}}\right),$$

where u_j includes the update step size. Then

$$Q_{\text{leaves}} \leq \frac{R_0 P_J}{R_J}, \quad W_{\text{add}} \leq C_0 B^4 \frac{P_J}{R_J} \left(2R_0 + \sum_{j=1}^J \frac{u_j(1+L_j)}{P_j} \right).$$

All rational radius lengths are included in B .

Proof idea. The direct origin draw gives the initial state mass R_0/R_J . The same cost recurrence starts with the initial radius R_0 in place of one.

Proof. Use origin weights $R_0, u_1, \dots, u_j$. The leaf count starts at one and its recurrence telescopes to $R_0 P_J / R_J$. For bit work the equality majorant for $R_j T_j$ starts at $2R_0 C_0 B^4$ and obeys

$$W_j = \left(1 + \frac{u_j m_j}{R_{j-1}} \right) W_{j-1} + C_0 B^4 u_j (1 + L_j).$$

This follows by subtracting the direct-origin equalities, whose nonrecursive term is $C_0 B^4 (R_j + R_0)$. Division by P_j and summation prove the display. All choices are the same exact mixtures as in the additive theorem, so its conditional charging and finite-depth termination proof apply. $\square$

With uniform $m_j \leq m$, $m \geq 1$, and $L_j \leq L$, the resulting bit bound is $O(B^4(2+L)(1+\log(R_J/R_0))(R_J/R_0)^{m-1})$. Indeed, $P_J \leq (R_J/R_0)^m$ and $\sum_j u_j/P_j \leq R_0 \log(R_J/R_0)$ for the uniform majorant. The chosen final head supplies its additional factor. These expressions preserve variation between layers. The carrier family's matching exponent applies to its specific construction; it does not assign an exponent to an arbitrary checkpoint.

E.8 Other positive cases

The finite-query-list theorem applies when all permitted queries are known before the table is built and the online input specifies one by its list identifier. Record the list size M and a certified cost $H_0(B+p)^c$ for enclosing all probabilities of one candidate, where $H_0 \geq V$ and $c \geq 2$ is fixed. The prefill and expected sample bounds are respectively

$$O(MH_0[B + \log(VH_0 + 2)]^c), \quad O([B + \log(VH_0 + 2)]^c).$$

The list and its table are specific to that cache. A vocabulary size alone does not provide a fixed query list across all histories.

The implicit rank-one query of Theorem B.24 needs the slopes of its one-dimensional score functions. A slope bound Λ gives $O(1 + \Lambda^2)$ source requests and $O((1 + \Lambda^2)B^4)$ auxiliary work for the stated attenuated value-sign interface. Its certified table construction is charged in prefill. This interface and the ordered rank-one index of Corollary B.20 cannot be substituted for each other without their final head and source costs.

For the fixed-first-layer-rank bound of Theorem A.67, fix the rank cap r independently of width. At logarithmic depth and parameter precision, that theorem gives $O_r(Dn^2 \log^6 n)$ preprocessing. Its constants depend on r .

F Guarantees of related attention methods

This appendix states the guarantees of three attention methods summarized in Section 10, in the notation of their papers. No proof in this paper uses them.

KDEformer. KDEformer gives a high-probability spectral approximation to an attention matrix product [44]. In its notation, with unnormalized attention matrix A , diagonal row-sum matrix D , and value matrix V , its guarantee is

$$\|D^{-1}AV - \tilde{D}^{-1}A\Pi^\top\Pi V\|_{\text{op}} \leq \epsilon\|D^{-1}A\|_{\text{op}}\|V\|_{\text{op}}.$$

For N rows, its sample count is $m = O(\epsilon^{-2} \log N [\text{srnk}(D^{-1}A) + \text{srnk}(V)])$, where $\text{srnk}(M) = \|M\|_F^2 / \|M\|_{\text{op}}^2$. Constructing Π and $\tilde{D}$ requires the KDE work; materializing the approximate product adds $O(Nmd)$ operations at width d . The theorem concerns all rows.

HyperAttention. HyperAttention proves a related operator-norm guarantee under conditions on the maximum column norm of normalized attention and on residual row masses after a heavy-entry mask is supplied [17]. Under its bounds $N \max_j \|(D^{-1}A)_{*,j}\|_2^2 \leq N^{o(1)}$ and residual row-mass ratio at most $N^{o(1)}$, with $\epsilon > N^{-o(1)}$, the time is $O(dN^{1+o(1)} + d \text{nnz}(\text{mask}))$ and the success probability is at least 0.98. Finding the mask requires its own assumptions. The guarantee supplies no certified interval for every output probability on every random run.

MagicPIG. MagicPIG uses a prefill-built locality-sensitive hash index for decoding [10]. Its ideal attention-sampling theorem assumes draws from the true attention distribution and gives an unbiased sample average. The implemented LSH construction instead uses inclusion probabilities u_i in the ratio

$$\hat{v} = \frac{\sum_{i \in S} e^{a_i} v_i / u_i}{\sum_{i \in S} e^{a_i} / u_i}.$$

For L tables of K hash bits, retrieving a key after at least two table collisions gives $u_i = 1 - (1 - \zeta_i^K)^L - L\zeta_i^K(1 - \zeta_i^K)^{L-1}$, where ζ_i is its one-bit collision probability. Such ratios are generally biased. For example, take weights $(1, 2)$, values $(0, 1)$, and one uniformly included index. The self-normalized estimate is the selected value, with mean $1/2$, while the true weighted mean is $2/3$. This distinction concerns exactness; empirical approximation quality requires separate measurements.

G Numerical checks and Lean formalization

G.1 Deterministic checks

The mathematical proofs do not use numerical premises. The accompanying scripts provide independent consistency checks for selected identities and inequalities. Exact rational arithmetic is used for the finite hypergeometric, operator, and matrix identities. The main analytic identity checks use 120 decimal digits, and positional reduction uses 450 digits. The new carrier recurrence checks use binary64 arithmetic with the tolerances recorded in their JSON report. These finite checks do not certify an inequality on a continuum and do not measure an implementation’s sampling runtime.

Construction	Recorded checks
Indexed lower bound	93,262 score, embedding, probability-gap, and exponent checks
Dynamic fixed-rank grid	15 coordinate charts; 7,425 cell-score checks; four rank-growth streams; one 200-record near-subspace case
Moment decoder	108 attention-coordinate checks; eight background replay schedules; 52 geometric-moment inequalities
Positional rotations	108 reduction cases, including 34 period-boundary cases with a different computed period quotient
Corrected Bernstein factory	624 hypergeometric moment cases; 4,900 ratio checks; 3,960 coefficient bounds; 800 degree-elevation identities
Signed scalar bottleneck	6,381 checks on fifteen signed networks through depth sixteen, including the complex strip and Taylor value/derivative bounds
Two-source Bernstein factory	117 operator identities; 3,159 coefficient bounds; 957 hypergeometric fourth-moment identities
Three-source factory and batching	32 operator identities; 14,336 coefficient bounds; 6,561 seven-product matrix identities
Column-envelope correction	27 antithetic identities; 1,728 covariance-pair identities; 324 third-moment entries; 90 tilted laws over 2,232 signed endpoints
Gaussian RMS factory	543 error-function, sine, Gaussian, and normalization identities
Scalar normalization law	1,380 geometric-majority, curvature, and radius checks
Additive and affine FFN families	Recorded rational recurrence and high-precision floor, amplification, and bounded-readout checks
New carrier gain law	108 binary64 scenarios; 207,100 reciprocal-square comparisons

Each numerical check script writes a JSON report with its precision, case counts, and limitations. For example, the largest recorded Taylor identity residual in the signed-bottleneck checks was $2.76 \cdot 10^{-121}$ after rounding upward for reporting. The associated truncation errors were below the analytic bounds in every tested case. This is numerical evidence consistent with the proof.

The two-source and three-source scripts use exact rational and integer arithmetic throughout, without floating-point tolerances. Every recorded case passed. In both coefficient tests, the largest ratio of the tested coefficient magnitude to its proposed envelope was $1/2$. These are finite polynomial test cases; this ratio is neither a proved extremum over the function class nor a replacement for the uniform coefficient bounds in the proofs.

The dynamic-grid checks recorded no old-record migrations in four streams that grow in affine rank after a long constant prefix. The largest recorded accepted-mass identity residual was below $6.06 \cdot 10^{-123}$. The moment checks compared exact integer counters with direct Taylor sums and exponential attention; the largest identity residual was below $1.74 \cdot 10^{-118}$. All eight tested background schedules used at most two historical-token replays per append, including a prefill of one million tokens. The positional cases used positions of up to 201 bits and checked the phase-error bound even when the approximate period quotient differed from the true quotient. These checks implement selected invariants, not an exact sampler or a timing benchmark.

G.2 Lean 4 formalization

The directory `formalization` of the repository <https://github.com/SamMausberg/exact-sampling-networks> contains a Lean 4 development with Mathlib. It has more than 2,400 theorems. It builds with warnings treated as errors, and an audit of every declaration finds only the axioms `propext`, `Classical.choice` and `Quot.sound`. The documentation of each theorem names the paper result it formalizes by its label, and a generated map lists the theorems for each label. Running `python3 verify.py` repeats the build, the axiom audit, and a check that every cited label exists in this paper or in the conference version.

Lean directory	Paper results
Conference	The proofs of the conference version: one bounded tanh row, bounded charts, the appendable grid index, the finite moment evaluator, depth-uniform prefix stability, and exact sampling from a maintained evaluator.
Tanh, Stopping	Scalar factories, radius recursions and rates, majority and arity bounds, product arity, and the stopping lemmas (Lemma A.10 , Lemma B.14).
Lower, Instance	Lemma A.17 , Theorem A.2 , the finite joint and critical laws, and Appendix D .
Critical	Theorems A.61 , A.67 and A.69 , and Section A.16 .
Residual	Residual networks and fused factories, including Theorem A.39 .
Attention	Appendix B .
Normalization	Appendix C .
Checkpoint	Appendix E .
Links	Statements that combine results of several modules.

The formalization covers the probability laws, exactness identities, stopping estimates and quantitative inequalities of these proofs, with the paper’s constants. It does not cover the bit-operation cost model, data structures, the certified arithmetic routines, or the complexity-theoretic reductions and hypotheses. When a proof uses a fact from outside the paper, such as the cosh product of [\[35\]](#) or a coefficient bound from complex analysis, the Lean statement takes that fact as an explicit hypothesis, and its module lists it. The lower bounds are proved for samplers given as finite mixtures of decision trees that answer repeated queries from a cache. The reduction from an almost surely terminating algorithm with an infinite random tape to this form is not formalized.